

Gauge-Covariant Variational Free Energy and Renormalization

Two Associated Statistical Bundles, One Exact Bound, and a
General-to-Gaussian Scale Theory

Robert C. Dennis

Independent Researcher

Date: July 2026



Contents

I	General Gauge-Variational Foundations	1
1	Scope, Claims, and Reading Map	2
2	The Geometric Setting	5
2.1	The contextual base	5
2.2	One principal bundle and two associated statistical bundles	5
2.3	Frame choices, gauge symmetry, and relative frames	7
2.4	Cross-associated-bundle morphisms	7
2.5	Chosen connections, parallel transport, and covariant defects	9
2.6	Agents and finite-moment pushforward	10
2.7	Pointwise frame comparisons	11
2.8	The Čech picture and bundle topology	11
2.9	Channel-specific graph links and trivializing subfamilies	12
2.10	Flat graph links and residuals	14
2.11	Common-frame and common-connection specializations	14
2.12	Optional product-gauge extension	15
3	Probability Semantics	16
3.1	The finite design and its evaluation	16
3.2	Structural data and measurable typing	17
3.3	Reference measures	18
3.4	Kernels first, densities second	19
3.5	The absolute-continuity convention	21
3.6	Recognition marginals and what they fail to determine	23
3.7	Continuum sections versus finite evaluation	23
3.8	What the finite construction does not supply	25
4	The Fixed Normalized Generative Law	26
4.1	Parameters, structure, and kernel types	26
4.2	Iterated composition and densities	27
4.3	The typing prohibition	28
4.4	Exact normalization	29
4.5	The normalized undirected alternative	29
4.6	Covariance under independent channel-frame re-expression	30

5	The Exact Single Evidence Bound	32
5.1	The correlated population recognition kernel	32
5.2	Hypotheses	34
5.3	The exact evidence identity	35
5.4	The general restriction principle	37
5.5	The factorwise decomposition	38
5.6	Exact coordinates and accepted proposals	40
5.7	What this chapter does not settle	43
6	Belief Fibers and Exponential Families	45
6.1	Law fibers before coordinates	45
6.2	Dominated and smooth tiers	46
6.3	Regular finite-dimensional exponential families	46
6.4	Operator and probability layers	47
6.5	Declared repairs	48
6.6	Relative entropy and affine information projection	49
6.7	Boundary of this chapter	50
II	General Coarse Maps and Renormalization	51
7	General Coarse-Graining	52
7.1	Markov coarse maps	52
7.2	KL contraction, equality, and recovery	53
7.3	Fisher contraction is local	54
7.4	The fixed-evidence ELBO consequence	54
7.5	Pushforward, energy precomposition, and identification	55
7.6	Graph-exponential energy closure	55
7.7	Galerkin operators and message passing	57
7.8	Gauge and permutation equivariance	57
7.9	Holonomy-conditioned marginal-law modes	58
7.10	Projective systems of laws	61
8	General Renormalization	62
8.1	The scale category and comparison data	62
8.2	The common-principal geometric state	63
8.3	Autonomous maps and cocycles	66
8.4	Scheme dependence and invariant candidates	67
8.5	Two-index thermodynamic and RG limits	67
8.6	Projective laws and refinement	68
8.7	Typed comparison with Bayesian renormalization	69

III The Multivariate-Gaussian Realization	71
9 A Linear-Gaussian Generative Realization	72
9.1 Directed forest and conditional laws	72
9.2 Independent channel-frame covariance	74
9.3 Common-frame specialization	75
10 The Gaussian Information Form and the Interaction Family	77
10.1 The recognition family in information form	77
10.2 What an off-diagonal precision block is	78
10.3 The declared interaction subfamily	80
10.4 The interaction form is a hypothesis and not a consequence	81
10.4.1 Trivialized coordinates and flat comparison	81
10.4.2 What a directed linear-Gaussian model actually contributes	82
10.4.3 Exactly when the form does hold	84
10.5 Nonemptiness, and a note on conditioning	86
10.6 Marginalization belongs to the Gaussian coarse chapter	87
11 Gaussian Recognition Restrictions	88
11.1 The block restriction and its exact optimum	88
11.2 Observation independence, and its exact scope	91
11.3 Comparing recognition structures at fixed model	93
12 Gaussian Information Geometry	97
12.1 Canonical and expectation coordinates	97
12.2 The chart correction	98
12.3 Sample space and parameter space	101
12.4 Gauge invariants of the interaction precision	103
12.5 What is not claimed	105
13 Multivariate-Gaussian Coarse-Graining	107
13.1 Hard aggregation of the interaction precision	107
13.2 Kron reduction: a counterexample and the closed cone	108
13.3 Nonflat links and effective support	110
13.4 Cut links and rectangular sheaf closure	111
13.5 Partial fixed-section aggregation	112
13.5.1 Constant-metric Gaussian relaxation	114
13.5.2 Gaussian marginal-law specialization	115
13.6 Quotient volume for hard membership	118
13.7 Recognition-side costs	118
13.8 The frame sector	119
14 Multivariate-Gaussian Operator Renormalization	121

14.1 The MVG operator arrow	121
14.2 The biadditive closed family	122
14.3 Cone dynamics: what Birkhoff does and does not prove	123
14.4 An exact finite-sector gate	125
14.5 Gauge-invariant spectral data	125
14.6 Finite stochastic refinement on matrix cones	127
14.7 Relations to Bayesian, Laplacian, network, and learned RG	128
14.8 Remaining MVG obligations	129
IV Boundaries, Evidence, and Interpretation	130
15 Obstructions	131
15.1 The flat unanchored reciprocal Gaussian fold is singular	131
15.2 A proper anchor restores definiteness and makes the global normalizer essential	133
15.3 Bayesian posterior flow is a different construction	135
15.4 What survives	135
15.5 The participatory content survives in the inference	137
15.6 In the reciprocal-pair architecture, the kernel is a flatness artifact	139
16 Empirical Commitments and Interpretation	142
16.1 What the base is, and what it is not	142
16.2 When a noumenal reading earns its keep	143
16.3 Which realism this is	143
16.4 Two agreements, and the cost of conflating them	144
16.5 The participatory reading and its exact cost	144
16.6 What would make the interpretation testable	145
A Typed Notation Contract	146
B Central Open-Obligation Ledger	148
B.1 Mathematical and probabilistic closure	148
B.2 General coarse maps and renormalization	148
B.3 Multivariate-Gaussian realization	149
B.4 Geometric and physical interpretation	149
C Numerical Provenance	151

List of Tables

1.1	Status tags used throughout.	4
-----	--------------------------------------	---

Part I

General Gauge-Variational Foundations

Chapter 1

Scope, Claims, and Reading Map

This document develops a gauge-covariant variational theory from typed probability objects and then asks which parts survive coarse description. Its ambient geometry consists of one principal gauge bundle $P \rightarrow \mathcal{C}$ with structure group G , and two associated law bundles $\mathcal{E}_b = P \times_{\hat{\rho}_b} \mathcal{B}_b$ and $\mathcal{E}_m = P \times_{\hat{\rho}_m} \mathcal{B}_m$. The representations ρ_b and ρ_m may be inequivalent and may act on fibers of different dimensions; the hats denote their induced actions on probability laws. Belief and model coordinates may use different local sections of P . Their unique relative principal-frame field is a G -valued comparison, not a map between the two associated fibers. The declared cross-channel morphisms

$$\Phi : \mathcal{E}_b \longrightarrow \mathcal{E}_m, \quad \tilde{\Phi} : \mathcal{E}_m \longrightarrow \mathcal{E}_b$$

cover $\text{id}_{\mathcal{C}}$; they are neither principal-bundle morphisms nor assumed inverses. Separate connections induce same-channel parallel transports Ω_γ on $\mathcal{E}_b$ and $\tilde{\Omega}_\gamma$ on $\mathcal{E}_m$. Parallelness of either cross map is an additional hypothesis, measured otherwise by its covariant defect. The two channels may use different connections on the same P ; neither a common local frame nor an equal connection is assumed. Independently declared graph links are denoted Θ_e^b, Θ_e^m on oriented edge copies (abbreviated by endpoints when the copy is unique), and are identified with base parallel transport only after a curve assignment and an equality hypothesis. These distinctions are the type contract of Chapter 2. **[DEFINITION]**

On a finite design the theory fixes one normalized generative joint $P_\theta(do, dY \mid X)$ and one correlated recognition law $Q_X(dY \mid o)$. Under the stated absolute-continuity and integrability hypotheses, there is one exact identity,

$$\log p_\theta(o \mid X) = \mathcal{L}(Q_X; X) + D_{\text{KL}}(Q_X \| P_\theta(\cdot \mid o, X)). \quad (1.1)$$

It is invariant under independent belief- and model-coordinate pushforwards. These passive re-expressions form a product of frame-coordinate changes; they must not be confused with replacing the ambient diagonal G -gauge symmetry by a principal $G \times G$ -bundle. Equality holds exactly when the recognition law is the posterior as a measure; matching coordinate marginals is insufficient. Excluding the posterior makes every pointwise gap positive, but does not by itself make the optimized infimum positive or attained. **[ESTABLISHED]**

The logical order is therefore

typed bundles and kernels $\longrightarrow$ normalized joint $\longrightarrow$ exact ELBO
 $\longrightarrow$ general coarse maps $\longrightarrow$ typed RG $\longrightarrow$ Gaussian realization.

The general coarse layer distinguishes three operations that are often conflated. A normalized Markov kernel pushes laws forward, preserves the observation evidence when it acts only on latent variables, and contracts relative entropy. Energy precomposition restricts an unnormalized energy along a sample-space map and requires a separately declared coarse reference measure and finite normalizer. Parameter aggregation is a conditional closure statement inside a selected graph-exponential family. A renormalization further requires comparison or rescaling maps that type different scales; without them there is only a changing-space coarse diagram, not a fixed point problem. [ESTABLISHED]

The multivariate Gaussian enters only after that general structure. It realizes the smooth statistical tier in information coordinates and supplies explicit determinant gaps, Fisher charts, matrix-weighted interaction operators, aggregation formulas, Kron counterexamples, and generalized spectra. The Laplacian-plus-self-term interaction cone is a declared subfamily, not a consequence of every linear-Gaussian directed model. [HYPOTHESIS] Within its exact domain, aggregation is closed, but closure alone does not select that cone or a nondegenerate partition. [ESTABLISHED] Unrestricted matrix-weighted Kron closure fails, whereas the later Gaussian coarse chapter records the exact closed congruence-diagonal domain and its qualified maximality boundary. [ESTABLISHED]

The RG conclusions are deliberately narrower. Rescaling is declared rather than derived. [DEFINITION] Bi-additive Gaussian parameters are closed under additive pushforward, and an identified homogeneous hierarchy gives an operator fixed ray, but the ray lies on the boundary of the full-space Gaussian probability domain. [ESTABLISHED] Infinite-hierarchy attraction, uniqueness within a basin, and blocking-scheme independence remain open. [OPEN] Finite numerical gates test specified diagnostics and controls; they do not prove those asymptotic claims. [NUMERICAL] Identifying any repaired limiting object with a physical law remains the semantic claim B.1; it requires a complete probability repair, a target system, a validation criterion, and discriminating evidence. [OPEN]

Epistemic contract

Every nontrivial statement carries one of the following tags. The central open-obligation appendix collects unresolved conjectures and problems once; established claims remain with their hypotheses or proofs, and numerical claims remain bound to the numerical-provenance appendix.

A gesture is either discharged by proof, calculation, or a checked source, or converted into an open obligation with exact hypotheses. Likewise, [NOT-CLAIMED] is never used for a refuted statement. The obstruction chapter contains exact negative results; the interpretation chapter marks readings and empirical commitments without promoting them to theorems.

Table 1.1: Status tags used throughout.

Tag	What it promises the reader
[ESTABLISHED]	Proved here, or a standard result cited to a source that has been checked.
[DEFINITION]	A declared type, construction, or convention. Nothing is being proved and the text says so.
[HYPOTHESIS]	A restriction the development adopts by choice. What it excludes, and where it is used, are stated.
[CONJECTURE]	Believed and precisely stated, but not proved. Stated sharply enough to be attacked.
[NUMERICAL]	Supported by computation only, with its measurement, seed, and control reported. Computation is not proof.
[OPEN]	Unsettled. What would settle it, and what obstructs it, are named.
[NOT-CLAIMED]	A statement the development deliberately declines to make. Declining is not refuting, and the text says which it is doing.

A physicist's fast path

A first reading may take the object and frame-covariance contract from Chapters 1 and 2, then read only the normalized-joint and ELBO identities in Chapters 3 to 5. The next pass should follow the three-operation distinction, Markov coarse theorem, graph-exponential closure, and typed scale diagram in the general coarse and RG chapters. The Gaussian spine then consists of the interaction cone, recognition costs and Fisher geometry in Chapters 10 to 12, followed by Gaussian aggregation, congruence-diagonal Kron closure, holonomy, fixed-ray qualifications, generalized spectra, and numerical gates in Chapters 13 and 14. Finally, Chapters 15 and 16 separate exact no-go results and repairs from empirical and interpretive commitments. Disintegration proofs, convex-duality details, long rational witnesses, ledger rows, and protocol metadata can be recovered through their semantic cross-references on a second reading.

Chapter 2

The Geometric Setting

The ambient geometry has one principal gauge bundle over the contextual base and two associated statistical bundles, one for belief laws and one for model laws. The two associated actions of the same group may be inequivalent and may act on fibers of different dimensions. Belief and model frames may nevertheless be chosen separately as local sections of the common principal bundle, and the two channels may carry different connections. Cross-channel maps are additional associated-bundle morphisms; neither a relative frame nor a shared structure group supplies such a map by itself. A product-gauge theory with two independent principal bundles is recorded only as the optional extension in Section 2.12.

2.1 The contextual base

Definition 2.1 (Contextual base). Let $\mathcal{C}$ be a finite-dimensional smooth, second countable, Hausdorff manifold whose points c are called *contexts*. [DEFINITION]

This declaration fixes a type and proves nothing. The base is neither the population index set $V = \{1, \dots, N\}$ nor its interaction graph, neither latent sample space, and not physical spacetime. Its smooth structure supports curves and differential forms. A connection on a bundle over $\mathcal{C}$ defines transport in that bundle along curves in $\mathcal{C}$; it does not, unless separately specified as a connection on $T\mathcal{C}$, define parallel transport of tangent vectors of the base. The finite design $D = \{c_a\}_{a=1}^M$ used later is a finite subset of $\mathcal{C}$, not a continuum limit or a discretization theorem.

2.2 One principal bundle and two associated statistical bundles

Definition 2.2 (Common principal gauge bundle). Let

$$\pi : P \longrightarrow \mathcal{C} \tag{2.1}$$

be a smooth principal right G -bundle for a Lie group G . For each agent i , choose an open support $\mathcal{C}_i \subseteq \mathcal{C}$ and two local principal-frame sections

$$u_i^b : \mathcal{C}_i \longrightarrow P, \quad u_i^m : \mathcal{C}_i \longrightarrow P. \tag{2.2}$$

[DEFINITION]

The superscripts distinguish two choices of local section, not two principal bundles. They may be unequal. Because each principal fiber is a right G -torsor, there is a unique smooth map

$$h_i : \mathcal{C}_i \longrightarrow G, \quad u_i^m = u_i^b \cdot h_i, \quad (2.3)$$

called the *relative principal-frame field*. Its existence uses only the common principal bundle and is independent of all associated representations. **[ESTABLISHED]**

Let $(K, \mathcal{K})$ and $(M, \mathcal{M})$ be the belief and model sample fibers in local coordinates, and declare normalized-law fibers

$$\mathcal{B}_b \subseteq \mathcal{P}(K), \quad \mathcal{B}_m \subseteq \mathcal{P}(M). \quad (2.4)$$

A law fiber is not a kernel fiber. A generative transition belongs to a separately declared collection of normalized Markov kernels between specified measurable spaces. No density chart, common dominating measure, affine structure, or Fisher metric follows from (2.4).

Suppose the same group acts measurably on the two sample fibers through representations

$$\rho_b : G \longrightarrow \text{Aut}(K), \quad \rho_m : G \longrightarrow \text{Aut}(M). \quad (2.5)$$

The representations need not be equivalent, faithful, or of equal dimension. Denote their induced actions on laws by

$$\hat{\rho}_b(g)q = (\rho_b(g))_{\#}q, \quad \hat{\rho}_m(g)s = (\rho_m(g))_{\#}s. \quad (2.6)$$

Thus $(\hat{\rho}_b(g)q)(A) = q(\rho_b(g)^{-1}A)$, and similarly in the model channel. The hats distinguish actions on laws from actions on samples. Multiplying a matrix into a density is not the pushforward in (2.6).

Hypothesis 2.3 (Selected smooth tier). When differential statements are made, $\mathcal{B}_b$ and $\mathcal{B}_m$ are finite-dimensional smooth parametrized-measure models, each is closed under its G -action, and

$$(g, q) \longmapsto (\rho_b(g))_{\#}q, \quad (g, s) \longmapsto (\rho_m(g))_{\#}s \quad (2.7)$$

are smooth. **[HYPOTHESIS]**

A finite-dimensional statistical interpretation additionally requires differentiability in quadratic mean, square-integrable scores, and the integrability required by each tensor. The Fisher tensor is a Riemannian metric only when it is nondegenerate. Stratified and infinite-dimensional fibers require their own categories and are not included silently.

Definition 2.4 (Belief and model associated bundles). On invariant law fibers define

$$\mathcal{E}_b = P \times_{\hat{\rho}_b} \mathcal{B}_b, \quad \mathcal{E}_m = P \times_{\hat{\rho}_m} \mathcal{B}_m, \quad (2.8)$$

using the quotient convention

$$(u \cdot g, \beta) \sim (u, g \cdot \beta). \quad (2.9)$$

[DEFINITION]

At the general tier these are associated measurable bundles. Under Hypothesis 2.3 they are smooth associated bundles. Smoothness of P alone does not upgrade an arbitrary law fiber.

The field h_i compares the two principal sections, even when ρ_b and ρ_m are inequivalent. It then produces two different represented operators, $\widehat{\rho}_b(h_i)$ on $\mathcal{B}_b$ and $\widehat{\rho}_m(h_i)$ on $\mathcal{B}_m$. It does not canonically produce an operator from $\mathcal{B}_b$ to $\mathcal{B}_m$. If “frame” instead means an arbitrary basis of an associated vector fiber that is not induced from a section of this same P , the existence of a G -valued relative field is not guaranteed. Moreover, when a representation is nonfaithful, represented frame data alone determine at most a coset modulo its kernel; the principal sections, rather than their represented images, are what make h_i unique.

2.3 Frame choices, gauge symmetry, and relative frames

Independent changes of the two local trivializations may be written

$$u_i^{b'} = u_i^b \cdot a_i, \quad u_i^{m'} = u_i^m \cdot b_i, \quad a_i, b_i : \mathcal{C}_i \rightarrow G. \quad (2.10)$$

They change associated coordinates by $\beta'_b = \widehat{\rho}_b(a_i)^{-1} \beta_b$ and $\beta'_m = \widehat{\rho}_m(b_i)^{-1} \beta_m$. The unique relative field becomes

$$h'_i = a_i^{-1} h_i b_i. \quad (2.11)$$

[ESTABLISHED]

These two passive coordinate choices must not be confused with a product gauge symmetry. The principal gauge group of the ambient theory is the single group of G -equivariant automorphisms of P over $\text{id}_{\mathcal{C}}$, and each such automorphism acts simultaneously on both associated bundles. For such an automorphism F , define its local functions by $F(u_i^b) = u_i^b \cdot k_i^b$ and $F(u_i^m) = u_i^m \cdot k_i^m$. Equivariance and (2.3) imply

$$k_i^m = h_i^{-1} k_i^b h_i. \quad (2.12)$$

Thus the two local descriptions of one gauge transformation are related, not independent. By contrast, (2.10) simply permits two atlases to be replaced independently. Applying one principal gauge automorphism to both sections leaves their relative principal element unchanged, since equivariance gives $F(u_i^m) = F(u_i^b \cdot h_i) = F(u_i^b) \cdot h_i$.

2.4 Cross-associated-bundle morphisms

Definition 2.5 (Cross morphisms). At the general measurable tier, declare fiber-preserving measurable maps

$$\Phi : \mathcal{E}_b \longrightarrow \mathcal{E}_m, \quad \widetilde{\Phi} : \mathcal{E}_m \longrightarrow \mathcal{E}_b, \quad \varpi_m \circ \Phi = \varpi_b, \quad \varpi_b \circ \widetilde{\Phi} = \varpi_m, \quad (2.13)$$

where ϖ_b and ϖ_m are the associated-bundle projections. Both maps cover id_C . **[DEFINITION]**

These maps are extra structure. They are not principal-bundle morphisms, gauge automorphisms, same-channel transports, or assumed inverses. Under Hypothesis 2.3, differential statements additionally assume that they are smooth. In the frames u_i^b, u_i^m , let ϕ_i and $\tilde{\phi}_i$ denote their local fiber representatives. The rechoices in (2.10) give

$$\phi'_i = \hat{\rho}_m(b_i)^{-1} \circ \phi_i \circ \hat{\rho}_b(a_i), \quad (2.14)$$

$$\tilde{\phi}'_i = \hat{\rho}_b(a_i)^{-1} \circ \tilde{\phi}_i \circ \hat{\rho}_m(b_i). \quad (2.15)$$

[ESTABLISHED]

Proof. Write $\beta_m = \phi_i(\beta_b)$ and substitute $\beta_b = \hat{\rho}_b(a_i)\beta'_b$ and $\beta'_m = \hat{\rho}_m(b_i)^{-1}\beta_m$. The second law follows by interchanging the channels. $\square$

Proposition 2.6 (Equivariant fiber maps induce cross morphisms). Suppose maps $\phi : \mathcal{B}_b \rightarrow \mathcal{B}_m$ and $\tilde{\phi} : \mathcal{B}_m \rightarrow \mathcal{B}_b$ satisfy, for every $g \in G$,

$$\phi \circ \hat{\rho}_b(g) = \hat{\rho}_m(g) \circ \phi, \quad \tilde{\phi} \circ \hat{\rho}_m(g) = \hat{\rho}_b(g) \circ \tilde{\phi}. \quad (2.16)$$

Then

$$[p, \beta_b] \mapsto [p, \phi(\beta_b)], \quad [p, \beta_m] \mapsto [p, \tilde{\phi}(\beta_m)] \quad (2.17)$$

are well-defined cross morphisms. **[ESTABLISHED]**

Proof. The quotient representatives $(p \cdot g, \beta)$ and $(p, g \cdot \beta)$ have the same image under (2.17) exactly when the corresponding equality in (2.16) holds. $\square$

Equivariance is sufficient for the representative-independent construction (2.17). A general bundle morphism is instead characterized by local maps satisfying the overlap laws

$$\begin{aligned} \phi_i \circ \hat{\rho}_b(T_{ij}^b) &= \hat{\rho}_m(T_{ij}^m) \circ \phi_j, \\ \tilde{\phi}_i \circ \hat{\rho}_m(T_{ij}^m) &= \hat{\rho}_b(T_{ij}^b) \circ \tilde{\phi}_j. \end{aligned} \quad (2.18)$$

When the selected regularity category equips the relevant map space with a suitable structure, this compatible family is equivalently a section of the associated map bundle with action $f \mapsto \hat{\rho}_m(g) \circ f \circ \hat{\rho}_b(g)^{-1}$. In neither description does h_i replace the intertwiner. For example, if Φ is induced by the first map in (2.17), its representative from the u_i^b coordinates to the u_i^m coordinates is $\hat{\rho}_m(h_i^{-1}) \circ \phi$. The factor involving h_i only converts model coordinates after ϕ has supplied the cross-fiber map.

2.5 Chosen connections, parallel transport, and covariant defects

Definition 2.7 (Two connections on the common principal bundle). Under Hypothesis 2.3, choose

$$\omega_b, \omega_m \in \Omega^1(P, \mathfrak{g}) \quad (2.19)$$

as principal connections on P . They induce horizontal distributions H^b, H^m on the two associated bundles and, for every piecewise-smooth $\gamma : [0, 1] \rightarrow \mathcal{C}$, parallel transports

$$\Omega_\gamma : (\mathcal{E}_b)_{\gamma(0)} \longrightarrow (\mathcal{E}_b)_{\gamma(1)}, \quad \tilde{\Omega}_\gamma : (\mathcal{E}_m)_{\gamma(0)} \longrightarrow (\mathcal{E}_m)_{\gamma(1)}. \quad (2.20)$$

[DEFINITION]

Explicitly, ω_x selects the unique horizontal lift of γ through a chosen point of $P_{\gamma(0)}$; carrying the associated-law coordinate along that lift defines the corresponding map in (2.20). This is transport *over* the base curve, not transport of vectors in TC . A tangent-vector transport would require a separately chosen affine connection on TC , unless P and its representation have explicitly been identified with the frame bundle of $\mathcal{C}$.

The two connections need not agree. Their difference $\omega_m - \omega_b$ is horizontal and Ad -equivariant, hence descends to an $\text{Ad}(P)$ -valued one-form on $\mathcal{C}$. In an associated vector-bundle realization, the induced horizontal distributions also define covariant derivatives ∇^b, ∇^m . No linear covariant derivative is assumed on a merely measurable law fiber.

Let $A_i^b = (u_i^b)^* \omega_b$ and $A_i^m = (u_i^m)^* \omega_m$. Under (2.10),

$$A_i^{b'} = \text{Ad}_{a_i^{-1}} A_i^b + a_i^{-1} da_i, \quad (2.21)$$

$$A_i^{m'} = \text{Ad}_{b_i^{-1}} A_i^m + b_i^{-1} db_i. \quad (2.22)$$

If endpoint coordinates are changed by a_0, a_1 in the belief frame and b_0, b_1 in the model frame, the represented transports obey

$$\Omega'_\gamma = \hat{\rho}_b(a_1)^{-1} \circ \Omega_\gamma \circ \hat{\rho}_b(a_0), \quad (2.23)$$

$$\tilde{\Omega}'_\gamma = \hat{\rho}_m(b_1)^{-1} \circ \tilde{\Omega}_\gamma \circ \hat{\rho}_m(b_0). \quad (2.24)$$

[ESTABLISHED]

These are the standard connection and endpoint laws for the quotient convention (2.9) [39, 50]. They also show why a same-point frame comparison and transport along a nonconstant curve are different objects.

Definition 2.8 (Covariant defects). Let $e \in \mathcal{E}_b$ and $f \in \mathcal{E}_m$ lie over $c \in \mathcal{C}$, and let $X \in T_c \mathcal{C}$. For smooth associated fibers, define the vertical-valued tensors

$$(\mathcal{D}\Phi)_e(X) = T_e \Phi(H_e^b X) - H_{\Phi(e)}^m X, \quad (2.25)$$

$$(\mathcal{D}\tilde{\Phi})_f(X) = T_f \tilde{\Phi}(H_f^m X) - H_{\tilde{\Phi}(f)}^b X. \quad (2.26)$$

[DEFINITION]

Each difference is vertical because its two terms project to the same X . The maps are parallel exactly when these defects vanish, equivalently when every base curve satisfies

$$\tilde{\Omega}_\gamma \circ \Phi_{\gamma(0)} = \Phi_{\gamma(1)} \circ \Omega_\gamma, \quad (2.27)$$

$$\Omega_\gamma \circ \tilde{\Phi}_{\gamma(0)} = \tilde{\Phi}_{\gamma(1)} \circ \tilde{\Omega}_\gamma. \quad (2.28)$$

[ESTABLISHED]

Proof. Vanishing defect says that the image of every horizontal lift is horizontal. Integration along γ gives the two commutative squares, and differentiation at the initial point gives the converse. $\square$

Only in a fiberwise-linear associated-vector-bundle specialization do the defects reduce to

$$D\Phi = \nabla^m \circ \Phi - \Phi \circ \nabla^b, \quad D\tilde{\Phi} = \nabla^b \circ \tilde{\Phi} - \tilde{\Phi} \circ \nabla^m. \quad (2.29)$$

If R_b, R_m denote the represented coordinate changes, their matrices obey

$$(D\Phi)' = R_m(D\Phi)R_b^{-1}, \quad (D\tilde{\Phi})' = R_b(D\tilde{\Phi})R_m^{-1}. \quad (2.30)$$

[ESTABLISHED] The matrix laws are not asserted for nonlinear law manifolds. The connections in (2.19) are chosen data; no curvature or transport is inferred from the agent frames.

2.6 Agents and finite-moment pushforward

Definition 2.9 (Agent). Agent i is the section-bearing object

$$\mathcal{A}^i = (\mathcal{C}_i; q_i, s_i, u_i^b, u_i^m), \quad (2.31)$$

where

$$q_i \in \Gamma(\mathcal{C}_i, \mathcal{E}_b|_{\mathcal{C}_i}), \quad s_i \in \Gamma(\mathcal{C}_i, \mathcal{E}_m|_{\mathcal{C}_i}). \quad (2.32)$$

[DEFINITION]

The sections are smooth only in the selected smooth tier. A sample

$$Y(c) = ((k_1(c), m_1(c)), \dots, (k_N(c), m_N(c))) \quad (2.33)$$

is a random element of a product of sample fibers, not a tuple of agents. The continuum sections above and the finite recognition law introduced later are separately declared objects; their compatibility is a hypothesis in the probability chapter.

Proposition 2.10 (Finite-second-moment congruence). Let a belief law q on $\mathbb{R}^K$ have finite mean μ and covariance Σ , and let $R = \rho_b(g)$ be invertible and linear for $g \in G$. Then

$$R_{\#}q \text{ has mean } R\mu \text{ and covariance } R\Sigma R^\top. \quad (2.34)$$

The model channel has the same statement in its own dimension. **[ESTABLISHED]**

Proof. For $X \sim q$, linearity gives $\mathbb{E}[RX] = R\mu$ and $\mathbb{E}[(RX - R\mu)(RX - R\mu)^\top] = R\Sigma R^\top$. $\square$

2.7 Pointwise frame comparisons

Fix $c \in \mathcal{C}_i \cap \mathcal{C}_j$. Freeness and transitivity give unique elements of the same group G such that

$$u_j^b(c) = u_i^b(c) \cdot T_{ij}^b(c), \quad u_j^m(c) = u_i^m(c) \cdot T_{ij}^m(c). \quad (2.35)$$

They act through $\rho_b(T_{ij}^b)$ and $\rho_m(T_{ij}^m)$ in the two associated fibers. Under (2.10),

$$T_{ij}^{b'} = a_i^{-1} T_{ij}^b a_j, \quad T_{ij}^{m'} = b_i^{-1} T_{ij}^m b_j. \quad (2.36)$$

Moreover, the relative fields force

$$T_{ij}^m = h_i^{-1} T_{ij}^b h_j. \quad (2.37)$$

[ESTABLISHED]

Proof. The transformation law follows by substitution and uniqueness in the principal fiber. The two expressions $u_j^m = u_j^b h_j = u_i^b T_{ij}^b h_j$ and $u_j^m = u_i^m T_{ij}^m = u_i^b h_i T_{ij}^m$ give (2.37). $\square$

The comparisons are same-point changes of coordinates. They involve no curve and no connection. The two group elements are related by (2.37), while their represented actions can be inequivalent.

2.8 The Čech picture and bundle topology

Let $\mathcal{C}_0 = \bigcup_i \mathcal{C}_i$, and let $\mathcal{N}$ be the nerve of this open cover:

$$\{i_0, \dots, i_r\} \in \mathcal{N} \iff \mathcal{C}_{i_0 \dots i_r} = \bigcap_{a=0}^r \mathcal{C}_{i_a} \neq \emptyset. \quad (2.38)$$

For either frame atlas $x \in \{b, m\}$, the maps $T_{ij}^x : \mathcal{C}_{ij} \rightarrow G$ obey

$$T_{ii}^x = e_G, \quad T_{ji}^x = (T_{ij}^x)^{-1}, \quad T_{ij}^x T_{jl}^x = T_{il}^x. \quad (2.39)$$

[ESTABLISHED]

Proof. On a triple overlap, composing the unique change from l to j with the unique change from j to i gives the unique change from l to i . $\square$

Let $\mathcal{G}$ be the sheaf of smooth G -valued maps. There is one principal bundle and hence one nonabelian Čech class:

$$[T^b] = [T^m] = [P|_{\mathcal{C}_0}] \in \check{H}^1(\{\mathcal{C}_i\}; \mathcal{G}). \quad (2.40)$$

Indeed, (2.37) is exactly the coboundary equivalence between the two cocycles. The class is distinguished exactly when $P|_{\mathcal{C}_0}$ is trivial, equivalently when, for either atlas, there are smooth $U_i^x : \mathcal{C}_i \rightarrow G$ such that

$$T_{ij}^x = U_i^x (U_j^x)^{-1}. \quad (2.41)$$

[ESTABLISHED]

Proof. If (2.41) holds, the rechoice $u_i^{x'} = u_i^x U_i^x$ makes all transitions identity, so the sections glue to a global section. A global section trivializes a principal bundle. Conversely, a global trivialization supplies the U_i^x . This is the standard global-section characterization of a trivial principal bundle [34, 39, 61]. $\square$

The single principal class does not make the two associated bundles isomorphic. Inequivalent representations can send the same principal cocycle to nonisomorphic associated bundles and to different induced characteristic data.

Hypothesis 2.11 (Common principal trivialization). Whenever global principal-frame coordinates are used, declare $P|_{\mathcal{C}_0}$ trivial and choose one reference section σ_0 such that

$$u_i^b = \sigma_0 \cdot (U_i^b)^{-1}, \quad u_i^m = \sigma_0 \cdot (U_i^m)^{-1}, \quad h_i = U_i^b (U_i^m)^{-1}. \quad (2.42)$$

[HYPOTHESIS]

This is one topological hypothesis, not two channelwise hypotheses. The general theory retains local frames and their common Čech class. A global section of an associated bundle does not in general trivialize P : if an associated fiber contains a G -fixed point β_0 , then $c \mapsto [p, \beta_0]$ is well defined for any P . **[ESTABLISHED]**

2.9 Channel-specific graph links and trivializing subfamilies

Definition 2.12 (Typed graph links). Let $\mathcal{J} = (V, E, \bar{\cdot})$ be a finite interaction multigraph declared independently of $\mathcal{C}$. Its oriented edge copies have a fixed-point-free reversal involution, and parallel copies are allowed. For $e : j \rightarrow i$, declare

$$\Theta_e^b, \Theta_e^m \in G, \quad \Theta_e^x = (\Theta_e^x)^{-1}. \quad (2.43)$$

Under independently changed channel coordinates at the vertices,

$$\Theta_e^{b'} = (a_i^b)^{-1} \Theta_e^b a_j^b, \quad \Theta_e^{m'} = (a_i^m)^{-1} \Theta_e^m a_j^m. \quad (2.44)$$

Here the $a_i^x \in G$ are passive vertex-coordinate choices. They are induced by smooth frame rechoices only after vertices are anchored at contexts $c_i \in \mathcal{C}_i$, in which case $a_i^b = a_i(c_i)$ and $a_i^m = b_i(c_i)$. Allowing two coordinate descriptions of two graph-link data sets does not enlarge the principal gauge group to $G \times G$. [DEFINITION]

The links are channel-specific data. Neither is determined by the other, by h_i , by Φ or $\tilde{\Phi}$, by pointwise frame comparison, or by a principal connection. For a closed edge-labeled walk $\gamma = (e_0, \dots, e_{r-1})$, where $e_a : i_{a+1} \rightarrow i_a$ and $i_r = i_0$, define

$$H^x(\gamma) = \prod_{a=0}^{r-1} \Theta_{e_a}^x. \quad (2.45)$$

Its conjugacy class and the condition $H^x(\gamma) = e_G$ are invariant under the corresponding coordinate changes. [ESTABLISHED]

Proposition 2.13 (Trivializing criterion). Fix a channel x , a subfamily $F \subseteq V$, and a G -valued comparison assignment Ξ_e^x on its internal oriented edge copies. The following are equivalent:

$$\begin{aligned} \exists V_i^x \in G \ (i \in F) \quad \Xi_e^x &= V_i^x (V_j^x)^{-1} \quad \text{for every internal } e : j \rightarrow i, \\ \prod_{a=0}^{r-1} \Xi_{e_a}^x &= e_G \quad \text{for every closed internal walk.} \end{aligned} \quad (2.46)$$

[ESTABLISHED]

Proof. The first condition telescopes around every closed walk. Conversely, choose a base vertex in each connected component and define V_i^x by an ordered path product. Two choices differ by a closed walk and hence agree. Appending $e : j \rightarrow i$ gives the required edge factorization. $\square$

A subfamily is *jointly trivializing* when the criterion holds separately for both link data sets. For T^x evaluated at one common context the criterion is automatic by the Čech cocycle law. Its nontrivial graph content concerns the independently declared Θ^x . Pointwise graph triviality is weaker than triviality of P , which requires smooth overlap data on the entire cover.

Hypothesis 2.14 (Graph links realized by base transport). Choose contexts $c_i \in \mathcal{C}_i$ and, for every $e : j \rightarrow i$, a piecewise-smooth curve γ_e from c_j to c_i . In the chosen endpoint frames impose

$$\hat{\rho}_b(\Theta_e^b) = \Omega_{\gamma_e}, \quad \hat{\rho}_m(\Theta_e^m) = \tilde{\Omega}_{\gamma_e}. \quad (2.47)$$

[HYPOTHESIS]

Without the curve assignment and these equalities, graph holonomy and base-connection holonomy are unrelated. Under the hypothesis, represented graph-loop products equal the corresponding parallel transports around concatenated base loops. This is a connection-holonomy statement, not a claim of nontrivial principal-bundle topology.

2.10 Flat graph links and residuals

Hypothesis 2.15 (Pointwise flat-link specialization). At one common context, declare separately for the two link data sets that

$$\Theta_e^b = T_{ij}^b = U_i^b (U_j^b)^{-1}, \quad \Theta_e^m = T_{ij}^m = U_i^m (U_j^m)^{-1}, \quad e : j \rightarrow i. \quad (2.48)$$

[HYPOTHESIS]

This excludes represented graph holonomy in either channel. It does not assert that either principal connection is flat or that the two channel frames coincide. Let μ_i be a belief-channel vector coordinate, set $z_i = \rho_b(U_i^b)^{-1} \mu_i$, and suppress ρ_b only in the defining representation. Then

$$\mu_i - \rho_b(\Theta_e^b) \mu_j = \rho_b(U_i^b) (z_i - z_j). \quad (2.49)$$

For an independent graph link,

$$\mu_i - \rho_b(\Theta_e^b) \mu_j = \rho_b(U_i^b) (z_i - H_e^b z_j), \quad H_e^b = \rho_b(U_i^b)^{-1} \rho_b(\Theta_e^b) \rho_b(U_j^b). \quad (2.50)$$

More generally, if $\Theta_e^b = V_i^b (V_j^b)^{-1}$ on a belief-trivializing subfamily and $z_i = \rho_b(V_i^b)^{-1} \mu_i$, then

$$\mu_i - \rho_b(\Theta_e^b) \mu_j = \rho_b(V_i^b) (z_i - z_j). \quad (2.51)$$

[ESTABLISHED]

These identities follow by substitution. Belief-link trivialization says nothing about model links, either cross morphism, or equality of the belief and model frames.

2.11 Common-frame and common-connection specializations

The one-principal-bundle construction is already the ambient theory. A *common-frame* specialization makes the stronger declaration

$$u_i^b = u_i^m = u_i, \quad h_i = e_G, \quad T_{ij}^b = T_{ij}^m = T_{ij}. \quad (2.52)$$

Even then the represented frame changes generally differ:

$$R_i^b = \rho_b(g_i), \quad R_i^m = \rho_m(g_i). \quad (2.53)$$

An independent *equal-connection* specialization declares $\omega_b = \omega_m = \omega$. The underlying principal holonomy is then common, while its actions on belief and model laws remain $\hat{\rho}_b$ and $\hat{\rho}_m$. A further common-link specialization declares $\Theta_e^b = \Theta_e^m = \Theta_e \in G$. None of these three stronger hypotheses follows from the others. **[HYPOTHESIS]**

The cross maps remain extra equivariant structure governed by (2.16); they are not forced to be inverses or parallel. Thus inequivalent representations do not obstruct a common principal frame, but they may obstruct the existence of a nontrivial linear intertwiner.

2.12 Optional product-gauge extension

If the application genuinely requires independently nonisomorphic principal bundles or two independent gauge symmetries, replace the ambient P by

$$\pi_b : P_b \rightarrow \mathcal{C}, \quad \pi_m : P_m \rightarrow \mathcal{C}, \quad Q = P_b \times_{\mathcal{C}} P_m, \quad (2.54)$$

where Q is a principal $G_b \times G_m$ -bundle. The belief and model associated bundles are then formed using the two projections. There is no G -valued relative field between frames in P_b and P_m unless a suitable principal-bundle identification or bridge is supplied. In a category with the required mapping object, a belief-to-model morphism is a section of

$$Q \times_{G_b \times G_m} \text{Map}(\mathcal{B}_b, \mathcal{B}_m), \quad (g_b, g_m) \cdot f = \hat{\rho}_m(g_m) \circ f \circ \hat{\rho}_b(g_b)^{-1}. \quad (2.55)$$

A constant fiber map descends only if it is fixed by every independent pair. A compatible principal bridge and an intertwining fiber map are sufficient, but a general mapping-bundle section need not arise that way. Thus the product bundle does not supply either cross morphism automatically.

[DEFINITION]

This extension carries additional topology and an independent product gauge freedom. It should therefore be invoked only when those degrees of freedom are intended, not merely because belief and model use different associated representations of one group.

Chapter 3

Probability Semantics

Chapter 2 fixed a geometry. This chapter fixes the measurable typing on which every probabilistic statement rests, and it fixes it before any density is written. The order is a substantive commitment: probability measures and normalized kernels are declared first, densities appear afterward as Radon-Nikodym derivatives against explicitly declared reference measures, and logarithms of density ratios appear only under an absolute-continuity convention stated in advance. That convention looks like bookkeeping and is not. It is what makes the exact evidence bound of Chapter 5 an identity rather than an inequality with hidden exceptions, and it is what later forbids degenerate recognition laws whose support is a null set of the posterior. A development that introduces densities first and normalizes later can write those degenerate objects down without noticing, and the resulting statements are false rather than merely imprecise.

Throughout, $\mathcal{P}(S, \mathcal{S})$ denotes the set of probability measures on a measurable space $(S, \mathcal{S})$; the calligraphic symbol with a measurable-space argument is unrelated to the total space P of the principal bundle in Chapter 2.

3.1 The finite design and its evaluation

Definition 3.1 (Finite design and incidence). A *design* is a finite set of distinct contexts

$$D = \{c_a\}_{a=1}^M \subseteq \mathcal{C}, \quad (3.1)$$

indexed by $a \in \{1, \dots, M\}$. The *incidence set* is $\mathfrak{A} = \{(i, a) : c_a \in \mathcal{C}_i\}$, where $\mathcal{C}_i$ is the support of agent i from Definition 2.2. For the reference presentation $D \subseteq \bigcap_{i \in V} \mathcal{C}_i$, so $\mathfrak{A} = V \times \{1, \dots, M\}$ is rectangular and every listed agent is defined at every design point. **[DEFINITION]**

Nothing is proved by this. The sparse-support case replaces the rectangular index set by $\mathfrak{A}$ throughout, and it is stated this way to make one point explicit: a sparse design is handled by indexing over declared incidences, never by writing a product over all (i, a) and then deleting the coordinates that happen to be undefined. Deleting coordinates from a product silently changes the sample space, and a bound computed on one sample space is not a bound on another.

Definition 3.2 (Sample spaces and product sigma-algebras). At each admissible (i, a) declare measurable state, model, and observation fibers $(K_{i,a}, \mathcal{K}_{i,a})$, $(M_{i,a}, \mathcal{M}_{i,a})$, and $(O_{i,a}, \mathcal{O}_{i,a})$. The finite

latent and observation spaces are

$$Y_D = \prod_{a=1}^M \prod_{i \in V} (K_{i,a} \times M_{i,a}), \quad \mathcal{Y}_D = \bigotimes_{a=1}^M \bigotimes_{i \in V} (\mathcal{K}_{i,a} \otimes \mathcal{M}_{i,a}), \quad (3.2)$$

$$O_D = \prod_{a=1}^M \prod_{i \in V} O_{i,a}, \quad \mathcal{O}_D = \bigotimes_{a=1}^M \bigotimes_{i \in V} \mathcal{O}_{i,a}. \quad (3.3)$$

The stacked latent is written $Y \in Y_D$ and the realized observation $o \in O_D$. In the Gaussian realization $K_{i,a} = \mathbb{R}^K$ and $M_{i,a} = \mathbb{R}^{d_m}$ with their Borel sigma-algebras. **[DEFINITION]**

Evaluation and section evaluation have different codomains, and the difference is the one Chapter 2 insisted on. For a sample-field realization $y = ((k_i, m_i))_{i \in V}$ defined on D , the evaluation map

$$\text{Ev}_D(y) = ((k_i(c_a), m_i(c_a))_{i \in V})_{a=1}^M \in Y_D \quad (3.4)$$

returns a point of the latent space. Evaluating a distribution-valued section returns something else entirely: $q_i^{o,X}(c_a)$ is a probability measure on $K_{i,a}$ and $s_i^{o,X}(c_a)$ is a probability measure on $M_{i,a}$, and neither is an element of Y_D . The finite random element Y takes values in Y_D ; the section values are laws. Section 3.7 is where the two are related, and it relates them by hypothesis.

Missing or aggregated observations are not represented by removing latent coordinates from (3.2). They require a separately declared observation map or kernel into a coarser observation space, and the declaration changes O_D rather than Y_D .

3.2 Structural data and measurable typing

Definition 3.3 (Structural configuration and kernel signatures). Fix a measurable configuration space $(X, \mathcal{X})$. Every $X \in X$ uses the same design D , the same agent index set V , the same incidence set $\mathfrak{A}$, and the same measurable identifications of the fibers in (3.2) and (3.3); therefore $(Y_D, \mathcal{Y}_D)$ and $(O_D, \mathcal{O}_D)$ do not vary with X . A configuration may carry the finite interaction complex $\mathfrak{I}$ of Definition 2.12 with its edge set, the frames U_i or the links L_{ij} , and the parameters of the observation and transition mechanisms. The generative and recognition objects have the signatures

$$P_\theta : (X, \mathcal{X}) \longrightarrow \mathcal{P}(O_D \times Y_D, \mathcal{O}_D \otimes \mathcal{Y}_D), \quad X \longmapsto P_\theta(\cdot \mid X), \quad (3.5)$$

$$Q : (O_D \times X, \mathcal{O}_D \otimes \mathcal{X}) \longrightarrow \mathcal{P}(Y_D, \mathcal{Y}_D), \quad (o, X) \longmapsto Q_X(\cdot \mid o), \quad (3.6)$$

each measurable in its argument: for every $A \in \mathcal{O}_D \otimes \mathcal{Y}_D$ the map $X \mapsto P_\theta(A \mid X)$ is $\mathcal{X}$ -measurable, and for every $B \in \mathcal{Y}_D$ the map $(o, X) \mapsto Q_X(B \mid o)$ is $\mathcal{O}_D \otimes \mathcal{X}$ -measurable. **[DEFINITION]**

Signature (3.5) is the typing on which several later results depend, and it is worth reading for what it omits. The arguments of the generative kernel are θ and X and nothing else. A recognition law, a recognition parameter, and a posterior are not among them, so no factor of the generative model may read one. This is a definitional requirement of the fixed-joint construction rather than a

modeling preference: the joint must be a single measure fixed before an observation is seen, and a factor that reads a posterior makes the joint a function of the object it is supposed to be conditioned into. Nothing is proved here beyond the type; the consequences are drawn in Chapter 4, where the joint is built, and again in Chapter 15, where a construction that violates the typing is examined and found inadmissible.

Later chapters may make X random, but only within this fixed measurable space, and at a different probability level. Doing so introduces a configuration prior in $\mathcal{P}(X, \mathcal{X})$ and a configuration recognition kernel from $(O_D, \mathcal{O}_D)$ into $\mathcal{P}(X, \mathcal{X})$; neither changes D , $\mathfrak{A}$, or the identified fibers, and neither is denoted by any symbol reserved above for a state-level object.

3.3 Reference measures

Definition 3.4 (Declared reference measures). Each continuous Euclidean coordinate carries its Borel sigma-algebra and Lebesgue base measure. Each discrete coordinate carries its power-set sigma-algebra and counting base measure. Each mixed real coordinate carries the measure declared in Proposition 3.5. Any other fiber carries an explicitly declared sigma-finite base measure. Writing $\nu_{i,a}^k$, $\nu_{i,a}^m$, and $\nu_{i,a}^o$ for the component measures, the finite product reference measures are

$$\nu_D^Y = \bigotimes_{a=1}^M \bigotimes_{i \in V} (\nu_{i,a}^k \otimes \nu_{i,a}^m), \quad \nu_D^O = \bigotimes_{a=1}^M \bigotimes_{i \in V} \nu_{i,a}^o. \quad (3.7)$$

[DEFINITION]

Sigma-finiteness of every component is required and is not decoration. It is the hypothesis of the Radon-Nikodym theorem, without which the densities of Section 3.4 need not exist, and it is the hypothesis of the Fubini-Tonelli theorem, without which the iterated integrals used to marginalize Y out of the joint need not agree with the product integral. A finite product of sigma-finite measures is sigma-finite, so (3.7) inherits the property from its factors; this is the reason the construction is stated on a finite design and is one of the reasons the continuum question of Section 3.8 stays open.

Mixed coordinates are the case where a declaration is genuinely needed rather than conventional, and the following statement supplies it.

Proposition 3.5 (Dominating measure for a mixed coordinate). Let μ be a Borel probability measure on $\mathbb{R}$ whose Lebesgue decomposition has no singular continuous part, so that $\mu = \mu_{ac} + \mu_d$ with $\mu_{ac} \ll \lambda$ and μ_d carried by a countable set A . Put

$$\nu = \lambda + \sum_{x \in A} \delta_x. \quad (3.8)$$

Then ν is sigma-finite and $\mu \ll \nu$. [ESTABLISHED]

Proof. For sigma-finiteness, write $\mathbb{R} = (\bigcup_{n \geq 1}([-n, n] \setminus A)) \cup \bigcup_{x \in A} \{x\}$, a countable union; $\nu([-n, n] \setminus A) = \lambda([-n, n] \setminus A) \leq 2n < \infty$ and $\nu(\{x\}) = \lambda(\{x\}) + 1 = 1 < \infty$. For absolute continuity, let $\nu(B) = 0$. Then $\lambda(B) = 0$ and $B \cap A = \emptyset$, since a single atom of A inside B would contribute 1. Hence $\mu_{ac}(B) = 0$ because $\mu_{ac} \ll \lambda$, and $\mu_d(B) = \mu_d(B \cap A) = 0$ because μ_d is carried by A . Therefore $\mu(B) = 0$. $\square$

The hypothesis excluding a singular continuous part is a real restriction and is stated rather than assumed. A coordinate whose law carries a singular continuous component, of which the Cantor distribution is the standard example, is not dominated by (3.8), and this development supplies no generic reference measure for it. Such a coordinate would require its own declaration, and the declaration would have to be made before any density on that coordinate is written.

3.4 Kernels first, densities second

Definition 3.6 (Normalized kernels; densities as derivatives). The objects (3.5) and (3.6) are probability kernels, hence normalized by type. When $P_\theta(\cdot | X) \ll \nu_D^O \otimes \nu_D^Y$, its density is the Radon-Nikodym derivative $p_\theta(o, y | X)$ determined by

$$P_\theta(do, dY | X) = p_\theta(o, y | X) \nu_D^O(do) \nu_D^Y(dy), \quad (3.9)$$

and the evidence is the normalized marginal

$$p_\theta(o | X) = \int_{Y_D} p_\theta(o, y | X) \nu_D^Y(dy). \quad (3.10)$$

When $Q_X(\cdot | o) \ll \nu_D^Y$, its density is written $q_X(y | o)$. Densities are determined only up to a null set of the declared reference measure, and any two versions agreeing outside such a set denote the same density. **[DEFINITION]**

The order in Definition 3.6 is the content. A normalized kernel is declared, and only then is a density extracted; the reverse order, in which a nonnegative function is written down and a normalizer is invoked afterward, is a different construction whose normalizer must be shown finite before anything it appears in denotes a probability. This document uses the first order everywhere, and where a later chapter writes an unnormalized energy it says so and exhibits the normalizer.

Regular-conditional convention. Whenever a later result conditions on a realized observation, assume in addition that $(O_D, \mathcal{O}_D)$ and $(Y_D, \mathcal{Y}_D)$ are standard Borel spaces. This holds for the finite products of Euclidean and countable discrete fibers used in the worked construction; it is an additional hypothesis for a general measurable fiber. For fixed (θ, X) , write

$$P_{\theta, X}^O(A) := P_\theta(A \times Y_D | X), \quad A \in \mathcal{O}_D, \quad (3.11)$$

for the observation marginal. The standard-Borel hypothesis supplies a regular conditional probability

$$\Pi_{\theta,X}(o, B) = P_{\theta}(B \mid o, X), \quad B \in \mathcal{Y}_D, \quad (3.12)$$

unique only for $P_{\theta,X}^O$ -almost every o .

When the domination in (3.9) holds, Tonelli's theorem makes the function in (3.10) a density of $P_{\theta,X}^O$ relative to ν_D^O . After altering it on a ν_D^O -null set if necessary, there is a measurable set $O_{\theta,X}^{\text{reg}}$ with

$$P_{\theta,X}^O(O_{\theta,X}^{\text{reg}}) = 1, \quad 0 < p_{\theta}(o \mid X) < \infty, \quad (3.13)$$

such that, for every $o \in O_{\theta,X}^{\text{reg}}$ and $B \in \mathcal{Y}_D$,

$$\Pi_{\theta,X}(o, B) = \frac{\int_B p_{\theta}(o, y \mid X) \nu_D^Y(dy)}{p_{\theta}(o \mid X)}. \quad (3.14)$$

The displayed version is obtained directly from the density on the positive-finite evidence set and completed arbitrarily off that set. Fubini's theorem shows why the qualification is unavoidable: changing a joint density on a $(\nu_D^O \otimes \nu_D^Y)$ -null set can change an entire slice at an exceptional observation, while it changes (3.14) only on an observation-marginal-null set. An exact witness is available on $\mathbb{R}^2$. Let ϕ be the standard normal density and let ψ be any different probability density. The functions

$$p(o, y) = \phi(o)\phi(y), \quad \tilde{p}(o, y) = \begin{cases} \phi(o)\phi(y), & o \neq 0, \\ \phi(0)\psi(y), & o = 0 \end{cases}$$

are versions of the same joint density because they differ only on the Lebesgue-null slice $\{0\} \times \mathbb{R}$. Both displayed evidence representatives equal $\phi(0)$ at $o = 0$, but their density-ratio conditionals there are $\phi(y)dy$ and $\psi(y)dy$. The joint measure therefore determines neither pointwise posterior at that exceptional observation. Accordingly, every later fixed-observation posterior, ELBO, or free-energy statement is asserted for $P_{\theta,X}^O$ -almost every o , represented by $o \in O_{\theta,X}^{\text{reg}}$. An everywhere pointwise statement instead requires a particular jointly measurable density, evidence representative, and regular-conditional version to be declared as part of the model data; its exceptional-point values then belong to that declaration and are not determined by the joint measure alone. **[HYPOTHESIS]**

The same order is what makes the group action of Chapter 2 well typed. The action (2.6) is defined on measures. A frame change that does not preserve the reference measure changes the density and the base-measure Jacobian together, and only their combination is invariant, so a rule that multiplies a matrix into a density and stops is not the represented action unless the Jacobian is identically one.

Proposition 3.7 (Kernel integration preserves measurability). Let κ be a probability kernel from $(E, \mathcal{E})$ to $(F, \mathcal{F})$ and let $f : E \times F \rightarrow [0, \infty]$ be $\mathcal{E} \otimes \mathcal{F}$ -measurable. Then

$$x \mapsto \int_F f(x, y) \kappa(x, dy) \quad (3.15)$$

is $\mathcal{E}$ -measurable. Consequently $X \mapsto p_\theta(o \mid X)$ and the Q_X -expectations appearing in the bound of Chapter 5 are measurable in (o, X) whenever their integrands are jointly measurable. [ESTABLISHED]

Proof. Let $\mathcal{D}$ be the collection of $C \in \mathcal{E} \otimes \mathcal{F}$ for which $x \mapsto \kappa(x, C_x)$ is $\mathcal{E}$ -measurable, where $C_x = \{y : (x, y) \in C\}$. Measurable rectangles $A \times B$ lie in $\mathcal{D}$, since $\kappa(x, (A \times B)_x) = \mathbf{1}_A(x) \kappa(x, B)$ and $x \mapsto \kappa(x, B)$ is measurable by the definition of a kernel; the rectangles form a π -system generating $\mathcal{E} \otimes \mathcal{F}$. The collection $\mathcal{D}$ contains $E \times F$, is closed under proper differences because $\kappa(x, \cdot)$ is a finite measure, and is closed under increasing limits by continuity from below, hence is a λ -system; Dynkin's lemma gives $\mathcal{D} = \mathcal{E} \otimes \mathcal{F}$. The claim therefore holds for indicators, hence for nonnegative simple functions by linearity, hence for all nonnegative measurable f by monotone approximation and monotone convergence. The full development of kernels and their integrals is standard [36]. $\square$

This proposition is stated because the measurability of the evidence and of the bound in their arguments is used implicitly whenever one writes an expectation over configurations, and an implicit use of an unproved measurability claim is a gap. It is discharged here rather than assumed.

3.5 The absolute-continuity convention

Definition 3.8 (Standing convention on logarithmic density ratios). Wherever this document writes a logarithmic density ratio, or an expectation of one, the measure appearing in the numerator is absolutely continuous with respect to the measure appearing in the denominator. Relative entropy is understood in the measure-theoretic sense,

$$D_{\text{KL}}(Q \parallel P) = \begin{cases} \int \log\left(\frac{dQ}{dP}(y)\right) Q(dy), & Q \ll P, \\ +\infty, & \text{otherwise,} \end{cases} \quad (3.16)$$

following the measure-theoretic formulations of Kullback and Leibler [40] and Csiszár [20]. [DEFINITION]

The convention is not a restriction on what may be written; the second branch of (3.16) assigns a value in every case. It is a statement about which of those values is finite, and it is load-bearing precisely because the infinite branch is not an edge case to be ignored but the mechanism by which whole classes of candidate recognition laws are excluded. The next statement makes the criterion checkable in terms of densities, which is the form in which it is applied later.

Proposition 3.9 (Density criterion for absolute continuity). Let Q and P be measures on a common measurable space with $Q \ll \nu$ and $P \ll \nu$ for a sigma-finite ν , with densities q and p . Write $Z = \{p = 0\}$. Then

$$Q \ll P \iff Q(Z) = 0 \iff q = 0 \quad \nu\text{-almost everywhere on } Z. \quad (3.17)$$

[ESTABLISHED]

Proof. First, $P(Z) = \int_Z p(x)\nu(dx) = 0$. If $Q \ll P$ then $Q(Z) = 0$ follows at once. Conversely suppose $Q(Z) = 0$ and let A satisfy $P(A) = 0$, that is $\int_A p(x)\nu(dx) = 0$ with $p \geq 0$; then $\nu(A \cap \{p > 0\}) = 0$, so $Q(A \cap \{p > 0\}) = \int_{A \cap \{p > 0\}} q(x)\nu(dx) = 0$, while $Q(A \cap Z) \leq Q(Z) = 0$. Adding gives $Q(A) = 0$. The second equivalence is immediate from $Q(Z) = \int_Z q(x)\nu(dx)$ and $q \geq 0$. $\square$

The consequence used later has two cases that must not be conflated. Proposition 3.9 applies when both measures are dominated by the same ν . If a candidate Q is singular relative to that reference measure, its failure of absolute continuity must instead be checked directly from sets. Namely, if there is a measurable S with

$$Q(S) = 1, \quad P(S) = 0, \quad (3.18)$$

then $Q \not\ll P$ by the definition of absolute continuity, and (3.16) gives $D_{\text{KL}}(Q\|P) = +\infty$. For example, let P be a nondegenerate Lebesgue-dominated law on $\mathbb{R}^d$ and let Q be supported on a proper affine subspace S . The subspace is Lebesgue null, hence $P(S) = 0$, whereas $Q(S) = 1$; the direct argument (3.18), not Proposition 3.9, proves the claim. This distinction matters because a subspace-supported Q has no Lebesgue density to which the density criterion could be applied. The specific instance, in which an identification of latent coordinates across a cluster is shown to be a generative construction rather than a restriction on the recognition law, is developed in Part II.

Hypothesis 3.10 (Regular observation, support, and log-integrability). For fixed X , the observation satisfies $o \in \mathcal{O}_{\theta, X}^{\text{reg}}$ under the regular-conditional convention above, and

$$0 < p_\theta(o \mid X) < \infty, \quad Q_X(\cdot \mid o) \ll \Pi_{\theta, X}(o, \cdot), \quad (3.19)$$

and

$$\mathbb{E}_{Q_X} \left[\left| \log p_\theta(o, Y \mid X) \right| + \left| \log q_X(Y \mid o) \right| \right] < \infty. \quad (3.20)$$

[HYPOTHESIS]

These are choices, and each excludes something identifiable. Membership in $\mathcal{O}_{\theta, X}^{\text{reg}}$ makes the posterior the selected regular-conditional version and makes the statement invariant under changes of density version outside a marginal-full set. The lower bound in (3.19) excludes observations the selected representative deems impossible, for which no bound on the log evidence exists because the log evidence is $-\infty$. The absolute-continuity requirement excludes exactly the degenerate laws

just described. Condition (3.20) excludes recognition laws whose tails make the two expectations individually infinite even when their difference would formally cancel, and it is what permits the decomposition in Chapter 5 to be rearranged term by term. Where a later result tolerates an extended-real relative entropy, the relaxation is stated at that point rather than absorbed into the notation here.

3.6 Recognition marginals and what they fail to determine

Definition 3.11 (Coordinate marginals of the recognition law). With $\text{pr}_{i,a}^k$ and $\text{pr}_{i,a}^m$ the coordinate projections out of Y_D , define the marginalization maps $\text{Marg}_{i,a}^k(R) = (\text{pr}_{i,a}^k)_\# R$ and $\text{Marg}_{i,a}^m(R) = (\text{pr}_{i,a}^m)_\# R$ on $\mathcal{P}(Y_D, \mathcal{Y}_D)$, and set

$$q_{i,a}^{o,X} = \text{Marg}_{i,a}^k(Q_X(\cdot | o)), \quad s_{i,a}^{o,X} = \text{Marg}_{i,a}^m(Q_X(\cdot | o)). \quad (3.21)$$

[DEFINITION]

The recognition law is not required to factor across agents, design points, or channels. In the Gaussian realization with at least two nondegenerate real coordinates, the marginals (3.21) are therefore a lossy summary, and the next statement exhibits the loss. No such nonidentifiability is asserted for an arbitrary collection of fibers: if the finite latent space is a singleton, its unique law is determined by every marginal.

Proposition 3.12 (Coordinate marginals do not determine a sufficiently rich joint). Suppose Y_D contains at least two nondegenerate real coordinates, as it does in the nondegenerate Gaussian realization. Then there exist distinct probability measures on Y_D with identical coordinate marginals. Consequently the family $\{q_{i,a}^{o,X}, s_{i,a}^{o,X}\}$ does not determine $Q_X(\cdot | o)$ under this richness hypothesis. Without it the conclusion can fail: for one agent and one design point with $K_{i,a} = M_{i,a} = \{0\}$, the space Y_D is a singleton and its unique law is determined by its marginals. [ESTABLISHED]

Proof. Take $Q^{(0)} = \mathcal{N}(0, I_2)$ and $Q^{(r)} = \mathcal{N}(0, \Sigma_r)$ with $\Sigma_r = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}$ for any $r \in (-1, 1)$ with $r \neq 0$; both are probability measures, since Σ_r has eigenvalues $1 \pm r > 0$. The coordinate marginals of a centered Gaussian $\mathcal{N}(0, \Sigma)$ on $\mathbb{R}^2$ are $\mathcal{N}(0, \Sigma_{11})$ and $\mathcal{N}(0, \Sigma_{22})$, so both measures have standard normal marginals in both coordinates. They are distinct because a Gaussian law is determined by its mean and covariance and the covariances differ. Embedding this pair in any two coordinates of Y_D and keeping the remaining coordinates fixed produces two distinct elements of $\mathcal{P}(Y_D, \mathcal{Y}_D)$ with the same family (3.21). $\square$

3.7 Continuum sections versus finite evaluation

Chapter 2 gave each agent continuum sections $q_i^{o,X}$ and $s_i^{o,X}$ over its support. This chapter has given the finite recognition law Q_X and its marginals. There are now two ways to obtain a probability

measure on $K_{i,a}$: sample the continuum section at c_a , or marginalize the finite law at (i, a) . Write the first as

$$\text{Samp}_{D,i}(q_i^{o,X}) = (q_i^{o,X}(c_a))_{a=1}^M, \quad \text{Samp}_{D,i}(s_i^{o,X}) = (s_i^{o,X}(c_a))_{a=1}^M. \quad (3.22)$$

The two constructions have different domains. The map (3.22) acts on sections in $\Gamma(\mathcal{C}_i, \mathcal{E}|_{\mathcal{C}_i})$; the maps of Definition 3.11 act on $\mathcal{P}(Y_D, \mathcal{Y}_D)$. Neither is defined in terms of the other, so their outputs coincide only if that is declared. The following statement shows that the declaration is substantive in one direction and empty in the other, so that calling it a consequence would be wrong in both.

Proposition 3.13 (Compatibility determines neither side). Suppose the compatibility relation

$$q_i^{o,X}(c_a) = q_{i,a}^{o,X}, \quad s_i^{o,X}(c_a) = s_{i,a}^{o,X} \quad \text{for every } (i, a) \in \mathfrak{A} \quad (3.23)$$

holds. Then (i), under the two-nondegenerate-coordinate richness hypothesis of Proposition 3.12, (3.23) does not determine $Q_X(\cdot | o)$; and (ii), if some support $\mathcal{C}_i$ contains a point outside the finite design D and the selected section class admits a nonzero smooth deformation supported away from D , the relation does not determine the sections $q_i^{o,X}$ and $s_i^{o,X}$. The positive-dimensional Gaussian tier supplies such deformations. If every support is zero-dimensional and exhausted by its design points, the finite values can instead determine a section pointwise. [ESTABLISHED]

Proof. (i) By Proposition 3.12 there are distinct laws in $\mathcal{P}(Y_D, \mathcal{Y}_D)$ with the same coordinate marginals, and (3.23) constrains Q_X only through those marginals; both such laws satisfy it against the same sections. (ii) Fix $c^* \in \mathcal{C}_i \setminus D$. In the positive-dimensional Gaussian realization, choose a neighborhood of c^* disjoint from the finite set D and a smooth bump function $\chi : \mathcal{C}_i \rightarrow [0, 1]$ supported there with $\chi(c^*) = 1$. The section obtained from $q_i^{o,X}$ by replacing $\mu_i(c)$ with $\mu_i(c) + \chi(c)v$ for any fixed $v \neq 0$ is a smooth section of the same bundle, differs from $q_i^{o,X}$ at c^* , and agrees with it on all of D ; it therefore satisfies (3.23) against the same marginals. By contrast, if $\mathcal{C}_i = D \cap \mathcal{C}_i$ as a zero-dimensional support, equality at those points is pointwise equality of the section, so the nonuniqueness conclusion has no such witness and can fail. $\square$

Hypothesis 3.14 (Sampling and marginalization are compatible). Where a result requires the continuum sections and the finite recognition law to describe the same coordinate laws, relation (3.23) is adopted. [HYPOTHESIS]

This is a choice and is named as one. It excludes configurations in which an agent's continuum section and the finite law disagree at a design point, which is a genuine possibility because the two objects are declared independently. It is invoked only where a statement about the finite law is transferred to the geometric layer, or the reverse; a result stated entirely about the finite construction does not need it. Proposition 3.13 records that, under its richness hypothesis, adopting compatibility does not reconstruct the joint from its marginals and, when the support extends beyond the design and the selected fiber is locally deformable, does not reconstruct a continuum section from finite

samples. Singleton finite fibers and zero-dimensional supports exhausted by the design are the explicit exceptions.

3.8 What the finite construction does not supply

Open Problem 3.15 (Continuum probabilistic theory). The finite construction supplies probability spaces and kernels associated with the declared design D . It supplies no probability measure on a space of sections over $\mathcal{C}$, no unique extension of the finite marginals to such a space, and no continuum limit. Whether one exists for this construction is not settled here. **[OPEN]**

What would close it can be listed exactly, and listing it is the point of declaring the gap rather than gesturing at it. One would need a refining family of designs; a compatible, that is projective, system of finite laws indexed by those designs, so that a Kolmogorov-type extension theorem applies [36]; a declared topology and sigma-algebra on the candidate space of sections, since the extension theorem produces a measure on a product space and further work is needed to place it on a space of smooth sections; tightness or compactness sufficient to control the limit; convergence of the functionals and observables that the finite theory computes; control of the gauge choices of Chapter 2 and of the reference measures of Section 3.3 under refinement, since neither is canonical; and a proof that any proposed continuum interaction energy is normalizable.

The last item is where the obstruction is concrete rather than merely laborious. The reference measures (3.7) are finite products, and sigma-finiteness was inherited from finitely many sigma-finite factors; the same construction does not extend to infinitely many non-probability factors, so a continuum theory cannot obtain its reference measure by the route used here and would have to declare a measure on the section space directly. Until that is done, a continuum energy has no density to be the exponent of, and the objects that the finite theory calls densities have no continuum counterparts. Every evidence-bound claim in this document is accordingly a finite-design claim, and none of them is asserted in a continuum limit.

Chapter 4

The Fixed Normalized Generative Law

This chapter fixes one normalized joint before inference begins. The latent channels inherit the two representations of the common principal gauge group from Chapter 2. Their local coordinates may be re-expressed independently, so the covariance formulas carry a passive product of frame changes without introducing a second principal bundle. The linear-Gaussian realization is deferred to Chapter 9.

4.1 Parameters, structure, and kernel types

Fix the design $D = \{c_a\}_{a=1}^M$, the agent set $V = \{1, \dots, N\}$, and the measurable spaces and sigma-finite reference measures of Chapter 3. Structural data $X \in \mathbf{X}$ contain a finite directed acyclic graph $\Gamma = (V, E)$, design incidence, and the geometric data. Numerical parameters that a learning coordinate may move are denoted θ . This split is declared rather than derived; all typing requirements below apply to both symbols. **[DEFINITION]**

Write $\text{pa}(i)$ for the parents of i and $\mathcal{R} = \{r : \text{pa}(r) = \emptyset\}$. At one design point, declare normalized Markov kernels

$$P_{\theta,i}^m(dm_i \mid m_{\text{pa}(i)}, X), \quad P_{\theta,i}^k(dk_i \mid k_{\text{pa}(i)}, m_i, X), \quad P_{\theta,i}^o(do_i \mid k_i, m_i, X). \quad (4.1)$$

At a root, the first factor is a proper model law and the second is a normalized bridge from the same-agent model coordinate to the belief coordinate. **[DEFINITION]**

The channels are asymmetric by declaration. At a root, write $K_i(m, A) = P_{\theta,i}^k(A \mid m, X)$. This kernel induces a map on laws,

$$\tilde{\phi}_i(s)(A) = \int_{\mathbf{M}} K_i(m, A) s(dm). \quad (4.2)$$

Only this pushforward-on-laws map can represent the local associated-bundle morphism $\tilde{\Phi} : (\mathcal{E}_m)_c \rightarrow (\mathcal{E}_b)_c$, and only if its contextual and gauge compatibility is declared. At a nonroot, the kernel is indexed by the parent belief coordinate; it does not define a map $\mathcal{E}_m \rightarrow \mathcal{E}_b$ unless that parent is fixed or the domain is enlarged. Otherwise it is a separately declared kernel-valued bridge. In

every case the generative dependence runs from model samples to belief samples. The geometric existence of $\Phi : \mathcal{E}_b \rightarrow \mathcal{E}_m$ neither inserts a reverse factor into (4.1) nor forces cyclic generative dependence. A model using both directions would need a separately normalized joint and cannot infer its factorization from the two associated-bundle morphisms.

4.2 Iterated composition and densities

Choose a topological ordering $v_1, \dots, v_N$ of Γ . The one-design-point law is the successive kernel composition

$$P_{\theta,a}(do_a, dY_a \mid X) = \prod_{\ell=1}^N P_{\theta,v_\ell}^m(dm_{v_\ell} \mid m_{\text{pa}(v_\ell)}, X) P_{\theta,v_\ell}^k(dk_{v_\ell} \mid k_{\text{pa}(v_\ell)}, m_{v_\ell}, X) P_{\theta,v_\ell}^o(do_{v_\ell} \mid k_{v_\ell}, m_{v_\ell}, X). \quad (4.3)$$

The product denotes ordered application of kernels, not multiplication of measures. For bounded measurable f , its meaning is

$$\int f dP_{\theta,a} = \int P_{\theta,v_1}^m(dm_{v_1} \mid X) \int P_{\theta,v_1}^k(dk_{v_1} \mid m_{v_1}, X) \int P_{\theta,v_1}^o(do_{v_1} \mid k_{v_1}, m_{v_1}, X) \cdots f(o_a, y_a). \quad (4.4)$$

Every conditioning coordinate has been introduced before its kernel is applied.

Construction 4.1 (Finite directed law). Equation (4.4) defines a probability kernel $X \mapsto P_{\theta,a}(\cdot \mid X)$. **[ESTABLISHED]**

Proof. Integrating a bounded measurable function against a probability kernel preserves boundedness and measurability in the remaining variables. Finite iteration therefore defines a measure, and $f \equiv 1$ gives total mass one. This is the finite kernel construction underlying directed graphical models [38, 43, 68]. $\square$

Distinct design points are conditionally independent given X :

$$P_\theta(do, dY \mid X) = \bigotimes_{a=1}^M P_{\theta,a}(do_a, dY_a \mid X). \quad (4.5)$$

This modeling hypothesis excludes residual cross-design dependence and is used in the finite-design normalization and factor count. **[HYPOTHESIS]**

When component kernels have densities relative to the declared component reference measures,

$$p_{\theta,a}(o_a, y_a \mid X) = \prod_{i \in V} p_{\theta,i}^m(m_i \mid m_{\text{pa}(i)}, X) p_{\theta,i}^k(k_i \mid k_{\text{pa}(i)}, m_i, X) p_{\theta,i}^o(o_i \mid k_i, m_i, X), \quad (4.6)$$

and

$$p_\theta(o, y \mid X) = \prod_{a=1}^M p_{\theta,a}(o_a, y_a \mid X), \quad P_\theta(do, dY \mid X) = p_\theta(o, y \mid X) \nu_D^O(do) \nu_D^Y(dy). \quad (4.7)$$

The measure P_θ is authoritative; p_θ denotes only its Radon–Nikodym density.

4.3 The typing prohibition

Requirement 4.2 (Typing prohibition). *Each factor in (4.1) is a function only of θ , X , and its displayed parent and same-agent latent coordinates. No generative kernel may take as an argument the recognition kernel Q_X , one of its marginals or parameters, or a posterior of the model.*

[DEFINITION]

This requirement defines the fixed-joint construction and proves nothing. A transition built from a live peer recognition marginal is a map from recognition laws to kernels, not a factor of a joint fixed before inference.

Proposition 4.3 (No distinguished target). Let $\mathcal{Q}$ be a nonempty family of recognition laws. If a factor in (4.3) is replaced by a nonconstant Q -indexed family, producing a normalized joint $P_{\theta,Q}$ for each $Q \in \mathcal{Q}$, assume at the displayed observation that $0 < p_{\theta,Q}(o \mid X) < \infty$, that $Q \ll P_{\theta,Q}(\cdot \mid o, X)$, and that the log-density terms defining the ELBO and KL divergence are integrable. Then for each fixed Q

$$\log p_{\theta,Q}(o \mid X) = \mathcal{L}(Q; X) + D_{\text{KL}}(Q \parallel P_{\theta,Q}(\cdot \mid o, X)). \quad (4.8)$$

The family itself selects no joint as “the model.” Writing $e(Q) = \log p_{\theta,Q}(o \mid X)$, two members satisfy

$$\begin{aligned} \mathcal{L}(Q'; X) - \mathcal{L}(Q; X) &= e(Q') - e(Q) \\ &\quad - [D_{\text{KL}}(Q' \parallel P_{\theta,Q'}(\cdot \mid o, X)) - D_{\text{KL}}(Q \parallel P_{\theta,Q}(\cdot \mid o, X))]. \end{aligned} \quad (4.9)$$

Thus a divergence decrease certifies improvement relative to one fixed reference value exactly when the two members lie on one level set of e . [ESTABLISHED]

Proof. Apply the exact ELBO identity to each fixed pair $(Q, P_{\theta,Q})$ and subtract the two instances. No member is distinguished because no selection rule was declared. $\square$

Proposition 4.4 (The stronger consequences fail). Nonconstant dependence of $P_{\theta,Q}$ on Q does not imply that e varies, that no ascent can improve a fixed quantity, or that no single member’s evidence bounds the family of bounds. [ESTABLISHED]

Proof. Take one binary latent y , one binary observation o , and $Q_\beta = \text{Ber}(\beta)$ for $\beta \in (0, 1)$. Let $g(\beta) = \beta/2 + 1/4$ and define

$$P_{\theta,Q_\beta}(o, y \mid X) = \frac{1}{2} \text{Ber}(g(\beta))(y). \quad (4.10)$$

The joint is normalized and varies injectively with β , but $e(Q_\beta) = \log(1/2)$ for every β . The posterior is $\text{Ber}(g(\beta))$, so the gap at $\beta = 9/10, 7/10, 1/2$ is respectively

$$\frac{9}{10} \log \frac{9}{7} + \frac{1}{10} \log \frac{1}{3} = 0.1163218 \dots, \quad \frac{7}{10} \log \frac{7}{6} + \frac{3}{10} \log \frac{3}{4} = 0.0216009 \dots, \quad 0. \quad (4.11)$$

The bound therefore increases to the fixed evidence, and the member at $\beta = 1/2$ has evidence equal to the supremum of all these bounds. $\square$

These are closed-form relative entropies, not numerical experiments. The development declines the three stronger statements as false in this witness, while retaining the absence of a distinguished member. **[NOT-CLAIMED]** The typing prohibition also does not ban population coupling through latents inside a fixed joint; it bans a generative factor from reading the posterior that the joint is supposed to determine.

4.4 Exact normalization

Proposition 4.5 (Normalization without a partition function). Under the domination assumptions of Chapter 3 and Construction 4.1,

$$P_\theta(O_D \times Y_D \mid X) = 1, \quad \int p_\theta(o, y \mid X) \nu_D^O(do) \nu_D^Y(dy) = 1 \quad (4.12)$$

for every admissible (θ, X) , without a global normalizing constant. **[ESTABLISHED]**

Proof. At one design point, Tonelli's theorem permits reverse-topological integration. Integrate each observation factor first. Then remove each nonroot in reverse topological order by integrating its normalized belief kernel and then its normalized model kernel; all descendant factors containing those coordinates have already disappeared. Finally integrate each root bridge and proper root law. Every integral contributes one. The same induction proves the measure statement without domination, and (4.5) multiplies M unit masses. $\square$

4.5 The normalized undirected alternative

An undirected cyclic model is valid when its global normalizer exists. For nonnegative node and edge potentials, define

$$\Psi_X(o, y) = \prod_{i \in V} \psi_i(o_i, k_i, m_i; X) \prod_{\{i, j\} \in E} \psi_{ij}(k_i, m_i, k_j, m_j; X), \quad (4.13)$$

$$Z_X = \int \Psi_X(o, y) \nu_D^O(do) \nu_D^Y(dy), \quad 0 < Z_X < \infty, \quad (4.14)$$

$$p_\theta^{\text{Gibbs}}(o, y \mid X) = Z_X^{-1} \Psi_X(o, y). \quad (4.15)$$

Finiteness and positivity of Z_X are model-existence conditions. **[DEFINITION]**

Proposition 4.6 (Local normalization is insufficient). Normalized node potentials and an everywhere positive, finite, continuous edge potential need not give finite Z_X . [ESTABLISHED]

Proof. On $\mathbb{R}^2$ take

$$\psi_i(y_i) = (2\pi)^{-1/2} e^{-y_i^2/2}, \quad \psi_{12}(y_1, y_2) = e^{cy_1 y_2}, \quad c \geq 1. \quad (4.16)$$

The quadratic precision is

$$\Lambda_{\text{ex}} = \begin{pmatrix} 1 & -c \\ -c & 1 \end{pmatrix}, \quad \text{spec } \Lambda_{\text{ex}} = \{1 - c, 1 + c\}. \quad (4.17)$$

In orthonormal sum-and-difference coordinates the integral factorizes, and the factor in the $1 - c$ direction diverges. $\square$

The document chooses the directed construction because its normalization does not depend on the parameters, not because normalized cyclic Gaussian Markov random fields are invalid.

4.6 Covariance under independent channel-frame re-expression

At each design point c_a and vertex, independently re-express the belief and model frame coordinates by $g_{a,i}^b = g_i^b(c_a) \in G$ and $g_{a,i}^m = g_i^m(c_a) \in G$, and write

$$R_{a,i}^b = \rho_b(g_{a,i}^b), \quad R_{a,i}^m = \rho_m(g_{a,i}^m). \quad (4.18)$$

With the passive-section convention of (2.10), this sample-coordinate transformation corresponds to the section rechoice $u_i^{x'} = u_i^x \cdot (g_{a,i}^x)^{-1}$. Thus the represented matrix $R_{a,i}^x$ multiplies sample coordinates, while its inverse multiplies the principal frame on the right. Within a formula for one design point, suppress the index a and write g_i^x, R_i^x . They act on latent coordinates as $k_i' = R_i^b k_i$ and $m_i' = R_i^m m_i$, while observations are gauge inert. To permit all such independent coordinate changes, the finite-design structural data carry graph-link copies $\Theta_{a,ij}^x$, one for each design incidence. At fixed a write Θ_{ij}^x ; their discrete vertex frame changes are $h_{a,i}^x = g_i^x(c_a)$, abbreviated h_i^x , and the suppressed one-design-point law is

$$\Theta_{ij}^{b'} = h_i^b \Theta_{ij}^b (h_j^b)^{-1}, \quad \Theta_{ij}^{m'} = h_i^m \Theta_{ij}^m (h_j^m)^{-1}. \quad (4.19)$$

If instead one structural link is constrained to be identical at every design point, an admissible frame rechoice must preserve that constraint: for every two design indices a, a' and every constrained edge,

$$h_{a,i}^x \Theta_{ij}^x (h_{a,j}^x)^{-1} = h_{a',i}^x \Theta_{ij}^x (h_{a',j}^x)^{-1}.$$

Context-independent rechoices are a sufficient subgroup, but they are not necessary; the full residual group is the stabilizer of the shared-link constraints. A larger transformation law requires

an explicitly declared parameter compensation. Across the unconstrained finite design the passive coordinate group is $\prod_{a,i}(G \times G)$. This product records two independently chosen trivializations of associated bundles induced from the same P ; it is not a second principal structure group. The active principal gauge group is $\text{Aut}_G(P)$ over id_G ; it acts diagonally on both associated bundles through ρ_b and ρ_m . It is not the principal right G -action itself. A context-independent frame re-expression is the subgroup for which $g_{a,i}^x$ does not depend on a . **[DEFINITION]**

Hypothesis 4.7 (Kernel covariance). The transformed parameters (θ', X') are declared so that every nonroot factor obeys

$$P_{\theta',i}^m(\cdot \mid R_{\text{pa}(i)}^m m_{\text{pa}(i)}, X') = (R_i^m)_\# P_{\theta,i}^m(\cdot \mid m_{\text{pa}(i)}, X), \quad (4.20)$$

$$P_{\theta',i}^k(\cdot \mid R_{\text{pa}(i)}^b k_{\text{pa}(i)}, R_i^m m_i, X') = (R_i^b)_\# P_{\theta,i}^k(\cdot \mid k_{\text{pa}(i)}, m_i, X), \quad (4.21)$$

$$P_{\theta',i}^o(\cdot \mid R_i^b k_i, R_i^m m_i, X') = P_{\theta,i}^o(\cdot \mid k_i, m_i, X), \quad (4.22)$$

with the analogous root identities. **[HYPOTHESIS]**

The second identity is the covariance requirement for the model-to-belief sample kernel. At a root it implies that the induced law map (4.2) obeys (2.15); at a nonroot it remains a parent-indexed kernel law unless an enlarged cross-bundle domain is declared. Neither route uses Φ .

Proposition 4.8 (Joint covariance and evidence invariance). Let $T(o, y) = (o, Ry)$, where $R = \bigoplus_{a,i}(R_{a,i}^b \oplus R_{a,i}^m)$. Under Hypothesis 4.7,

$$P_{\theta'}(\cdot \mid X') = T_\# P_\theta(\cdot \mid X), \quad p_{\theta'}(o, y' \mid X') = |\det R|^{-1} p_\theta(o, R^{-1}y' \mid X), \quad (4.23)$$

when the local actions are linear and densities exist. The observation marginals agree:

$$p_{\theta'}(o \mid X') = p_\theta(o \mid X) \quad \text{for } \nu_D^O\text{-almost every } o. \quad (4.24)$$

[ESTABLISHED]

Proof. Apply (4.20)–(4.22) along the topological ordering. Each factor is the receiving-coordinate pushforward of the original factor, so the composed law is $T_\# P_\theta$. The density formula is change of variables. Integrating over y' cancels its Jacobian because T is the identity on observations. $\square$

Evidence invariance is a redundancy constraint, not an empirical prediction: the action changes latent coordinates while leaving the observation law fixed. The linear-Gaussian chapter verifies Hypothesis 4.7 parameter by parameter and then displays the stronger common-frame reduction.

The passive coordinate action on the full parameter space can have nontrivial stabilizers, so its orbit space need not be a manifold. Subtracting the dimension of the coordinate group from a parameter count is unjustified without freeness and properness. Determining generic stabilizers and a regular quotient remains open; nothing later depends on that quotient. **[OPEN]**

Chapter 5

The Exact Single Evidence Bound

The previous chapter fixed one normalized joint. This chapter introduces one recognition law for the entire population and derives the single exact identity that relates the two. There is one bound in this document, it bounds one number, and the difference between the bound and that number is one relative entropy between two measures on the same space. Everything that follows about restricted families, coarse-graining, and the flow of couplings is a statement about this identity or about objects it defines.

The chapter is organized around three things that are easy to get wrong. The recognition law is a joint law and is not recoverable from its coordinate marginals, so a calculation that substitutes marginals into an entropy is computing something else and the difference has a name and a sign. The identity holds under stated hypotheses, one of which, absolute continuity with respect to the posterior, is a genuine restriction and is exactly what later forbids degenerate recognition laws. And the exact coordinates of the alternating scheme are two specific objects; an optimizer direction is a proposal that must pass an acceptance test before it is entitled to the conclusions that attach to a coordinate.

There is also a measure-theoretic qualification that applies to every fixed-observation formula in this chapter. For fixed (θ, X) , let $P_{\theta, X}^O$, $\Pi_{\theta, X}$, and the marginal-full set $O_{\theta, X}^{\text{reg}}$ be the observation marginal, regular conditional probability, and regular-observation set of Chapter 3. Unless an everywhere-defined density and conditional version has been declared as part of the model data, every posterior and ELBO statement below is a $P_{\theta, X}^O$ -almost-everywhere statement, represented by selecting $o \in O_{\theta, X}^{\text{reg}}$. Changing a density on a null slice can change exceptional point values but cannot change these almost-everywhere statements.

5.1 The correlated population recognition kernel

Definition 5.1 (Correlated population recognition kernel). Fix the finite design D , the observed value $o \in O_D$, and the structural value $X \in \mathcal{X}$ of Chapter 3. Recognition is represented by one normalized Markov kernel

$$Q : (O_D \times \mathcal{X}, \mathcal{O}_D \otimes \mathcal{X}) \longrightarrow \mathcal{P}(\mathcal{Y}_D, \mathcal{Y}_D), \quad (o, X) \longmapsto Q_X(dY \mid o), \quad (5.1)$$

so that $Q_X(Y_D | o) = 1$ and, for every $B \in \mathcal{Y}_D$, the map $(o, X) \mapsto Q_X(B | o)$ is $\mathcal{O}_D \otimes \mathcal{X}$ -measurable.

[DEFINITION] The law is permitted to correlate agents with one another, the state channel with the model channel, and one design point with another. No factorization is imposed here; the restrictions that impose one are the subject of Chapter 11, and they are restrictions on this object rather than replacements for it.

Write $q_{i,a}^{o,X}$ and $s_{i,a}^{o,X}$ for the coordinate marginals defined in Definition 3.11. By Proposition 3.12, these marginals do not reconstruct $Q_X(\cdot | o)$ whenever the latent space contains at least two nondegenerate real coordinates.

The proposition is a statement about a single pair of coordinates, and the general situation is worse in a specific way. The set of joint laws with a prescribed family of marginals is the Fréchet class of those marginals; it is convex, it contains the product law, and in continuous Euclidean coordinates it is parameterized by the dependence structure, the copula, which the marginals leave entirely free [52]. When second moments exist, every cross-agent, cross-channel, and cross-design covariance block is unspecified by the marginals. The consequence for this chapter is not stylistic. Marginal recognition laws cannot be substituted for the population law in an exact entropy or evidence calculation, and the following makes the error exact rather than merely warning against it.

Proposition 5.2 (Total correlation and the two signs). Partition the coordinates of Y into blocks indexed by b , let Q_X have density q_X with respect to ν_D^Y and block marginal densities q_b , and suppose the joint and all block differential entropies are finite. Let

$$\text{TC}(Q_X) = D_{\text{KL}} \left(Q_X \left\| \bigotimes_b Q_{X,b} \right. \right) \geq 0 \quad (5.2)$$

be the total correlation of the blocks. Then

$$\mathbb{E}_{Q_X} [\log q_X(Y | o)] = \sum_b \mathbb{E}_{Q_X} [\log q_b(Y_b | o)] + \text{TC}(Q_X), \quad (5.3)$$

with equality of the two log-density expectations exactly when the blocks are independent under Q_X . Define the substituted free energy and its negative pseudo-ELBO by

$$\tilde{\mathcal{F}} := \mathbb{E}_{Q_X} \left[-\log p_\theta(o, Y | X) + \sum_b \log q_b(Y_b | o) \right], \quad (5.4)$$

$$\tilde{\mathcal{L}} := -\tilde{\mathcal{F}}. \quad (5.5)$$

Then

$$\tilde{\mathcal{F}} = \mathcal{F} - \text{TC}(Q_X), \quad \tilde{\mathcal{L}} = \mathcal{L}(Q_X; X) + \text{TC}(Q_X). \quad (5.6)$$

Thus it is the *substituted free energy* that is smaller by total correlation, while its negative *pseudo-ELBO* is larger by total correlation and need not be a lower bound on the log evidence. In the

Gaussian case with joint covariance C and diagonal blocks C_{bb} ,

$$\text{TC}(Q_X) = \frac{1}{2} \left(\sum_b \log \det C_{bb} - \log \det C \right). \quad (5.7)$$

[ESTABLISHED]

Proof. Expanding the definition, $\text{TC}(Q_X) = \mathbb{E}_{Q_X}[\log q_X] - \sum_b \mathbb{E}_{Q_X}[\log q_b]$, which is (5.3). Non-negativity and the equality case are those of relative entropy [20, 40]. Subtracting (5.4) from $\mathcal{F} = \mathbb{E}_{Q_X}[-\log p_\theta + \log q_X]$ gives $\text{TC}(Q_X)$, and negation gives the second identity in (5.6). For (5.7), substitute the Gaussian differential entropy $\frac{1}{2} \log \det(2\pi e C)$ for the joint and for each block; the dimensional constants cancel because the block dimensions sum to d . $\square$

The sign in Proposition 5.2 matters and is the reason the point is made here rather than in a remark. Substituting marginal entropies produces a smaller free-energy functional and, after negation, a larger pseudo-ELBO that is not a bound at all. A scheme that reports the latter as an evidence lower bound is reporting a quantity that can exceed the log evidence, and by an amount that grows with exactly the correlation the recognition law was introduced to represent.

5.2 Hypotheses

Hypothesis 5.3 (Evidence-domain hypotheses). The identity of Section 5.3 holds at a selected (o, X) under the following four hypotheses. The selection is subject to the standing qualification $o \in \mathcal{O}_{\theta, X}^{\text{reg}}$; equivalently, the theorem holds for $P_{\theta, X}^O$ -almost every observation. [HYPOTHESIS]

(H1) Types and conditional version. The joint kernel $P_\theta(\cdot | X)$ of Chapter 4 and the recognition kernel $Q_X(\cdot | o)$ of Definition 5.1 are normalized and measurable as declared, the densities $p_\theta(o, y | X)$ and $q_X(y | o)$ exist relative to the declared reference measures, and $P_\theta(\cdot | o, X)$ denotes the regular conditional probability $\Pi_{\theta, X}(o, \cdot)$ represented by (3.14) on $\mathcal{O}_{\theta, X}^{\text{reg}}$. This is what makes the objects in the statement well typed. An everywhere pointwise version may replace the almost-everywhere convention only when that version is declared as model data.

(H2) Positive finite evidence. $0 < p_\theta(o | X) < \infty$. This excludes observed values that the model assigns zero density, at which the conditional posterior is undefined and there is nothing to approximate. It is used to divide by the evidence in (5.13).

(H3) Absolute continuity. $Q_X(\cdot | o) \ll \Pi_{\theta, X}(o, \cdot) = P_\theta(\cdot | o, X)$. This is a genuine restriction on the recognition family and not a technicality; Proposition 5.4 below makes the exclusion it enforces precise. It is used to form the Radon–Nikodym derivative in (5.14).

(H4) Log-integrability.

$$\mathbb{E}_{Q_X} [|\log p_\theta(o, Y | X)| + |\log q_X(Y | o)|] < \infty. \quad (5.8)$$

This is not implied by (H1) through (H3), and a witness settles the point rather than leaving it asserted: take $d = 1$ with a standard Gaussian conditional posterior and a standard Cauchy

recognition law. Both have strictly positive Lebesgue densities, so (H1) through (H3) hold; but $\log p_\theta(o, y \mid X)$ differs from $-y^2/2$ by a constant, and the Cauchy law has no second moment, so (5.8) fails. Relative entropy remains well defined in $[0, \infty]$ without (H4). What fails without it is the split of the identity into two separately finite expectations: the difference defining the ELBO can be an undefined $\infty - \infty$. The hypothesis is used to write the expectation of a difference as a difference of expectations.

Hypothesis (H3) deserves its own statement, because it is the one with consequences later.

Proposition 5.4 (Subspace support violates absolute continuity). Suppose the conditional posterior $P_\theta(\cdot \mid o, X)$ is dominated by Lebesgue measure on $\mathbb{R}^d$, as it is in the linear-Gaussian realization of Chapter 4 with the positive definite covariances (9.9), and suppose $Q_X(\cdot \mid o)$ is supported on a proper affine subspace $\mathcal{S} \subsetneq \mathbb{R}^d$, meaning $Q_X(\mathcal{S} \mid o) = 1$ with $\dim \mathcal{S} < d$. Then (H3) fails and

$$D_{\text{KL}}(Q_X(\cdot \mid o) \parallel P_\theta(\cdot \mid o, X)) = +\infty. \quad (5.9)$$

[ESTABLISHED]

Proof. A proper affine subspace of $\mathbb{R}^d$ is Lebesgue null, so $P_\theta(\mathcal{S} \mid o, X) = 0$ while $Q_X(\mathcal{S} \mid o) = 1$. Absolute continuity fails on the set $\mathcal{S}$. By the definition of relative entropy for measures that are not absolutely continuous, the divergence is $+\infty$ [20, 40]. $\square$

This is the exact mechanism that forbids subspace-supported recognition laws, and its scope should be stated exactly. It does not say that constraining the recognition law is illegitimate; it says that a constraint which collapses the support onto a lower-dimensional set puts the law outside the family for which the bound is finite, so the resulting cost is not a large finite number to be traded against something else but an infinite one. Chapter 11 takes this up in the case that matters, where a coarse-graining is expressed by tying groups of coordinates to a common value: the tied law is supported on the range of an aggregation matrix, hence exactly of the form excluded here, and the admissible finite operation constrains the recognition mean instead, with a cost that the same chapter computes exactly. The forward reference is deliberate; the present chapter supplies the reason, not the application.

5.3 The exact evidence identity

Under Hypothesis 5.3 and for $o \in \mathcal{O}_{\theta, X}^{\text{reg}}$, define the evidence lower bound at the selected (o, X) by

$$\mathcal{L}(Q_X; X) = \mathbb{E}_{Q_X} [\log p_\theta(o, Y \mid X) - \log q_X(Y \mid o)]. \quad (5.10)$$

The notation suppresses the fixed observation o and the generative parameter θ ; the parameter subscript is restored as $\mathcal{L}_\theta$ whenever two parameter values are compared. The variational free energy is the opposite sign,

$$\mathcal{F}[Q_X; X, o] = -\mathcal{L}(Q_X; X). \quad (5.11)$$

Theorem 5.5 (Exact evidence identity). Under Hypothesis 5.3, for $P_{\theta,X}^O$ -almost every observation o (equivalently, for the selected $o \in O_{\theta,X}^{\text{reg}}$),

$$\log p_{\theta}(o \mid X) = \mathcal{L}(Q_X; X) + D_{\text{KL}}(Q_X(\cdot \mid o) \parallel P_{\theta}(\cdot \mid o, X)). \quad (5.12)$$

[ESTABLISHED]

Proof. For $B \in \mathcal{Y}_D$, the regular-observation construction (3.14), together with (H1) and (H2), gives the dominated conditional posterior

$$P_{\theta}(B \mid o, X) = \frac{1}{p_{\theta}(o \mid X)} \int_B p_{\theta}(o, y \mid X) \nu_D^Y(dy), \quad \frac{dP_{\theta}(\cdot \mid o, X)}{d\nu_D^Y}(y) = \frac{p_{\theta}(o, y \mid X)}{p_{\theta}(o \mid X)}, \quad (5.13)$$

which is normalized because the denominator is the evidence. Writing $q_X(y \mid o) = dQ_X(\cdot \mid o)/d\nu_D^Y$ and using (H3) with the Radon–Nikodym chain rule gives, Q_X -almost surely,

$$\frac{dQ_X(\cdot \mid o)}{dP_{\theta}(\cdot \mid o, X)}(y) = \frac{q_X(y \mid o)p_{\theta}(o \mid X)}{p_{\theta}(o, y \mid X)}. \quad (5.14)$$

Hypothesis (H4) permits integration of its logarithm term by term, so

$$\begin{aligned} D_{\text{KL}}(Q_X(\cdot \mid o) \parallel P_{\theta}(\cdot \mid o, X)) &= \int_{\mathcal{Y}_D} Q_X(dy \mid o) \log \frac{dQ_X(\cdot \mid o)}{dP_{\theta}(\cdot \mid o, X)}(y) \\ &= \mathbb{E}_{Q_X} [\log q_X(Y \mid o) - \log p_{\theta}(o, Y \mid X)] + \log p_{\theta}(o \mid X) \\ &= -\mathcal{L}(Q_X; X) + \log p_{\theta}(o \mid X). \end{aligned} \quad (5.15)$$

Rearranging gives (5.12). □

The almost-everywhere qualifier is part of the theorem, not a regularity detail suppressed by notation. If $\tilde{p}_{\theta}$ is another joint-density version, Fubini's theorem gives equality of the two slices for ν_D^Y -almost every y outside a ν_D^O -null set of observations, and the two evidence and posterior versions therefore agree outside a $P_{\theta,X}^O$ -null set. At an exceptional observation, the values of (5.13) and (5.10) can be changed without changing the joint measure. The theorem consequently makes no version-independent claim there.

Every corollary and coordinate statement below inherits this regular-observation qualification. For a comparison of finitely many parameter values, the observation must lie in the intersection of their marginal-full regular sets. An optimization over an uncountable parameter space requires either a common full-measure set proved for that family or jointly measurable pointwise density and conditional versions declared as part of the parameterized model; separate almost-everywhere versions for each parameter do not by themselves define a pointwise objective on the whole parameter space.

Corollary 5.6 (Lower bound and exact equality condition). Under Hypothesis 5.3, $\mathcal{L}(Q_X; X) \leq \log p_\theta(o | X)$, and equality holds if and only if

$$Q_X(\cdot | o) = P_\theta(\cdot | o, X) \quad \text{as probability measures on } (Y_D, \mathcal{Y}_D). \quad (5.16)$$

Equality of all coordinate marginals is not sufficient. [ESTABLISHED]

Proof. The inequality is nonnegativity of relative entropy. For the equality case, strict convexity of $u \mapsto u \log u$ gives that the divergence vanishes exactly when the Radon–Nikodym derivative (5.14) equals one $P_\theta(\cdot | o, X)$ -almost surely, which is equality of the two measures [20, 40]. Insufficiency of marginal agreement is Proposition 3.12, whose Gaussian witnesses have the same coordinate marginals but are distinct measures. $\square$

5.4 The general restriction principle

Fix (θ, X) , select $o \in \mathcal{O}_{\theta, X}^{\text{reg}}$ with $0 < p_\theta(o | X) < \infty$, and write $P^\star = P_\theta(\cdot | o, X)$. For every recognition law Q in the domain of Theorem 5.5,

$$\log p_\theta(o | X) = \mathcal{L}(Q; X) + D_{\text{KL}}(Q \| P^\star). \quad (5.17)$$

Proposition 5.7 (Restriction principle). Let $\mathcal{Q}' \subseteq \mathcal{Q}$ be sets of recognition laws whose members satisfy Hypothesis 5.3. Then

$$\sup_{Q \in \mathcal{Q}'} \mathcal{L}(Q; X) = \log p_\theta(o | X) - \inf_{Q \in \mathcal{Q}'} D_{\text{KL}}(Q \| P^\star), \quad (5.18)$$

and

$$\sup_{\mathcal{Q}} \mathcal{L} - \sup_{\mathcal{Q}'} \mathcal{L} = \inf_{\mathcal{Q}'} D_{\text{KL}}(Q \| P^\star) - \inf_{\mathcal{Q}} D_{\text{KL}}(Q \| P^\star) \geq 0. \quad (5.19)$$

The identities do not require attainment of the suprema or infima. [ESTABLISHED]

Proof. Equation (5.17) holds pointwise in Q , and its left side does not depend on Q . Taking suprema converts the constant minus a divergence into the constant minus its infimum. Subtracting the two instances gives (5.19); set inclusion gives the inequality. $\square$

Constancy of the evidence is load bearing: the generative kernel is fixed once (θ, X) is fixed and does not read Q . If a proposed model changes with the recognition law, Proposition 4.3 and Proposition 4.4 show that no common evidence comparison follows. Thus every later closed-form “restriction cost” inherits its meaning from the fixed-target hypothesis.

Infinite-cost restrictions. Equation (5.19) is finite only on its declared domain. If each $Q \in \mathcal{Q}'$ has a measurable set S_Q with

$$Q(S_Q) = 1, \quad P^\star(S_Q) = 0, \quad (5.20)$$

then $Q \not\ll P^*$ and $D_{\text{KL}}(Q \| P^*) = +\infty$. A proper closed support subset alone is insufficient: the uniform law on $[0, \frac{1}{2}]$ is absolutely continuous with respect to the uniform law on $[0, 1]$ and has divergence $\log 2$. The null-set condition, not proper closedness, is the obstruction. [ESTABLISHED]

Orientation, separation, and attainment. The infimum is a reverse information projection because Q is the first argument of the divergence. Excluding P^* does not ensure a positive or attained optimum: the family $\{\mathcal{N}(\mu, 1) : \mu > 0\}$ excludes $\mathcal{N}(0, 1)$, but its gaps $\mu^2/2$ have unattained infimum zero. A positive optimized gap requires $\inf_{Q \in \mathcal{Q}} D_{\text{KL}}(Q \| P^*) > 0$. Nor does uniqueness follow from convexity of a product family, because mixtures of product laws need not be products. The Gaussian restrictions of Chapter 11 instead prove convexity and coercivity in the moment parameters they optimize. [ESTABLISHED]

Corollary 5.8 (Independent channel-frame invariance). Let $T(o, y) = (o, Ry)$ be an independent belief–model frame re-expression as in Chapter 4, where $R = \bigoplus_{a,i} (R_{a,i}^b \oplus R_{a,i}^m)$ is the assembled represented bijection of Y_D , and transport the recognition law as $Q'_{X'}(\cdot | o) = R_{\#} Q_X(\cdot | o)$. Then

$$D_{\text{KL}}(Q'_{X'} \| P_{\theta'}(\cdot | o, X')) = D_{\text{KL}}(Q_X \| P_{\theta}(\cdot | o, X)), \quad \mathcal{L}'(Q'_{X'}; X') = \mathcal{L}(Q_X; X). \quad (5.21)$$

[ESTABLISHED]

Proof. By the joint pushforward identity of Chapter 4 and the fact that T acts as the identity on the observation coordinate, the conditional posterior transforms as $P_{\theta'}(\cdot | o, X') = R_{\#} P_{\theta}(\cdot | o, X)$. Relative entropy is invariant under a common bimeasurable bijection, because the Radon–Nikodym derivative of the pushforwards is the derivative of the originals composed with R^{-1} and the integral is a change of variables. This gives the first identity. The evidence is unchanged by the same chapter’s invariance result, so subtracting the first identity from (5.12) gives the second. $\square$

Corollary 5.8 is the statement that the bound is a bound on a frame-independent quantity, computed by a frame-independent procedure. It is used in Chapter 7 to rule out selection criteria that change under reframing, and it is the reason that any invariant offered later is stated in generalized-eigenvalue form rather than in terms of absolute eigenvalues.

5.5 The factorwise decomposition

Hypothesis 5.9 (Factorwise log-integrability). Restore the design index a suppressed in Chapter 4 and let $\mathcal{R} \subseteq V$ be the root set. To display the factor expectations separately, assume in this section that the absolute logarithm of every component density in the finite directed factorization is Q_X -integrable. [HYPOTHESIS] This is strictly stronger than (H4), which controls only the complete log joint, and it is what excludes cancellation between individually undefined extended-real summands; a decomposition displayed without it can be a formal rearrangement of divergent quantities.

Proposition 5.10 (Exact factorwise decomposition). Under Hypotheses 5.3 and 5.9, substituting the normalized directed density (4.7) into (5.11) yields exactly one decomposition,

$$\begin{aligned}
 \mathcal{F}[Q_X; X, o] = & \sum_{a=1}^M \sum_{r \in \mathcal{R}} \mathbb{E}_{Q_X} \left[-\log p_{\theta, r, a}^m(m_{r, a} \mid X) \right] \\
 & + \sum_{a=1}^M \sum_{r \in \mathcal{R}} \mathbb{E}_{Q_X} \left[-\log p_{\theta, r, a}^k(k_{r, a} \mid m_{r, a}, X) \right] \\
 & + \sum_{a=1}^M \sum_{i \in V \setminus \mathcal{R}} \mathbb{E}_{Q_X} \left[-\log p_{\theta, i, a}^m(m_{i, a} \mid m_{\text{pa}(i), a}, X) \right] \\
 & + \sum_{a=1}^M \sum_{i \in V \setminus \mathcal{R}} \mathbb{E}_{Q_X} \left[-\log p_{\theta, i, a}^k(k_{i, a} \mid k_{\text{pa}(i), a}, m_{i, a}, X) \right] \\
 & + \sum_{a=1}^M \sum_{i \in V} \mathbb{E}_{Q_X} \left[-\log p_{\theta, i, a}^o(o_{i, a} \mid k_{i, a}, m_{i, a}, X) \right] \\
 & + \mathbb{E}_{Q_X} [\log q_X(Y \mid o)].
 \end{aligned} \tag{5.22}$$

The generative sectors contain $3MN$ factors in total, and the final line is one joint recognition log-density, equivalently the negative joint differential entropy relative to ν_D^Y when that entropy is finite. It is not a sum of marginal entropies. **[ESTABLISHED]**

Proof. Write $\ell_{\theta, a}$ for the one-design-point log density obtained by taking the logarithm of (4.6), which is legitimate factor by factor under Hypothesis 5.9, and note that (4.7) gives $\log p_{\theta}(o, Y \mid X) = \sum_{a=1}^M \ell_{\theta, a}(Y_a; o_a, X)$. Taking $\mathbb{E}_{Q_X}$ of the negative of this and adding $\mathbb{E}_{Q_X}[\log q_X]$ reproduces the right side of (5.22) term by term. Denoting that right side by $\mathcal{F}^{\text{fact}}$, the residual is

$$\begin{aligned}
 \mathcal{F}[Q_X; X, o] - \mathcal{F}^{\text{fact}} = & \mathbb{E}_{Q_X} [\log q_X(Y \mid o) - \log p_{\theta}(o, Y \mid X)] \\
 & - \left\{ \mathbb{E}_{Q_X} [\log q_X(Y \mid o)] - \mathbb{E}_{Q_X} \left[\sum_{a=1}^M \ell_{\theta, a}(Y_a; o_a, X) \right] \right\} = 0,
 \end{aligned} \tag{5.23}$$

where the same joint recognition term appears once on each side and cancels as a whole, and no step replaces it by marginal entropies. For the count, the root sector contributes $2M|\mathcal{R}|$ latent factors, the nonroot sector $2M(N - |\mathcal{R}|)$, and the observation sector MN , summing to $2MN + MN = 3MN$. Root state bridges replace nonroot state transitions at roots; they are not additional factors. $\square$

Two readings of (5.22) must be resisted. The first is the substitution warned against in Proposition 5.2: the last line is a single number determined by the joint law, and replacing it by $\sum_b \mathbb{E}_{Q_X} [\log q_b]$ inflates the bound by the total correlation. The second is that the decomposition licenses independent optimization of its terms. Every expectation is taken under the same joint Q_X , so the terms are coupled through their common measure; the decomposition is an exact rewriting of one functional, not a decoupling of it into separately minimizable pieces.

5.6 Exact coordinates and accepted proposals

All coordinate statements below refer to the same normalized family P_θ of Chapter 4. Because Y_D is a finite product of Euclidean and discrete spaces it is a standard Borel space, so regular conditional distributions of the posterior exist and the relative entropy chain rule is available [38].

Proposition 5.11 (Exact E-coordinate with finite-value hypotheses). Let $\mathcal{B}$ be a block of latent coordinates with complement $-\mathcal{B}$, put

$$P := P_\theta(\cdot \mid o, X), \quad P_- := P_{\theta, -\mathcal{B}}(\cdot \mid o, X), \quad R := Q_{X, -\mathcal{B}}(\cdot \mid o),$$

and assume

$$R \ll P_-, \quad D_{\text{KL}}(R \parallel P_-) < \infty. \quad (5.24)$$

Choose a regular conditional version $P(dY_{\mathcal{B}} \mid Y_{-\mathcal{B}})$, which is defined P_- -almost everywhere and hence R -almost everywhere. Then among *all* probability laws with complement marginal R , the extended-real divergence from P has the unique minimum $D_{\text{KL}}(R \parallel P_-)$, attained by

$$Q_X^{\text{new}}(dY_{\mathcal{B}}, dY_{-\mathcal{B}} \mid o) = P_\theta(dY_{\mathcal{B}} \mid Y_{-\mathcal{B}}, o, X) R(dY_{-\mathcal{B}}). \quad (5.25)$$

If, in addition, this new law satisfies the log-integrability condition

$$\mathbb{E}_{Q_X^{\text{new}}} [|\log p_\theta(o, Y \mid X)| + |\log q_X^{\text{new}}(Y \mid o)|] < \infty, \quad (5.26)$$

then it satisfies (H3)–(H4) and maximizes $\mathcal{L}$ among the laws with complement marginal R that satisfy Hypothesis 5.3. Neither absolute continuity in (5.24) nor finite complement relative entropy by itself implies (5.26). **[ESTABLISHED]**

Proof. For any probability law Q_X with complement marginal R , choose a regular conditional law $Q_{X, \mathcal{B} \mid -\mathcal{B}}$. The relative-entropy chain rule on a standard Borel product holds in the extended nonnegative reals, including when one or both sides are $+\infty$, and gives

$$D_{\text{KL}}(Q_X \parallel P) = D_{\text{KL}}(R \parallel P_-) + \mathbb{E}_R [D_{\text{KL}}(Q_{X, \mathcal{B} \mid -\mathcal{B}} \parallel P_\theta(\cdot \mid Y_{-\mathcal{B}}, o, X))]. \quad (5.27)$$

The first term is the finite constant in (5.24). The second is an integral of nonnegative extended-real quantities and is zero exactly when the two conditional laws agree for R -almost every $Y_{-\mathcal{B}}$. This gives (5.25), the minimum value, and uniqueness as a joint measure. Moreover, if $r = dR/dP_-$, direct disintegration gives

$$\frac{dQ_X^{\text{new}}}{dP} = r(Y_{-\mathcal{B}}) \quad P\text{-almost everywhere,}$$

so $Q_X^{\text{new}} \ll P$ and its relative entropy is exactly $\int \log r dR = D_{\text{KL}}(R \parallel P_-)$. Thus (H3) holds. Under the separate assumption (5.26), Theorem 5.5 applies both to the new law and to every compared admissible law, and converts minimization of the divergence into maximization of the bound. $\square$

The two extra obligations cannot be recovered from complement absolute continuity. First take P_- to be standard Gaussian and R standard Cauchy on $\mathbb{R}$. Then $R \ll P_-$, but $\log(dR/dP_-)(x)$ contains $x^2/2$ up to logarithmic and constant terms, and the Cauchy second moment is infinite; hence $D_{\text{KL}}(R||P_-) = +\infty$. Second, finite relative entropy still does not imply (H4). On the countable space $\{2, 3, \dots\}$, let

$$P_-(\{n\}) = R(\{n\}) = \frac{c}{n(\log n)^2},$$

where c normalizes the series. Then $D_{\text{KL}}(R||P_-) = 0$, but for all sufficiently large n , $|\log P_-(\{n\})| \geq \frac{1}{2} \log n$, and therefore

$$\sum_{n \geq 2} P_-(\{n\}) |\log P_-(\{n\})| \geq \frac{c}{2} \sum_{n \text{ large}} \frac{1}{n \log n} = +\infty.$$

With a trivial observation and conditional block this is precisely a $Q_X^{\text{new}} = P$ of zero relative entropy for which the separate log expectations in (H4) are not integrable. Equation (5.26) is therefore an independent hypothesis, not a consequence of the KL chain rule.

The exact full E-step is the special case $\mathcal{B}$ equal to all coordinates, giving $Q_X^{\text{new}} = P_\theta(\cdot | o, X)$ and, when (H4) holds for that posterior, a zero gap by Corollary 5.6. A product-family coordinate is a different object entirely: it optimizes within a restricted family, is not given by (5.25), and leaves a gap that Chapter 11 quantifies.

Proposition 5.12 (Exact M-coordinate). Hold Q_X fixed. Suppose the parameterized family supplies the common regular-observation set or the jointly measurable pointwise versions required above, and suppose Hypothesis 5.3 holds for Q_X at every candidate parameter being compared. Specifically,

$$\mathbb{E}_{Q_X} [|\log q_X(Y | o)|] < \infty, \quad \mathbb{E}_{Q_X} [|\log p_\vartheta(o, Y | X)|] < \infty$$

for every such ϑ . If a maximizer exists in the declared parameter space, then any

$$\theta^{\text{new}} \in \operatorname{argmax}_{\vartheta} \mathbb{E}_{Q_X} [\log p_\vartheta(o, Y | X)] \quad (5.28)$$

also maximizes $\mathcal{L}_\vartheta(Q_X; X)$ over the same space, because $\mathbb{E}_{Q_X} [\log q_X(Y | o)]$ does not depend on ϑ . [ESTABLISHED]

The finiteness assumption is load-bearing. On $\{2, 3, \dots\}$, let

$$q_n = \frac{c}{n(\log n)^2}, \quad h_n = \mathbf{1}_{\{n=2\}} - q_2, \quad r_n = 1 + \varepsilon h_n,$$

where c normalizes q and $0 < \varepsilon < 1/q_2$. Then r is bounded, strictly positive, nonconstant, and $\sum_n q_n r_n = 1$. With a trivial observation, set $p_0 = q$ and $p_1 = qr$. The entropy of q is infinite, so

$$\mathbb{E}_q[\log p_0] = \mathbb{E}_q[\log p_1] = -\infty.$$

An extended-real argmax of the expected complete log-likelihood therefore treats both candidates as maximizers. The separately integrated ELBO of (5.10) is undefined because (H4) fails, while its canonical relative-log extension distinguishes them:

$$\mathbb{E}_q \left[\log \frac{p_0}{q} \right] = 0, \quad \mathbb{E}_q \left[\log \frac{p_1}{q} \right] = \mathbb{E}_q [\log r] = -D_{\text{KL}}(q \| p_1) < 0.$$

Thus cancellation of a parameter-independent infinite entropy does not justify the M-coordinate claim.

The existence proviso in (5.28) is a hypothesis and not a formality. **[HYPOTHESIS]** The covariance parameters of Chapter 4 range over an open cone, and the expected complete log-likelihood is unbounded above along sequences that drive a conditional covariance to the boundary whenever the corresponding expected residual second moment is singular. In that situation the supremum is not attained and (5.28) defines nothing. Under a nondegenerate Q_X the expected residual second moments are positive definite and the maximizer exists, so the hypothesis is mild in the Gaussian realization; it is stated because it is the hypothesis, not because it usually fails.

Definition 5.13 (Generalized M-coordinate acceptance). A generalized M-coordinate need not attain the maximum, but it must pass an acceptance condition:

$$\mathcal{L}_{\theta^{\text{new}}}(Q_X; X) \geq \mathcal{L}_{\theta^{\text{old}}}(Q_X; X). \quad (5.29)$$

[DEFINITION] The condition is a declaration about what counts as a step; it is checkable, and it refers to the same functional (5.10) that the identity bounds. It is meaningful only when Hypothesis 5.3, including (H4), holds at both the old and new parameters for this fixed Q_X .

Proposition 5.14 (Evidence monotonicity). Suppose Hypothesis 5.3 holds for the fixed Q_X at both θ^{old} and θ^{new} ; thus o lies in both regular-observation sets, or the corresponding pointwise versions have been declared as model data, and (H4) holds at both parameter values. If the E-coordinate is exact at the old parameter, so that $Q_X = P_{\theta^{\text{old}}}(\cdot \mid o, X)$, and the M-coordinate satisfies (5.29), then

$$\log p_{\theta^{\text{new}}}(o \mid X) \geq \mathcal{L}_{\theta^{\text{new}}}(Q_X; X) \geq \mathcal{L}_{\theta^{\text{old}}}(Q_X; X) = \log p_{\theta^{\text{old}}}(o \mid X). \quad (5.30)$$

[ESTABLISHED] The first inequality is Corollary 5.6 at θ^{new} , the second is (5.29), and the final equality is Corollary 5.6 at θ^{old} with a zero gap. This is the classical conclusion [22, 51], reproduced here to fix what it requires: an exact E-coordinate at the old parameter and an accepted M-coordinate, both referred to the same joint.

A natural-gradient direction and an ordinary optimizer direction are proposals, not accepted coordinates. The distinction is not pedantic, and the following states exactly what is and is not guaranteed.

Proposition 5.15 (Finite natural-gradient steps need not be monotone). Let $G \succ 0$ be a preconditioner, write $\nabla \mathcal{L}$ for the gradient of $\vartheta \mapsto \mathcal{L}_\vartheta(Q_X; X)$, and consider the step $\vartheta^+ = \vartheta + \alpha G^{-1} \nabla \mathcal{L}$. Define $\varphi(\alpha) = \mathcal{L}_{\vartheta^+}(Q_X; X)$. At a nonstationary point the direction is an ascent direction, since

$$\varphi'(0) = \nabla \mathcal{L}^\top G^{-1} \nabla \mathcal{L} > 0. \quad (5.31)$$

This is an infinitesimal statement and does not extend to finite α . Suppose $\mathcal{L}_\xi = \mathcal{L}^* - \frac{1}{2}(\xi - \theta^*)^\top H(\xi - \theta^*)$ with $H \succ 0$ throughout a neighborhood $\mathcal{U}$, let $d_{\max}$ be the largest generalized eigenvalue of the pair (H, G) , and let v be a corresponding generalized eigenvector normalized so that $v^\top G v = 1$. Then at any $\vartheta = \theta^* + cv \in \mathcal{U}$ with $c \neq 0$, for every step whose endpoint ϑ^+ also lies in $\mathcal{U}$,

$$\mathcal{L}_{\vartheta^+} - \mathcal{L}_\vartheta = \frac{1}{2} d_{\max} c^2 \left[1 - (1 - \alpha d_{\max})^2 \right] < 0 \quad \text{whenever} \quad \alpha > \frac{2}{d_{\max}}. \quad (5.32)$$

[ESTABLISHED]

Proof. Equation (5.31) is the chain rule with G^{-1} positive definite. For the second part write $e = \vartheta - \theta^*$, so $\nabla \mathcal{L} = -He$ and $e^+ = (I - \alpha G^{-1} H)e$. With $Hv = d_{\max} Gv$, $v^\top G v = 1$, and $e = cv$ one gets $e^+ = (1 - \alpha d_{\max})cv$ and $\mathcal{L}_\vartheta = \mathcal{L}^* - \frac{1}{2} d_{\max} c^2$. The endpoint hypothesis permits the same quadratic formula to be evaluated at ϑ^+ . Substituting gives (5.32), and the bracket is negative exactly when $|1 - \alpha d_{\max}| > 1$, that is when $\alpha d_{\max} > 2$. $\square$

The content of Proposition 5.15 is that monotonicity at finite step size is an additional requirement, supplied by a line search satisfying a sufficient-increase condition, by a trust region with an acceptance test, or by any other rule that verifies (5.29) on the same functional. Absent such a test the proposal is not entitled to (5.30), and the failure is not exotic: it occurs for a quadratic objective at any step length exceeding twice the reciprocal of the largest preconditioned curvature. Nor is the difficulty removed by improving the preconditioner, since (5.32) is stated in the preconditioned spectrum. Separately, optimizing a different objective, for instance a population energy that is not (5.10), supplies no guarantee for the fixed-joint evidence at all: two functionals with different stationary points can move in opposite directions under the same update, and only a bound with matched stationarity would transfer the conclusion.

5.7 What this chapter does not settle

Three limits of the development are recorded so that later chapters do not inherit more than was proved. Existence of a maximizer of (5.10) over a restricted recognition family is not established here and is not implied by Theorem 5.5; the theorem is an identity at a fixed pair of measures and says nothing about the attainability of any optimum. Tightness of the bound for any particular family is likewise not claimed, and the gap for the restrictions this document uses is computed in Chapter 11 rather than bounded here.

The third limit is a genuine open problem.

Open Problem 5.16 (Convergence of the alternating scheme). Whether the alternating scheme built from block E-coordinates (5.25) and accepted M-coordinates (5.29) converges to a stationary point of $\mathcal{L}$ under the present hypotheses is not settled. [OPEN] What (5.30) gives is a nondecreasing sequence of evidence values, which converges when the evidence is bounded above; convergence of the parameter sequence is a separate question requiring regularity conditions of the type identified for the classical algorithm [71], and global optimality does not follow even when they hold. Closing the problem in the present setting requires either verifying those conditions for the declared parameter space, which is an open cone so that compactness of the relevant level sets must be checked rather than assumed, or exhibiting a counterexample within the declared family. Nothing in Chapter 7 or Chapter 8 depends on the answer, because those chapters analyze the structure of the objective and its transformation under aggregation rather than the trajectory of any optimizer.

Chapter 6

Belief Fibers and Exponential Families

The objects carried by the belief and model associated bundles of Chapter 2 are normalized laws, while generative fibers carry kernels. Coordinates, smoothness, domination, and exponential-family structure are additional choices. This chapter states that hierarchy before selecting the finite-dimensional exponential subclass used by the general coarse theory.

6.1 Law fibers before coordinates

Definition 6.1 (Law and kernel fibers). For a standard Borel space $(Z, \mathcal{Z})$, let

$$\mathcal{B}(Z) \subseteq \mathcal{P}(Z) \tag{6.1}$$

be a declared collection of normalized probability laws. For standard Borel spaces X, Y , let

$$\mathcal{K}(X, Y) \subseteq \text{Markov}(X, Y) \tag{6.2}$$

be a declared collection of normalized Markov kernels. A law is not a kernel, and neither declaration supplies a common density, affine chart, manifold, Fisher metric, or convex cone. **[DEFINITION]**

The belief and model bundles may use different law fibers. Cross-bundle maps $\Phi : \mathcal{E}_b \rightarrow \mathcal{E}_m$ and $\tilde{\Phi} : \mathcal{E}_m \rightarrow \mathcal{E}_b$ act between those associated bundles and are not iterations on either fiber. To prevent collision with that reserved notation, a generic endomorphism of a parameter set will be written Ψ .

Definition 6.2 (Partial realization). A parametrized model consists of a parameter space Θ , a domain $\Theta_{\text{law}} \subseteq \Theta$, and a map

$$\tau : \Theta_{\text{law}} \longrightarrow \mathcal{B}(Z), \quad \vartheta \longmapsto Q_{\vartheta}. \tag{6.3}$$

Outside Θ_{law} , a parameter may still denote an algebraic operator, but it does not denote a law through τ . **[DEFINITION]**

A bimeasurable gauge action $g : Z \rightarrow Z$ acts on laws by

$$Q \longmapsto g_{\#}Q, \tag{6.4}$$

and on kernels by

$$(g_{\text{out}} \cdot K \cdot g_{\text{in}}^{-1})(x, B) = K(g_{\text{in}}^{-1}x, g_{\text{out}}^{-1}B). \quad (6.5)$$

[DEFINITION] These formulas type the action; closure of a declared fiber under it is a separate hypothesis.

6.2 Dominated and smooth tiers

Domination is not automatic. An uncountable family of Dirac laws on an uncountable standard Borel space admits no common sigma-finite dominating measure: any such measure would have to assign positive mass to uncountably many disjoint atoms, which a sigma-finite measure cannot do. **[ESTABLISHED]** Hence density charts cannot be inferred from the general law fiber.

The differential tier used below is a finite-dimensional smooth parametrized-measure model. Differentiability in quadratic mean, square-integrable scores, and the interchange of differentiation and integration are hypotheses. The Fisher form is a Riemannian metric only after nondegeneracy is proved. Parametrized-measure models supply an intrinsic framework for this tier [7]. Stratified and infinite-dimensional tiers require their own tangent and integration theories; no universal global Fisher metric or natural chart is claimed. **[HYPOTHESIS][NOT-CLAIMED]**

6.3 Regular finite-dimensional exponential families

Definition 6.3 (Canonical family). Let ν be a nonzero sigma-finite measure on $(Z, \mathcal{Z})$, let $T : Z \rightarrow T$ be measurable into a finite-dimensional inner-product space, and define

$$A(\theta) = \log \int_Z e^{\langle \theta, T(z) \rangle} \nu(dz), \quad N = \{\theta : A(\theta) < \infty\}. \quad (6.6)$$

For $\theta \in N$,

$$q_\theta(z) = \exp\{\langle \theta, T(z) \rangle - A(\theta)\}, \quad Q_\theta(dz) = q_\theta(z)\nu(dz). \quad (6.7)$$

Nothing is asserted for $\theta \notin N$. **[DEFINITION]**

Proposition 6.4 (Convexity and lower semicontinuity). A is convex and lower semicontinuous as an extended-real function, and N is convex. **[ESTABLISHED]**

Proof. For $0 < t < 1$, Hölder's inequality gives

$$\int e^{\langle t\theta + (1-t)\vartheta, T \rangle} d\nu \leq \left(\int e^{\langle \theta, T \rangle} d\nu \right)^t \left(\int e^{\langle \vartheta, T \rangle} d\nu \right)^{1-t}, \quad (6.8)$$

and taking logarithms proves convexity and convexity of the effective domain. Fatou's lemma applied to a convergent parameter sequence proves lower semicontinuity. $\square$

Proposition 6.5 (Cumulants and Fisher information). At every interior parameter where local exponential domination permits differentiation under the integral,

$$\nabla A(\theta) = \mathbb{E}_\theta[T], \quad \nabla^2 A(\theta) = \text{Cov}_\theta(T). \quad (6.9)$$

The Hessian is the Fisher information in the natural chart. **[ESTABLISHED]**

Proof. Differentiate the normalizer once and twice. The score is $T - \mathbb{E}_\theta T$, so its second moment is the covariance in (6.9). Local exponential domination follows by enclosing the parameter in a compact subset of the interior and applying the elementary bound

$$|x|^r \leq C_{r,\varepsilon} (e^{\varepsilon x} + e^{-\varepsilon x}) \quad (6.10)$$

to each coordinate of T . $\square$

Hypothesis 6.6 (Regularity and minimality). The family is regular, so N is open, and minimal, so no nonzero $a \in T$ makes $\langle a, T \rangle$ constant ν -almost everywhere. Regularity excludes natural-domain boundary points; minimality excludes affine redundancy. **[HYPOTHESIS]**

Proposition 6.7 (Minimality and nondegenerate information). Under the preceding hypothesis, $\nabla^2 A(\theta)$ is positive definite, A is strictly convex, and $\theta \mapsto Q_\theta$ is injective. Failure of minimality makes the Hessian singular. **[ESTABLISHED]**

Proof. For every a ,

$$a^\top \nabla^2 A(\theta) a = \text{Var}_{Q_\theta}(\langle a, T \rangle).$$

The variance vanishes exactly when the displayed statistic is almost surely constant. Since $q_\theta > 0$, Q_θ and ν have the same null sets, so minimality gives positive definiteness. Strict convexity follows. Equality of two densities makes $\langle \theta - \vartheta, T \rangle$ constant almost everywhere, and minimality then gives $\theta = \vartheta$. $\square$

6.4 Operator and probability layers

Definition 6.8 (Operator layer). Let $\mathcal{C} \subseteq T$ be a separately declared closed convex cone preserved by the algebraic maps under study. Its probability layer is $\mathcal{C} \cap N$. The cone is chosen for algebraic closure, not normalization, and no such cone is asserted for an arbitrary law fiber. **[DEFINITION]**

Proposition 6.9 (A natural-domain boundary carries no law). Under regularity, if $\theta^* \in \partial N$, then

$$\int e^{\langle \theta^*, T(z) \rangle} \nu(dz) = +\infty. \quad (6.11)$$

For $\theta_\tau = (1 - \tau)\theta_0 + \tau\theta^*$, with $\theta_0 \in N$ and $w = \theta^* - \theta_0$,

$$A(\theta_\tau) \rightarrow +\infty, \quad \langle w, \nabla A(\theta_\tau) \rangle \rightarrow +\infty, \quad \int_0^1 \text{Var}_{Q_{\theta_\tau}}(\langle w, T \rangle) d\tau = +\infty. \quad (6.12)$$

[ESTABLISHED]

Proof. Regularity puts θ^* outside N , proving (6.11). Convexity puts every θ_τ , $\tau < 1$, in N , while lower semicontinuity forces the first limit. For $g(\tau) = A(\theta_\tau)$, convexity makes g' nondecreasing. If g' were bounded above, then g would remain bounded, a contradiction. The identities

$$g' = \langle w, \nabla A \rangle, \quad g'' = \text{Var}_{Q_{\theta_\tau}}(\langle w, T \rangle)$$

and integration of g'' prove the remaining assertions. $\square$

Thus the declared representation supplies no finite log partition, mean parameter, Fisher metric, relative entropy from the nonexistent endpoint law, or instance of the exact ELBO at θ^* . [ESTABLISHED] This does not rule out a quotient, pin, new reference measure, or boundary stratum; each is a different declared object. [NOT-CLAIMED]

Proposition 6.10 (An operator fixed point need not be a law). Let $\Psi : \mathcal{C} \rightarrow \mathcal{C}$ be continuous. Every iterate may lie in $\mathcal{C} \cap N$ while the unique fixed point lies on ∂N . For $Z = \mathbb{R}$, $T(z) = -z^2/2$, and Lebesgue ν ,

$$A(\theta) = -\frac{1}{2} \log \theta + \frac{1}{2} \log 2\pi, \quad N = (0, \infty), \quad \mathcal{C} = [0, \infty), \quad (6.13)$$

the map $\Psi(\theta) = \theta/2$ has this property. [ESTABLISHED]

Proof. Starting at one gives $\theta_\ell = 2^{-\ell} > 0$, while $\theta_\ell \rightarrow 0$, the unique fixed point, and $A(\theta_\ell) \rightarrow +\infty$. The constant kernel at zero has infinite Lebesgue integral. $\square$

6.5 Declared repairs

Definition 6.11 (Pinning). Choose $\theta_m \in N$ and replace $\mathcal{C}$ by $\theta_m + \mathcal{C}$; the pin does not flow. [DEFINITION]

Proposition 6.12 (Pinning under a recession condition). If $\theta_m \in \text{int } N$ and every $\eta \in \mathcal{C}$ is a recession direction of N , then $\theta_m + \mathcal{C} \subseteq \text{int } N$. [ESTABLISHED]

Proof. Translate an open ball around θ_m by η . The recession hypothesis keeps the translated ball inside N , so its center is interior. $\square$

The pin changes the family, may change the fixed set, and introduces a parameter whose removal need not commute with the long-scale limit. Whether the mass sector is relevant, marginal, or irrelevant is an open spectral question for the declared scale map. [OPEN]

Definition 6.13 (Quotienting). Suppose $Z = \mathbb{R}^n$, ν is Lebesgue measure, and the energy is invariant along a nonzero linear subspace $\mathcal{V}$. Declare the new sample space $\mathcal{W} = \mathcal{V}^\perp$, its Lebesgue measure, and the restricted statistic. [DEFINITION]

Proposition 6.14 (Quotient normalization is not full-space normalization). Writing $z = v + w$ in the orthogonal splitting,

$$\int_{\mathbb{R}^n} e^{\langle \theta, T(z) \rangle} dz = \lambda_{\mathcal{V}}(\mathcal{V}) \int_{\mathcal{W}} e^{\langle \theta, T(w) \rangle} dw = +\infty. \quad (6.14)$$

The quotient law is normalized exactly when the second factor is finite. [ESTABLISHED]

Proof. The change of variables has unit Jacobian and the integrand is independent of v . Tonelli's theorem gives (6.14); the first factor is infinite. $\square$

Definition 6.15 (Retaining a mass sector). Enlarge the state by a declared coercive parameter that remains in the natural domain under the scale map. This is neither pinning nor quotienting: the mass is a running coordinate and its scaling law must be stated. [DEFINITION]

A Moore–Penrose inverse and pseudodeterminant inserted into an interior full-space density formula do not repair normalization. They compute the finite quotient expression, whose dimension and reference measure differ, while the original integral remains infinite by (6.14). [ESTABLISHED]

6.6 Relative entropy and affine information projection

For a regular minimal family, let $\tau_\theta = \nabla A(\theta)$ and let A^* be the convex conjugate. Direct expectation of the log-density ratio gives

$$D_{\text{KL}}(Q_\vartheta \| Q_\theta) = A(\theta) - A(\vartheta) - \langle \nabla A(\vartheta), \theta - \vartheta \rangle = D_A(\theta, \vartheta), \quad (6.15)$$

and Legendre duality gives

$$D_{\text{KL}}(Q_\vartheta \| Q_\theta) = D_{A^*}(\tau_\vartheta, \tau_\theta). \quad (6.16)$$

[ESTABLISHED] This is the exact exponential-family divergence; in general it is not a finite-displacement Fisher quadratic [2, 17, 20].

Proposition 6.16 (Affine mean constraints attain a unique projection). Assume the extended A is Legendre, as in a regular minimal canonical family, and let $M = \text{int dom } A^*$. If C is a closed affine subspace with $C \cap M \neq \emptyset$, then for every $\tau_\theta \in M$ there is a unique $\hat{\tau} \in C \cap M$ minimizing

$$\tau \longmapsto D_{A^*}(\tau, \tau_\theta). \quad (6.17)$$

The corresponding natural parameter is uniquely $\hat{\vartheta} = \nabla A^*(\hat{\tau})$. [ESTABLISHED]

Proof. The finite-dimensional Legendre Bregman projection theorem gives existence [8], uniqueness, and interiority on every nonempty closed convex feasible set meeting the interior domain. Strict convexity gives uniqueness, and essential smoothness excludes boundary attainment. For $C = \{\tau : B\tau = b\}$, the solution equivalently satisfies

$$B\hat{\tau} = b, \quad \nabla A^*(\hat{\tau}) - \theta + B^\top \lambda = 0, \quad (6.18)$$

which also proves the stated natural-parameter formula. $\square$

Feasibility, closedness, and convexity are load bearing. For a Bernoulli mean domain $(0, 1)$, $C = \{0\}$ is infeasible; an open feasible interval may have a finite unattained infimum; and a closed nonconvex two-point set can have two minimizers. Gauge compatibility and commutation of this projection with a coarse parameter map remain open and require an intertwining theorem; existence alone does not provide them. [OPEN]

6.7 Boundary of this chapter

This chapter supplies the general law-fiber hierarchy, the exponential-family interior, its operator boundary, and the Bregman projection theorem. It does not contain graph aggregation, Gaussian determinants, Schur complements, or matrix pencils. Graph-exponential energy closure is developed in Chapter 7; the multivariate-Gaussian realization and its quadratic boundary rates appear later. An operator RG additionally requires a preserved parameter category, scale comparisons, rescaling, and topology. A law-level RG additionally requires normalized laws or kernels, compatible reference measures, and finite normalizers at every scale. A Fisher or ELBO reading additionally requires the hypotheses of the corresponding earlier chapters. [HYPOTHESIS]

Part II

General Coarse Maps and Renormalization

Chapter 7

General Coarse-Graining

Coarse-graining is not one operation. A normalized Markov channel pushes probability laws forward and obeys data processing. Identification of sample variables precomposes an unnormalized energy and requires a new reference measure and normalizer. A Galerkin map restricts an operator and becomes a propagator only after a reference form and signal maps are supplied. This chapter separates these types before any Gaussian matrix formula is used.

7.1 Markov coarse maps

Definition 7.1 (Coarse channel). For standard Borel spaces X, Y , a coarse channel is a parameter-independent Markov kernel

$$K : X \rightsquigarrow Y.$$

It sends $P \in \mathcal{P}(X)$ to

$$(PK)(B) = \int_X K(x, B)P(dx). \quad (7.1)$$

A measurable statistic $c : X \rightarrow Y$ is the deterministic kernel $K_c(x, B) = \mathbf{1}_B(c(x))$. [DEFINITION]

Proposition 7.2 (Markov kernels form a category). If $K : X \rightsquigarrow Y$ and $L : Y \rightsquigarrow Z$, then

$$(KL)(x, C) = \int_Y L(y, C)K(x, dy) \quad (7.2)$$

is a Markov kernel, composition is associative, the deterministic identity is an identity, and

$$P(KL) = (PK)L. \quad (7.3)$$

[ESTABLISHED]

Proof. Kernel integration gives measurability, and monotone convergence gives countable additivity. Tonelli's theorem gives, for measurable C ,

$$[P(KL)](C) = \int_X \int_Y L(y, C)K(x, dy)P(dx) = \int_Y L(y, C)(PK)(dy),$$

which proves (7.3); applying the identity to a third kernel proves associativity. $\square$

A statistical experiment $\mathcal{E} = (X, \{P_\theta\}_{\theta \in \Theta})$ is therefore sent functorially to $\mathcal{E}K = (Y, \{P_\theta K\}_{\theta \in \Theta})$. Parameter independence is load bearing: a family K_θ is a parameter-dependent model transformation, not one channel between experiments. **[ESTABLISHED]**

7.2 KL contraction, equality, and recovery

Theorem 7.3 (Data processing with its exact equality condition). Let $P, Q \in \mathcal{P}(X)$, $P \ll Q$, and $D_{\text{KL}}(P\|Q) < \infty$. For a Markov kernel $K : X \rightsquigarrow Y$, form

$$\mathbb{Q}(dx, dy) = Q(dx)K(x, dy), \quad r = \frac{dP}{dQ}, \quad \bar{r}(y) = \mathbb{E}_{\mathbb{Q}}[r(X) \mid Y = y].$$

Then

$$\frac{d(PK)}{d(QK)} = \bar{r}, \quad D_{\text{KL}}(PK\|QK) \leq D_{\text{KL}}(P\|Q), \quad (7.4)$$

and equality holds exactly when

$$r(X) = \bar{r}(Y) \quad \mathbb{Q}\text{-almost surely.} \quad (7.5)$$

[ESTABLISHED]

Proof. For bounded measurable f ,

$$\int f d(PK) = \mathbb{E}_{\mathbb{Q}}[f(Y)r(X)] = \mathbb{E}_{\mathbb{Q}}[f(Y)\bar{r}(Y)],$$

which proves the Radon–Nikodym formula. Conditional Jensen for the strictly convex function $\phi(t) = t \log t$ gives

$$\mathbb{E}_{\mathbb{Q}}\phi(\bar{r}(Y)) \leq \mathbb{E}_{\mathbb{Q}}\phi(r(X)).$$

Finite divergence and strict convexity make equality equivalent to (7.5). This is the information-loss theorem of Csiszár [20], Kullback and Leibler [40] in kernel form. $\square$

Corollary 7.4 (Pairwise Bayes recovery). Let $R_Q(y, dx)$ be a regular conditional law of X given $Y = y$ under $\mathbb{Q}$. Then $QKR_Q = Q$, and under (7.5), $PKR_Q = P$. Conversely, any one kernel R recovering both P and Q forces equality in (7.4). **[ESTABLISHED]**

Proof. Disintegration gives the first identity. Under equality,

$$(PKR_Q)(A) = \int \bar{r}(y)R_Q(y, A)(QK)(dy) = \iint \mathbf{1}_A(x)r(x)\mathbb{Q}(dx, dy) = P(A).$$

The converse follows by applying data processing through K and then R . $\square$

This recovery is pairwise. It may depend on the chosen reference Q . For a dominated experiment, one common dominating probability P_0 and simultaneous finite equalities

$$D_{\text{KL}}(P_\theta K \| P_0 K) = D_{\text{KL}}(P_\theta \| P_0) \quad (\theta \in \Theta)$$

make the single Bayes kernel R_{P_0} recover every P_θ . Without those simultaneous hypotheses, experiment-level sufficiency means directly that one parameter-independent R satisfies $P_\theta K R = P_\theta$ for all θ . Pairwise equality does not prove that stronger statement. **[ESTABLISHED]**

7.3 Fisher contraction is local

Theorem 7.5 (Score projection and Fisher loss). Let $\{P_\theta\}_{\theta \in \Theta \subseteq \mathbb{R}^d}$ be differentiable in quadratic mean at θ_0 , with score $\ell_{\theta_0} \in L_0^2(P_{\theta_0}; \mathbb{R}^d)$, and let K be parameter independent. Under the joint law $\mathbb{P}_{\theta_0}(dx, dy) = P_{\theta_0}(dx)K(x, dy)$, the coarse family is differentiable in quadratic mean with score

$$\bar{\ell}_{\theta_0}(y) = \mathbb{E}_{\mathbb{P}_{\theta_0}}[\ell_{\theta_0}(X) \mid Y = y]. \quad (7.6)$$

Its Fisher information obeys

$$I_Y(\theta_0) \preceq I_X(\theta_0), \quad I_X - I_Y = \mathbb{E} \text{Cov}(\ell_{\theta_0}(X) \mid Y) \succeq 0. \quad (7.7)$$

Equality holds exactly when the score is Y -measurable. **[ESTABLISHED]**

Proof. Attach the parameter-independent channel to obtain the joint family. Its score is $\ell_{\theta_0}(X)$. Projection onto Y preserves quadratic mean differentiability and replaces a score by its conditional expectation; this is the L^2 -projection theorem for parametrized-measure models [7]. Taking second moments and using total covariance proves (7.7). Conditional expectation is an orthogonal projection, so equality holds precisely on its range. $\square$

Fisher equality at one parameter is local score sufficiency, not global recovery. Let $X = (A, B)$ with independent Bernoulli coordinates,

$$\Pr_\theta(A = 1) = \frac{1}{2} + \frac{\theta}{4}, \quad \Pr_\theta(B = 1) = \frac{1}{2} + \frac{\theta^2}{4},$$

and let K discard B . At $\theta = 0$, the B -score vanishes, so the fine and coarse Fisher informations agree and are nonzero. Yet $\Pr_\theta(B = 1 \mid A)$ depends on θ , so no parameter-independent reverse kernel can recover the whole experiment. **[ESTABLISHED]**

7.4 The fixed-evidence ELBO consequence

Theorem 7.6 (Evidence-preserving coarse channel). Fix one normalized joint $P(do, dx)$, a regular observation o , posterior P_o , recognition law Q_o , and normalized channel $K : X \rightsquigarrow Y$ that

does not read Q_o . Define

$$\bar{P}(do, dy) = \int K(x, dy)P(do, dx), \quad \bar{Q}_o = Q_o K. \quad (7.8)$$

Then $\bar{P}^O = P^O$, $\bar{P}_o = P_o K$ almost surely, $Q_o \ll P_o$ implies $\bar{Q}_o \ll \bar{P}_o$, and the extended ELBOs satisfy

$$\bar{\mathcal{L}}^{\text{ext}}(\bar{Q}_o; o) = \log p(o) - D_{\text{KL}}(Q_o K \| P_o K) \geq \log p(o) - D_{\text{KL}}(Q_o \| P_o) = \mathcal{L}^{\text{ext}}(Q_o; o). \quad (7.9)$$

[ESTABLISHED]

Proof. Normalization of K preserves the observation marginal. Disintegration commutes with the parameter-independent channel. Absolute continuity and (7.9) follow from the preceding data-processing theorem. $\square$

Thus different latent inventories are not by themselves a prohibition. ELBOs are comparable under an evidence-preserving common Markov pushforward applied to the posterior and recognition law. Absent that relation, separately declared models may have different evidences, and an ELBO difference mixes evidence with variational tightness. [ESTABLISHED] The increase in (7.9) reflects discarded information; it is not model improvement or EM evidence ascent. [NOT-CLAIMED]

7.5 Pushforward, energy precomposition, and identification

Let $\iota : Z \rightarrow X$ be a measurable injection. The formula

$$\bar{E}(z) = E(\iota z), \quad \bar{P}(dz) = \frac{e^{-\bar{E}(z)}}{\bar{Z}} \bar{\nu}(dz) \quad (7.10)$$

defines a new model only after $\bar{\nu}$ and a finite $\bar{Z}$ are declared. It is not a channel from X to Z , and its pushforward can be supported on a fine-measure-null image. Therefore KL data processing, evidence preservation, and (7.9) do not follow. [DEFINITION]

Exact marginalization is a special Markov pushforward. Hard identification is energy precomposition. Restricting a recognition family is neither: it keeps the fixed joint and optimizes the same ELBO over a subset. A proof about any one of these operations cannot be transferred to another by calling all three coarse-graining. [ESTABLISHED]

7.6 Graph-exponential energy closure

Definition 7.7 (Graph-exponential energy). Let V be finite, Z_0 standard Borel with sigma-finite ν_0 , and let $t : Z_0 \rightarrow T_0$ and $u : Z_0^2 \rightarrow T_1$ be measurable, with u symmetric. Set

$$\mathcal{E}_\theta(z) = \sum_{i \in V} \langle \alpha_i, t(z_i) \rangle + \sum_{\{i,j\}} \langle \beta_{ij}, u(z_i, z_j) \rangle. \quad (7.11)$$

This is an unnormalized energy until its integral relative to $\nu_0^{\otimes V}$ is finite. **[DEFINITION]**

For a partition $\mathcal{P}$, define

$$\iota_{\mathcal{P}} : Z_0^{\mathcal{P}} \rightarrow Z_0^V, \quad (\iota_{\mathcal{P}} \bar{z})_i = \bar{z}_{I(i)}.$$

Assume the diagonal-affinity hypothesis

$$u(z, z) = \Upsilon t(z) + u_0 \quad (7.12)$$

for a linear $\Upsilon : T_0 \rightarrow T_1$. **[HYPOTHESIS]** This excludes pair statistics whose diagonal restriction cannot be represented by the declared node statistic.

Theorem 7.8 (Graph-exponential trace closure). Under (7.12),

$$\mathcal{E}_{\theta}(\iota_{\mathcal{P}} \bar{z}) = \sum_I \langle \alpha_I, t(\bar{z}_I) \rangle + \sum_{I < J} \langle \beta_{IJ}, u(\bar{z}_I, \bar{z}_J) \rangle + c_{\mathcal{P}}(\theta), \quad (7.13)$$

where

$$\alpha_I = \sum_{i \in I} \alpha_i + \Upsilon^* \sum_{\{i,j\} \subseteq I} \beta_{ij}, \quad \beta_{IJ} = \sum_{\substack{i \in I \\ j \in J}} \beta_{ij}, \quad c_{\mathcal{P}}(\theta) = \left\langle \sum_I \sum_{\{i,j\} \subseteq I} \beta_{ij}, u_0 \right\rangle. \quad (7.14)$$

The induced parameter map is linear. A normalized coarse law exists if and only if

$$\bar{Z}_{\mathcal{P}}(\theta) = \int e^{\mathcal{E}_{\theta}(\iota_{\mathcal{P}} \bar{z})} \nu_0^{\otimes \mathcal{P}}(d\bar{z}) < \infty. \quad (7.15)$$

[ESTABLISHED]

Proof. Substitute the identified variables into (7.11). Cut edges sum into β_{IJ} . Internal edges use (7.12); the linear part joins α_I and the constant joins $c_{\mathcal{P}}$. This gives (7.13)–(7.14). The proof is algebraic and does not prove (7.15). $\square$

If $u(z, z)$ is constant, one may choose $\Upsilon = 0$, so internal edges are annihilated and node parameters add. Translation invariance implies this constant-diagonal property, but the converse is false because diagonal data do not determine u off the diagonal. Uniqueness of the coordinate choice $\Upsilon = 0$ requires minimal effective node statistics or quotienting null parameter directions.

[ESTABLISHED]

This theorem states conditional energy closure, not normalized-law closure. Its reference measure, normalizer, and any gauge action must be propagated separately. A compatible normalized family must satisfy

$$P_{\theta} K = \bar{P}_{\rho(\theta)}$$

for a declared channel K and parameter map ρ ; energy precomposition alone does not prove that equation. **[HYPOTHESIS]**

7.7 Galerkin operators and message passing

Let $P : V_c \rightarrow V_f$ be injective, let $R \succ 0$ be a fine reference form, and let L be a fine bilinear form. Define

$$R_c = P^\top R P, \quad C = R_c^{-1} P^\top R, \quad H = R^{-1} L, \quad H_c = C H P = R_c^{-1} P^\top L P. \quad (7.16)$$

Then $CP = I$. The form $P^\top L P$ is the Galerkin coarse form, but H_c is the typed coarse propagator. **[DEFINITION]**

Proposition 7.9 (Exact message-passing residual). The reduction and propagation obey

$$CH - H_c C = CH(I - PC). \quad (7.17)$$

Hence exact coarse message passing holds on all fine signals exactly when the right side vanishes. If H is self-adjoint in the R -inner product, $\text{range } P$ being H -invariant is a sufficient and, for orthogonal projection PC , equivalent condition. **[ESTABLISHED]**

Proof. Substitute $H_c = C H P$ and use $CP = I$:

$$CH - H_c C = CH - CHPC = CH(I - PC).$$

For an R -self-adjoint operator, invariance of a closed subspace is equivalent to reduction by its R -orthogonal projector. $\square$

Thus exact quadratic-form restriction, generalized spectral approximation, and message-passing preservation are different statements. The first follows from Galerkin algebra; the other two require additional invariant-subspace or error hypotheses. A neural pooling layer also needs a signal map and a partition rule. No GNN or oversmoothing theorem follows from energy closure alone. **[NOT-CLAIMED]**

7.8 Gauge and permutation equivariance

Proposition 7.10 (Equivariant Markov channels). Let a group G act bimeasurably on X, Y , and assume

$$K(gx, B) = K(x, g^{-1}B). \quad (7.18)$$

Then

$$(g_\# P)K = g_\#(PK), \quad (7.19)$$

and composites of equivariant kernels are equivariant. **[ESTABLISHED]**

Proof. For measurable B ,

$$[(g_{\#}P)K](B) = \int K(gx, B)P(dx) = \int K(x, g^{-1}B)P(dx) = [g_{\#}(PK)](B).$$

Substitution in (7.2) proves closure under composition. $\square$

For independently chosen belief and model frame coordinates, the passive coordinate group is $G \times G$, although the underlying principal bundle has structure group G . A joint coarse channel must intertwine the two represented coordinate actions. Under a genuine principal gauge transformation, those actions are the ρ_b - and ρ_m -images of the same element of G . KL equality does not automatically make its Bayes recovery equivariant; an equivariant conditional version is an additional hypothesis or theorem, especially for noncompact groups. [\[OPEN\]](#)

A data-dependent hard partition selector $S(X)$ must also be natural under node relabeling. If P permutes fine nodes and Q permutes coarse labels,

$$S(P \cdot X) = PS(X)Q^{\top} \quad (7.20)$$

is the required law. Gauge-carrying inputs must be reduced to invariant assignment scores, while represented messages transform equivariantly. These conditions do not select a unique partition; they only type an admissible selector. Constructing a selector satisfying them and the later holonomy conditions remains open. [\[OPEN\]](#)

7.9 Holonomy-conditioned marginal-law modes

Fix a channel $x \in \{b, m\}$, a finite connected cluster I , and one declared internal path groupoid. The paths may be edge-labeled graph paths for typed graph links or base curves for the selected channel connection; the two cases are not identified. Let each represented transport $T_{\gamma}^x : Z_i^x \rightarrow Z_j^x$, for $\gamma : i \rightarrow j$, be a bimeasurable bijection satisfying composition and inversion. Assume that the parent-law family $\mathcal{M}_i^x \subseteq \mathcal{P}(Z_i^x)$ is equivariant: $(T_{\gamma}^x)_{\#}\mathcal{M}_i^x = \mathcal{M}_j^x$. Under a passive frame rechoice g_i^x , also assume that the represented coordinate map $\chi_i^x = \rho_x(g_i^x)$ gives $\mathcal{M}_i^{x'} = ((\chi_i^x)^{-1})_{\#}\mathcal{M}_i^x$. [\[HYPOTHESIS\]](#) Bimeasurability excludes lossy transports, and family equivariance excludes parent classes tied to one root or frame; both hypotheses are used in the invariance proof below.

Choose a root $r \in I$, paths $\gamma_i : i \rightarrow r$, marginal laws $q_i^x \in \mathcal{P}(Z_i^x)$, and weights $w_i > 0$ with $\sum_{i \in I} w_i = 1$. Define

$$P_i^x = (T_{\gamma_i}^x)_{\#}q_i^x, \quad \mathfrak{H}_I^x(r) = \{T_{\lambda}^x : \lambda : r \rightarrow r\}, \quad (7.21)$$

$$\mathcal{Q}_{I, \text{fix}}^x(r) = \{Q \in \mathcal{M}_r^x : (H)_{\#}Q = Q \text{ for every } H \in \mathfrak{H}_I^x(r)\}, \quad (7.22)$$

and the extended-real forward-KL score

$$\mathfrak{D}_I^x = \inf_{Q \in \mathcal{Q}_{I, \text{fix}}^x(r)} \sum_{i \in I} w_i D_{\text{KL}}(P_i^x \| Q), \quad \inf \emptyset := +\infty. \quad (7.23)$$

Any minimizer, when one exists, is a holonomy-conditioned marginal-law mode. These are definitions, not existence assertions. **[DEFINITION]**

Theorem 7.11 (Invariant holonomy-conditioned marginal-law mode). The value in (7.23) is independent of the chosen paths, root, and passive gauge coordinates. If the infimum is attained, then

$$\mathfrak{D}_I^x = 0 \iff \text{there is } Q_\star \in \mathcal{Q}_{I,\text{fix}}^x(r) \text{ such that } (T_{\gamma_i}^x)_\# q_i^x = Q_\star \text{ for every } i \in I. \quad (7.24)$$

The right side is precisely a holonomy-stabilized parallel marginal-law section. If $\mathcal{Q}_{I,\text{fix}}^x(r) = \emptyset$, then $\mathfrak{D}_I^x = +\infty$. If $\mathfrak{D}_I^x = 0$ but the infimum is not attained, the conclusion is only forward-KL approximability by admissible fixed parent laws; no member $Q_\star$ of that family has been produced. **[ESTABLISHED]**

Proof. For a bimeasurable bijection A , if $P \not\ll Q$, then $A_\# P \not\ll A_\# Q$, so both divergences below are infinite. If $P \ll Q$, then

$$\frac{d(A_\# P)}{d(A_\# Q)} = \frac{dP}{dQ} \circ A^{-1} \quad A_\# Q\text{-almost surely}, \quad (7.25)$$

and integration gives

$$D_{\text{KL}}(A_\# P \| A_\# Q) = D_{\text{KL}}(P \| Q), \quad (7.26)$$

as an identity in $[0, +\infty]$. If γ'_i is another path from i to r , then $H_i = T_{\gamma'_i}^x \circ (T_{\gamma_i}^x)^{-1}$ belongs to $\mathfrak{H}_I^x(r)$. Hence, for every fixed parent Q ,

$$D_{\text{KL}}((H_i)_\# P_i^x \| Q) = D_{\text{KL}}((H_i)_\# P_i^x \| (H_i)_\# Q) = D_{\text{KL}}(P_i^x \| Q). \quad (7.27)$$

Thus every objective value, and therefore its infimum, is path independent.

For a new root r' and a path $\delta : r \rightarrow r'$, transport conjugates $\mathfrak{H}_I^x(r)$ onto $\mathfrak{H}_I^x(r')$. Equivariance of $\mathcal{M}^x$ therefore makes $Q \mapsto (T_\delta^x)_\# Q$ a bijection between the two fixed parent families, while the paths $\delta \circ \gamma_i$ send every P_i^x by that same bijection. Equation (7.26) preserves every term. Path independence then covers arbitrary paths to the new root.

Under passive frame rechoices g_i^x , write $\chi_i^x = \rho_x(g_i^x)$ for the represented bimeasurable coordinate map. Then

$$T_{\gamma_i}^{x'} = (\chi_r^x)^{-1} \circ T_{\gamma_i}^x \circ \chi_i^x, \quad q_i^{x'} = ((\chi_i^x)^{-1})_\# q_i^x. \quad (7.28)$$

Consequently all transported marginals are pushed forward by $(\chi_r^x)^{-1}$, the holonomy group is conjugated by the same map, and parent-family equivariance carries the fixed family bijectively. A final application of (7.26) proves gauge invariance.

Relative entropy is nonnegative and vanishes exactly when its two probability measures agree. If a minimizer $Q_\star$ exists, positivity of every w_i makes its weighted sum zero exactly when $P_i^x = Q_\star$ for all i , which proves (7.24). Holonomy-fixity makes the reconstructed laws $((T_{\gamma_i}^x)^{-1})_\# Q_\star$ independent

of path, so they form the stated parallel section. The empty-family assertion is the convention in (7.23). Finally, if the zero infimum is not attained, its definition supplies, for every $n \geq 1$, some $Q_n \in \mathcal{Q}_{I,\text{fix}}^x(r)$ with $\sum_i w_i D_{\text{KL}}(P_i^x \| Q_n) < 1/n$, and supplies no minimizer. $\square$

The score matches marginal laws after transport. The cluster and its weights are inputs, so it does not select a partition. It supplies neither a Markov kernel nor a normalized coarse channel, and it does not recover the dependence structure of a joint law. These stronger conclusions are not claimed. **[NOT-CLAIMED]** For a statistical experiment $\{P_\theta\}_{\theta \in \Theta}$, exact common recovery instead requires parameter-independent Markov kernels C and R satisfying

$$P_\theta C R = P_\theta \quad \text{for every } \theta \in \Theta. \quad (7.29)$$

Here C, R are local kernel notation, not the Galerkin reduction and reference form in (7.16). This is the common-recovery condition, not a consequence of (7.23). **[DEFINITION]** Under the richness hypothesis of Proposition 3.12, distinct joint laws can have identical coordinate marginals. Identical transported marginals therefore do not imply (7.29). **[ESTABLISHED]**

Corollary 7.12 (Parallel cross morphisms preserve fixed marginal-law sections). Let $\Phi : \mathcal{E}_b \rightarrow \mathcal{E}_m$ be a cross-associated-bundle morphism on law fibers that is parallel for the chosen path system. Writing $\hat{T}_\gamma^x = (T_\gamma^x)_\#$, this means

$$\hat{T}_\gamma^m \circ \Phi_i = \Phi_j \circ \hat{T}_\gamma^b \quad (\gamma : i \rightarrow j). \quad (7.30)$$

For base-connection paths this is ordinary parallelness. For independently declared graph paths, (7.30) is an additional link-intertwining hypothesis and does not follow from smooth parallelness. Assume also $\Phi_r(\mathcal{M}_r^b) \subseteq \mathcal{M}_r^m$. **[HYPOTHESIS]** Then

$$\Phi_r(\mathcal{Q}_{I,\text{fix}}^b(r)) \subseteq \mathcal{Q}_{I,\text{fix}}^m(r). \quad (7.31)$$

Moreover, applying Φ_i fiberwise to a holonomy-stabilized parallel belief-law section gives such a model-law section. If $\tilde{\Phi} : \mathcal{E}_m \rightarrow \mathcal{E}_b$ is likewise parallel for the chosen path system and $\tilde{\Phi}_r(\mathcal{M}_r^m) \subseteq \mathcal{M}_r^b$, then

$$\tilde{\Phi}_r(\mathcal{Q}_{I,\text{fix}}^m(r)) \subseteq \mathcal{Q}_{I,\text{fix}}^b(r), \quad (7.32)$$

and the analogous model-to-belief statement holds. **[ESTABLISHED]**

Proof. If $Q \in \mathcal{Q}_{I,\text{fix}}^b(r)$ and $\lambda : r \rightarrow r$, then

$$\hat{T}_\lambda^m(\Phi_r(Q)) = \Phi_r(\hat{T}_\lambda^b(Q)) = \Phi_r(Q). \quad (7.33)$$

Parent-family compatibility proves (7.31). If the belief marginals transport to Q , the same commutative square shows that their Φ_i -images transport to $\Phi_r(Q)$. Interchanging b, m and replacing Φ by $\tilde{\Phi}$ proves (7.32). $\square$

This corollary does not identify $\mathcal{E}_b$ with $\mathcal{E}_m$, does not require equal fiber dimensions or equivalent representations ρ_b, ρ_m . No KL-preservation hypothesis is imposed, so no equality or ordering between $\mathfrak{D}_I^b$ and $\mathfrak{D}_I^m$ is claimed. Existence, parallelness, and parent-family compatibility of each cross morphism are additional structure. **[NOT-CLAIMED]**

7.10 Projective systems of laws

Theorem 7.13 (Cylinder-level projective law). Let $\mathcal{I}$ be an index set and each coordinate space Z_i standard Borel. For every finite $D \subset \mathcal{I}$, let P_D be a probability law on $Z_D = \prod_{i \in D} Z_i$. If

$$(\pi_{E,D})_{\#} P_E = P_D \quad (D \subset E), \quad (7.34)$$

then there is a unique law on the product cylinder sigma-algebra with finite-dimensional marginals P_D . Compatible finite-dimensional gauge equivariance passes to the limit. **[ESTABLISHED]**

Proof. This is the Kolmogorov extension theorem for standard Borel coordinate spaces [36]. The transformed and original laws have identical finite-dimensional marginals under compatible equivariance, so uniqueness gives the last assertion. $\square$

Consistency alone does not put the limit on continuous or smooth sections, define a continuum reference density, justify an ELBO limit, or produce an RG flow. Those conclusions require support regularity, compatible kernels and rescalings, convergence or uniform integrability of the functionals, and a declared topology. **[OPEN]**

Physicist summary. A general coarse-graining is a noisy detector. Detectors compose, and they cannot increase KL distinguishability or Fisher information. Equality in KL means that a reverse detector recovers the particular laws; equality in Fisher means only that the infinitesimal score survived. Restricting an energy to block-constant fields is a different construction: it is a new model until its measure and normalizer are supplied. Galerkin restriction is exact for the quadratic form, but propagation needs a mass form and has the explicit residual (7.17). Holonomy determines the admissible parallel marginal-law modes. When the infimum is attained, transported forward KL tests whether the actual marginals occupy one; without attainment, a zero infimum establishes only approximability. Neither construction selects a partition or proves joint-law recovery. These distinctions are the general spine that the later multivariate-Gaussian chapters realize.

Chapter 8

General Renormalization

Coarse arrows alone form a scale diagram. They become a renormalization dynamics only after comparison and rescaling data identify what is being compared across levels. In the present theory the state includes one principal gauge bundle, two associated statistical bundles, their possibly distinct connections, and both cross-bundle morphisms. A scalar normalization of one operator cannot stand for that full state.

8.1 The scale category and comparison data

Definition 8.1 (Scale diagram). Let S be a category whose objects are resolutions ℓ . Choose a target category $\mathcal{K}$: sets and maps for a deterministic state, standard Borel spaces and Markov kernels for a stochastic state, an appropriate linear category for an operator state, or an explicitly declared product of such categories for a mixed state. A state functor $F : S \rightarrow \mathcal{K}$ assigns an object $\mathfrak{X}_\ell = F(\ell)$ and morphisms

$$C_{\ell k} \in \text{Hom}_{\mathcal{K}}(\mathfrak{X}_\ell, \mathfrak{X}_k), \quad C_{kr} C_{\ell k} = C_{\ell r}. \quad (8.1)$$

The arrow denotes the appropriate composition in $\mathcal{K}$, including kernel composition in the stochastic case and componentwise composition in a product category. One symbol is never asked to be simultaneously a map, a kernel, and an operator. **[DEFINITION]**

Comparison requires more data. Either choose levelwise re-embeddings $J_\ell : \mathfrak{X}_{\ell+1} \rightarrow \mathfrak{X}_\ell$, or identify each level with a reference object by declared isomorphisms $I_\ell : \mathfrak{X}_\ell \xrightarrow{\sim} \mathfrak{X}_\star$ in $\mathcal{K}$. Field, base, time, unit, and reference-measure rescalings are part of these maps. Only the isomorphism option supplies the inverse used in the reference-space transformations

$$\widehat{\mathcal{R}}_\ell = I_{\ell+1} C_{\ell, \ell+1} I_\ell^{-1} : \mathfrak{X}_\star \longrightarrow \mathfrak{X}_\star. \quad (8.2)$$

[DEFINITION] A positive scalar ζ_ℓ can be one component of this comparison; it cannot by itself identify changing fields, measures, or bundle objects.

If probability laws are present, every scale declares its sample space and reference measure. A law-level arrow is a normalized Markov kernel as in Chapter 7. An energy-level arrow additionally owes the coarse reference measure, its Radon–Nikodym or Jacobian rule, and a finite normalizer. Algebraic closure does not supply those data. **[HYPOTHESIS]**

8.2 The common-principal geometric state

Definition 8.2 (Geometric RG state). At level ℓ , declare a smooth base $\mathcal{C}_\ell$, a Lie group G_ℓ , and one principal right G_ℓ -bundle $\pi_\ell : P_\ell \rightarrow \mathcal{C}_\ell$. Let $\mathcal{B}_{\ell,b}$ and $\mathcal{B}_{\ell,m}$ be statistical fibers with possibly different left actions $\hat{\rho}_{\ell,b}$ and $\hat{\rho}_{\ell,m}$ of the same group. The belief and model bundles are

$$\mathcal{E}_{\ell,b} = P_\ell \times_{\hat{\rho}_{\ell,b}} \mathcal{B}_{\ell,b}, \quad \mathcal{E}_{\ell,m} = P_\ell \times_{\hat{\rho}_{\ell,m}} \mathcal{B}_{\ell,m}. \quad (8.3)$$

Choose two principal connections $\omega_{\ell,b}, \omega_{\ell,m} \in \Omega^1(P_\ell, \mathfrak{g}_\ell)$. They induce horizontal distributions H_ℓ^b, H_ℓ^m and parallel transports $\Omega_{\ell,\gamma}$ and $\tilde{\Omega}_{\ell,\gamma}$ on the two associated bundles. Thus the general geometric state contains

$$\mathfrak{X}_\ell = \left(\mathcal{C}_\ell; G_\ell, P_\ell; \mathcal{E}_{\ell,b}, H_\ell^b; \mathcal{E}_{\ell,m}, H_\ell^m; \Phi_\ell, \tilde{\Phi}_\ell; \dots \right). \quad (8.4)$$

Here

$$\Phi_\ell : \mathcal{E}_{\ell,b} \rightarrow \mathcal{E}_{\ell,m}, \quad \tilde{\Phi}_\ell : \mathcal{E}_{\ell,m} \rightarrow \mathcal{E}_{\ell,b}$$

are separately declared associated-bundle morphisms over the identity. They need not be invertible or mutual inverses. A fixed fiber map $f_\ell^{bm} : \mathcal{B}_{\ell,b} \rightarrow \mathcal{B}_{\ell,m}$ induces such a morphism only when it intertwines the actions,

$$f_\ell^{bm} \circ \hat{\rho}_{\ell,b}(g) = \hat{\rho}_{\ell,m}(g) \circ f_\ell^{bm} \quad \text{for every } g \in G_\ell, \quad (8.5)$$

and analogously in the reverse direction. Hence a common principal bundle does not manufacture either cross map. The ellipsis in (8.4) denotes law, kernel, or operator components declared by a realization. **[DEFINITION]**

Local belief and model frames may nevertheless be chosen separately. On a common trivializing neighborhood, two sections $u_{\ell,i}^b, u_{\ell,i}^m : U_i \rightarrow P_\ell$ determine a unique smooth relative principal-frame field

$$u_{\ell,i}^m = u_{\ell,i}^b \cdot h_{\ell,i}, \quad h_{\ell,i} : U_i \rightarrow G_\ell. \quad (8.6)$$

Existence and uniqueness follow because each principal fiber is a free and transitive G_ℓ -torsor; the statement is independent of the representations. If the sections are re-chosen as $u_{\ell,i}^b a_{\ell,i}^b$ and $u_{\ell,i}^m a_{\ell,i}^m$, then $h_{\ell,i} \mapsto (a_{\ell,i}^b)^{-1} h_{\ell,i} a_{\ell,i}^m$. When the representations differ, $\hat{\rho}_{\ell,b}(h_{\ell,i})$ and $\hat{\rho}_{\ell,m}(h_{\ell,i})$ act on different fibers; there is no matrix quotient of the associated frames and no cross-fiber identification without extra intertwining data such as (8.5). If only associated-frame data are given and either action is nonfaithful, the principal element $h_{\ell,i}$ need not be reconstructible without chosen lifts to P_ℓ .

[ESTABLISHED]

Scale change must also be typed. Let $c_\ell : \mathcal{C}_\ell \rightarrow \mathcal{C}_{\ell+1}$ be smooth, let $\kappa_\ell : G_\ell \rightarrow G_{\ell+1}$ be a Lie-group homomorphism, and let $\mathcal{P}_\ell : P_\ell \rightarrow P_{\ell+1}$ cover c_ℓ and obey

$$\mathcal{P}_\ell(p \cdot g) = \mathcal{P}_\ell(p) \cdot \kappa_\ell(g). \quad (8.7)$$

For $s \in \{b, m\}$, a fiber map $q_{\ell,s} : \mathcal{B}_{\ell,s} \rightarrow \mathcal{B}_{\ell+1,s}$ descends to a bundle map only if

$$q_{\ell,s} \circ \widehat{\rho}_{\ell,s}(g) = \widehat{\rho}_{\ell+1,s}(\kappa_{\ell}(g)) \circ q_{\ell,s}. \quad (8.8)$$

It then induces

$$C_{\ell,s} : \mathcal{E}_{\ell,s} \rightarrow \mathcal{E}_{\ell+1,s}, \quad C_{\ell,s}[p, z] = [\mathcal{P}_{\ell}(p), q_{\ell,s}(z)], \quad (8.9)$$

covering c_{ℓ} . More general associated-bundle maps may be declared directly, but they owe the same transition compatibility. For a genuine multi-step scale functor, the maps c , κ , $\mathcal{P}$, and $q_{\ell,s}$ also obey their respective composition laws; adjacent maps without these laws form only a sequence of coarse steps. If local comparison functions a_{ℓ}^s satisfy

$$\mathcal{P}_{\ell}(u_{\ell}^s(x)) = u_{\ell+1}^s(c_{\ell}x) \cdot a_{\ell}^s(x),$$

then equivariance of $\mathcal{P}_{\ell}$ forces

$$a_{\ell}^b \kappa_{\ell}(h_{\ell}) = (h_{\ell+1} \circ c_{\ell}) a_{\ell}^m. \quad (8.10)$$

Thus the two represented frame channels remain compatible even when their fiber dimensions or actions differ. **[ESTABLISHED]**

For general nonlinear associated fibers, the cross-scale morphism defects are the ordered pairs of maps

$$\mathbf{D}_{\ell}^{\Phi} = (C_{\ell,m} \circ \Phi_{\ell}, \Phi_{\ell+1} \circ C_{\ell,b}), \quad \mathbf{D}_{\ell}^{\tilde{\Phi}} = (C_{\ell,b} \circ \tilde{\Phi}_{\ell}, \tilde{\Phi}_{\ell+1} \circ C_{\ell,m}). \quad (8.11)$$

The defect vanishes when the two components of the corresponding pair are equal. For a base curve $\gamma : x \rightarrow y$, the cross-scale connection defects are likewise the pairs

$$\begin{aligned} \mathbf{D}_{\ell,\gamma}^b &= (C_{\ell,b,y} \circ \Omega_{\ell,\gamma}, \Omega_{\ell+1,c_{\ell}\gamma} \circ C_{\ell,b,x}), \\ \mathbf{D}_{\ell,\gamma}^m &= (C_{\ell,m,y} \circ \tilde{\Omega}_{\ell,\gamma}, \tilde{\Omega}_{\ell+1,c_{\ell}\gamma} \circ C_{\ell,m,x}). \end{aligned} \quad (8.12)$$

Independently, within-scale nonparallelness is measured by

$$\begin{aligned} \mathbf{A}_{\ell,\gamma}^{\Phi} &= (\tilde{\Omega}_{\ell,\gamma} \circ \Phi_{\ell,x}, \Phi_{\ell,y} \circ \Omega_{\ell,\gamma}), \\ \mathbf{A}_{\ell,\gamma}^{\tilde{\Phi}} &= (\Omega_{\ell,\gamma} \circ \tilde{\Phi}_{\ell,x}, \tilde{\Phi}_{\ell,y} \circ \tilde{\Omega}_{\ell,\gamma}). \end{aligned} \quad (8.13)$$

[DEFINITION] Subtraction of the two components is reserved for a declared vector or affine realization. A sufficient principal-level condition for $\mathbf{D}_{\ell,\gamma}^s$ to vanish is

$$\mathcal{P}_{\ell}^* \omega_{\ell+1,s} = d\kappa_{\ell} \circ \omega_{\ell,s}, \quad s \in \{b, m\}. \quad (8.14)$$

If a representation is not faithful, vanishing of its represented transport defect need not imply this principal-level identity.

Proposition 8.3 (Covariance of the defect diagrams). Under admissible local-frame changes in the belief and model associated bundles, both components of every pair in (8.11)–(8.13) acquire

the same endpoint coordinate changes. Consequently the equality locus of each pair is gauge invariant. In a fiberwise-linear vector-bundle realization, subtracting the two components gives the familiar Hom-bundle defect with the same left–right law. **[ESTABLISHED]**

Proof. Let $\mathcal{G}_{\ell,b,x}$ and $\mathcal{G}_{\ell,m,x}$ denote the fiber coordinate changes induced by the selected belief and model frame re-trivializations at x , represented through $\hat{\rho}_{\ell,b}$ and $\hat{\rho}_{\ell,m}$. For example,

$$\Phi_{\ell,x} \mapsto \mathcal{G}_{\ell,m,x} \circ \Phi_{\ell,x} \circ (\mathcal{G}_{\ell,b,x})^{-1}, \quad \Omega_{\ell,\gamma} \mapsto \mathcal{G}_{\ell,b,y} \circ \Omega_{\ell,\gamma} \circ (\mathcal{G}_{\ell,b,x})^{-1},$$

and a cross-scale map obeys

$$C_{\ell,b,x} \mapsto \mathcal{G}_{\ell+1,b,c_\ell(x)} \circ C_{\ell,b,x} \circ (\mathcal{G}_{\ell,b,x})^{-1},$$

with analogous model and endpoint laws. Substitution shows that both components of $\mathbf{A}_{\ell,\gamma}^\Phi$ acquire the same model coordinate change at y and inverse belief coordinate change at x . The equality of the two components is therefore independent of the chosen frames. The other pairs are identical calculations. In a linear realization, subtraction factors out the common endpoint actions. $\square$

The defect data may be retained as running fields; the theory does not require them to vanish. A numerical size requires a declared metric or divergence on each target fiber. The compositions $\tilde{\Phi}_\ell \Phi_\ell$ and $\Phi_\ell \tilde{\Phi}_\ell$ are independent running endomorphisms and are not identities unless imposed. Smooth parallel transport remains distinct from separately declared graph links $\Theta_{ij}^b, \Theta_{ij}^m$. **[ESTABLISHED]**

The default common-principal construction does not impose a common local frame or a common connection. The stronger common-frame hypothesis is $u_{\ell,i}^b = u_{\ell,i}^m$, equivalently $h_{\ell,i} = e$, on the chosen patches. Independently, the equal-connection hypothesis is $\omega_{\ell,b} = \omega_{\ell,m}$. Even then, different representations generally produce different associated transports on different fibers: they represent the same principal holonomy but are not literally the same map. If a further associated-bundle isomorphism $\Xi_\ell : \mathcal{E}_{\ell,b} \xrightarrow{\sim} \mathcal{E}_{\ell,m}$ is declared, its existence is additional global data; equal fiber type is necessary but not sufficient in general. Its parallel and RG-preservation conditions are

$$\tilde{\Omega}_{\ell,\gamma} \Xi_{\ell,x} = \Xi_{\ell,y} \Omega_{\ell,\gamma}, \quad C_{\ell,m} \Xi_\ell = \Xi_{\ell+1} C_{\ell,b}.$$

The optional Ξ_ℓ is unavailable when the associated bundles are nonisomorphic, as can occur when ranks or fiber types differ. No such isomorphism is needed to define Φ_ℓ and $\tilde{\Phi}_\ell$. **[HYPOTHESIS]**

Optional product-gauge extension. If an application truly requires independent gauge groups, independent principal-bundle topology, or independently acting gauge automorphisms, one may instead introduce

$$P_{\ell,b} \rightarrow \mathcal{C}_\ell, \quad P_{\ell,m} \rightarrow \mathcal{C}_\ell, \quad Q_\ell = P_{\ell,b} \times_{\mathcal{C}_\ell} P_{\ell,m},$$

where Q_ℓ is principal for $G_{\ell,b} \times G_{\ell,m}$. Each associated bundle then uses its own factor. This extension carries extra topological and gauge data and is not part of the minimal theory. In a

category admitting the required mapping object, its belief-to-model maps are sections of

$$Q_\ell \times_{G_{\ell,b} \times G_{\ell,m}} \text{Map}(\mathcal{B}_{\ell,b}, \mathcal{B}_{\ell,m}), \quad (g_b, g_m) \cdot f = \hat{\rho}_{\ell,m}(g_m) \circ f \circ \hat{\rho}_{\ell,b}(g_b)^{-1}.$$

A constant fiber map descends only if it is fixed by this action, a generally restrictive condition for independently varying (g_b, g_m) . Cross maps therefore still have to be supplied, and a diagonal reduction to one group requires additional group and bundle-identification data. **[DEFINITION]**

8.3 Autonomous maps and cocycles

Definition 8.4 (Dynamical type). The sequence $\hat{\mathcal{R}}_\ell$ in (8.2) is: autonomous when it is one map $\hat{\mathcal{R}}$; periodic when $\hat{\mathcal{R}}_{\ell+p} = \hat{\mathcal{R}}_\ell$; stationary random when it is a measurable stationary cocycle; and general otherwise. **[DEFINITION]**

The correct invariant object depends on this type. An autonomous fixed object satisfies $\hat{\mathcal{R}}x = x$. A periodic scheme has a fixed object of its monodromy or an attracting p -cycle. A stationary random scheme may have a random invariant section. A general cocycle may have pullback or forward attracting sections. The term *fixed ray* additionally requires a positive cone and projectivization. **[DEFINITION]**

Proposition 8.5 (Uniform contraction need not give one fixed ray). Uniform Hilbert-projective contraction of a nonautonomous positive cocycle can erase initial conditions without selecting one scale-independent ray. **[ESTABLISHED]**

Proof. On $\mathbb{R}_{>0}^2$, let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Both are strictly positive and have a uniform projective contraction coefficient below one. Their Perron rays differ. Under the alternating cocycle, even iterates converge to the Perron ray $[1, \sqrt{2}]$ of

$$BA = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix},$$

while odd iterates converge to the distinct ray $A[1, \sqrt{2}]$. The attractor is period two, not one fixed ray. $\square$

For maps satisfying $\tau(\hat{\mathcal{R}}_\ell) \leq q < 1$, Birkhoff contraction gives

$$d_H(\hat{\mathcal{R}}_{\ell-1} \cdots \hat{\mathcal{R}}_0 x, \hat{\mathcal{R}}_{\ell-1} \cdots \hat{\mathcal{R}}_0 y) \leq q^\ell d_H(x, y), \quad (8.15)$$

where defined [13, 54]. This proves forgetting of initial rays, not time independence of the attracting section.

8.4 Scheme dependence and invariant candidates

Let $g(t)$ be running coordinates. Under a smooth reparameterization $g' = f(g)$ and scale change $t' = ct$,

$$\beta'(g') = \frac{1}{c} Df(g) \beta(g). \quad (8.16)$$

[ESTABLISHED] Hence raw couplings, beta-function components, and a chosen blocking statistic are scheme dependent. They remain valid diagnostics when their scheme is stated, but are not universal observables.

For a differentiable homogeneous positive endomorphism at a fixed ray, let μ_0 be the radial eigenvalue and μ_a a transverse eigenvalue. The projective ratios and block-normalized dimensions are

$$\rho_a = \frac{\mu_a}{\mu_0}, \quad y_a = \frac{\log |\rho_a|}{\log b}, \quad (8.17)$$

for declared block factor b . [ESTABLISHED] They are invariant under smooth coordinate conjugacy at the ray once gauge, normalization, and other redundant directions have been quotiented and the same scale normalization is used.

Candidate cross-scheme quantities include conjugacy classes, invariant manifolds, projective eigenvalue ratios, dimensionless amplitude ratios, and matched spectral or heat exponents. Their existence, convergence, and completeness must be proved in the chosen realization. Universality is a statement about basins and invariant observables, not algebraic closure alone. [HYPOTHESIS]

8.5 Two-index thermodynamic and RG limits

A finite-volume RG family has two indices:

$$X_{n,\ell}, \quad 0 \leq \ell \leq L(n), \quad (8.18)$$

where n controls system size and ℓ RG depth. Fixed-depth, maximal-depth, and diagonal limits are different:

$$\lim_{\ell \rightarrow \infty} \lim_{n \rightarrow \infty} X_{n,\ell}, \quad \lim_{n \rightarrow \infty} X_{n,L(n)}, \quad \lim_{n \rightarrow \infty} X_{n,\ell(n)}. \quad (8.19)$$

[DEFINITION]

Proposition 8.6 (The two limits need not commute). For path graphs with hard blocks of size b , one coarse step maps P_{b^n} to $P_{b^{n-1}}$, up to endpoint convention. At each fixed ℓ , $n \rightarrow \infty$ yields the infinite path, whose lower spectral edge has counting behavior $N(\lambda) - N(0) \asymp \lambda^{1/2}$. Along $\ell = n$, every graph collapses to one vertex and the spectral measure is δ_0 . [ESTABLISHED]

Proof. Hard aggregation of consecutive path vertices preserves the path incidence pattern. At fixed ℓ , the number of remaining vertices tends to infinity; at $\ell = n$, it is one. The finite path

eigenvalues have the standard small-wave-number quadratic behavior, yielding the square-root counting edge. The one-vertex operator is zero. $\square$

Therefore a thermodynamic RG claim must specify a triangular family, projective consistency at fixed depth, uniform control of the coarse error, and a diagonal scaling $\ell(n)$. Operator, law, connection, and cross-morphism convergence are separate obligations. Proving an exchange of limits or a nontrivial diagonal limit remains open. **[OPEN]**

If an infinite-volume integrated density satisfies

$$N(\lambda) - N(0) \sim c\lambda^\alpha$$

and a Tauberian hypothesis supplies the corresponding heat asymptotic, then the nonzero heat trace and unnormalized spectral entropy obey

$$Z(t) \sim c\Gamma(\alpha + 1)t^{-\alpha}, \quad -\frac{dS}{d\log t} = t^2 \text{Var}_t(\lambda) \longrightarrow \alpha. \quad (8.20)$$

[ESTABLISHED] Dividing entropy by $\log |V_n|$ divides the susceptibility by the same factor and generally destroys a nonzero thermodynamic plateau. Zero-mode atoms and normalization must therefore be declared before interpreting a plateau as spectral dimension.

8.6 Projective laws and refinement

Compatible finite-dimensional laws obey the projective theorem of Section 7.10. An energy recursion alone does not provide those laws. A probabilistic RG needs normalized kernels K_{ℓ_k} with

$$\mu_k = \mu_\ell K_{\ell_k}, \quad K_{\ell_k} K_{\ell_r} = K_{\ell_r},$$

and corresponding rules for the two bundle sectors and their reference measures. Smooth-section support, continuum ELBO convergence, and connection convergence remain additional open obligations. **[OPEN]**

If the coarse map Q is noninjective, no deterministic two-sided inverse exists. A refinement kernel R can at most be a stochastic right section,

$$Q_\# R(c, \cdot) = \delta_c, \quad (8.21)$$

which redraws a compatible microstate. It cannot recover every discarded realized microstate. **[ESTABLISHED]** Infinite divisibility may construct such refinements in distribution, but does not turn a many-to-one coarse-graining into a group inverse.

8.7 Typed comparison with Bayesian renormalization

Berman, Klinger, and Stapleton build on the dynamical Bayesian-inference flow of Berman et al. [10], using a centered-KL posterior flow on model-parameter space. In the regular late-data regime they approximate the posterior by a Gaussian with covariance $T^{-1}I^{-1}$. Set $\tau = 1/T$, and consider an inverse problem with a differentiable forward map G . The delta method gives the leading local predictive approximation

$$G(\theta) \sim \mathcal{N}\left(G(\mu_\tau), \quad \tau J_G(\mu_\tau) I(\mu_\tau)^{-1} J_G(\mu_\tau)^\top\right).$$

It is exact when G is affine, but a nonlinear Gaussian pushforward is not in general Gaussian. Accordingly, the pushed inverse Fisher cometric

$$D_\tau = J_G(\mu_\tau) I(\mu_\tau)^{-1} J_G(\mu_\tau)^\top \quad (8.22)$$

is the local diffusion tensor used in their ERG-form Fokker–Planck equation; the corresponding small- τ transition covariance is τD_τ [11]. Their continuation of this tensor beyond the late-data, small- τ regime is a renormalization construction, not an exact nonlinear pushforward theorem or a Bayesian asymptotic theorem. [ESTABLISHED]

This is an exact, carefully scoped literature statement. It does not identify an arbitrary recognition Fisher tensor with the BKS generative-model Fisher metric, transfer their stiff/sloppy eigenspaces to another operator sector, or prove that a graph aggregation runs in the opposite direction. [NOT-CLAIMED] In the small- τ approximation, covariance broadens like τI^{-1} if I is held fixed, whereas a later Gaussian sufficient-statistic construction may add precision. That is a typed local contrast. A theorem of opposite flow directions would require a common state space, an explicit scale map, and a transported monotone including

$$\frac{d}{d\tau} [\tau I(\mu_\tau)^{-1}] = I^{-1} - \tau I^{-1} \frac{dI}{d\tau} I^{-1}.$$

No such bridge is asserted. [OPEN]

Theorem 7.11 gives a separate admissibility statement: represented holonomy restricts the marginal-law modes available inside an already proposed block. The BKS Fisher scale instead orders model-parameter directions by statistical distinguishability; it neither chooses node partitions nor tests holonomy. [ESTABLISHED] Constructing a nondegenerate partition selector compatible with these separate structures remains open. [OPEN]

Mehta and Schwab give an exact variational-RG/deep-network correspondence under their RBM kernel and trace condition [49]. It is a historical exact precedent inside that construction, not a theorem that deterministic energy precomposition is a neural layer. Likewise, scalar Laplacian RG provides a heat-spectrum scale diagnostic and a heuristic real-space blocker; lifting its entrywise blocker covariantly to the two represented frame channels requires a new invariant statistic. These are constraints on applications, not definitions of the present general RG. [ESTABLISHED]

Physicist summary. Blocking removes detail; renormalization adds the maps that say how the blocked theory is compared with the original units. Here that comparison must carry one principal bundle, its two differently represented statistical bundles, two connection choices that may coincide, and two separately supplied cross morphisms. Separate belief and model frames are local views of the same principal geometry; their relative element lies in G_ℓ , even when its two representations act on unlike spaces. Failures to commute with blocking are covariant defect fields. Once all scales are identified, the dynamics may be autonomous, periodic, random, or a general cocycle, and each case has a different invariant object. System size and RG depth then form a two-index limit. Fixed points, universality, and continuum laws are meaningful only after these types and limits have been declared.

Part III

The Multivariate-Gaussian Realization

Chapter 9

A Linear-Gaussian Generative Realization

This chapter realizes the fixed directed law of Chapter 4 in two associated bundles induced from the same principal G -bundle, using possibly different representations and separately chosen local frames. The Gaussian is a tractable subclass, not the ambient theory. The specialization assumes that at a root the model-to-belief kernel induces a map from model laws to belief laws that may realize $\tilde{\Phi} : \mathcal{E}_m \rightarrow \mathcal{E}_b$. The sample matrix inside that kernel is not itself the associated-bundle morphism. At a nonroot the kernel is parent-indexed unless its domain is enlarged. The existence of $\Phi : \mathcal{E}_b \rightarrow \mathcal{E}_m$ creates no reverse conditional.

9.1 Directed forest and conditional laws

Take

$$k_i \in \mathbb{R}^K, \quad m_i \in \mathbb{R}^{d_m}, \quad o_i \in \mathbb{R}^{d_{o,i}}, \quad (9.1)$$

and specialize the directed acyclic graph to a forest, so each nonroot i has one parent $j = \text{par}(i)$. This is a displayed subclass; the general DAG replaces each parent term by a finite sum.

[DEFINITION]

Fix one design point c_a . Let $\Theta_{ij}^b := \Theta_{a,ij}^b \in G$ and $\Theta_{ij}^m := \Theta_{a,ij}^m \in G$ abbreviate the design-indexed same-channel graph-link copies of Section 4.6. A submodel that constrains one common link across all design points retains the stabilizer of that constraint, namely the frame rechoices for which $h_{a,i}^x \Theta_{ij}^x (h_{a,j}^x)^{-1}$ is independent of a . Context-independent rechoices form a sufficient, but not generally maximal, subgroup; any larger law must be declared explicitly. For a nonroot,

$$m_i \mid m_j, X \sim \mathcal{N} \left(F_i^m \rho_m(\Theta_{ij}^m) m_j + \eta_i^m, \Sigma_i^m \right), \quad (9.2)$$

$$k_i \mid k_j, m_i, X \sim \mathcal{N} \left(F_i^b \rho_b(\Theta_{ij}^b) k_j + B_i m_i + \eta_i^b, \Sigma_i^b \right), \quad (9.3)$$

$$o_i \mid k_i, m_i, X \sim \mathcal{N} \left(H_i^b k_i + H_i^m m_i + \eta_i^o, \Sigma_i^o \right). \quad (9.4)$$

The parameter shapes are

$$\rho_b(\Theta_{ij}^b), F_i^b \in \mathbb{R}^{K \times K}, \quad \rho_m(\Theta_{ij}^m), F_i^m \in \mathbb{R}^{d_m \times d_m}, \quad B_i \in \mathbb{R}^{K \times d_m}, \quad (9.5)$$

$$H_i^b \in \mathbb{R}^{d_{o,i} \times K}, \quad H_i^m \in \mathbb{R}^{d_{o,i} \times d_m}, \quad \eta_i^b \in \mathbb{R}^K, \quad \eta_i^m \in \mathbb{R}^{d_m}, \quad \eta_i^o \in \mathbb{R}^{d_{o,i}}. \quad (9.6)$$

The represented links are invertible. The conditional endomorphisms F_i^b, F_i^m need not be: positive definiteness of the receiving covariance, not invertibility of the mean map, normalizes a Gaussian kernel.

The bridge matrix $B_i : \mathbb{R}^{d_m} \rightarrow \mathbb{R}^K$ acts on sample coordinates. It is one parameter of a Gaussian kernel, not a map between probability-law fibers. At a nonroot, the kernel also depends on the parent belief coordinate. Only after fixing that parent, or enlarging the domain to include it, does integration against a model law define a belief law. The left-right covariance law below is a parameter transformation that makes this induced law map covariant; it does not identify B_i with $\tilde{\Phi}$. No reading supplies a belief-to-model factor.

Each root r has the proper laws

$$m_r \mid X \sim \mathcal{N}(\bar{m}_r, \Sigma_{r,0}^m), \quad (9.7)$$

$$k_r \mid m_r, X \sim \mathcal{N}(B_r m_r + \eta_{r,0}^b, \Sigma_{r,0}^b), \quad (9.8)$$

followed by (9.4). Every receiving covariance is positive definite:

$$\Sigma_i^m, \Sigma_{r,0}^m \in \mathbb{S}_{++}^{d_m}, \quad \Sigma_i^b, \Sigma_{r,0}^b \in \mathbb{S}_{++}^K, \quad \Sigma_i^o \in \mathbb{S}_{++}^{d_{o,i}}. \quad (9.9)$$

[HYPOTHESIS]

Proposition 9.1 (Normalization and root moments). Under (9.9), every kernel above is a nondegenerate normalized density and the assembled directed law is a probability measure. The root belief coordinate has

$$\bar{k}_r = B_r \bar{m}_r + \eta_{r,0}^b, \quad \Sigma_{r,0}^{b,\text{marg}} = \Sigma_{r,0}^b + B_r \Sigma_{r,0}^m B_r^\top. \quad (9.10)$$

[ESTABLISHED]

More generally, if a root model law is $s = \mathcal{N}(\mu_m, C_m)$, then the law map induced by the root kernel is

$$\tilde{\phi}_r(s) = \mathcal{N}\left(B_r \mu_m + \eta_{r,0}^b, \quad \Sigma_{r,0}^b + B_r C_m B_r^\top\right). \quad (9.11)$$

This map, rather than B_r alone, is the candidate local realization of $\tilde{\Phi}$. [ESTABLISHED]

Proof. A Gaussian density with positive definite covariance integrates to one for every mean, so the normalization proposition of Chapter 4 applies. The tower property gives the first moment and the law of total covariance gives the second identity. Its added term is positive semidefinite, so the marginal covariance dominates the conditional covariance. $\square$

The Gaussian observation kernel may be replaced by any normalized Markov kernel from $\mathbb{R}^K \times \mathbb{R}^{d_m}$ to the observation space. That changes the likelihood but not the exact ELBO identity, which uses only the fixed normalized joint. [ESTABLISHED]

9.2 Independent channel-frame covariance

For the global coordinate formulas in this section, invoke Hypothesis 2.11 on a declared region $\mathcal{C}_0$ containing the contexts under study. Choose $\sigma_0 : \mathcal{C}_0 \rightarrow P|_{\mathcal{C}_0}$ and restrict each local frame to $\mathcal{C}_i \cap \mathcal{C}_0$. Then write the independent frame coordinates as

$$U_i^b : \mathcal{C}_i \cap \mathcal{C}_0 \rightarrow G, \quad U_i^m : \mathcal{C}_i \cap \mathcal{C}_0 \rightarrow G, \quad u_i^x = \sigma_0 \cdot (U_i^x)^{-1}. \quad (9.12)$$

Thus σ_0 is common only on the explicitly trivialized region; no global triviality of P is inferred. The corresponding relative principal-frame field is

$$h_i = U_i^b (U_i^m)^{-1}, \quad u_i^m = u_i^b h_i. \quad (9.13)$$

This G -valued field exists even when ρ_b and ρ_m are inequivalent or have different dimensions. Its represented actions $\rho_b(h_i)$ and $\rho_m(h_i)$ live in different endomorphism spaces and do not define the bridge B_i or either associated-bundle morphism. An independent pair of channel-frame re-expressions is $(g_i^b, g_i^m) \in G \times G$, with represented matrices $R_i^b = \rho_b(g_i^b)$ and $R_i^m = \rho_m(g_i^m)$ as in (4.18). The pair records two choices of local trivialization of bundles associated to the same P ; it does not replace the principal structure group G by $G \times G$. In the convention of (2.10), the corresponding principal-section rechoices are $u_i^{b'} = u_i^b \cdot (g_i^b)^{-1}$ and $u_i^{m'} = u_i^m \cdot (g_i^m)^{-1}$. This accounts for the opposite-sided inverse in the geometric local-map law and makes the formulas below identical to (2.15).

The same-channel links obey (4.19). The conditional maps and offsets transform as

$$F_i^{b'} = R_i^b F_i^b (R_i^b)^{-1}, \quad F_i^{m'} = R_i^m F_i^m (R_i^m)^{-1}, \quad (9.14)$$

$$B_i' = R_i^b B_i (R_i^m)^{-1}, \quad \eta_i^{b'} = R_i^b \eta_i^b, \quad \eta_i^{m'} = R_i^m \eta_i^m. \quad (9.15)$$

The left-right law for B_i , together with the covariance and offset laws, makes the induced root law map (9.11) obey the local law of $\tilde{\Phi}$. The parameter B_i alone is not that morphism, and the two frames remain independent. Receiving covariances transform by their own congruences:

$$\Sigma_i^{b'} = R_i^b \Sigma_i^b (R_i^b)^\top, \quad \Sigma_i^{m'} = R_i^m \Sigma_i^m (R_i^m)^\top. \quad (9.16)$$

For either channel,

$$\Lambda_i = \Sigma_i^{-1}, \quad \Sigma_i' = R_i \Sigma_i R_i^\top, \quad \Lambda_i' = R_i^{-\top} \Lambda_i R_i^{-1}. \quad (9.17)$$

At a root,

$$\bar{m}'_r = R_r^m \bar{m}_r, \quad \Sigma_{r,0}^{m'} = R_r^m \Sigma_{r,0}^m (R_r^m)^\top, \quad (9.18)$$

$$B'_r = R_r^b B_r (R_r^b)^{-1}, \quad \eta_{r,0}^{b'} = R_r^b \eta_{r,0}^b, \quad \Sigma_{r,0}^{b'} = R_r^b \Sigma_{r,0}^b (R_r^b)^\top. \quad (9.19)$$

Gauge-inert observations require

$$H_i^{b'} = H_i^b (R_i^b)^{-1}, \quad H_i^{m'} = H_i^m (R_i^m)^{-1}, \quad \eta_i^{o'} = \eta_i^o, \quad \Sigma_i^{o'} = \Sigma_i^o. \quad (9.20)$$

[DEFINITION]

Proposition 9.2 (Gaussian kernel covariance). The transformation laws above satisfy the three kernel pushforward identities (4.20)–(4.22), and hence the normalized joint and its evidence obey (4.23)–(4.24) under all independent channel-frame re-expressions. [ESTABLISHED]

Proof. For the belief kernel, the transformed conditional mean evaluated at transformed inputs is

$$F_i^{b'} \rho_b(\Theta_{ij}^{b'}) (R_j^b k_j) + B'_i (R_i^m m_i) + \eta_i^{b'} = R_i^b \left(F_i^b \rho_b(\Theta_{ij}^b) k_j + B_i m_i + \eta_i^b \right). \quad (9.21)$$

The model-channel calculation is the same without a bridge. Congruence (9.16) therefore makes each transformed Gaussian the receiving-frame pushforward of the original. At the observation kernel,

$$H_i^{b'} (R_i^b k_i) + H_i^{m'} (R_i^m m_i) + \eta_i^{o'} = H_i^b k_i + H_i^m m_i + \eta_i^o, \quad (9.22)$$

and its covariance is fixed. The root calculation is identical. Composition now invokes the frame-covariance result of Chapter 4. $\square$

Equation (9.10) transforms by belief-channel congruence, because both of its summands do. Positive definiteness is preserved independently in each channel.

Group-valued frame coordinates remain primary. An expression $U = \exp \phi$ is valid only inside an exponential chart; for example, $\text{diag}(-1, -2) \in \text{GL}^+(2)$ has no real logarithm because each negative eigenvalue occurs in one Jordan block [21]. [ESTABLISHED]

9.3 Common-frame specialization

Under the common-frame specialization of Section 2.11, impose $u_i^b = u_i^m$ and use one frame re-expression $g_i \in G$ with

$$R_i^b = \rho_b(g_i), \quad R_i^m = \rho_m(g_i). \quad (9.23)$$

If the graph links are also identified, Θ_{ij}^b and Θ_{ij}^m are the same group link seen through the two representations. Equations (9.14)–(9.20) then reduce to the familiar common-frame, two-representation formulas. [ESTABLISHED]

This reduction ties the gauge choices but does not identify the latent dimensions, make B_i invertible, turn Φ and $\tilde{\Phi}$ into inverses, or add a reverse generative factor. Compatibility of a cross map with the

two connections remains the parallelness condition of Chapter 2; an equal connection is a further hypothesis.

Finally, the assembled Gaussian precision need not have the Laplacian-plus-self-terms form used later. A general directed transition can contribute an off-diagonal block that is neither symmetric nor sign-correct. The interaction family is therefore a separately declared Gaussian operator subfamily, not a consequence of linear-Gaussian directedness. [NOT-CLAIMED]

Chapter 10

The Gaussian Information Form and the Interaction Family

The preceding chapters first allow general belief-law and model-law fibers, with smooth manifolds selected only at the differential tier, and then identify regular exponential families as a tractable dominated subclass. Nothing in that construction selects the multivariate Gaussian. This chapter makes that additional choice. It develops one finite-dimensional Gaussian realization in information form, where the primitive coordinates are a vector and a matrix of natural parameters rather than a mean and a covariance, and it then isolates the still narrower subfamily of Gaussian precisions on which the renormalization half of this document acts. None of the matrix identities below is asserted for an arbitrary belief fiber.

Two things are worth separating at the outset, because conflating them is the standing hazard of the whole subject. Writing a Gaussian in information form is a change of coordinates and costs nothing. Restricting the precision to the Laplacian-plus-self-terms shape of Section 10.3 is a substantive restriction that costs a great deal, and it does not follow from the generative constructions of the preceding part. Section 10.4 proves that it does not, exhibits the exact obstruction, and then states precisely the parameter condition under which the shape is recovered. The restriction is carried forward as a declared hypothesis, and every later result that uses it says so.

Throughout, the development is written for the state sector, with agents $i \in V = \{1, \dots, N\}$, state latents $k_i \in \mathbb{R}^K$, and stacked latent Y of dimension $n := NK$. The model sector, with latents $m_i \in \mathbb{R}^{d_m}$, is identical with K replaced by d_m , and a partition mixing the two sectors is admissible wherever a partition is required. The design $D = \{c_a\}_{a=1}^M$ plays no role in the algebra below; either fix one design point, or read the index i as ranging over agent-design pairs, and every statement stands unchanged.

10.1 The recognition family in information form

Let ν be Lebesgue measure on $\mathbb{R}^n$. The Gaussian family in *information form* is parameterized by a vector $h \in \mathbb{R}^n$ and a symmetric positive definite matrix $J \in \mathbb{S}_{++}^n$ through the density

$$q(y \mid h, J) = \exp \left(h^\top y - \frac{1}{2} y^\top J y - A(h, J) \right), \quad y \in \mathbb{R}^n. \quad (10.1)$$

The pair (h, J) are the natural parameters, J is the precision, and A is the log normalizer. The symbol A is reserved for the log normalizer and never for a self term; the self terms of Section 10.3 are written A_i with a subscript and are $K \times K$ matrices.

Proposition 10.1 (Log normalizer, moments, and convexity). For every $(h, J) \in \mathbb{R}^n \times \mathbb{S}_{++}^n$ the density in Equation (10.1) is normalizable, with

$$A(h, J) = \frac{1}{2} h^\top J^{-1} h - \frac{1}{2} \log \det J + \frac{n}{2} \log 2\pi. \quad (10.2)$$

Its differential is

$$dA = \mu^\top dh - \frac{1}{2} \text{Tr} \left[(C + \mu\mu^\top) dJ \right], \quad \mu = J^{-1}h, \quad C = J^{-1}, \quad (10.3)$$

so the mean and covariance of $q(\cdot \mid h, J)$ are μ and C , and the Hessian of A in h at fixed J is C . Regarded as a function of the natural parameter pair (h, Θ) with $\Theta = -\frac{1}{2}J$, the log normalizer is strictly convex on the open convex domain $\mathbb{R}^n \times \{\Theta \prec 0\}$. **[ESTABLISHED]**

Proof. Complete the square in the exponent,

$$h^\top y - \frac{1}{2} y^\top J y = -\frac{1}{2} (y - J^{-1}h)^\top J (y - J^{-1}h) + \frac{1}{2} h^\top J^{-1}h, \quad (10.4)$$

and integrate the Gaussian kernel, whose value is $(2\pi)^{n/2}(\det J)^{-1/2}$ because $J \succ 0$. Exponentiating and taking logarithms gives Equation (10.2). Differentiating Equation (10.2) termwise, $d(\frac{1}{2}h^\top J^{-1}h) = h^\top J^{-1}dh - \frac{1}{2} \text{Tr}(J^{-1}hh^\top J^{-1}dJ)$ and $d(-\frac{1}{2} \log \det J) = -\frac{1}{2} \text{Tr}(J^{-1}dJ)$, which is Equation (10.3). The identification of $\nabla_h A$ with the mean and $\nabla_h^2 A$ with the covariance is the standard exponential-family cumulant property, valid here because the family is regular: the domain is open and differentiation under the integral is licensed on the interior of the natural domain [17]. Convexity is the convexity of a log-Laplace transform, and strictness follows because the family is minimal, the statistics $(y, yy^\top)$ being affinely independent [17, 68]. Since $J \mapsto -\frac{1}{2}J$ is affine, convexity in (h, Θ) transfers to convexity in (h, J) on $\{J \succ 0\}$. $\square$

Two consequences of Equation (10.3) organize everything that follows. The first is that (h, J) and (μ, C) are two coordinate systems on one family, related by the Legendre correspondence of a minimal exponential family, so no information is lost by working in either. The second is that the Fisher information of the family in the natural parameter h is exactly $C = J^{-1}$, while the Fisher information in the mean parameter μ at fixed covariance is exactly J . The precision is therefore simultaneously a coupling matrix and a metric, and the renormalization chapters use both readings.

10.2 What an off-diagonal precision block is

Let $\mathfrak{B}$ be a finite partition of the n coordinates into blocks, with block dimensions n_b summing to n , and write J_{bc} for the corresponding submatrices. The meaning of J_{bc} for $b \neq c$ is fixed by the

following statement, which is standard but is stated here in full because the rest of the document depends on reading it correctly.

Proposition 10.2 (Conditional reading of the precision). Let $Y \sim \mathcal{N}(\mu, J^{-1})$ with $J \succ 0$. For any block b ,

$$\mathbb{E}[Y_b \mid Y_{-b} = y_{-b}] = \mu_b - J_{bb}^{-1} \sum_{c \neq b} J_{bc} (y_c - \mu_c), \quad \text{Cov}(Y_b \mid Y_{-b}) = J_{bb}^{-1}. \quad (10.5)$$

For any two distinct blocks b and c , the variables Y_b and Y_c are conditionally independent given all remaining coordinates if and only if $J_{bc} = 0$. **[ESTABLISHED]**

Proof. Fix y_{-b} and read the exponent of Equation (10.1) as a function of y_b alone. It is $-\frac{1}{2}y_b^\top J_{bb}y_b$ plus a linear term, so the conditional law is Gaussian with precision J_{bb} , and completing the square in y_b around μ gives the displayed conditional mean. For the second claim, condition on the coordinates outside $b \cup c$ and repeat the argument: the conditional law of (Y_b, Y_c) is Gaussian with precision $\begin{pmatrix} J_{bb} & J_{bc} \\ J_{cb} & J_{cc} \end{pmatrix}$, and a Gaussian density factorizes into a function of y_b times a function of y_c if and only if the quadratic form has no cross term, that is if and only if $J_{bc} = 0$. This is the Gaussian case of the pairwise Markov property [43, Ch. 5]; the same statement organizes the Gaussian Markov random field literature [59, Ch. 2]. $\square$

Equation (10.5) also fixes the sign convention that Section 10.3 will exploit. A negative semidefinite off-diagonal block $J_{bc} \preceq 0$ makes the conditional mean of Y_b move *toward* y_c as y_c departs from its own mean, so negative off-diagonal precision is attractive coupling. The interaction family is the family in which every off-diagonal block is attractive in this sense.

Precision blocks are not covariance values. A zero off-diagonal precision block is a statement about conditional dependence and carries no implication about the marginal covariance. The tridiagonal example

$$J = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}, \quad J^{-1} = \frac{1}{4} \begin{pmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{pmatrix}, \quad (10.6)$$

has $J_{13} = 0$ and $(J^{-1})_{13} = \frac{1}{4} \neq 0$. In general the marginal covariance of a block is the inverse of a Schur complement,

$$(J^{-1})_{bb} = \left(J_{bb} - J_{b,-b} J_{-b,-b}^{-1} J_{-b,b} \right)^{-1}, \quad (10.7)$$

so it records the effect of every path through the remaining coordinates, whereas J_{bb}^{-1} records only the conditional dispersion of Equation (10.5) [73]. The two coincide exactly when $J_{b,-b} = 0$. **[ESTABLISHED]**

The distinction exhibited by (10.6) is the reason this document works in information form. A restriction imposed on the recognition family is a restriction on conditional dependence, and it is stated cleanly only in coordinates in which conditional dependence is a coordinate. The exact price is computed in Chapter 11.

10.3 The declared interaction subfamily

Definition 10.3 (Interaction family). Let $\Lambda \in \mathbb{S}^{NK}$ be partitioned into $K \times K$ blocks indexed by agents. Say that Λ belongs to the *interaction family* $\mathcal{I}(V)$ when there exist matrices $A_i \in \mathbb{S}_+^K$ and $W_{ij} = W_{ji} \in \mathbb{S}_+^K$ for $i \neq j$ with

$$\Lambda_{ii} = A_i + \sum_{j \neq i} W_{ij}, \quad \Lambda_{ij} = -W_{ij} \quad (i \neq j). \quad (10.8)$$

The A_i are the *self terms*, the W_{ij} the *edge weights*, and $L := \Lambda - \text{blockdiag}(A_1, \dots, A_N)$ the *Laplacian part*. Nothing is proved by this definition; it declares a type. [DEFINITION]

The edge weights are matrices, so L is a matrix-weighted graph Laplacian rather than a scalar-weighted one. That object, and the ways in which it fails to behave like its scalar counterpart, are the subject of the matrix-weighted consensus literature [62], and one of those failures appears immediately below.

Proposition 10.4 (Energy and kernel). For $\Lambda \in \mathcal{I}(V)$ and any $y = (y_1, \dots, y_N)$,

$$y^\top \Lambda y = \sum_{i \in V} y_i^\top A_i y_i + \sum_{i < j} (y_i - y_j)^\top W_{ij} (y_i - y_j). \quad (10.9)$$

Consequently $\Lambda \succeq 0$ for every admissible choice of parameters; the consensus subspace $\mathbf{1} \otimes \mathbb{R}^K$ lies in $\ker L$; and $\Lambda \succ 0$ if and only if the only y satisfying $A_i y_i = 0$ for all i together with $W_{ij}(y_i - y_j) = 0$ for all $i < j$ is $y = 0$. A necessary condition for $\Lambda \succ 0$ is $\sum_i A_i \succ 0$. [ESTABLISHED]

Proof. Expand the right-hand side of Equation (10.9). The self sum contributes $\sum_i y_i^\top A_i y_i$ to the diagonal blocks. Each unordered pair contributes $y_i^\top W_{ij} y_i + y_j^\top W_{ij} y_j - 2y_i^\top W_{ij} y_j$, which places W_{ij} on the ii and jj diagonal blocks and $-W_{ij}$ on the ij and ji off-diagonal blocks. Summing reproduces Equation (10.8). Every term of Equation (10.9) is nonnegative because $A_i \succeq 0$ and $W_{ij} \succeq 0$, so $\Lambda \succeq 0$; and a sum of nonnegative terms vanishes only when each vanishes, which for a positive semidefinite matrix A and vector v means $v^\top A v = 0$ if and only if $A v = 0$. Taking $y = \mathbf{1} \otimes v$ makes every difference vanish, so the consensus subspace lies in $\ker L$, and evaluating Equation (10.9) there gives $v^\top (\sum_i A_i) v$, whence the necessary condition. $\square$

The kernel criterion in Proposition 10.4 is where the matrix-weighted setting departs from the scalar one. In the scalar case a connected interaction graph forces $\ker L$ to be exactly the consensus subspace. With matrix weights, a singular W_{ij} leaves the difference $y_i - y_j$ free along $\ker W_{ij}$, so $\ker L$ can be strictly larger than the consensus subspace even on a connected graph, and connectivity alone is not a definiteness criterion [62]. The criterion stated in the proposition is exact and replaces it.

Proposition 10.5 (The interaction family is thin). For $K \geq 2$ and $N \geq 2$, $\mathcal{I}(V)$ is contained in a proper linear subspace of $\mathbb{S}^{NK}$ of codimension at least $\binom{N}{2} \frac{K(K-1)}{2}$, hence has Lebesgue measure zero in $\mathbb{S}^{NK}$. [ESTABLISHED]

Proof. Symmetry of a matrix Λ already forces $\Lambda_{ji} = \Lambda_{ij}^\top$ for every ordered pair, so this identity carries no information about the family. Membership additionally requires each off-diagonal block Λ_{ij} to be symmetric as a $K \times K$ matrix, since $W_{ij} = W_{ji}$ and $W_{ji} = W_{ij}^\top$ together give $W_{ij} = W_{ij}^\top$. That is $\frac{K(K-1)}{2}$ independent linear equations per unordered pair, and there are $\binom{N}{2}$ pairs. The remaining requirements are conic inequalities, which can only shrink the set further. $\square$

Proposition 10.5 is the quantitative form of the claim that the interaction shape is a restriction in a fixed product chart rather than a normalization. It is not invariant under arbitrary independent block-diagonal coordinate changes, so membership may be created or destroyed by such a change and no coordinate-orbit exclusion follows from thinness. An exact witness starts from the interaction precision with $N = 2$, $K = 2$, $A_1 = A_2 = I_2$, and $W_{12} = I_2$. Congruence by $G = \text{diag}(I_2, S)$, $S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, changes the off-diagonal block from $-I_2$ to $-S$, which is not symmetric and hence is outside the family; applying G^{-1} to that transformed precision creates membership again. The invariant statement is only that a declared trivialization must be fixed before Equation (10.8) is imposed. Section 10.4 asks when the generative construction supplies such a distinguished chart.

10.4 The interaction form is a hypothesis and not a consequence

10.4.1 Trivialized coordinates and flat comparison

In this Gaussian belief-channel subsection, write $\Theta_{ij} := \rho_b(\Theta_{ij}^b) \in \text{GL}^+(K)$ for the represented graph link. This abbreviation does not identify it with a base-manifold parallel transport.

The state latents carry frames. A residual between two agents is not $k_i - k_j$, which compares vectors in different fibers, but $k_i - \Theta_{ij}k_j$, where Θ_{ij} is the represented belief-channel graph link. The interaction shape of Definition 10.3 involves plain differences, so it can only be a statement in coordinates in which the graph links have been removed. When that removal is possible is settled exactly by the following.

Proposition 10.6 (Simultaneous trivialization is exactly coboundary). Let $G = (V, E)$ be a connected graph and let $\{\Theta_{ij}\}_{(i,j) \in E} \subset \text{GL}^+(K)$ satisfy $\Theta_{ji} = \Theta_{ij}^{-1}$. There exist frames $\{U_i\} \subset \text{GL}^+(K)$ with $U_i^{-1}\Theta_{ij}U_j = I$ for every edge if and only if $\Theta_{ij} = U_iU_j^{-1}$ for those frames, and this holds if and only if the ordered product of graph links around every cycle of G is the identity. When it holds, the frames are unique up to a common right factor $U_i \mapsto U_i g$ with $g \in \text{GL}^+(K)$ independent of i . [ESTABLISHED]

Proof. The condition $U_i^{-1}\Theta_{ij}U_j = I$ is literally $\Theta_{ij} = U_iU_j^{-1}$, which gives the first equivalence. For the second, suppose such frames exist and let $i_0, i_1, \dots, i_r = i_0$ be a cycle. Then $\prod_t \Theta_{i_t i_{t+1}} = \prod_t U_{i_t} U_{i_{t+1}}^{-1} = U_{i_0} U_{i_0}^{-1} = I$ by telescoping. Conversely, assume every cycle product is the identity, fix

a spanning tree and a root r , set $U_r = I$, and define U_i as the ordered product of graph links along the tree path from r to i . The relation $\Theta_{ij} = U_i U_j^{-1}$ then holds on every tree edge by construction, and on a non-tree edge it holds precisely because the cycle formed by that edge and the tree path between its endpoints has trivial product. For uniqueness, if $U_i U_j^{-1} = U'_i U'^{-1}_j$ on every edge then $U'^{-1}_i U_i = U'^{-1}_j U_j$ for adjacent i, j , so this quantity is constant on a connected graph. $\square$

Hypothesis 10.7 (Flat comparison). The declared graph links are a coboundary, $\Theta_{ij} = U_i U_j^{-1}$, so that the trivialization $z_i = U_i^{-1} \mu_i$ of the means, and the same map $z_i = U_i^{-1} k_i$ applied to the latents themselves, turn every residual into a plain difference,

$$\mu_i - \Theta_{ij} \mu_j = U_i (z_i - z_j). \quad (10.10)$$

This is a choice, not a derivation. By Proposition 10.6 it excludes exactly the graph-link assignments with nontrivial holonomy around some cycle, which are permitted in Regime II. It is used in two places: it makes the interaction shape of Definition 10.3 a statement in one fixed coordinate system, and it supplies a clean sufficient domain for closure under aggregation. In the flat chart the represented link $\hat{\Theta}_{ij} := U_i^{-1} \Theta_{ij} U_j$ equals I , so merging two agents by $z_i = z_j = z$ annihilates their internal residual. More generally, in an arbitrary common chart the internal edge energy becomes

$$z^\top (I - \hat{\Theta}_{ij})^\top W_{ij} (I - \hat{\Theta}_{ij}) z,$$

which vanishes for every z exactly when $W_{ij}^{1/2} (I - \hat{\Theta}_{ij}) = 0$. Positive-definite W_{ij} , or faithfulness of the weight on every represented direction, reduces this condition to $\hat{\Theta}_{ij} = I$; with merely positive-semidefinite weights, a nontrivial represented link can survive in invisible null directions. Thus Proposition 10.6 remains the exact coboundary theorem, while the converse from energy annihilation to a trivial represented link requires the stated nondegeneracy. The coarse-graining chapter carries out the weighted computation and separates the sufficient trivializing domain from degenerate exceptions. **[HYPOTHESIS]**

Two remarks fix the status of Equation (10.10). First, the change of variables $y = Uz$ with $U = \text{blockdiag}(U_1, \dots, U_N)$ acts on precisions by congruence, $\Lambda^{(y)} = U^{-\top} \Lambda^{(z)} U^{-1}$, which preserves inertia by Sylvester's law [32, Ch. 4] but does not preserve membership in $\mathcal{I}(V)$, since the transformed off-diagonal block $U_i^{-\top} \Lambda_{ij}^{(z)} U_j^{-1}$ need not be symmetric when $U_i \neq U_j$. Membership in the interaction family is therefore a frame-dependent assertion. Second, that frame dependence is exactly why the invariants of the renormalization chapters are stated as generalized eigenvalues of the pair (L, Λ) rather than as eigenvalues of either matrix alone: the pair transforms by the same congruence and the generalized spectrum is unchanged. **[ESTABLISHED]**

10.4.2 What a directed linear-Gaussian model actually contributes

Consider the state sector of a directed linear-Gaussian model of the kind used in the preceding part: each non-root agent i has a single parent $j = \text{pa}(i)$ and receiving kernel

$$k_i \mid k_j \sim \mathcal{N}(G_i \Theta_{ij} k_j, \Lambda_i^{-1}), \quad \Lambda_i \in \mathbb{S}_{++}^K, \quad G_i \in \mathbb{R}^{K \times K}, \quad (10.11)$$

and each root r carries a proper prior $\mathcal{N}(0, \Lambda_r^{-1})$, a nonzero prior mean shifting only the information vector h and not the precision. The gain is written G_i rather than A_i because the latter symbol is reserved for the self terms of Definition 10.3. Unlike the represented graph link Θ_{ij} , the conditional mean map G_i need not be invertible: positive definiteness of the innovation covariance normalizes the receiving kernel for every matrix G_i , exactly as Chapter 4 declares. A single subscript on Λ denotes the innovation precision of one receiving kernel; a double subscript denotes a block of the assembled precision. When an invertible-gain submodel is wanted it will be named explicitly.

Proposition 10.8 (Assembled precision blocks). Write $T_i := G_i \Theta_{i \text{pa}(i)}$ and $\text{ch}(i) = \{\ell : \text{pa}(\ell) = i\}$. The joint precision of the model above has blocks

$$\Lambda_{i \text{pa}(i)} = -\Lambda_i T_i, \quad \Lambda_{ii} = \Lambda_i + \sum_{\ell \in \text{ch}(i)} T_\ell^\top \Lambda_\ell T_\ell, \quad (10.12)$$

and $\Lambda_{ij} = 0$ whenever i and j are not adjacent in the parent graph. [ESTABLISHED]

Proof. The negative log joint density is $\frac{1}{2} \sum_i (k_i - T_i k_{\text{pa}(i)})^\top \Lambda_i (k_i - T_i k_{\text{pa}(i)})$ plus terms not depending on k , with the parent term absent at a root. Expanding one summand gives $\frac{1}{2} k_i^\top \Lambda_i k_i - k_i^\top \Lambda_i T_i k_{\text{pa}(i)} + \frac{1}{2} k_{\text{pa}(i)}^\top T_i^\top \Lambda_i T_i k_{\text{pa}(i)}$. The precision is the Hessian of this function, and collecting the three contributions of each factor gives Equation (10.12). $\square$

By Equation (10.12), membership in the interaction family requires the matrix

$$W_{i \text{pa}(i)} := \Lambda_i T_i = \Lambda_i G_i \Theta_{i \text{pa}(i)} \quad (10.13)$$

to be symmetric and positive semidefinite. Neither property is automatic, and both fail on explicit parameter values.

Two sources of failure must be kept apart, and a convention does it. Under Hypothesis 10.7 the trivialized residual is $z_i - \tilde{T}_i z_j$ with

$$\tilde{T}_i := U_i^{-1} T_i U_j = U_i^{-1} G_i U_i, \quad \tilde{\Lambda}_i := U_i^\top \Lambda_i U_i, \quad (10.14)$$

the second equality in the first display using $\Theta_{ij} = U_i U_j^{-1}$, so that (10.12) holds verbatim in trivialized coordinates with (Λ_i, T_i) replaced by $(\tilde{\Lambda}_i, \tilde{T}_i)$. The graph links have then already been removed and any remaining obstruction is attributable to the gain alone. For the rest of this section the tildes are dropped and (Λ_i, T_i) denote the trivialized quantities.

Proposition 10.9 (Two global-condition witnesses). There exist admissible parameters for which no interaction-family representation of the assembled precision exists.

$$\Lambda_i = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \succ 0, \quad T_i = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in \text{GL}^+(2) \quad \implies \quad -\Lambda_i T_i = \begin{pmatrix} 1 & -2 \\ 3 & -1 \end{pmatrix}, \quad (10.15)$$

which differs from its own transpose by a matrix of Frobenius norm $5\sqrt{2}$, so the off-diagonal block is not symmetric and no W exists at all. Separately, taking $K = 1$ with $G_i = -1$ and $\Theta_{ij} = \Lambda_i = 1$

gives a symmetric block with $W = -1 < 0$, so the sign requirement fails; the same construction with $G_i = -I$, $\Theta_{ij} = I$ and $\Lambda_i = I$ gives $W = -I \prec 0$ at every K . [ESTABLISHED]

Proof. Direct computation. For the first, $\Lambda_i T_i = \begin{pmatrix} -1 & 2 \\ -3 & 1 \end{pmatrix}$, so $-\Lambda_i T_i$ is as displayed and $-\Lambda_i T_i - (-\Lambda_i T_i)^\top = \begin{pmatrix} 0 & -5 \\ 5 & 0 \end{pmatrix}$ with Frobenius norm $\sqrt{50}$. Positive definiteness of Λ_i follows from $\det = 5 > 0$ and trace $5 > 0$; invertibility of T_i from $\det T_i = 1 > 0$. For the second, (10.13) evaluates to $\Lambda_i G_i \Theta_{ij} = -I$, and $-I$ is not positive semidefinite for any $K \geq 1$. $\square$

Proposition 10.9 settles the status of the interaction shape. The gain matrix G_i is a free generative parameter in the construction of the preceding part, not required to be symmetric, positive, or even close to the identity, and for the displayed values the assembled precision has no representation of the form of Equation (10.8). Whatever else may be said for the interaction family, it is not a consequence of the general linear-Gaussian directed model.

10.4.3 Exactly when the form does hold

Having shown that the form fails, the obligation is to say when it holds. Two statements are available, one global and one local, and they answer different questions.

Proposition 10.10 (Exact global interaction condition). The assembled precision of Equation (10.12) lies in $\mathcal{I}(V)$ with the edge assignment of Equation (10.13) if and only if, for every non-root i ,

$$W_i := \Lambda_i T_i \in \mathbb{S}_+^K, \quad (10.16)$$

and, for every agent i ,

$$A_i := \Lambda_i - W_i + \sum_{\ell \in \text{ch}(i)} (W_\ell \Lambda_\ell^{-1} W_\ell - W_\ell) \succeq 0, \quad (10.17)$$

with the term $-W_i$ omitted at a root. [ESTABLISHED]

Proof. Condition (10.16) is exactly the requirement that the off-diagonal blocks have the form $-W$ with W symmetric positive semidefinite. Given it, $T_i = \Lambda_i^{-1} W_i$ and $T_\ell^\top \Lambda_\ell T_\ell = W_\ell \Lambda_\ell^{-1} \Lambda_\ell \Lambda_\ell^{-1} W_\ell = W_\ell \Lambda_\ell^{-1} W_\ell$, using symmetry of W_ℓ . The self term is then the residual $A_i = \Lambda_{ii} - \sum_{j \sim i} W_{ij}$, where the edges at i are the one to its parent, of weight W_i , and one to each child ℓ , of weight W_ℓ . Substituting Equation (10.12) gives Equation (10.17), and family membership requires exactly that this residual be positive semidefinite. $\square$

Condition (10.17) is a balance at each node and permits cancellation: a parent deficit $\Lambda_i - W_i \prec 0$ may be offset by child surpluses. That freedom disappears if one asks the stronger and more natural question, namely whether each generative factor separately lies in the family, so that the interaction structure can be read off the model factor by factor rather than after a global cancellation. The answer is sharp.

The cancellation is not merely formal. Take one scalar receiving factor $j \rightarrow i$ with $\Lambda_i = 1$, $T_i = \frac{1}{2}$, and root precision $\Lambda_j = 1$. In the block order (i, j) , Equation (10.12) gives

$$\Lambda_{\text{joint}} = \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & \frac{5}{4} \end{pmatrix} = \begin{pmatrix} A_i + W & -W \\ -W & A_j + W \end{pmatrix}, \quad W = \frac{1}{2}, \quad A_i = \frac{1}{2}, \quad A_j = \frac{3}{4}. \quad (10.18)$$

Thus the assembled precision belongs to $\mathcal{I}(\{i, j\})$, although $T_i^2 = \frac{1}{4} \neq \frac{1}{2} = T_i$, so the receiving factor does not belong locally: its parent-side residual is $T_i^2 \Lambda_i - W = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4}$, and the root prior offsets it to $A_j = 1 - \frac{1}{4} = \frac{3}{4}$. The matrix is positive definite because its leading principal minors are 1 and 1. This exact witness proves that factorwise membership is sufficient but not necessary for global membership. **[ESTABLISHED]**

Proposition 10.11 (Edge-local characterization with singular gains allowed). Consider a single receiving factor $j \rightarrow i$ in isolation, contributing Λ_i to the ii block, $T_i^\top \Lambda_i T_i$ to the jj block, and $-\Lambda_i T_i$ to the ij block. This factor lies in $\mathcal{I}(\{i, j\})$ if and only if

$$X := \Lambda_i^{-1/2} (\Lambda_i T_i) \Lambda_i^{-1/2} \text{ is an orthogonal projection,} \quad (10.19)$$

equivalently if and only if $\Lambda_i T_i$ is symmetric and T_i is idempotent, equivalently if and only if T_i is a Λ_i -orthogonal projection. In that case the self terms are $A_i = \Lambda_i^{1/2} (I - X) \Lambda_i^{1/2}$ and $A_j = 0$. The full-rank case $X = I$ is $T_i = I$, in which the factor is the pure agreement factor whose mean is the parent's own value in the trivialized coordinates and whose edge weight is $W = \Lambda_i$. **[ESTABLISHED]**

Proof. Membership requires three things: $W := \Lambda_i T_i \succeq 0$ and symmetric; the i -side residual $\Lambda_i - W \succeq 0$; and the j -side residual $T_i^\top \Lambda_i T_i - W \succeq 0$. Assume W symmetric and set $X = \Lambda_i^{-1/2} W \Lambda_i^{-1/2}$, which is symmetric. Then $W \succeq 0$ reads $X \succeq 0$; $\Lambda_i - W \succeq 0$ reads $X \preceq I$; and, using $T_i = \Lambda_i^{-1} W$ so that $T_i^\top \Lambda_i T_i = W \Lambda_i^{-1} W = \Lambda_i^{1/2} X^2 \Lambda_i^{1/2}$, the third reads $X^2 \succeq X$. Now $0 \preceq X \preceq I$ implies $X^2 \preceq X$, since X and X^2 are simultaneously diagonalizable and $\lambda^2 \leq \lambda$ for $\lambda \in [0, 1]$. Combining, $X^2 = X$, so X is a symmetric idempotent, that is an orthogonal projection. Conversely any orthogonal projection satisfies all three conditions, and then $T_i = \Lambda_i^{-1/2} X \Lambda_i^{1/2}$ satisfies $T_i^2 = T_i$ and $\Lambda_i T_i = \Lambda_i^{1/2} X \Lambda_i^{1/2}$ is symmetric positive semidefinite. The self terms follow by substitution, the j -side residual being $\Lambda_i^{1/2} (X^2 - X) \Lambda_i^{1/2} = 0$. $\square$

Corollary 10.12 (Invertible-gain factor). If the receiving gain T_i is required to be invertible, the isolated factor lies in $\mathcal{I}(\{i, j\})$ if and only if $T_i = I$. Indeed, Proposition 10.11 makes T_i idempotent, and multiplying $T_i^2 = T_i$ by T_i^{-1} gives $T_i = I$; the converse is the full-rank case of that proposition. **[ESTABLISHED]**

Proposition 10.11 describes the stronger factor-local requirement, not global family membership. With singular gains allowed, a proper projection constrains only the components of the residual inside $\text{range}(X)$ and leaves the complement to the receiving-side self term. In the invertible-gain submodel, Corollary 10.12 removes every proper projection and leaves only pure agreement. The deterministic replacement check `CHK-GAUSS-PROJECTION` tests 4000 generic gains and exact

controls of ranks 0 through 3 under a fully specified protocol; its current output corroborates the characterization but does not prove generic measure zero. [NUMERICAL]

Hypothesis 10.13 (Global interaction form). The renormalization half of this document acts on recognition precisions that belong globally to Equation (10.8), in trivialized coordinates, and this is adopted as a declared restriction. For a precision assembled from the directed model of Equation (10.11), the restriction is exactly Proposition 10.10; it does *not* require every receiving factor to satisfy Proposition 10.11, because the balance terms in Equation (10.17) may cancel as Equation (10.18) proves. It is used in the aggregation closure statement, in the definition of the flow, and in the fixed-point analysis; each of those results names it. The document does not claim that it is forced by anything, and Proposition 10.5 records that it is a measure-zero restriction on the space of Gaussian recognition laws. [HYPOTHESIS]

Hypothesis 10.14 (Optional factorwise-agreement realization). When the interaction parameters are required to be readable from the generative model one factor at a time, every receiving factor is separately assumed to satisfy Proposition 10.11. This is a strictly stronger hypothesis than global membership: it permits Λ_i -orthogonal projections in the relaxed singular-gain model and only $T_i = I$ in the invertible-gain model, while it excludes the globally admissible witness (10.18). The operator-level renormalization results require Hypothesis 10.13, not this stronger factorwise realization, unless they explicitly invoke a factor interpretation. [HYPOTHESIS]

10.5 Nonemptiness, and a note on conditioning

A declared subfamily is worthless if it is empty or numerically degenerate, so both should be checked. Nonemptiness is a construction, not an experiment.

Proposition 10.15 (Constructive nonemptiness). Under Hypothesis 10.7, let each edge of a graph on V carry an agreement factor with residual $\mu_i - \Theta_{ij}\mu_j$ and link covariance $R_{ij} \in \mathbb{S}_{++}^K$, and let each agent carry a proper prior of precision $A_i \succeq 0$ in the trivialized coordinates. Then the assembled trivialized energy is

$$E(z) = \frac{1}{2} \sum_i z_i^\top A_i z_i + \frac{1}{2} \sum_{i < j} (z_i - z_j)^\top W_{ij} (z_i - z_j), \quad W_{ij} = U_i^\top R_{ij}^{-1} U_i, \quad (10.20)$$

so the associated precision lies in $\mathcal{I}(V)$, with each W_{ij} symmetric positive definite by construction. [ESTABLISHED]

Proof. By (10.10) the residual is $U_i(z_i - z_j)$, so the quadratic form $(\mu_i - \Theta_{ij}\mu_j)^\top R_{ij}^{-1}(\mu_i - \Theta_{ij}\mu_j)$ equals $(z_i - z_j)^\top U_i^\top R_{ij}^{-1} U_i (z_i - z_j)$. The matrix $U_i^\top R_{ij}^{-1} U_i$ is symmetric and positive definite because R_{ij}^{-1} is and U_i is invertible. Comparing with (10.9) identifies the parameters. $\square$

Proposition 10.15 also makes the role of Hypothesis 10.7 concrete by contrast. Without trivialization the same agreement factor contributes the off-diagonal block $-R_{ij}^{-1}\Theta_{ij}$, which is symmetric only

in special position. The deterministic replacement check `CHK-GAUSS-TRIVIALIZATION` emits its matrices and compares that asymmetric raw block with the symmetric positive-definite trivialized weight. It corroborates the proposition and is not part of its proof. [NUMERICAL]

Conditioning is a different question and admits only protocol-specific computational evidence. The deterministic replacement check `CHK-GAUSS-CONDITIONING` specifies and runs a 200-draw interaction ensemble together with a zero-self consensus control and 200 unstructured positive-definite controls. Its output concerns only that sampler. Positive definiteness is proved by Proposition 10.4 and thinness by Proposition 10.5; no universal conditioning theorem is claimed. [NUMERICAL]

10.6 Marginalization belongs to the Gaussian coarse chapter

Aggregation identifies variables, whereas exact Gaussian marginalization eliminates them by a Schur complement. The interaction cone is closed under the former and not under unrestricted matrix-weighted Kron reduction. The exact positive-definite $N = 3, K = 2$ rational counterexample, including the asymmetric manufactured coupling, is stated and proved once in Chapter 13. [ESTABLISHED]

The same chapter gives the positive result in its sharp current form: every fixed congruence-diagonal cone $\mathcal{C}_H = \{H D H^\top : D \succeq 0 \text{ diagonal}\}$ is closed under every defined node-block Schur elimination. This strictly contains the common-orthogonal-eigenbasis case because congruence-diagonal matrices need not commute. Maximality is proved only for rich, independently parameterized closed convex product cones with an SPD order unit and closure under every defined one-node elimination. Correlated, nonconvex, variable-chart, rank-changing, and nonflat families remain outside that theorem. [ESTABLISHED]

Chapter 11

Gaussian Recognition Restrictions

Fix the general restriction principle of Proposition 5.7. This chapter specializes it to nondegenerate multivariate-Gaussian laws, computes the exact costs of block factorization and mean restriction, and states when comparisons between recognition structures are meaningful.

11.1 The block restriction and its exact optimum

For the rest of this chapter the exact conditional is Gaussian and nondegenerate,

$$P^* = \mathcal{N}(\mu, J^{-1}), \quad J = J^\top \succ 0, \quad \mu = J^{-1}h, \quad (11.1)$$

on $\mathbb{R}^n$ with $n = NK$ in the state sector of the previous chapter. A partition $\mathfrak{B}$ of the n coordinates into blocks of dimensions n_b is declared, and the restricted family is

$$\mathcal{Q}_{\mathfrak{B}} = \{\mathcal{N}(\mu_Q, C_Q) : \mu_Q \in \mathbb{R}^n, \quad C_Q = \text{blockdiag}_{b \in \mathfrak{B}}(C_b), \quad C_b \in \mathbb{S}_{++}^{n_b}\}. \quad (11.2)$$

The blocks are whatever the declared partition says they are. A block may collect one agent's coordinates, may split one agent's state and model channels, or may collect several agents; nothing below identifies a matrix block with an agent, and every statement is relative to $\mathfrak{B}$.

Proposition 11.1 (Gaussian reverse relative entropy). For $Q = \mathcal{N}(\mu_Q, C_Q)$ with $C_Q \succ 0$ and P^* as in Equation (11.1),

$$D_{\text{KL}}(Q \| P^*) = \frac{1}{2} \left[\text{Tr}(JC_Q) + (\mu_Q - \mu)^\top J (\mu_Q - \mu) - n - \log \det J - \log \det C_Q \right]. \quad (11.3)$$

[ESTABLISHED]

Proof. Write $D_{\text{KL}}(Q \| P^*) = \mathbb{E}_Q[\log q] - \mathbb{E}_Q[\log p^*]$. The first term is the negative differential entropy of a Gaussian, $-\frac{1}{2} \log \det(2\pi e C_Q)$. For the second, $-\log p^*(y) = \frac{1}{2}(y - \mu)^\top J (y - \mu) + \frac{n}{2} \log 2\pi - \frac{1}{2} \log \det J$, and taking the expectation under Q uses $\mathbb{E}_Q[(Y - \mu)^\top J (Y - \mu)] = \text{Tr}(JC_Q) + (\mu_Q - \mu)^\top J (\mu_Q - \mu)$, which is the standard expansion of a quadratic form about a shifted center. Adding the two and canceling the $\frac{n}{2} \log 2\pi$ terms gives Equation (11.3). $\square$

Equation (11.3) separates: the mean enters only through the quadratic term and the covariance only through the trace and the log determinant, with no cross term. That separation is used repeatedly

below and is the reason the two kinds of restriction studied in this chapter have exactly the costs they have.

Theorem 11.2 (Exact block optimum). Over the family of Equation (11.2) the objective of Equation (11.3) has a unique minimizer, given by

$$\mu_Q^* = \mu, \quad C_b^* = J_{bb}^{-1} \quad \text{for every } b \in \mathfrak{B}. \quad (11.4)$$

The optimal recognition mean is exact even though the recognition covariance is restricted. [ESTABLISHED]

Proof. On the block-diagonal set, $\text{Tr}(JC_Q) = \sum_b \text{Tr}(J_{bb}C_b)$, because only the diagonal blocks of J meet a block-diagonal C_Q under the trace, and $\log \det C_Q = \sum_b \log \det C_b$. The objective is therefore

$$D_{\text{KL}} = \frac{1}{2} (\mu_Q - \mu)^\top J (\mu_Q - \mu) + \frac{1}{2} \sum_{b \in \mathfrak{B}} [\text{Tr}(J_{bb}C_b) - \log \det C_b] + \text{const}, \quad (11.5)$$

a sum of a strictly convex quadratic in μ_Q , since $J \succ 0$, and one term per block. Each block term is strictly convex on $\mathbb{S}_{++}^{n_b}$, since $C_b \mapsto \text{Tr}(J_{bb}C_b)$ is linear and $C_b \mapsto -\log \det C_b$ is strictly convex on the positive definite cone [16, §3.1.5]. The parameter set is convex, being $\mathbb{R}^n$ times a product of positive definite cones, so the minimizer is unique if it exists. Differentiating on the open cone gives

$$\nabla_{\mu_Q} D_{\text{KL}} = J (\mu_Q - \mu), \quad \nabla_{C_b} D_{\text{KL}} = \frac{1}{2} (J_{bb} - C_b^{-1}), \quad (11.6)$$

whose simultaneous zero is Equation (11.4), admissible because $J_{bb} \succ 0$ as a principal submatrix of a positive definite matrix. $\square$

The same optimum arrives from coordinate ascent, which is worth recording because the two derivations fail in different ways and agreeing is informative. Holding all factors but b fixed, the exact coordinate maximizer of the bound is Gaussian with covariance J_{bb}^{-1} and mean

$$\mu_b^{\text{new}} = J_{bb}^{-1} \left(h_b - \sum_{c \neq b} J_{bc} \mu_c \right), \quad (11.7)$$

which is Equation (10.5) read as an update rule. Any fixed point of the full sweep satisfies $J\mu_Q = h$, hence $\mu_Q = \mu$ [14, 15]. The covariance is pinned in one step and the mean only at a fixed point, which is the standard asymmetry of Gaussian coordinate ascent.

Theorem 11.3 (Determinant gap). Let $\mathcal{L}_{\mathfrak{B}}^*$ denote the bound at the optimum of Theorem 11.2. Then

$$\log p_\theta(o \mid X) - \mathcal{L}_{\mathfrak{B}}^* = \frac{1}{2} \left[\sum_{b \in \mathfrak{B}} \log \det J_{bb} - \log \det J \right] \geq 0, \quad (11.8)$$

with equality if and only if $J_{bc} = 0$ for every pair of distinct blocks of $\mathfrak{B}$. [ESTABLISHED]

Proof. At the optimum, $\text{Tr}(JC_Q^*) = \sum_b \text{Tr}(J_{bb}J_{bb}^{-1}) = \sum_b n_b = n$ and $\log \det C_Q^* = -\sum_b \log \det J_{bb}$, while the quadratic mean term vanishes. Substituting into Equation (11.3) leaves

$$\inf_{Q \in \mathcal{Q}_{\mathfrak{B}}} D_{\text{KL}}(Q \| P^*) = \frac{1}{2} \left[\sum_{b \in \mathfrak{B}} \log \det J_{bb} - \log \det J \right], \quad (11.9)$$

and Equation (5.17) converts this into Equation (11.8).

For nonnegativity and the equality condition, two lemmas suffice. The first is that $0 \prec S \preceq T$ implies $\det S \leq \det T$, with equality only when $S = T$. To see it, set $R := T^{-1/2}ST^{-1/2}$, which is symmetric with eigenvalues in $(0, 1]$ and with $\det R = \det S / \det T$. That ratio is a product of numbers in $(0, 1]$, hence at most one, and it equals one only when every eigenvalue equals one, which for a symmetric matrix forces $R = I$ and so $S = T$. The second lemma is that, for $J \succ 0$ partitioned into a first block and its complement,

$$\det J = \det J_{11} \cdot \det (J_{22} - J_{21}J_{11}^{-1}J_{12}), \quad (11.10)$$

with the Schur complement positive definite [73]. Since $J_{21}J_{11}^{-1}J_{12} \succeq 0$, that Schur complement is $\preceq J_{22}$, so the first lemma gives $\det J \leq \det J_{11} \det J_{22}$, with equality if and only if $J_{21}J_{11}^{-1}J_{12} = 0$, which since $J_{11}^{-1} \succ 0$ holds if and only if $J_{11}^{-1/2}J_{12} = 0$, that is if and only if $J_{12} = 0$. Applying this with the first block of $\mathfrak{B}$ against the union of the rest, and then inducting on the number of blocks, gives Equation (11.8) and the stated equality condition. This is Fischer's determinant inequality for positive definite matrices [32, §7.8]. $\square$

Proposition 11.4 (The optimum is dominated by the exact marginal). For every block b ,

$$C_b^* = J_{bb}^{-1} \preceq (J^{-1})_{bb}, \quad (11.11)$$

with equality if and only if $J_{b,-b} = 0$. [ESTABLISHED]

Proof. By Equation (10.7), $(J^{-1})_{bb} = S_b^{-1}$ with $S_b = J_{bb} - J_{b,-b}J_{-b,-b}^{-1}J_{-b,b}$, and $S_b \preceq J_{bb}$ because the subtracted matrix is positive semidefinite. Inversion reverses the Loewner order on positive definite matrices: if $0 \prec S \preceq T$ then $T^{-1/2}ST^{-1/2} \preceq I$, so the eigenvalues of $T^{1/2}S^{-1}T^{1/2}$ are at least one, giving $S^{-1} \succeq T^{-1}$ [32, §7.7], [12, Ch. V]. Applying this to $S_b \preceq J_{bb}$ gives Equation (11.11). Equality forces $S_b = J_{bb}$, hence $J_{b,-b}J_{-b,-b}^{-1}J_{-b,b} = 0$, hence $J_{b,-b} = 0$ as in the proof of Theorem 11.3. $\square$

Proposition 11.4 is the exact form of a statement usually made loosely, that a factorized reverse-relative-entropy approximation understates posterior uncertainty [14, 15, 63]. Here it is a Loewner inequality with a proof and an equality condition, and it holds blockwise rather than merely in some scalar summary. It also fixes what the restriction is and is not.

Conditional dependence, not marginal covariance. The family of Equation (11.2) constrains the recognition covariance, hence the recognition precision, to be block diagonal. By Proposition 10.2 that is a statement that the recognition law makes distinct blocks conditionally independent given

the rest. It is not, and does not imply, the assertion that the target's marginal covariance blocks $(J^{-1})_{bc}$ vanish; (10.6) shows that those can be nonzero while $J_{bc} = 0$. The reverse-relative-entropy projection changes only the candidate Q : it sets $C_{bc}^* = 0$ for $b \neq c$ and $C_{bb}^* = J_{bb}^{-1} \preceq (J^{-1})_{bb}$. It neither leaves nor replaces any covariance block of the fixed target $P = \mathcal{N}(\mu, J^{-1})$. Reading the block restriction as a claim about the target's marginal correlation is a category error, and it is the error that makes the determinant gap look like a correlation-strength measurement rather than what Theorem 11.3 says it is, a coupling-structure measurement. [ESTABLISHED]

11.2 Observation independence, and its exact scope

Theorem 11.3 expresses the cost as a pure log-determinant functional of J . Whether that makes the cost independent of the realized observation depends on where J comes from, and the answer differs for the two kinds of restriction this document uses. Getting the difference right matters because a later chapter compares recognition structures across data and would draw the wrong conclusion from the wrong version.

Proposition 11.5 (Conditional precision is observation free). Suppose the joint law of (o, Y) given X is Gaussian with precision $\mathcal{J}$ and information vector η , both functions of (θ, X) alone. Then the exact conditional law of Y given o has precision and mean

$$J = \mathcal{J}_{YY}, \quad \mu = J^{-1}(\eta_Y - \mathcal{J}_{Yo}o), \quad (11.12)$$

so J does not depend on the realized value of o while μ is an affine function of it. [ESTABLISHED]

Proof. Read the exponent of the joint density as a function of y at fixed o . The quadratic part is $-\frac{1}{2}y^\top \mathcal{J}_{YY}y$ and the linear part is $(\eta_Y - \mathcal{J}_{Yo}o)^\top y$, so by Equation (10.1) the conditional law is Gaussian in information form with the displayed natural parameters, and $\mu = J^{-1}h$. $\square$

Corollary 11.6 (Observation independence of the determinant gap). Under the hypothesis of Proposition 11.5, the gap of Theorem 11.3 is a function of (θ, X) and the declared partition alone, and does not depend on the realized observation. [ESTABLISHED]

Proof. The gap depends on Q and o only through J , by Theorem 11.3, and through nothing else, because the mean sector contributes exactly zero at the optimum by Theorem 11.2. Proposition 11.5 makes J observation free. $\square$

Two hypotheses are doing the work in Corollary 11.6 and both are load bearing. The first is joint Gaussianity with an observation-independent joint precision, which excludes, for instance, a model whose observation noise covariance is itself a function of the realized observation; without it Equation (11.12) fails and J carries data. The second, and the one more easily overlooked, is that the restriction is a *factorization*, so that the optimal recognition mean coincides with the target mean and the quadratic term in Equation (11.3) vanishes identically at the optimum. A restriction

that constrains the mean does not have that property, and the next proposition computes exactly what survives.

Proposition 11.7 (Cost of a mean restriction). Let $[B \ B_\perp]$ be an orthogonal matrix with $B \in \mathbb{R}^{n \times r}$, and restrict the recognition family to

$$\mathcal{Q}_B = \{\mathcal{N}(\mu_Q, C_Q) : \mu_Q \in \text{range}(B), \ C_Q \in \mathbb{S}_{++}^n\}, \quad (11.13)$$

with the covariance unrestricted. Writing $\mu_\perp := B_\perp^\top \mu$ for the component of the target mean orthogonal to the retained subspace, the exact minimum of the relative entropy over $\mathcal{Q}_B$ is

$$\inf_{Q \in \mathcal{Q}_B} D_{\text{KL}}(Q \| P^*) = \frac{1}{2} \mu_\perp^\top \left(B_\perp^\top J^{-1} B_\perp \right)^{-1} \mu_\perp, \quad (11.14)$$

attained at $C_Q^* = J^{-1}$ and $\mu_Q^* = \mu - J^{-1} B_\perp (B_\perp^\top J^{-1} B_\perp)^{-1} \mu_\perp$. The matrix $(B_\perp^\top J^{-1} B_\perp)^{-1}$ is the *marginal* precision of the restricted directions, that is the inverse of their exact marginal covariance. **[ESTABLISHED]**

Proof. By the separation noted after Equation (11.3), the covariance sector is unconstrained and its minimum is attained at $C_Q^* = J^{-1}$, where it contributes zero. What remains is to minimize $\frac{1}{2}(\mu_Q - \mu)^\top J(\mu_Q - \mu)$ subject to $B_\perp^\top \mu_Q = 0$. Put $v = \mu_Q - \mu$, so the constraint reads $B_\perp^\top v = -\mu_\perp$. Stationarity of the Lagrangian gives $Jv = B_\perp \lambda$, hence $v = J^{-1} B_\perp \lambda$; imposing the constraint gives $\lambda = -(B_\perp^\top J^{-1} B_\perp)^{-1} \mu_\perp$, which is well defined because $B_\perp$ has full column rank and $J^{-1} \succ 0$. The objective at that point is $\frac{1}{2} v^\top J v = \frac{1}{2} \lambda^\top (B_\perp^\top J^{-1} B_\perp) \lambda$, which is Equation (11.14), and the minimizer is as displayed. The final identification holds because $\text{Cov}(B_\perp^\top Y) = B_\perp^\top J^{-1} B_\perp$ under P^* . $\square$

Corollary 11.8 (A mean restriction carries the data). Under the hypothesis of Proposition 11.5, the cost in Equation (11.14) is a quadratic function of the realized observation, since $\mu_\perp = B_\perp^\top J^{-1}(\eta_Y - \mathcal{J}_{Y|o})$ is affine in o . It is observation free if and only if $B_\perp^\top J^{-1} \mathcal{J}_{Y|o} = 0$. Thus the observation independence of Corollary 11.6 does not extend to restrictions of the mean, and a construction that reports one cost as though it were the other is reporting a data-dependent quantity as a structural one. **[ESTABLISHED]**

Proposition 11.9 (Marginal precision against restricted precision). With $[B \ B_\perp]$ as above,

$$B_\perp^\top J B_\perp - \left(B_\perp^\top J^{-1} B_\perp \right)^{-1} = B_\perp^\top J B \left(B^\top J B \right)^{-1} B^\top J B_\perp \succeq 0, \quad (11.15)$$

so the marginal precision of the restricted directions is dominated in the Loewner order by the restriction $B_\perp^\top J B_\perp$ of the precision to those directions, the difference has rank at most r , and the two coincide if and only if $B^\top J B_\perp = 0$, that is if and only if the retained and restricted subspaces are J -orthogonal. **[ESTABLISHED]**

Proof. Since $[B \ B_\perp]$ is orthogonal, conjugating by it gives $\tilde{J} := [B \ B_\perp]^\top J [B \ B_\perp]$ and also $[B \ B_\perp]^\top J^{-1} [B \ B_\perp] = \tilde{J}^{-1}$. Hence $B_\perp^\top J^{-1} B_\perp$ is the lower right block of $\tilde{J}^{-1}$, and by the block inversion formula its inverse is the Schur complement

$$\tilde{J}_{\perp\perp} - \tilde{J}_{\perp\parallel} \tilde{J}_{\parallel\parallel}^{-1} \tilde{J}_{\parallel\perp}, \quad (11.16)$$

where $\parallel$ and $\perp$ index the two groups of columns [73]. Subtracting Equation (11.16) from $\tilde{J}_{\perp\perp} = B_\perp^\top J B_\perp$ leaves exactly the displayed product, which is positive semidefinite and of rank at most $\text{rank}(B^\top J B_\perp) \leq r$. It vanishes if and only if $(B^\top J B)^{-1/2} B^\top J B_\perp = 0$, that is if and only if $B^\top J B_\perp = 0$. $\square$

Proposition 11.9 is the reason Equation (11.14) must be written with the marginal precision and not with the restriction of the precision. The two are different matrices, ordered but not equal, and substituting one for the other overstates the cost of a mean restriction by the rank- r correction of Equation (11.15) except in the J -orthogonal case. The deterministic replacement check CHK-RESTRICTION-SCHUR independently reconstructs the block-inversion and constrained-optimization endpoints and includes the orthogonal equality control. It corroborates the proved identities and is not evidence for them. [NUMERICAL]

Corollary 11.10 (Costs of combined restrictions add). Restricting the mean to $\text{range}(B)$ and the covariance to the block-diagonal set of Equation (11.2) simultaneously has exact cost

$$\frac{1}{2} \left[\sum_{b \in \mathfrak{B}} \log \det J_{bb} - \log \det J \right] + \frac{1}{2} \mu_\perp^\top \left(B_\perp^\top J^{-1} B_\perp \right)^{-1} \mu_\perp, \quad (11.17)$$

the first term structural and the second data dependent. [ESTABLISHED]

Proof. Equation (11.3) has no term coupling μ_Q to C_Q , and the two constraint sets act on disjoint parameter blocks, so the minimizations are independent and their values add. Each value is supplied by Theorem 11.3 and Proposition 11.7 respectively. $\square$

11.3 Comparing recognition structures at fixed model

Equation (11.8) assigns a number to a partition, which invites the question of whether one recognition structure is better than another. The question is well posed only under conditions that are easy to violate, and this section states both the affirmative result and its boundary.

Proposition 11.11 (Refinement monotonicity). Let $\mathfrak{B}'$ refine $\mathfrak{B}$, meaning every block of $\mathfrak{B}'$ is contained in a block of $\mathfrak{B}$. Then $\mathcal{Q}_{\mathfrak{B}'} \subseteq \mathcal{Q}_{\mathfrak{B}}$ and

$$\mathcal{L}_{\mathfrak{B}}^* - \mathcal{L}_{\mathfrak{B}'}^* = \sum_{b \in \mathfrak{B}} \frac{1}{2} \left[\sum_{b' \in \mathfrak{B}': b' \subseteq b} \log \det J_{b'b'} - \log \det J_{bb} \right] \geq 0, \quad (11.18)$$

each summand being the determinant gap of the positive definite matrix J_{bb} under the partition induced on b , and vanishing exactly when J_{bb} is block diagonal under that induced partition. The coarser structure therefore always dominates the finer one at fixed model, and the mechanism is the Loewner ordering of Theorem 11.3: within each coarse block, a Schur complement is dominated by the corresponding diagonal block. [ESTABLISHED]

Proof. A block-diagonal matrix under $\mathfrak{B}'$ is block diagonal under $\mathfrak{B}$ when $\mathfrak{B}'$ refines $\mathfrak{B}$, giving the inclusion of families. Subtracting two instances of Equation (11.8) cancels $\log \det J$ and regroups the remaining terms by coarse block, giving Equation (11.18). Each bracket is Theorem 11.3 applied to $J_{bb} \succ 0$ with the induced partition, hence nonnegative with the stated equality condition. $\square$

Proposition 11.12 (Non-nested structures are not ordered). Refinement is a partial order, and outside it the bounds are not ordered by any structural feature such as the number or the sizes of the blocks. For $n = 3$ take

$$J(a, b) = \begin{pmatrix} 1 & a & 0 \\ a & 1 & b \\ 0 & b & 1 \end{pmatrix}, \quad a^2 + b^2 < 1, \quad (11.19)$$

which is positive definite, and compare the two partitions $\mathfrak{B}_1 = \{\{1, 2\}, \{3\}\}$ and $\mathfrak{B}_2 = \{\{1\}, \{2, 3\}\}$, which have identical block-size multisets and neither of which refines the other. Their gaps are $\frac{1}{2} \log \frac{1-a^2}{1-a^2-b^2}$ and $\frac{1}{2} \log \frac{1-b^2}{1-a^2-b^2}$, so the ordering is decided by whether $|a|$ exceeds $|b|$ and reverses when they are exchanged. At $(a, b) = (0.8, 0.2)$ the values are 0.0589 and 0.5493; at $(0.2, 0.8)$ they are exchanged. [ESTABLISHED]

Proof. Positive definiteness follows from the leading minors 1 , $1 - a^2$ and $1 - a^2 - b^2$. The gaps follow from Equation (11.8) with $\det J = 1 - a^2 - b^2$, $\det J_{\{1,2\}\{1,2\}} = 1 - a^2$, $\det J_{\{2,3\}\{2,3\}} = 1 - b^2$ and the remaining diagonal entries equal to one. The numerical values are $\frac{1}{2} \log(0.36/0.32)$ and $\frac{1}{2} \log(0.96/0.32)$. $\square$

Proposition 11.12 says that a recognition structure is well chosen relative to a given coupling matrix, not well chosen in the abstract, and it is what makes any partition-selection rule an optimization over structures rather than a reading off of a scalar. It also disposes of the temptation to treat block count as a proxy for quality.

Proposition 11.13 (The gap does not select a structure). Over the lattice of partitions of the n coordinates, the gap of Equation (11.8) is nonincreasing under coarsening by Proposition 11.11, and attains its minimum value zero at the trivial partition with one block, for every $J \succ 0$. Minimizing the gap therefore selects the unrestricted family and returns no information about structure. [ESTABLISHED]

Proof. For the one-block partition, $\sum_b \log \det J_{bb} = \log \det J$ and the gap is zero; nonnegativity is Theorem 11.3 and monotonicity is Proposition 11.11. $\square$

Proposition 11.14 (ELBO values do not order unrelated joints). Let P_θ and $P_{\theta'}$ be normalized joints over the same observation but possibly different latent inventories, with recognition families $\mathcal{Q}$ and $\mathcal{Q}'$ on their respective latent spaces. Assume the selected o belongs to the regular-observation set of each joint, or that pointwise versions have been declared for both. If no evidence-preserving channel relating the two joints is given, then

$$\mathcal{L}^* - \mathcal{L}'^* = [\log p_\theta(o \mid X) - \log p_{\theta'}(o \mid X)] - \left[\inf_{\mathcal{Q}} D_{\text{KL}} - \inf_{\mathcal{Q}'} D_{\text{KL}} \right], \quad (11.20)$$

which combines an evidence difference, a model-comparison quantity, with a tightness difference, a recognition-quality quantity. Neither bracket is identifiable from the value of the left side, so the sign of that value orders neither the models nor the recognition families. [ESTABLISHED]

Proof. Apply Equation (5.18) to each model and subtract. To show that both failures occur within normalized models, let $\ell \leq 0$, $g \geq 0$, take $O \in \{o, \bar{o}\}$, $Y \in \{0, 1\}$, and define a normalized joint by

$$P_{\ell,g}(O = o, Y = 1) = e^{\ell-g}, \quad P_{\ell,g}(O = o, Y = 0) = e^\ell(1 - e^{-g}), \quad P_{\ell,g}(O = \bar{o}, Y = 0) = 1 - e^\ell,$$

with the remaining atom assigned mass zero. Then

$$\log P_{\ell,g}(O = o) = \ell, \quad P_{\ell,g}(Y = 1 \mid O = o) = e^{-g}.$$

For the singleton recognition family $\mathcal{Q} = \{\delta_1\}$, the minimized gap is g , so the optimized bound is $\ell - g$. To make the two latent inventories literally different, append an unused deterministic binary coordinate $Z = 0$ to either second model and replace its recognition law by $\delta_1 \otimes \delta_0$; taking a product with this point mass changes neither normalization, evidence, relative entropy, nor the optimized ELBO. The pairs

$$(\ell_1, g_1) = (-1, 0), \quad (\ell_2, g_2) = (0, 2)$$

give $\mathcal{L}_1^* = -1 > \mathcal{L}_2^* = -2$ although the evidence ordering is $\ell_1 < \ell_2$. The pairs

$$(\ell_1, g_1) = (0, 1), \quad (\ell_2, g_2) = (-2, 0)$$

again give $\mathcal{L}_1^* = -1 > \mathcal{L}_2^* = -2$, although the first recognition family has the larger gap and hence the less tight bound. Thus a bound ordering determines neither bracket in Equation (11.20). $\square$

Proposition 11.14 is a statement about unrelated joints, not about latent inventory by itself. If a normalized Markov kernel $K : x \rightsquigarrow z$ defines the coarse joint $\bar{P} = (\text{id}_O \otimes K)_\# P$ and the coarse recognition law is $\bar{Q} = QK$, then the observation marginal and evidence are unchanged and data processing gives

$$\bar{\mathcal{L}}(\bar{Q}; o) = \log p(o) - D_{\text{KL}}(QK \| P_o K) \geq \log p(o) - D_{\text{KL}}(Q \| P_o) = \mathcal{L}(Q; o).$$

Thus different latent spaces can carry ordered bounds when both laws are related by the same evidence-preserving channel. Absent such a relation, an ELBO difference mixes the evidence difference with the tightness difference as in (11.20). The general hypotheses and equality conditions are stated in Chapter 7. [ESTABLISHED]

Open Problem 11.15 (An intrinsic criterion). Propositions 11.13 and 11.14 leave a gap that this document does not close. There is no criterion here, internal to a single evidence bound at a fixed model, that selects a nondegenerate recognition structure: the determinant gap is minimized at the trivial partition, the mean-restriction cost of Equation (11.14) is a cost at fixed structure, and an arbitrary replacement joint supplies no common evidence. Evidence-preserving Markov pushforwards are the important exception, but their data-processing order still does not select a nondegenerate partition. What would settle the problem positively is a functional on admissible channels or partitions, expressible relative to one fixed joint and not monotone to an endpoint, whose optimizer is nondegenerate. A negative result would have to prove endpoint degeneracy for a precisely stated class of functionals. Absent either, the choice of structure remains a model-selection or RG-scheme declaration. [OPEN]

Chapter 12

Gaussian Information Geometry

Chapter 6 distinguishes general law fibers from smooth statistical manifolds, regular exponential families, and the multivariate-Gaussian realization. This chapter works only in the last tier. It identifies the dual Gaussian charts, computes their Fisher metrics, and separates frame-invariant generalized spectral data from ordinary chart-dependent eigenvalues. It does not assert that an RG flow contracts any information-geometric quantity.

12.1 Canonical and expectation coordinates

Use the nondegenerate Gaussian family and log normalizer of Proposition 10.1. For the stacked latent $Y \in \mathbb{R}^n$, $n = NK$, write

$$q(y) = \exp \left(h^\top y - \frac{1}{2} y^\top J y - A(h, J) \right), \quad h \in \mathbb{R}^n, \quad J \in \mathbb{S}_{++}^n, \quad (12.1)$$

where

$$A(h, J) = \frac{1}{2} h^\top J^{-1} h - \frac{1}{2} \log \det J + \frac{n}{2} \log 2\pi. \quad (12.2)$$

For sufficient statistic $T(y) = (y, yy^\top)$, the canonical natural and expectation coordinates are

$$\eta = \left(h, -\frac{1}{2} J \right), \quad (12.3)$$

$$\tau = \nabla_\eta A = (\mu, M_2), \quad M_2 = C + \mu\mu^\top = \mathbb{E}_q[YY^\top], \quad (12.4)$$

The information-form chart (h, J) ranges over $\mathbb{R}^n \times \mathbb{S}_{++}^n$, whereas the canonical matrix coordinate $-J/2$ is negative definite. The family is regular and minimal, so the Legendre map is a bijection on its natural domain [17]. **[ESTABLISHED]**

The Fisher matrix in natural coordinates is the Hessian of the potential and equals the covariance of the sufficient statistic,

$$g_\eta = \nabla_\eta^2 A(\eta) = \text{Cov}_q(T(Y)), \quad (12.5)$$

which is the standard exponential-family fact [2, 3, 17]. Because $\tau = \nabla_\eta A$, the Jacobian of the coordinate change is $\partial\tau/\partial\eta = g_\eta$, and the Fisher metric expressed in the expectation coordinates is

$$g_\tau = \left(\frac{\partial\eta}{\partial\tau} \right)^\top g_\eta \left(\frac{\partial\eta}{\partial\tau} \right) = g_\eta^{-1}, \quad (12.6)$$

the Hessian of the convex conjugate $A^*(\tau) = \sup_{\eta} (\langle \eta, \tau \rangle - A(\eta))$, which for this family is the negative differential entropy of q [3, 53]. [ESTABLISHED]

Four coordinate presentations must be kept apart. The canonical natural chart is $\eta = (h, -J/2)$; the affinely equivalent information-form chart is (h, J) ; the expectation chart is $\tau = (\mu, M_2)$; and the moment chart is (μ, C) . The last is useful for sampling and covariance transport but is not Legendre dual to the natural chart, because $C = M_2 - \mu\mu^\top$. [DEFINITION]

Symmetric-matrix coordinates carry one further convention. Derivatives in the matrix argument are taken with respect to the Frobenius pairing $\langle X, Z \rangle = \text{Tr}(X^\top Z)$ on the space of symmetric matrices. Concretely this is the chart whose coordinates are the diagonal entries together with the off-diagonal entries scaled by $\sqrt{2}$, for which the Euclidean inner product of coordinate vectors equals the Frobenius pairing of the matrices they represent. With the unscaled half-vectorization instead, every off-diagonal coordinate derivative acquires a factor of two and the matrix representing (12.6) changes accordingly. All matrix blocks displayed below are in the Frobenius-paired chart. [DEFINITION]

When the recognition precision has the interaction form of Chapter 10, $J = \Lambda$, with $\Lambda_{ii} = A_i + \sum_{j \neq i} W_{ij}$, $\Lambda_{ij} = -W_{ij}$, and $C = \Lambda^{-1}$. Below, Λ emphasizes the interaction operator and J the natural parameter; they are the same matrix.

12.2 The chart correction

It is often asserted that the mean sector of the Fisher metric of a Gaussian family *is* the precision. The assertion is true, and it is the standard statement of Gaussian information geometry, but it is true in the moment chart. It is false in the expectation chart, which is the chart the exponential-family duality of Section 12.1 actually delivers. Both facts are proved below, and the discrepancy between them is not a small correction.

Proposition 12.1 (Fisher metric in the moment chart). *In the chart (μ, C) the Fisher metric of the Gaussian family is block diagonal, and for tangent vectors (u, A) and (v, B) with $u, v \in \mathbb{R}^n$ and A, B symmetric,*

$$g_{(\mu, C)}((u, A), (v, B)) = u^\top \Lambda v + \frac{1}{2} \text{Tr}(\Lambda A \Lambda B), \quad \Lambda = C^{-1}. \quad (12.7)$$

[ESTABLISHED]

Proof. Write $e = y - \mu$ and $\log q(y) = -\frac{1}{2}e^\top \Lambda e + \frac{1}{2} \log \det \Lambda - \frac{n}{2} \log 2\pi$. A displacement (u, A) in the moment chart moves μ by u and C by A , hence Λ by $-\Lambda A \Lambda$, and the directional score is

$$\partial_{(u, A)} \log q = u^\top \Lambda e + \frac{1}{2} e^\top \Lambda A \Lambda e - \frac{1}{2} \text{Tr}(\Lambda A). \quad (12.8)$$

Under $e \sim \mathcal{N}(0, C)$ the first term has second moment $u^\top \Lambda C \Lambda u = u^\top \Lambda u$. The cross term between the linear and quadratic pieces is a third central moment of a Gaussian and vanishes, which is

exactly why the metric is block diagonal in this chart. Setting $S = \Lambda A \Lambda$, the remaining piece is $\frac{1}{4} \text{Var}(e^\top S e) = \frac{1}{2} \text{Tr}(S C S C) = \frac{1}{2} \text{Tr}(\Lambda A \Lambda A)$. Polarization gives (12.7). $\square$

Expression (12.7) is the classical Fisher–Rao metric of the multivariate normal model; it is the metric whose geodesic equations and Rao distance are studied by Skovgaard [60] and by Atkinson and Mitchell [4], and it appears in the survey of Pinele et al. [58] as their equation (7), in the general reparameterized form $g_{ij} = \partial_i \mu^\top \Sigma^{-1} \partial_j \mu + \frac{1}{2} \text{Tr}(\Sigma^{-1} \partial_i \Sigma \Sigma^{-1} \partial_j \Sigma)$. The printed equation in that reference carries a typographical slip, repeating the index i in place of j in the second factor of the trace; the intended and correct expression is the one displayed here, and it agrees with (12.7) on setting $\theta = (\mu, C)$. That reference also records, as its equation (8), that the affine action $(\mu, \Sigma) \mapsto (Q\mu + c, Q\Sigma Q^\top)$ is an isometry of this metric for every $Q \in \text{GL}_n(\mathbb{R})$ and $c \in \mathbb{R}^n$, a fact used again in Section 12.4.

Proposition 12.2 (Fisher metric in the expectation chart). *In the chart (μ, M_2) with $M_2 = C + \mu\mu^\top$, the Fisher metric has mean block*

$$g_{[\mu\mu]} = \left(1 + \mu^\top \Lambda \mu\right) \Lambda + \Lambda \mu \mu^\top \Lambda, \quad (12.9)$$

cross block

$$g_{[\mu M_2]}(u, B) = -u^\top \Lambda B \Lambda \mu, \quad (12.10)$$

and second block $g_{[M_2 M_2]}(A, B) = \frac{1}{2} \text{Tr}(\Lambda A \Lambda B)$, unchanged from (12.7). Hence the metric is not block diagonal in this chart unless $\mu = 0$. [ESTABLISHED]

Proof. The chart change is $C = M_2 - \mu\mu^\top$, so a displacement (u, B) in the expectation chart induces

$$\delta\mu = u, \quad \delta C = B - (u\mu^\top + \mu u^\top), \quad (12.11)$$

and the metric in the new chart is the pullback of (12.7) along (12.11). For the mean block take $B = 0$, so $\delta C = -(u\mu^\top + \mu u^\top)$. Then

$$\begin{aligned} \text{Tr}(\Lambda \delta C \Lambda \delta C) &= \text{Tr} \left[\left(\Lambda u \mu^\top + \Lambda \mu u^\top \right) \left(\Lambda u \mu^\top + \Lambda \mu u^\top \right) \right] \\ &= 2 \left(u^\top \Lambda \mu \right)^2 + 2 \left(\mu^\top \Lambda \mu \right) \left(u^\top \Lambda u \right), \end{aligned} \quad (12.12)$$

each of the four traces being evaluated by cycling. Adding the mean contribution $u^\top \Lambda u$ from (12.7) and halving (12.12) gives the quadratic form

$$u^\top \Lambda u + \left(\mu^\top \Lambda \mu \right) \left(u^\top \Lambda u \right) + \left(u^\top \Lambda \mu \right)^2 = u^\top \left[\left(1 + \mu^\top \Lambda \mu \right) \Lambda + \Lambda \mu \mu^\top \Lambda \right] u, \quad (12.13)$$

which is (12.9). For the cross block pair $(u, 0)$ with $(0, B)$: the mean parts do not meet, and the covariance parts contribute $\frac{1}{2} \text{Tr}(\Lambda [-(u\mu^\top + \mu u^\top)] \Lambda B) = -u^\top \Lambda B \Lambda \mu$, which is (12.10). For the second block take $u = 0$, so $\delta C = B$ and nothing changes. $\square$

Corollary 12.3 (Size of the discrepancy and the exact equality case). Write $v = \Lambda^{1/2}\mu$. Then

$$g_{[\mu\mu]} = \Lambda^{1/2} \left[(1 + \|v\|^2) I_n + vv^\top \right] \Lambda^{1/2}, \quad (12.14)$$

so the eigenvalues of $\Lambda^{-1}g_{[\mu\mu]}$ are $1 + \mu^\top \Lambda \mu$ with multiplicity $n - 1$ and $1 + 2\mu^\top \Lambda \mu$ with multiplicity one. Consequently $g_{[\mu\mu]} \succeq \Lambda$ always, and $g_{[\mu\mu]} = \Lambda$ if and only if $\mu = 0$. [ESTABLISHED]

Proof. Equation (12.14) is (12.9) rewritten. The bracket is $(1 + \|v\|^2) I + vv^\top$, whose eigenvalues are $1 + \|v\|^2 + \|v\|^2$ along v and $1 + \|v\|^2$ on $v^\perp$; since $\Lambda^{-1}g_{[\mu\mu]} = \Lambda^{-1/2} [\cdot] \Lambda^{1/2}$ is similar to the bracket, its spectrum is that list. For the equality case, $g_{[\mu\mu]} - \Lambda = (\mu^\top \Lambda \mu) \Lambda + \Lambda \mu \mu^\top \Lambda$ is a sum of two positive semidefinite matrices, so it vanishes only if both vanish; the first forces $\mu^\top \Lambda \mu = 0$ and hence $\mu = 0$ because $\Lambda \succ 0$. The converse is immediate. $\square$

The discrepancy is therefore governed by the squared Mahalanobis norm of the mean, $\mu^\top \Lambda \mu$, and it is unbounded. It is not a normalization constant and it does not disappear in any limit that keeps the mean away from the origin.

The mechanism behind the discrepancy is worth stating plainly, because it is what makes the error easy to commit. Both charts call their first coordinate μ , but they do not have the same coordinate vector fields. The vector $\partial/\partial\mu$ at fixed C moves the mean while holding the covariance fixed, so M_2 changes; the vector $\partial/\partial\mu$ at fixed M_2 moves the mean and compensates in C . These are different tangent vectors at the same point of the manifold, and a metric assigns them different lengths. Writing $g_{[\mu\mu]}$ without saying which of the two is meant does not denote anything.

The same observation disposes of the phrase “the mean sector” in the expectation chart. A metric tensor has a well defined restriction to a subspace of the tangent space only once that subspace is declared; what is not chart independent is the *complement*. In the moment chart the covariance directions are Fisher-orthogonal to the mean directions, by the vanishing third-moment argument in the proof of Proposition 12.1, so “the mean sector” can be taken to mean either the restriction to the mean directions or the orthogonal quotient, and the two agree. In the expectation chart (12.10) is nonzero, the mean and second-moment directions are not Fisher-orthogonal, and the restriction (12.9) differs from the quotient obtained by projecting out the second-moment directions. Absent a declared splitting, “the mean sector” in that chart names no unique object.

The chart discipline. A coordinate identity that mixes the mean and covariance sectors of a Gaussian family must name the chart in which it holds. Identity (12.7) holds in (μ, C) ; identities (12.9) and (12.10) hold in (μ, M_2) ; by Corollary 12.3 the two mean blocks agree only at $\mu = 0$. [ESTABLISHED]

Verification

The statements are proved above in general dimension; the machine checks reported here test those proofs and are not a substitute for them. Two symbolic routes and two numerical routes were run, all in exact or double precision on the CPU.

The first symbolic route is independent of Proposition 12.1: the Hessian of (12.2) was formed by computer algebra in the natural chart $(h, \text{svec}(-J/2))$ with the Frobenius-paired symmetric coordinates, evaluated at exact rational (μ, Λ) , inverted exactly as in (12.6), and its blocks compared with (12.9) and (12.10). At dimensions $n = 1, 2, 3$ the difference was the exact zero matrix in both blocks, and at $n = 2$ the cross block became exactly zero when $\mu = 0$ was substituted and was nonzero otherwise, as Proposition 12.2 requires. The second symbolic route carries general symbols: at dimensions $n = 2$ and $n = 3$, with μ a vector of free symbols and Λ a general symmetric matrix of free symbols, the pullback of (12.7) along (12.11) was simplified to zero difference against all three blocks of Proposition 12.2. Those two runs are proofs at those dimensions, not samples.

The deterministic numerical route `CHK-IG-EXPECTATION-METRIC` uses the closed-form Gaussian relative entropy and its Fisher quadratic limit in six directions at a fully specified base point. It compares the expectation-chart block with Equation (12.9), uses the moment-chart precision as a negative control against chart conflation, and checks the predicted generalized spectrum. The computation is reproducible corroboration, not a proof. [NUMERICAL]

12.3 Sample space and parameter space

The renormalization chapters will import vocabulary from a literature that computes Fisher metrics of likelihoods with respect to *model* parameters and reads their spectra as a scale [11]. The object developed above is not that object, and keeping the two typed apart is a precondition for importing anything at all.

Three spaces are in play. The latent sample space is $\mathbb{R}^n$, where Y takes its values; the interaction energy $y \mapsto \frac{1}{2}y^\top \Lambda y$ is a quadratic form on it. The recognition parameter manifold is the set of laws (12.1), coordinatized by any of the four charts of Section 12.1; the Fisher metric (12.7) is a Riemannian metric on it. The model parameter space is where θ lives, indexing the generative kernel; the Fisher metric of the likelihood $\theta \mapsto p_\theta(o \mid X)$ is a Riemannian metric on *that*, and its dimension is the dimension of θ , which has no relation to $n = NK$.

What makes the confusion easy is an exact but scoped component identity in this family, and stating it precisely is more useful than warning against it. The recognition family is a location family in its mean, so the mean parameter takes values in the same vector space $\mathbb{R}^n$ as the sample variable, and the mean block (12.7) of the parameter-space metric has the same components as the sample-space form Λ . The equality is of components in matched charts, not of objects: one is a metric on the tangent spaces of a manifold of probability laws, the other is a bilinear form on the latent vector space. For a family that is not a location family the two have different dimensions and no comparison arises. The identity is therefore a feature of the Gaussian mean sector and cannot be used as evidence that a construction defined on one space transfers to the other. [ESTABLISHED]

There is a corresponding exact but scoped relation to the interaction Laplacian L . If transported Gaussian laws have covariance fixed within the mean submodel and parallel along each edge, and the edge quadratic uses that transported precision, their pairwise KL is exactly the associated

mean-sector Fisher quadratic; summing over edges gives the connection Laplacian form. For a regular non-Gaussian family, freezing the Fisher tensor at one parallel base law gives the analogous form only as the local second-order term of transported KL [3]. Outside these Gaussian or frozen-local hypotheses, a sample-space L is not a recognition Fisher tensor or a natural-gradient operator. [ESTABLISHED]

The distinction has a sharp consequence for coarse-graining, which the next chapters will need. Restricting the recognition mean to a subspace and marginalizing onto that subspace are different operations and give different metrics.

Proposition 12.4 (Pullback versus pushforward). *Let $\Lambda \succ 0$ on $\mathbb{R}^n$, let $B \in \mathbb{R}^{n \times k}$ have orthonormal columns, and let $B_\perp$ span the orthogonal complement. The Fisher metric of the submodel obtained by constraining the mean to $\text{range}(B)$ is the restriction $B^\top \Lambda B$; the Fisher metric of the mean of the pushforward law under $y \mapsto B^\top y$ is the marginal precision $(B^\top \Lambda^{-1} B)^{-1}$. They satisfy*

$$B^\top \Lambda B - (B^\top \Lambda^{-1} B)^{-1} = (B^\top \Lambda B_\perp) (B_\perp^\top \Lambda B_\perp)^{-1} (B_\perp^\top \Lambda B) \succeq 0, \quad (12.15)$$

with equality if and only if $B^\top \Lambda B_\perp = 0$, that is if and only if the retained and discarded subspaces are Λ -orthogonal. [ESTABLISHED]

Proof. The matrix $[B \ B_\perp]$ is orthogonal, so in that basis Λ has blocks $\Lambda_{11} = B^\top \Lambda B$, $\Lambda_{12} = B^\top \Lambda B_\perp$, $\Lambda_{22} = B_\perp^\top \Lambda B_\perp$. The block inversion formula gives $B^\top \Lambda^{-1} B = (\Lambda_{11} - \Lambda_{12} \Lambda_{22}^{-1} \Lambda_{21})^{-1}$ [32], and inverting again yields (12.15). The right side is a congruence of $\Lambda_{22}^{-1} \succ 0$ and is therefore positive semidefinite, and it vanishes exactly when $\Lambda_{12} = 0$. The identification of the two sides as pullback and pushforward metrics is the Gaussian conditional and marginal precision respectively [43]. $\square$

Restriction is a pullback along an injective map of parameters, and it makes the metric larger in the Loewner order; marginalization is a pushforward along a map of the sample space, and it makes the metric smaller. The monotonicity theorems of information geometry govern the second operation, not the first, a point taken up in Section 12.5. The deterministic replacement check CHK-IG-PULLBACK-PUSHFORWARD tests the Schur identity and Loewner order over a specified ensemble and includes the Λ -orthogonal equality control. Computation is not proof and the proof is above. [NUMERICAL]

Whether the scale-setting construction of the model-space literature transfers to Λ is therefore not settled by observing that both are called Fisher metrics. The transfer requires a declaration that makes the coarse recognition mean a parameter of the generative law fitted against the bound, after which the relevant Fisher metric is the one with respect to that parameter and the import becomes a statement rather than an analogy. This section records the requirement and does not discharge it. What would close it is that declaration together with a check that the resulting parameter-space metric is nondegenerate, since the construction in that literature is written throughout with an inverted metric and does not treat the singular case. [OPEN]

12.4 Gauge invariants of the interaction precision

A local reframing is a change of local trivialization of the fibers, and it changes the components of every object while changing no law. Let

$$T = \text{diag}(T_1, \dots, T_N), \quad T_i \in \text{GL}^+(K), \quad (12.16)$$

act on the assembled latent coordinate by $y \mapsto Ty$. Every quadratic form in y then transforms by inverse congruence,

$$\Lambda \mapsto T^{-\top} \Lambda T^{-1}, \quad L \mapsto T^{-\top} L T^{-1}, \quad A \mapsto T^{-\top} A T^{-1}, \quad (12.17)$$

where L is the Laplacian part of the interaction precision and $A = \text{diag}(A_1, \dots, A_N)$ collects the self terms, so that $\Lambda = L + A$. Two hypotheses are being used and should be visible. First, the splitting $\Lambda = L + A$ is a declared decomposition of the assembled precision, made once in the frame-dependent chart; (12.17) then says that congruence acts on the two summands separately, which is immediate from linearity. Second, the reframing is block diagonal with respect to the agent partition, which is what a change of local trivialization is; a coordinate change mixing agents would not preserve the block structure that makes A block diagonal. **[HYPOTHESIS]**

Under a nonconstant reframing the image $T^{-\top} L T^{-1}$ need not itself be a matrix-weighted Laplacian, since the row sums that define that normal form are not preserved when neighboring blocks are reframed differently. This is a statement about normal forms, not about the object: L is a bilinear form on the latent space, and (12.17) is how a bilinear form reads in a new chart. The invariants below compare forms and are therefore unaffected.

Proposition 12.5 (Frame invariance of the generalized spectrum). *Let $\Lambda \succ 0$ and $L = L^\top$, and let T be invertible. The generalized eigenvalue problem*

$$Lx = d\Lambda x \quad (12.18)$$

*and the reframed problem $(T^{-\top} L T^{-1}) x' = d' (T^{-\top} \Lambda T^{-1}) x'$ have the same eigenvalues, with eigenvectors related by $x' = Tx$. **[ESTABLISHED]***

Proof. Suppose $Lx = d\Lambda x$ and set $x' = Tx$. Then $T^{-\top} L T^{-1} x' = T^{-\top} Lx = dT^{-\top} \Lambda x = d(T^{-\top} \Lambda T^{-1}) x'$. The map $x \mapsto Tx$ is a bijection, so the correspondence is onto, and the argument run with T^{-1} gives the converse. $\square$

Because $\Lambda \succ 0$, the pair (L, Λ) is a symmetric-definite, hence regular, pencil: it is simultaneously diagonalizable by congruence, all n generalized eigenvalues are real, and they obey a Rayleigh characterization [28, 32]. The positive-definiteness hypothesis is load bearing. Congruence covariance alone also holds for a singular pencil, but it does not imply that the determinant polynomial has degree n or even that it is nonzero. For example, at a common singular ray

$(L, \Lambda) = (L, L)$ with $\ker L \neq \{0\}$,

$$\det(L - d\Lambda) = \det((1 - d)L) \equiv 0,$$

so the full-space determinant does not define an n -element generalized spectrum. A quotient, pin, or mass repair must first be declared and the resulting pencil proved regular before the conclusions below can be invoked. Thus Propositions 12.5 to 12.7 apply to the Gaussian probability interior $\Lambda \succ 0$ (or to a separately proved regular repair), not automatically to an operator-layer boundary point. That qualification is the pencil-level instance of the hierarchy in Chapter 6.

Proposition 12.6 (Localization and endpoints). *If in addition $L \succeq 0$ and $A = \Lambda - L \succeq 0$, then every generalized eigenvalue of (12.18) lies in $[0, 1]$. The value 0 is attained exactly on $\ker L$, whose dimension is at least K because every configuration constant across agents lies in it, and the value 1 exactly on $\ker A$. [ESTABLISHED]*

Proof. By the Rayleigh characterization, each generalized eigenvalue is a value of $d(x) = x^\top Lx / x^\top \Lambda x$ at a stationary point, and $x^\top \Lambda x = x^\top Lx + x^\top Ax$ with both terms nonnegative and the sum positive. Hence $0 \leq d(x) \leq 1$ for all $x \neq 0$, with $d(x) = 0$ iff $x^\top Lx = 0$, which for $L \succeq 0$ is $x \in \ker L$, and $d(x) = 1$ iff $x \in \ker A$. That the constant configurations $\mathbf{1} \otimes w$, $w \in \mathbb{R}^K$, lie in $\ker L$ follows from the difference form $x^\top Lx = \sum_{i < j} (x_i - x_j)^\top W_{ij} (x_i - x_j)$. $\square$

The generalized eigenvalue is therefore readable, and its reading is frame independent: d_k is the fraction of the total precision along the k -th pencil direction that is supplied by interaction rather than by self terms. The consensus directions sit at $d = 0$ because interaction supplies no precision there; a direction that no self term constrains sits at $d = 1$.

Proposition 12.7 (Frame dependence of absolute spectra and subspace determinants). (i) *For $T = cI_n$ with $c > 0$, which is a legitimate reframing since $\det(cI_K) = c^K > 0$, the ordinary spectrum obeys $\text{spec}(T^{-\top} \Lambda T^{-1}) = c^{-2} \text{spec}(\Lambda)$. (ii) Let $U \subset \mathbb{R}^n$ have orthonormal basis B , let $U' = TU$ have orthonormal basis B' , and let $\Lambda' = T^{-\top} \Lambda T^{-1}$. Then*

$$\log \det(B'^\top \Lambda' B') = \log \det(B^\top \Lambda B) - \log \det(B^\top T^\top T B), \quad (12.19)$$

so a subspace log-determinant is frame independent if and only if T restricted to U has unit determinant in absolute value; every orthogonal T satisfies this condition. [ESTABLISHED]

Proof. Part (i) is immediate. For (ii), write $TB = B'S$ with S invertible, so $B' = TB S^{-1}$; orthonormality of B' gives $S^{-\top} (B^\top T^\top T B) S^{-1} = I$ and hence $\det(S)^2 = \det(B^\top T^\top T B)$. Then $B'^\top \Lambda' B' = S^{-\top} B^\top T^\top T^{-\top} \Lambda T^{-1} T B S^{-1} = S^{-\top} (B^\top \Lambda B) S^{-1}$, and taking log determinants gives (12.19). $\square$

The consequence is a prohibition, and it is the reason this section exists. A reframing changes no law; it changes only the components in which the law is written. A quantity computed from those components that is not invariant under (12.17) is therefore a property of the chart. By

Proposition 12.7(i), any threshold criterion applied to the absolute eigenvalues of Λ can be made to accept or to reject any given direction by choosing c , so no such criterion states anything about the model. Any admissible criterion must be phrased in generalized-eigenvalue form, that is as a statement about a pencil in which both members transform by the same congruence. By Proposition 12.7(ii) the same warning applies to determinant criteria evaluated on a subspace: they are invariant under orthogonal reframings and not otherwise, so a construction that uses one must either declare its frame or carry the Jacobian term in (12.19) explicitly. [ESTABLISHED]

Verification

Proposition 12.5 is proved above and needs no support. The deterministic replacement check `CHK-GENERALIZED-SPECTRUM` measures the generalized spectrum over twelve fully specified block congruences and uses the moving ordinary spectrum as its negative control. It establishes numerical usability only for that protocol; congruence invariance is proved above. [NUMERICAL]

12.5 What is not claimed

The Fisher metric is used here as a declared choice of geometry on the recognition manifold, and the classical uniqueness results are not strong enough to make that choice forced in this setting. On a finite sample space the Fisher metric is, up to a positive scale, the unique Riemannian metric invariant under Markov morphisms; this is Chentsov's theorem [66], with the extension to non-normalized measures due to Campbell [18]. [ESTABLISHED] The latent space here is $\mathbb{R}^n$, so that theorem does not apply directly. Extensions to general sample spaces exist and are established, but they carry their own hypotheses: Ay et al. [5] and Lê [44] obtain uniqueness under a strong-continuity assumption on the information metric, and the monograph treatment collects the required regularity [6]. [ESTABLISHED] The development therefore adopts the Fisher metric because it is the Hessian of the exponential-family potential (12.5) and because that is the structure the calculations use, and it does not claim that no other invariant metric is available in the continuous setting. [HYPOTHESIS]

No Markov/data-processing monotonicity theorem is asserted for the deterministic Galerkin aggregation map. [NOT-CLAIMED] The established general statement is that the Fisher metric contracts under a Markov kernel: pushing a family forward through a stochastic map cannot increase Fisher information [3, 6]. [ESTABLISHED] That theorem does not cover the aggregation operation of the following chapters, because by Proposition 12.4 the coarse operator is a restriction, a pullback along an inclusion of mean submodels, and not a pushforward along a Markov kernel; the two differ by the positive semidefinite Schur term in (12.15), and the restriction is the larger of the two in the Loewner order. [ESTABLISHED] Whether a monotone functional decreases under the full declared aggregation/RG map is open. Closing it requires either exhibiting a Markov kernel whose pushforward realizes the map, after which the classical theorem applies verbatim, or exhibiting a Lyapunov functional and proving its decrease directly. [OPEN] The later fixed-covariance Gaussian relaxation in Section 13.5.1 is a different continuous-time dynamics: its quadratic energy has an

exact dissipation identity, and that scoped flow is a Fisher natural-gradient flow. [ESTABLISHED] It does not turn the coarse map itself into a Markov contraction. [NOT-CLAIMED]

Nothing here should be read as importing a uniqueness statement from the quantum setting. There, the metrics monotone under completely positive trace-preserving maps form a family parameterized by operator monotone functions rather than a single metric [57], so the phrase “the monotone metric” does not denote in that context, and an argument that relies on uniqueness cannot be transported from it. [ESTABLISHED] No Markov-morphism contraction theorem for the deterministic aggregation maps is asserted. [NOT-CLAIMED]

Finally, this section supplies a metric and not a dynamics. No natural-gradient identification should be inferred merely from the appearance of Λ as a preconditioner: an update is a natural gradient only when its metric is the Fisher metric of the family being updated, in a named chart, and with the objective differential typed in the matching chart [1, 47]. The fixed-covariance Gaussian mean flow in Section 13.5.1 satisfies those requirements and is identified there. [ESTABLISHED] No broader identification for the remaining updates is claimed. [NOT-CLAIMED]

Chapter 13

Multivariate-Gaussian Coarse-Graining

This chapter realizes the three operations of Chapter 7 in the multivariate-Gaussian interaction family. It does not repeat the Markov or graph-exponential theory. Its claims concern hard identification, Gaussian marginalization and recognition restrictions, all at finite dimension. Unless a channel index is displayed, the symbols $K, \Theta, \Lambda, L, A, W$ in this chapter denote the belief/recognition channel. A model-channel realization is separate and uses its own $K_m, \Theta^m, \Lambda_m, L_m, A_m, W_m$; no equality of the two channels is implied. **[DEFINITION]** Within the connection-Laplacian realization, Proposition 13.7 applies only to invertible represented transports with inverse reverse transports and positive-definite represented internal weights. Under those hypotheses, trivial represented internal holonomy is the criterion for a connected cluster to retain a full K -dimensional, state-independent structural fixed section; nontrivial holonomy may still leave a lower-dimensional fixed sector. **[ESTABLISHED]** Marginal similarity is not identified with this structural criterion. **[NOT-CLAIMED]**

13.1 Hard aggregation of the interaction precision

Assume the declared interaction family of Chapter 10,

$$\Lambda_{ii} = A_i + \sum_{j \neq i} W_{ij}, \quad \Lambda_{ij} = -W_{ij} \quad (i \neq j), \quad A_i \succeq 0, \quad W_{ij} = W_{ji} \succeq 0. \quad (13.1)$$

In flat trivialized coordinates its energy is

$$\mathcal{E}(z) = \sum_i z_i^\top A_i z_i + \sum_{i < j} (z_i - z_j)^\top W_{ij} (z_i - z_j). \quad (13.2)$$

[HYPOTHESIS] This form is a Gaussian subfamily, not a consequence of an arbitrary linear-Gaussian directed model.

For a hard partition $\mathcal{P}$, let

$$S = \hat{S} \otimes I_K, \quad \hat{S}_{iI} = \mathbf{1}_{\{i \in I\}}, \quad \Lambda_c = S^\top \Lambda S.$$

The columns of S are orthogonal but not normalized:

$$S^\top S = \text{diag}(n_I) \otimes I_K. \quad (13.3)$$

[DEFINITION]

Proposition 13.1 (Exact Gaussian aggregation). For every hard partition,

$$(\Lambda_c)_{IJ} = - \sum_{\substack{i \in I \\ j \in J}} W_{ij} \quad (I \neq J), \quad (\Lambda_c)_{II} = \sum_{i \in I} A_i + \sum_{\substack{i \in I \\ j \notin I}} W_{ij}. \quad (13.4)$$

Thus $A_I = \sum_{i \in I} A_i$, $W_{IJ} = \sum_{i \in I, j \in J} W_{ij}$, and every internal edge is annihilated. **[ESTABLISHED]**

Proof. Substitute one coordinate z_I for every z_i in a cluster. An internal edge contributes zero. A cut edge contributes its difference energy between z_I and z_J , and self terms add. Finite sums preserve positive semidefiniteness; independently, $\Lambda_c \succeq 0$ follows by congruence. $\square$

This is the Galerkin coarse operator with a piecewise-constant prolongator, not a new numerical construction [45, 65]. The parameter map is exact for hard identification. A separately fitted coupling is outside that exact trace construction, while learned or soft pooling is a different approximate model. **[NOT-CLAIMED]**

13.2 Kron reduction: a counterexample and the closed cone

Gaussian marginalization eliminates nodes by a Schur complement. It is not the aggregation above and is not closed on the unrestricted matrix-weighted interaction family.

Proposition 13.2 (Unrestricted Kron reduction leaves the family). Take $N = 3$, $K = 2$, $A_1 = A_2 = A_3 = I_2$, and

$$W_{12} = 0, \quad W_{13} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad W_{23} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \quad (13.5)$$

Then the assembled Λ is positive definite. Eliminating node 3 manufactures

$$-(\Lambda^{\text{Kron}})_{12} = W_{13} \Lambda_{33}^{-1} W_{23} = \frac{1}{19} \begin{pmatrix} 9 & 3 \\ 4 & 14 \end{pmatrix}, \quad (13.6)$$

which is not symmetric and hence is not a positive-semidefinite interaction weight. **[ESTABLISHED]**

Proof. $\Lambda_{33} = \begin{pmatrix} 4 & 1 \\ 1 & 5 \end{pmatrix}$ has determinant 19, and direct multiplication gives (13.6). The leading principal minors of the full precision are 2, 6, 18, 48, 102, 233, so Sylvester's criterion gives $\Lambda \succ 0$. The full Schur complement is symmetric; its two mirrored off-diagonal blocks are transposes. Membership in (13.1) additionally requires each individual block to be symmetric, which fails. $\square$

The scalar loopy-Laplacian family is closed under every defined elimination [23]. The matrix analog is closed on a larger subfamily than common orthogonal diagonalization.

Definition 13.3 (Congruence-diagonal cone). For $H \in \text{GL}(K)$, define

$$\mathcal{C}_H = \{H D H^\top : D = \text{diag}(d_1, \dots, d_K), d_a \geq 0\}. \quad (13.7)$$

[DEFINITION]

Theorem 13.4 (Congruence-diagonal Kron closure). If every A_i and W_{ij} lies in one fixed $\mathcal{C}_H$, then every defined node-block Schur complement remains in the interaction family with all reduced parameters in $\mathcal{C}_H$. [ESTABLISHED]

Proof. Factor

$$\Lambda = (I_N \otimes H) \Lambda_D (I_N \otimes H^\top), \quad (13.8)$$

where, after grouping coordinates by channel, Λ_D is a direct sum of K scalar loop Laplacians. For retained nodes R and eliminated nodes E ,

$$\text{Sc}_E(\Lambda) = (I_R \otimes H) \left[(\Lambda_D)_{RR} - (\Lambda_D)_{RE} (\Lambda_D)_{EE}^{-1} (\Lambda_D)_{ER} \right] (I_R \otimes H^\top). \quad (13.9)$$

The middle operator eliminates channel by channel. Each scalar Schur complement is another loop Laplacian: its off-diagonals remain nonpositive because the eliminated block is a nonsingular M -matrix, and its row sums remain nonnegative. Reassembling the channels gives reduced self terms and weights in $\mathcal{C}_H$. $\square$

This family strictly contains the common-orthogonal-eigenbasis family. For

$$H = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad X = H \text{diag}(1, 2) H^\top, \quad Y = H \text{diag}(3, 1) H^\top,$$

one has $XY - YX = \begin{pmatrix} 0 & -5 \\ 5 & 0 \end{pmatrix} \neq 0$, so X, Y cannot be simultaneously orthogonally diagonalized, while they belong to the same $\mathcal{C}_H$. [ESTABLISHED]

Theorem 13.5 (Qualified maximality). Let $\mathcal{C} \subseteq \mathbb{S}_+^K$ be a closed convex cone with an SPD order unit $M \in \text{relint } \mathcal{C}$. Assume that anchors and incident edge coefficients range independently over $\mathcal{C}$, arbitrary positive edge scaling is allowed, every defined one-node elimination must remain in the family, and every manufactured off-diagonal block must be minus a symmetric member of $\mathcal{C}$. Then $\mathcal{C} \subseteq \mathcal{C}_H$ for some H , and a maximal cone under these assumptions equals $\mathcal{C}_H$. [ESTABLISHED]

Proof. For $X, Y \in \mathcal{C}$, relative interior gives an $\varepsilon > 0$ such that

$$A_e = M - \varepsilon(X + Y) \in \mathcal{C}.$$

Use $\varepsilon X, \varepsilon Y$ as the two incident weights and M as retained anchors. The eliminated diagonal is M , and the manufactured coupling is $\varepsilon^2 X M^{-1} Y$. Its required symmetry forces

$$X M^{-1} Y = Y M^{-1} X.$$

After whitening by $M^{-1/2}$, every pair in the whitened cone commutes. Commuting real symmetric matrices are simultaneously orthogonally diagonalizable, so

$$\mathcal{C} \subseteq \{M^{1/2} O D O^\top M^{1/2} : D \succeq 0 \text{ diagonal}\} = \mathcal{C}_H.$$

The preceding theorem proves that $\mathcal{C}_H$ itself is closed, yielding maximality. $\square$

The assumptions exclude correlated or nonconvex admissible sets, cones without an SPD order unit, parameter-dependent constraints, variable congruence charts, and nonflat link data. Classification in those domains remains open. **[OPEN]**

13.3 Nonflat links and effective support

For a graph edge $e = (i, j)$, write the twisted energy as

$$(z_i - \Theta_e z_j)^\top W_e (z_i - \Theta_e z_j). \quad (13.10)$$

An internal edge under hard identification contributes

$$(I - \Theta_e)^\top W_e (I - \Theta_e), \quad (13.11)$$

which vanishes exactly when

$$W_e^{1/2} (I - \Theta_e) = 0. \quad (13.12)$$

[ESTABLISHED] Positive-definite W_e forces $\Theta_e = I$; a singular weight may hide a nonidentity twist.

For $N_e = \ker W_e$ and quotient map $q_e : \mathbb{R}^K \rightarrow \mathbb{R}^K / N_e$,

$$W_e^{1/2} (I - \Theta_e) = 0 \iff q_e \Theta_e = q_e. \quad (13.13)$$

[ESTABLISHED] This is edge-local. A single common quotient $\mathbb{R}^K / N$ on which every induced weight is positive definite exists exactly when

$$\ker W_e = N, \quad \Theta_e N = N \quad \text{for every relevant edge.} \quad (13.14)$$

On that quotient ordinary group-valued holonomy is defined. When the kernels differ, there is no canonical product of the edge-local quotients and no single “represented support.” **[ESTABLISHED]**

A triangle makes the boundary exact:

$$\begin{aligned} W_{12} &= \text{diag}(1, 0), & \Theta_{12} &= \text{diag}(1, 2), \\ W_{23} &= \text{diag}(0, 1), & \Theta_{23} &= \text{diag}(2, 1), \\ W_{31} &= I, & \Theta_{31} &= I. \end{aligned}$$

Every edge satisfies (13.12), yet the loop product is $2I$, fixing no nonzero vector. The local kernels differ, so no common effective quotient exists. **[ESTABLISHED]**

13.4 Cut links and rectangular sheaf closure

For positive-definite cut weights between two already trivialized clusters, define

$$C = \sum W_e, \quad B = \sum W_e \Theta_e, \quad A = \sum \Theta_e^\top W_e \Theta_e.$$

One group-valued coarse pair reproduces the cut exactly if and only if all cut twists coincide. Indeed, with $\bar{\Theta} = C^{-1}B$,

$$\Delta = A - B^\top C^{-1}B = \sum_e (\Theta_e - \bar{\Theta})^\top W_e (\Theta_e - \bar{\Theta}) \succeq 0, \quad (13.15)$$

and $\Delta = 0$ exactly under common twists. **[ESTABLISHED]** The weighted mean has the correct gauge law but can be singular; equal weights with twists I_2 and $-I_2$ give $\bar{\Theta} = 0$.

The exact closed category is larger than one invertible link.

Theorem 13.6 (Rectangular endpoint-map closure). Choose a minimal factor

$$W_e = C_e^\top C_e, \quad C_e : \mathbb{R}^K \rightarrow E_e.$$

Then

$$(z_i - \Theta_e z_j)^\top W_e (z_i - \Theta_e z_j) = \|R_{i,e} z_i - R_{j,e} z_j\|_{E_e}^2, \quad R_{i,e} = C_e, \quad R_{j,e} = C_e \Theta_e. \quad (13.16)$$

For partial injections $S_{iI} : V_I \rightarrow \mathbb{R}^K$, the coarse endpoint maps are

$$R_{I,e}^c = R_{i,e} S_{iI}, \quad R_{J,e}^c = R_{j,e} S_{jJ}. \quad (13.17)$$

They reproduce $S^\top \Lambda S$ exactly, parallel cut edges compress losslessly by the direct sum of their E_e , and nested blocking is associative. **[ESTABLISHED]**

Proof. Expanding (13.16) gives the original pair form. Precomposition gives (13.17); its cross block is

$$-(R_{I,e}^c)^\top R_{J,e}^c = -S_{iI}^\top W_e \Theta_e S_{jJ},$$

and the diagonal blocks agree by the same expansion. Vertical stacking in $\bigoplus_e E_e$ sums the energies without loss. A second injection T gives $(RS)T = R(ST)$, proving repeated closure. $\square$

This is the degree-zero cellular-sheaf Laplacian category [29]. Vertex spaces may have unequal dimensions and automorphism groups $\text{GL}(V_I)$; edge basis changes act on both endpoint maps and the edge metric. It is gauge covariant and strictly larger than the fixed- K group-link family. Open questions are canonical minimal compression, characterization of compression to one invertible transporter and one SPD weight, and compatibility with normalized laws and RG rescaling. **[OPEN]**

13.5 Partial fixed-section aggregation

For each undirected internal edge, choose one orientation $e = (i, j)$, so Θ_e transports from j to i . Assume $\Theta_e \in \text{GL}(K)$, $\Theta_{\bar{e}} = \Theta_e^{-1}$, and $W_e \succ 0$. The cluster connection Laplacian L_I is defined by

$$z^\top L_I z = \sum_{e=(i,j) \in E_I} (z_i - \Theta_e z_j)^\top W_e (z_i - \Theta_e z_j). \quad (13.18)$$

[HYPOTHESIS] Positive definiteness is the nondegeneracy assumption used below. With semidefinite weights, the exact condition remains the edgewise constraint in (13.12), and the resulting effective-support problem is not replaced by the following proposition.

Let $(I_\alpha)_\alpha$ be the connected components of the internal graph. Choose a root r_α and a spanning tree in each component. For a tree path from r_α to v , let $P_{v \leftarrow r_\alpha}$ be the ordered product of its represented transports. Let Hol_{r_α} be the group of represented transport products around all closed walks based at r_α , and define

$$\text{Fix}(\text{Hol}_{r_\alpha}) = \{u \in \mathbb{R}^K : Hu = u \text{ for every } H \in \text{Hol}_{r_\alpha}\}. \quad (13.19)$$

[DEFINITION]

Proposition 13.7 (Kernel–holonomy fixed-space isomorphism). On every connected component I_α , root evaluation is a linear isomorphism

$$\text{ev}_{r_\alpha} : \ker L_{I_\alpha} \xrightarrow{\sim} \text{Fix}(\text{Hol}_{r_\alpha}), \quad z \mapsto z_{r_\alpha}. \quad (13.20)$$

Consequently,

$$\ker L_I \cong \bigoplus_\alpha \text{Fix}(\text{Hol}_{r_\alpha}), \quad f_I := \dim \ker L_I = \sum_\alpha \dim \text{Fix}(\text{Hol}_{r_\alpha}). \quad (13.21)$$

[ESTABLISHED]

Proof. By (13.18) and $W_e \succ 0$, a vector is in $\ker L_{I_\alpha}$ exactly when $z_i = \Theta_e z_j$ on every edge. Such a parallel section is determined by its root value, so root evaluation is injective. Transporting its value around any rooted closed walk returns the same value, placing its image in $\text{Fix}(\text{Hol}_{r_\alpha})$.

Conversely, take $u \in \text{Fix}(\text{Hol}_{r_\alpha})$ and define $z_v = P_{v \leftarrow r_\alpha} u$ along the spanning tree. Tree-edge constraints hold by construction. For a non-tree edge $e = (i, j)$, the tree path from r_α to j , followed by e , followed by the reverse tree path from i to r_α , has holonomy

$$H_e = P_{r_\alpha \leftarrow i} \Theta_e P_{j \leftarrow r_\alpha}. \quad (13.22)$$

Since $H_e u = u$ and $P_{r_\alpha \leftarrow i} = P_{i \leftarrow r_\alpha}^{-1}$, multiplication by $P_{i \leftarrow r_\alpha}$ gives $\Theta_e z_j = z_i$. Thus every edge constraint holds, proving surjectivity. The full Laplacian is the direct sum over connected components, which yields (13.21). $\square$

For a connected cluster, $f_I = K$ exactly when every represented holonomy element is the identity. This is the full fixed sector. The case $0 < f_I < K$ is a partial fixed sector, while $f_I = 0$ leaves no retained cluster degree. A nonfaithful representation can make represented holonomy trivial even when principal holonomy is not. **[ESTABLISHED]** This rank is a state-independent structural admissibility criterion. It assigns no partition cost, singletons remain admissible, and it does not require the constituent belief means to coincide. **[NOT-CLAIMED]**

Define the abstract fixed fiber

$$F_I = \bigoplus_{\alpha} \text{Fix}(\text{Hol}_{r_{\alpha}}). \quad (13.23)$$

After choosing bases, let $\iota_I : F_I \hookrightarrow (\mathbb{R}^K)^I$ be the full-column matrix that extends each root value by the spanning-tree construction in the proof. Its range is independent of those choices and equals $\ker L_I$. For a partition $\mathcal{P}$, put $\iota_{\mathcal{P}} = \text{blkdiag}_{I \in \mathcal{P}} \iota_I$. Then

$$L_I \iota_I = 0, \quad \Lambda_c = \iota_{\mathcal{P}}^{\top} \Lambda \iota_{\mathcal{P}} \succeq 0, \quad \Lambda_c \succ 0 \iff \text{range}(\iota_{\mathcal{P}}) \cap \ker \Lambda = \{0\}. \quad (13.24)$$

[ESTABLISHED] Internal edges vanish, while cut edges become the rectangular endpoint data of (13.17). An ordinary smooth vector bundle over varying parameters additionally requires locally constant f_I ; rank jumps produce strata rather than a smooth bundle. **[ESTABLISHED]**

Proposition 13.8 (Nested fixed sections compose exactly). Let a larger cluster A be a union of first-level clusters, and let

$$\iota_{A \leftarrow \mathcal{P}} = \text{blkdiag}_{I \subset A} \iota_I, \quad L_A^{(1)} = \iota_{A \leftarrow \mathcal{P}}^{\top} L_A \iota_{A \leftarrow \mathcal{P}}. \quad (13.25)$$

Then the fine and first-level kernels obey

$$\ker L_A = \iota_{A \leftarrow \mathcal{P}} \ker L_A^{(1)}. \quad (13.26)$$

If $\iota_A^{(2)} : F_A \hookrightarrow \bigoplus_{I \subset A} F_I$ has range $\ker L_A^{(1)}$, then the fine-level fixed-section injection is

$$\iota_A^{\text{fine}} = \iota_{A \leftarrow \mathcal{P}} \iota_A^{(2)}. \quad (13.27)$$

[ESTABLISHED]

Proof. Every zero mode of L_A has zero energy in each positive-semidefinite edge summand. Its restriction to every first-level cluster therefore belongs to $\ker L_I$, so it lies in $\text{range}(\iota_{A \leftarrow \mathcal{P}})$. Write it as $\iota_{A \leftarrow \mathcal{P}} y$. Since $L_A \succeq 0$,

$$y \in \ker L_A^{(1)} \iff y^{\top} L_A^{(1)} y = 0 \iff \iota_{A \leftarrow \mathcal{P}} y \in \ker L_A. \quad (13.28)$$

This proves (13.26); substituting the range of $\iota_A^{(2)}$ gives (13.27). Matrix composition is associative, so the same statement iterates through further levels. $\square$

Thus the vertex fixed spaces and sheaf endpoint maps are exactly functorial under nested partitions. **[ESTABLISHED]** Probability normalization, rescaling, and minimal group-link compression remain separate obligations. **[OPEN]**

13.5.1 Constant-metric Gaussian relaxation

Proposition 13.9 (Relaxation onto the fixed sector). Let $R = R^\top \succ 0$ and $L = L^\top \succeq 0$ be constant matrices on $\mathbb{R}^d$. If the columns of N_0 form a basis of $\ker L$, define the R -orthogonal projector

$$\Pi_0^R = N_0(N_0^\top R N_0)^{-1} N_0^\top R, \quad (13.29)$$

and set $\Pi_0^R = 0$ when $\ker L = \{0\}$. The flow

$$R\dot{z} = -Lz, \quad z(0) = z_0, \quad (13.30)$$

converges to $\Pi_0^R z_0$. If the positive spectrum is nonempty and

$$\lambda_+ = \lambda_{\min}^+(R^{-1/2} L R^{-1/2}), \quad (13.31)$$

then

$$\|z(t) - \Pi_0^R z_0\|_R \leq e^{-\lambda_+ t} \|z_0 - \Pi_0^R z_0\|_R. \quad (13.32)$$

For $\mathcal{E}(z) = \frac{1}{2} z^\top L z$, the exact dissipation identity is

$$\frac{d}{dt} \mathcal{E}(z(t)) = -\dot{z}^\top R \dot{z} = -z^\top L R^{-1} L z \leq 0. \quad (13.33)$$

[ESTABLISHED]

Proof. Set $y = R^{1/2} z$ and $A = R^{-1/2} L R^{-1/2} \succeq 0$. Then $\dot{y} = -A y$. The spectral theorem gives convergence to the Euclidean projection of y_0 onto $\ker A = R^{1/2} \ker L$, with transverse norm decaying at the smallest positive eigenvalue of A . Transforming back gives (13.29) and (13.32). Direct differentiation gives (13.33). $\square$

For a constant passive reframing $z' = T z$, $T \in \text{GL}(d)$, covariance of the equation requires the congruence laws

$$R' = T^{-\top} R T^{-1}, \quad L' = T^{-\top} L T^{-1}, \quad \Pi_0^{R'} = T \Pi_0^R T^{-1}. \quad (13.34)$$

[ESTABLISHED] The choice $R = I_d$ is therefore equivariant only under orthogonal reframings. If $T = T(t)$, the ordinary derivative acquires the term $\dot{T} T^{-1} z'$. Covariance requires a time covariant derivative; when the original temporal connection is zero, it is

$$D'_t z' = \dot{z}' - \dot{T} T^{-1} z' = T \dot{z}, \quad R' D'_t z' = -L' z'. \quad (13.35)$$

[ESTABLISHED]

The information-geometric identification is exact on the fixed-covariance Gaussian mean family $q_z = \mathcal{N}(z, R^{-1})$. Its mean Fisher matrix is R , and

$$D_{\text{KL}}(q_z \| q_w) = \frac{1}{2}(z - w)^\top R(z - w). \quad (13.36)$$

Thus (13.30) is exactly the Fisher natural-gradient flow of $\mathcal{E}$ on that mean family. For a general regular statistical family, only the local expansion

$$D_{\text{KL}}(q_\theta \| q_{\theta+d\theta}) = \frac{1}{2}d\theta^\top F(\theta)d\theta + o(\|d\theta\|^2), \quad (13.37)$$

is available under the standard regularity hypotheses [3]. [ESTABLISHED] Freezing $F(\theta_0) = R$ yields a local linearization. The development does not identify a generic, state-dependent Fisher flow with the constant global projector and rate above. [NOT-CLAIMED]

13.5.2 Gaussian marginal-law specialization

Specialize Theorem 7.11 after transporting every constituent marginal to one root. Suppose the admissible parent family is the full nondegenerate Gaussian family, as it is under trivial represented holonomy. Write the transported marginals as $q_i^{(r)} = \mathcal{N}(m_i, C_i)$, with $C_i \succ 0$, and choose weights $a_i > 0$ satisfying $\sum_i a_i = 1$. [HYPOTHESIS]

Proposition 13.10 (Full-Gaussian forward-KL barycenter). The unique minimizer of

$$\mathcal{D}_I(m, C) = \sum_i a_i D_{\text{KL}}(\mathcal{N}(m_i, C_i) \| \mathcal{N}(m, C)), \quad (13.38)$$

matches the first two moments:

$$\bar{m} = \sum_i a_i m_i, \quad \bar{C} = \sum_i a_i \left[C_i + (m_i - \bar{m})(m_i - \bar{m})^\top \right]. \quad (13.39)$$

Its minimized score is

$$\mathcal{D}_I^* = \frac{1}{2} \left[\log \det \bar{C} - \sum_i a_i \log \det C_i \right]. \quad (13.40)$$

[ESTABLISHED]

Proof. The Gaussian KL formula gives

$$\begin{aligned} 2\mathcal{D}_I(m, C) &= \text{Tr}(C^{-1}M(m)) - K + \log \det C - \sum_i a_i \log \det C_i, \\ M(m) &= \sum_i a_i \left[C_i + (m_i - m)(m_i - m)^\top \right]. \end{aligned} \quad (13.41)$$

Completing the weighted square gives $M(m) = \bar{C} + (m - \bar{m})(m - \bar{m})^\top$, so $m = \bar{m}$ is the unique minimizer. At that value, $\text{Tr}(C^{-1}\bar{C}) - \log \det(C^{-1}\bar{C}) \geq K$, with equality exactly when $C = \bar{C}$. Substitution proves (13.40). $\square$

Under nontrivial represented holonomy, the moment-matched Gaussian need not belong to the holonomy-fixed parent family. [ESTABLISHED] The applicable optimization is then the constrained one in Theorem 7.11. The development does not claim that (13.39) removes that constraint. [NOT-CLAIMED]

For an explicit witness, let $K = 2$, $\mathcal{H} = \{I_2, -I_2\} \subset \text{SO}(2)$, and take the one-law ensemble $\mathcal{N}(u, I_2)$ with $u \neq 0$. Its unconstrained moment-matched Gaussian is itself and is not $\mathcal{H}$ -invariant. The Haar-constrained optimizer below is instead $\mathcal{N}(0, I_2 + uu^\top)$, with score $\frac{1}{2} \log(1 + \|u\|^2) > 0$. Thus the holonomy constraint can be active even when there is no pairwise disagreement among the transported constituents.

Corollary 13.11 (Compact-holonomy Gaussian barycenter). Let $\mathcal{H} \subset \text{GL}(K)$ be a compact represented holonomy group with normalized Haar probability measure $\mu_{\mathcal{H}}$. Define

$$a = \sum_i a_i m_i, \quad M_2 = \sum_i a_i (C_i + m_i m_i^\top), \quad (13.42)$$

and their Haar projections

$$a_{\mathcal{H}} = \int_{\mathcal{H}} h a d\mu_{\mathcal{H}}(h), \quad M_{2,\mathcal{H}} = \int_{\mathcal{H}} h M_2 h^\top d\mu_{\mathcal{H}}(h), \quad C_{\mathcal{H}} = M_{2,\mathcal{H}} - a_{\mathcal{H}} a_{\mathcal{H}}^\top. \quad (13.43)$$

Then $C_{\mathcal{H}} \succ 0$, and $\mathcal{N}(a_{\mathcal{H}}, C_{\mathcal{H}})$ is the unique minimizer of (13.38) over all $\mathcal{H}$ -invariant nondegenerate Gaussians. Its score retains the log-determinant form

$$\mathcal{D}_{I,\mathcal{H}}^* = \frac{1}{2} \left[\log \det C_{\mathcal{H}} - \sum_i a_i \log \det C_i \right]. \quad (13.44)$$

[ESTABLISHED]

Proof. For every $\mathcal{H}$ -invariant candidate q , invariance of KL under a common invertible pushforward gives

$$\sum_i a_i D_{\text{KL}}(q_i^{(r)} \| q) = \int_{\mathcal{H}} \sum_i a_i D_{\text{KL}}(h_{\#} q_i^{(r)} \| q) d\mu_{\mathcal{H}}(h). \quad (13.45)$$

The orbit-averaged source ensemble on the right has first moment $a_{\mathcal{H}}$ and second moment $M_{2,\mathcal{H}}$. Haar invariance makes both moments $\mathcal{H}$ -invariant, so its moment-matched Gaussian is an admissible candidate. Its covariance is positive definite because it contains the Haar average of the positive-definite matrices $h C_i h^\top$. The calculation in Proposition 13.10 therefore proves uniqueness. Every element of a compact matrix group has $|\det h| = 1$, so $\log \det(h C_i h^\top) = \log \det C_i$, which yields (13.44). $\square$

Compactness cannot be dropped from this averaging formula. If the represented holonomy contains $2I_K$, an invariant Gaussian would require $m = 2m$ and $C = 4C$; hence $m = 0$ and $C = 0$, contradicting nondegeneracy. The generated expanding holonomy group therefore admits no invariant SPD Gaussian. **[ESTABLISHED]**

There is a separate exact identity when every transported constituent and the parent are restricted to one fixed covariance $C_0 \succ 0$. Define the symmetrized convention

$$D_{\text{SKL}}(p, q) := \frac{1}{2} [D_{\text{KL}}(p||q) + D_{\text{KL}}(q||p)]. \quad (13.46)$$

For $q_i^{(r)} = \mathcal{N}(m_i, C_0)$ and $\bar{q}_{C_0} = \mathcal{N}(\bar{m}, C_0)$, the fixed-covariance barycenter score satisfies

$$\sum_i a_i D_{\text{KL}}(q_i^{(r)} || \bar{q}_{C_0}) = \sum_{i < j} a_i a_j D_{\text{SKL}}(q_i^{(r)}, q_j^{(r)}) = \frac{1}{2} \sum_{i, j} a_i a_j D_{\text{SKL}}(q_i^{(r)}, q_j^{(r)}). \quad (13.47)$$

[ESTABLISHED] This follows from the weighted variance identity in the C_0^{-1} inner product. It is not the free-covariance optimization of Proposition 13.10, whose covariance includes the between-mean term in (13.39).

Ordinary pairwise KL between untransported local marginals is not a cross-fiber quantity. After choosing a path, a second path can change the transported law by a loop-holonomy action. Stabilization by that action guarantees path independence; without it the two KL values need not agree. For example, using the holonomy H in (13.49) and $q_i = q_j = \mathcal{N}(e_2, I_3)$, the identity path gives KL zero, whereas a path differing by H gives KL equal to 2. Moreover, an unpenalized sum of nonnegative within-cluster pairwise KL values is minimized by the singleton partition, where it vanishes. It therefore supplies no nondegenerate partition selector by itself. It also contains only marginal information and does not test the common recovery-kernel condition for joint recovery in Section 7.2. **[ESTABLISHED]**

Two elementary examples separate the structural and marginal statements. On a two-vertex tree with identity transport and $W = I_K$, represented holonomy is trivial, yet the marginals $\mathcal{N}(-ae_1, I_K)$ and $\mathcal{N}(ae_1, I_K)$ have transported divergence

$$D_{\text{KL}}(\mathcal{N}(-ae_1, I_K) || \mathcal{N}(ae_1, I_K)) = 2a^2, \quad (13.48)$$

which is arbitrarily large. Conversely, take a triangle with $K = 3$, unit edge weights, identity transports on two edges, and closing transport

$$H = \text{diag}(1, -1, -1) \in \text{SO}(3). \quad (13.49)$$

Its represented holonomy is nontrivial and $\text{Fix}(H) = \text{span}(e_1)$, so the structural fixed sector has $f_I = 1 < K$. Nevertheless, H preserves $\mathcal{N}(0, \sigma^2 I_3)$; assigning that marginal at every vertex makes all transported marginal KL values zero. **[ESTABLISHED]** Transported KL therefore measures marginal occupation of the available law modes, not the dimension of the state-independent structural fixed section.

13.6 Quotient volume for hard membership

Let $P = \widehat{S} \in \{0, 1\}^{n \times m}$, with nonempty cluster sizes $s_1, \dots, s_m$, $D = \text{diag}(s_i)$, $s = (s_i)$, and $n = \sum_i s_i$. On the translation quotient, the metric induced from the fine Euclidean quotient is

$$H = D - \frac{ss^\top}{n}, \quad \ker H = \text{span}(\mathbf{1}_m), \quad \text{pdet } H = \frac{m \prod_i s_i}{n}. \quad (13.50)$$

[ESTABLISHED]

Proof. For a coarse class $[x]$,

$$\min_a \|Px + a\mathbf{1}_n\|^2 = x^\top \left(D - \frac{ss^\top}{n} \right) x.$$

Deleting row and column r , the matrix determinant lemma gives the principal cofactor $\prod_i s_i/n$. Since a rank- $(m-1)$ Laplacian with kernel $\mathbf{1}_m$ has pseudodeterminant m times any principal cofactor, (13.50) follows. $\square$

For K channels, standard quotient Lebesgue measure and fine-induced Hausdorff measure are related by

$$\mu_{\text{ind}} = J_{\mathcal{P}} \mu_{\text{std}}, \quad J_{\mathcal{P}} = \left(\frac{m \prod_i s_i}{n} \right)^{K/2}. \quad (13.51)$$

If

$$\text{range}(S) \cap \ker \Lambda = \mathbf{1}_n \otimes \mathbb{R}^K,$$

the quotient Gaussian is proper and its generalized determinant satisfies

$$\det(G^{-1}Q) = \frac{\text{pdet } \Lambda_c}{(\text{pdet } H)^K} = \frac{\text{pdet } \Lambda_c}{J_{\mathcal{P}}^2}. \quad (13.52)$$

Consequently $Z_{\text{ind}} = J_{\mathcal{P}} Z_{\text{std}}$, and the factor cancels from the normalized law because the measure changes by the same amount. [ESTABLISHED] Raw partition functions differ by $\log J_{\mathcal{P}}$, so a partition score needs a fixed volume convention, an exact correction, or a compensating prior.

13.7 Recognition-side costs

A recognition law supported on $\text{range } S$ is singular against a nondegenerate full-space Gaussian posterior. Its relative entropy is $+\infty$. Giving the transverse directions variance ε yields

$$\mathcal{G}_\varepsilon = \frac{r}{2} \log \frac{1}{\varepsilon} - \frac{r}{2} + \frac{1}{2} \log \det(B_\perp^\top \Lambda^{-1} B_\perp) + \mathcal{G}_{\text{tie}} + O(\varepsilon), \quad (13.53)$$

where $r = NK - mK$. [ESTABLISHED] Thus identification is a generative construction, not a finite recognition restriction.

The admissible mean-only family leaves covariance unrestricted. For $P = \mathcal{N}(\mu, \Lambda^{-1})$, its exact optimized cost is

$$\mathcal{G}_{\text{tie}} = \frac{1}{2} m_{\perp}^{\top} \left(B_{\perp}^{\top} \Lambda^{-1} B_{\perp} \right)^{-1} m_{\perp}, \quad m_{\perp} = B_{\perp}^{\top} \mu. \quad (13.54)$$

[ESTABLISHED]

Proof. Optimizing the Gaussian KL over covariance gives Λ^{-1} , leaving $\frac{1}{2}(\nu - \mu)^{\top} \Lambda(\nu - \mu)$. Minimizing over $\nu = Ba$ gives the Λ -projection of μ . In the orthogonal basis $[B, B_{\perp}]$, its remaining block is the Schur complement

$$\Lambda_{22} - \Lambda_{21} \Lambda_{11}^{-1} \Lambda_{12} = \left(B_{\perp}^{\top} \Lambda^{-1} B_{\perp} \right)^{-1}.$$

This proves (13.54). The regularized formula follows by splitting the covariance determinant and trace into retained and transverse blocks and optimizing the retained block. $\square$

The marginal precision in (13.54) obeys

$$\left(B_{\perp}^{\top} \Lambda^{-1} B_{\perp} \right)^{-1} \preceq B_{\perp}^{\top} \Lambda B_{\perp}, \quad (13.55)$$

with equality exactly when $B^{\top} \Lambda B_{\perp} = 0$. Using the restriction precision instead therefore overstates the cost except under Λ -orthogonality. [ESTABLISHED]

If only covariance factorization across a partition is imposed, the exact Gaussian optimum has gap

$$\mathcal{G}_{\text{fact}} = \frac{1}{2} \left[\sum_I \log \det \Lambda_{II} - \log \det \Lambda \right]. \quad (13.56)$$

[ESTABLISHED] The mean-tie cost is nonnegative and vanishes at the finest partition; the factorization gap decreases under merging by Fischer's determinant inequality and vanishes at the coarsest partition. The quotient volume is convention dependent across partitions. Therefore these terms do not supply a nondegenerate intrinsic scale without an additional coefficient or a different nonmonotone functional. Whether such a selector exists remains open. [OPEN]

The corrected cross-inventory boundary is the general theorem of Section 7.4: a common evidence-preserving Markov pushforward makes ELBOs comparable and ordered. The hard energy trace analyzed here is not such a pushforward unless a separate theorem proves it, so its cross-partition difference can mix evidence and tightness. [ESTABLISHED]

13.8 The frame sector

If a coarse Gaussian (U_I, μ_I, Σ_I) is sent to constituent i through $U_i U_I^{-1}$, then

$$\mathcal{N}(U_i U_I^{-1} \mu_I, U_i U_I^{-1} \Sigma_I U_I^{-\top} U_i^{\top}) = \mathcal{N}(U_i \bar{m}, U_i \bar{S} U_i^{\top}), \quad (13.57)$$

with $\bar{m} = U_I^{-1}\mu_I$ and $\bar{S} = U_I^{-1}\Sigma_I U_I^{-\top}$. Thus the coarse frame cancels. With faithful representation the level set is exactly one gauge orbit; without faithfulness only unidentifiability follows. **[ESTABLISHED]**

For $K \geq 2$, $n \geq 2$, no map $F : \text{GL}^+(K)^n \rightarrow \text{GL}^+(K)$ is both left equivariant and permutation symmetric. Choose an order- n rotation $v \in \text{SO}(K)$ and inputs $(I, v, \dots, v^{n-1})$. Left multiplication by v cyclically permutes the tuple. Symmetry keeps F fixed, while equivariance sends it to vF ; hence $(I - v)F = 0$, making F singular. **[ESTABLISHED]**

Physicist summary. Hard Gaussian aggregation is exact Galerkin restriction: internal springs disappear and cut springs add. Gaussian marginalization is a different operation and mixes matrix channels; it fails in general but closes exactly when all parameters decouple in one congruence metric. Nonflat or rank-changing links close in the larger sheaf category of endpoint measurement ports. With positive-definite represented internal weights, holonomy determines the available parallel modes: trivial holonomy retains the full K -dimensional sector, while nontrivial holonomy can retain only a partial sector or none. When the infimum is attained, transported KL measures whether the marginals occupy an available mode; without attainment, a zero infimum establishes only approximability. It is not a structural-holonomy test or a partition selector, and joint recovery remains separate. Quotient volumes, mean tying, and determinant gaps are explicit, but none selects an intrinsic blocking scale by itself.

Chapter 14

Multivariate-Gaussian Operator Renormalization

Chapter 8 separates a genuine scale diagram from a sequence of differently sized objects. This chapter instantiates that diagram on the multivariate-Gaussian (MVG) interaction sector of Chapter 13. The flowed object is an operator parameter, not automatically a normalized Gaussian law: a law-level flow also needs a coarse reference measure, an information-vector rule, and a finite normalizer at every scale. Likewise, the hard aggregation below is only one admissible coarse arrow. The effective-support, sheaf, and Kron reductions of Chapter 13 give other arrows on narrower domains. Unless a channel index is displayed, this chapter flows the belief/recognition operator sector; a model-channel flow is a separate typed construction with its own dimension, links, and parameters. [DEFINITION]

14.1 The MVG operator arrow

At level ℓ , let $\pi_\ell = ((A_i^{(\ell)}), (W_{ij}^{(\ell)}))$ with $A_i^{(\ell)}, W_{ij}^{(\ell)} \in \mathbb{S}_+^K$, and assemble

$$(\Lambda_\ell)_{ii} = A_i^{(\ell)} + \sum_{j \neq i} W_{ij}^{(\ell)}, \quad (\Lambda_\ell)_{ij} = -W_{ij}^{(\ell)}. \quad (14.1)$$

For a hard partition, write $S_\ell = \widehat{S}_\ell \otimes I_K$. Its unrescaled aggregation map is

$$\mathcal{A}_{S_\ell}(\Lambda_\ell) = S_\ell^\top \Lambda_\ell S_\ell, \quad \mathcal{A}_{S_\ell}(\pi_\ell) = \left(\left(\sum_{i \in I} A_i^{(\ell)} \right)_I, \left(\sum_{\substack{i \in I \\ j \in J}} W_{ij}^{(\ell)} \right)_{I \neq J} \right). \quad (14.2)$$

[DEFINITION] The second identity is exact on the hard-aggregation domain proved in Section 13.1; nonflat data must first satisfy the effective-support or rectangular endpoint conditions of Sections 13.3 and 13.4. [ESTABLISHED]

Proposition 14.1 (Aggregation is a positive semigroup action). If S_1 maps fine nodes to intermediate clusters and S_2 maps those clusters to coarse clusters, then

$$\mathcal{A}_{S_2} \mathcal{A}_{S_1} = \mathcal{A}_{S_1 S_2}. \quad (14.3)$$

Each $\mathcal{A}_S$ is linear and positive on the product cone of node and edge weights. It is generally noninjective because every edge internal to a cluster is annihilated. [ESTABLISHED]

Proof. Congruence gives $S_2^\top (S_1^\top \Lambda S_1) S_2 = (S_1 S_2)^\top \Lambda (S_1 S_2)$. The parameter formula follows by associativity of finite sums. Sums preserve positive semidefiniteness, so the map is positive. Changing an internal weight while holding every cut weight and self term fixed leaves $\mathcal{A}_S(\pi)$ unchanged, proving noninjectivity.

An RG arrow additionally declares a positive rescaling:

$$\Lambda_{\ell+1} = \zeta_\ell^{-1} \mathcal{A}_{S_\ell}(\Lambda_\ell), \quad \zeta_\ell > 0. \quad (14.4)$$

This is the MVG component of the comparison data required by Section 8.1. The source and target usually have different dimensions. A fixed ray is therefore typed only after comparison maps identify a repeated hierarchy with one fixed parameter space. Without that identification, (14.4) is a cocycle of positive maps, not an autonomous dynamical system.

Equal-block spectral bounds. If every cluster has b nodes, then $S^\top S = bI$. For every symmetric Λ ,

$$b\lambda_{\min}(\Lambda) \leq \lambda_{\min}(S^\top \Lambda S), \quad \lambda_{\max}(S^\top \Lambda S) \leq b\lambda_{\max}(\Lambda). \quad (14.5)$$

Indeed, congruence preserves the Loewner inequalities $\lambda_{\min}(\Lambda)I \preceq \Lambda \preceq \lambda_{\max}(\Lambda)I$. These are finite-level bounds. Divergence or convergence as $\ell \rightarrow \infty$ additionally requires an infinite hierarchy and a declared normalization. [ESTABLISHED]

14.2 The biadditive closed family

Let each node carry $x_i \geq 0$, and fix $A, M \in \mathbb{S}_+^K$. The family

$$A_i = x_i A, \quad W_{ij} = x_i x_j M \quad (14.6)$$

is exactly closed under every hard partition: with $x_I = \sum_{i \in I} x_i$,

$$A_I = x_I A, \quad W_{IJ} = x_I x_J M. \quad (14.7)$$

This follows from distributivity and proves closure, not attraction. [ESTABLISHED]

Proposition 14.2 (Positive additive rules are forced to be linear). Suppose $\alpha : \mathbb{R}_+ \rightarrow \mathbb{S}_+^K$ is additive and $w : \mathbb{R}_+^2 \rightarrow \mathbb{S}_+^K$ is symmetric and separately additive. Then there are $A, M \in \mathbb{S}_+^K$ such that

$$\alpha(x) = xA, \quad w(x, y) = xyM. \quad (14.8)$$

Thus (14.6) is the maximal positive family among rules depending only on additive scalar node masses and obeying those additivity hypotheses. [ESTABLISHED]

Proof. If $y \geq x$, additivity gives $\alpha(y) - \alpha(x) = \alpha(y - x) \succeq 0$; hence α is Loewner monotone. For every v , the scalar function $x \mapsto v^\top \alpha(x) v$ is additive, nonnegative, and therefore monotone. A monotone additive function on $\mathbb{R}_+$ is linear, so $v^\top \alpha(x) v = x v^\top \alpha(1) v$. Polarization yields $\alpha(x) = x \alpha(1)$. For fixed y , the same argument gives $w(x, y) = x w(1, y)$; applying it to the second argument gives $w(1, y) = y w(1, 1)$. Take $A = \alpha(1)$ and $M = w(1, 1)$.

On a homogeneous complete graph with equal blocks of size b , the unrescaled map acts on the common pair (A, W) by

$$A \mapsto bA, \quad W \mapsto b^2 W. \quad (14.9)$$

The dense normalization $\zeta = b^2$ fixes the coupling sector and contracts the self sector by $1/b$. Consequently the candidate limiting ray has $A = 0$. It is a boundary operator: its consensus directions lie in the kernel, so it does not define a full-space normalized Gaussian density. Pinning, quotienting the kernel, and retaining an external positive mass are different repairs and define different probability problems. A pseudoinverse inserted into the original Lebesgue density is not a fourth repair.

The exponent 2 is not topology independent. If a typical coarse cut contains $c_b \asymp b^s$ fine edges, a nondegenerate coupling normalization scales as b^s , while the self sector scales as b^{1-s} . Dense complete graphs have $s = 2$; consecutive blocks in a nearest-neighbor chain have $s = 0$. Relevance of the self term is therefore a property of the declared graph family and scheme, not of (14.4) alone.

14.3 Cone dynamics: what Birkhoff does and does not prove

Let

$$\mathcal{K}_N = \left(\prod_i \mathbb{S}_+^K \right) \times \left(\prod_{i < j} \mathbb{S}_+^K \right) \quad (14.10)$$

be the full MVG parameter cone. On the interior of a proper cone $\mathcal{K}$, Hilbert's projective metric is

$$d_H(x, y) = \log \left(\frac{\inf\{\beta > 0 : x \preceq_{\mathcal{K}} \beta y\}}{\sup\{\alpha > 0 : \alpha y \preceq_{\mathcal{K}} x\}} \right). \quad (14.11)$$

A positive endomorphism of finite projective diameter contracts this metric; a primitive finite-dimensional positive matrix is the familiar special case [13, 54]. The hypotheses fail for sum-only MVG aggregation on the full cone.

Proposition 14.3 (The full-cone route is blocked). After a hierarchical identification makes $\mathcal{A}_S$ an endomorphism, sum-only aggregation is not primitive on $\mathcal{K}_N$. Moreover, the dense candidate ray with $A = 0$ lies on $\partial \mathcal{K}_N$, whereas a full-cone Birkhoff contraction has an interior Perron ray. Hence Birkhoff's full-cone theorem cannot prove attraction to that candidate. [ESTABLISHED]

Proof. An absent coarse cut remains a zero edge coordinate, so the map can meet the boundary even from positive data on the represented fine edges. More sharply, define for a subspace $U \subseteq \mathbb{R}^K$

$$\mathcal{F}_U = \{(W_e)_e : \text{ran } W_e \subseteq U \text{ for every } e\}. \quad (14.12)$$

Each $\mathcal{F}_U$ is an invariant face because the range of a sum of positive semidefinite matrices contained in U is still contained in U . Proper invariant faces rule out primitivity. Finally, $A = 0$ is a boundary coordinate of $\prod_i \mathbb{S}_+^K$, while the Perron ray of a strict interior contraction is interior.

The coupling cone alone removes the vanishing self coordinate but not the matrix obstruction. Suppose an identified spatial map $B \geq 0$ has a positive eigenvector $c = (c_e)_e$. Because aggregation only adds weights, for every $M \in \mathbb{S}_+^K$

$$W_e = c_e M \quad (14.13)$$

is an eigenray with the same spatial eigenvalue. Different matrix rays are not mixed; on the raw product cone their projective separation can be infinite. Thus sum-only aggregation cannot select a fiber direction. It can at most contract a spatial cone after the fiber ray has been declared.

Conjecture 14.4 (Fixed- B scalarized attraction). Fix $M_0 \succ 0$ and restrict to

$$\mathcal{K}_{M_0}^W = \{(c_e M_0)_e : c_e \geq 0\}. \quad (14.14)$$

If a hierarchical identification yields one fixed spatial endomorphism B , some power B^q is strictly positive, and its Perron vector factorizes as $c_{ij} = x_i x_j$, then Birkhoff's theorem gives a unique attracting spatial ray $W_{ij} = x_i x_j M_0$ on this scalarized cone. [CONJECTURE]

This statement freezes every matrix direction. It is not a theorem about the full coupling cone, a nonautonomous sequence B_ℓ , or the common-principal RG state of Section 8.2. Alternating positive maps may contract projective distances while approaching a periodic orbit, as Section 8.3 shows. The missing obligations are a fixed hierarchical identification, primitivity of the resulting B , and the factorization of its Perron vector.

Proposition 14.5 (Residual fiber label). For $M \in \mathbb{S}_+^K$, the orbit under residual global reframing $M \mapsto h^\top M h$, $h \in \text{GL}(K)$, is determined exactly by $\text{rank } M$. Hence the biadditive ansatz has $K + 1$ internal congruence classes and no continuous internal modulus. [ESTABLISHED]

Proof. Sylvester's law of inertia classifies real symmetric matrices under congruence [32]. A positive semidefinite matrix has no negative eigenvalues, so its inertia is determined by its rank. In particular, every $M \succ 0$ is congruent to I_K , witnessed by $h = M^{-1/2}$.

Rank does not classify the spatial graph, the limiting spectral measure, corrections to scaling, basin, or scheme. Nor does the proposition extend unchanged to a nonflat pair (W, Θ) , whose residual action combines congruence on W with conjugation on Θ .

14.4 An exact finite-sector gate

The homogeneous calculation (14.9) is exact. On $\mathbb{S}^K \oplus \mathbb{S}^K$, its eigenvalues are b^2 on the coupling sector and b on the self sector, each with multiplicity $K(K+1)/2$. The spectral ratio outside the dominant eigenspace is therefore

$$\frac{b}{b^2} = \frac{1}{b}, \quad (14.15)$$

equal to 0.5 at $b = 2$. Comparing the two largest entries of the sorted spectrum would incorrectly return one because the dominant eigenvalue is degenerate. [ESTABLISHED]

Equation (14.15) is the complete finite-sector result. Attraction, uniqueness, and scheme independence remain open because they require a declared invariant cone, hierarchy, and limiting protocol. [OPEN]

14.5 Gauge-invariant spectral data

Let L be the interaction Laplacian obtained from (14.1) by setting all $A_i = 0$. Both L and $\Lambda = L + A$ are bilinear forms. A local reframing $G = \text{diag}(g_1, \dots, g_N)$ sends both by congruence. Ordinary eigenvalues of either matrix are therefore frame dependent. The generalized spectrum of the pair is invariant when the pencil is regular.

Proposition 14.6 (Regular-pencil invariance). Suppose $\det(L - d\Lambda)$ is not the zero polynomial. Then (L, Λ) and $(G^\top LG, G^\top \Lambda G)$ have the same generalized eigenvalues with the same multiplicities. If Λ is invertible, there are exactly NK finite roots. [ESTABLISHED]

Proof.

$$\det(G^\top LG - dG^\top \Lambda G) = \det(G)^2 \det(L - d\Lambda). \quad (14.16)$$

The nonzero factor does not change roots or regularity. When Λ is invertible, the coefficient of d^{NK} is $(-1)^{NK} \det \Lambda \neq 0$.

Proposition 14.7 (Regularity criterion). For $L, A \succeq 0$ and $\Lambda = L + A$, the pencil (L, Λ) is regular if and only if

$$\ker L \cap \ker A = \{0\}. \quad (14.17)$$

In that case $\Lambda \succ 0$. [ESTABLISHED]

Proof. Positive semidefiniteness gives $\ker(L + A) = \ker L \cap \ker A$. A nonzero common-kernel vector annihilates $L - d\Lambda$ for every d , so the determinant is identically zero. Conversely, if the intersection is trivial then $L + \Lambda = 2L + A \succ 0$; evaluating at $d = -1$ proves that the determinant polynomial is not identically zero. The same intersection identity gives $\ker \Lambda = \{0\}$.

At the dense candidate ray $A = 0$, one has $\Lambda = L$ and $\det(L - dL) \equiv 0$. Passing to a complement of $\ker L$ produces the constant generalized spectrum $\{1\}$, not a graph-sensitive label. Pinning and quotienting therefore do not preserve the same spectral object.

Proposition 14.8 (Mass-pencil transfer). Let $R \succ 0$, $a > 0$, and $\Lambda = L + aR$. Generalized eigenvalues of (L, R) and (L, Λ) are related by

$$d = \frac{\lambda}{\lambda + a}, \quad \lambda = \frac{ad}{1 - d}. \quad (14.18)$$

The correspondence preserves eigenvectors, multiplicities, and ordering. Consequently,

$$N_d(t) = N_\lambda\left(\frac{at}{1-t}\right), \quad 0 \leq t < 1. \quad (14.19)$$

If $N_\lambda(s) - N_\lambda(0) \sim Cs^\alpha$ near zero, then at fixed $a > 0$, $N_d(t) - N_d(0) \sim Ca^\alpha t^\alpha$. **[ESTABLISHED]**

Proof. The equation $Lv = d(L + aR)v$ is equivalent, for $d \neq 1$, to $Lv = (ad/(1-d))Rv$. The case $d = 1$ would require $Rv = 0$, which is impossible. Monotonicity of the fractional map gives the counting identity and its asymptotic consequence.

Mass removal and the infrared limit do not commute:

$$\lim_{a \downarrow 0} \lim_{\lambda \downarrow 0} \frac{\lambda}{\lambda + a} = 0, \quad \lim_{\lambda \downarrow 0} \lim_{a \downarrow 0} \frac{\lambda}{\lambda + a} = 1. \quad (14.20)$$

Thus a positive mass regularizes each finite problem but does not supply a mass-independent fixed-ray spectrum.

Thermodynamic spectral labels. For a declared sequence (V_n, L_n, R_n) , define

$$N_n(t) = \frac{1}{|V_n|K} \# \{k : \lambda_k^{(n)} \leq t\}. \quad (14.21)$$

A cumulative exponent $N(t) - N(0) \sim Ct^\alpha$ is meaningful only after proving convergence $N_n \rightarrow N$ in a stated mode. A fixed finite graph has an atomic measure, so a density exponent is not defined there. RG depth ℓ and thermodynamic size n form the two-index family of Section 8.5; neither the existence nor the commutation of those limits follows from the finite congruence identity. **[OPEN]**

Equality of one exponent is not equality of universality classes. The full limiting measure, amplitudes, corrections to scaling, invariant faces, basin, and blocking scheme may still differ. Projective eigenvalue ratios, block-normalized scaling dimensions, and matched heat/IDS exponents are candidate invariant observables; the raw running matrices are scheme-dependent coordinates.

14.6 Finite stochastic refinement on matrix cones

Section 8.6 proves the general typing statement: a stochastic right section samples a fine preimage and is not an inverse of the realized coarse map. The matrix cone sharpens both the obstruction and the available finite construction.

The central Wishart family has Laplace transform $\det(I + \sigma\Theta)^{-p}$. Apart from the degenerate law δ_0 , it is infinitely divisible exactly when $\text{rank } \sigma = 1$ [48]. Therefore a full-rank Wishart prior cannot provide convolution roots of every order for $K \geq 2$; the surviving rank-one case is an embedded scalar Gamma law on a proper face. This does not mean full-cone convolution semigroups are absent. Define the trace-one cone

$$\mathbb{S}_{+,1}^K = \{U \succeq 0 : \text{tr } U = 1\}.$$

Let α be a finite positive Borel measure on $\mathbb{S}_{+,1}^K$, and let $\beta : \mathbb{S}_{+,1}^K \rightarrow (0, \infty)$ be measurable with

$$\int_{\mathbb{S}_{+,1}^K} \log(1 + \beta(U)^{-1}) \alpha(dU) < \infty.$$

For $\Theta \succeq 0$, the corresponding matrix-Gamma law on $\mathbb{S}_+^K$ has Laplace transform

$$\mathcal{L}_{\alpha,\beta}(\Theta) = \exp \left[- \int_{\mathbb{S}_{+,1}^K} \log \left(1 + \frac{\text{tr}(U\Theta)}{\beta(U)} \right) \alpha(dU) \right], \quad (14.22)$$

and, for fixed β , replacing α by $t\alpha$, $t \geq 0$, produces a convolution semigroup; $t = 0$ is δ_0 [56]. Generic full-cone draws need not lie in one Kron-closed chart, however.

Theorem 14.9 (Congruence-diagonal Gamma–Dirichlet bridge). Fix the Kron-closed cone $\mathcal{C}_H$ from Section 13.2, channel parameters $a_r, \beta_r > 0$, and intensities $t_e > 0$. Independently draw

$$X_{e,r} \sim \text{Gamma}(t_e a_r, \text{rate } \beta_r), \quad W_e = H \text{diag}(X_{e,1}, \dots, X_{e,K}) H^\top. \quad (14.23)$$

For a coarse coordinate f formed from constituents E_f , $\sum_{e \in E_f} W_e$ has the same form with intensity $\sum_{e \in E_f} t_e$. Conditioned on

$$W_f = H \text{diag}(c_1, \dots, c_K) H^\top,$$

sample independently over channels

$$(P_{e,r})_{e \in E_f} \sim \text{Dirichlet}((t_e a_r)_{e \in E_f}), \quad X_{e,r} = c_r P_{e,r}. \quad (14.24)$$

Then every $W_e \in \mathcal{C}_H$ and $\sum_{e \in E_f} W_e = W_f$ exactly. Applied independently to coarse self terms and cut edges, this is a stochastic right section of any fixed finite aggregation map on the represented coordinates. [ESTABLISHED]

Proof. Scalar Gamma variables with a common rate add by addition of their shape parameters. Conditional on their sum, their proportions have the Dirichlet law in (14.24). The identities hold channelwise, and multiplication by the fixed congruence $H(\cdot)H^\top$ preserves both $\mathcal{C}_H$ and addition.

Internal fine edges are absent from the coarse coordinates and require a separate prior. The theorem neither recovers the discarded realization nor constructs a two-sided inverse. An infinite graph-level construction still owes compatible internal-edge priors, nested bridge kernels, gauge-covariant chart data, cylinder consistency, and tightness or another projective-limit theorem. [OPEN]

14.7 Relations to Bayesian, Laplacian, network, and learned RG

Bayesian renormalization. For a Gaussian mean family with fixed precision Λ , the Fisher metric in the mean coordinates is Λ , and the inverse Fisher cometric is Λ^{-1} . This local identity makes an MVG comparison with Bayesian renormalization natural. Berman, Klinger, and Stapleton, however, begin with a posterior over model parameters. In a late-data local Gaussian regime, the delta method pushes an inverse Fisher cometric through a forward map to obtain a sample-space diffusion tensor, exactly for an affine map and to leading local order for a nonlinear map [9, 11]. Howard, Klinger, Maiti, and Stapleton subsequently apply Bayesian RG to neural network field theories [33]. [ESTABLISHED] Hard energy precomposition $\Lambda \mapsto S^\top \Lambda S$ is not that posterior flow. [ESTABLISHED] No theorem identifies their scale orientation, connection data, or global continuation with (14.4); the comparison therefore supplies no “opposite-flow” theorem between this operator coarsening and Bayesian learning. [NOT-CLAIMED]

Laplacian RG. With a positive reference form R , the generalized heat generator $R^{-1}L$ transforms by similarity, so its spectrum and heat trace are gauge invariant. The scalar Laplacian-RG construction motivates the entropy scale diagnostic and the positive susceptibility

$$C(\tau) = -\frac{dS}{d \log \tau} \quad (14.25)$$

[67]. Raw off-diagonal heat-kernel blocks transform as $K_{ij} \mapsto g_i^{-1} K_{ij} g_j$, however, and are not invariant scalar node affinities. [ESTABLISHED] A gauge-invariant real-space block selector, zero-mode treatment, and entropy normalization remain open. The heat scale diagnostic does not by itself prove a unique or scheme-independent partition. [OPEN]

Network RG. Exact closure occurs in the independent-edge multiscale family of Garuccio, Lalli, and Garlaschelli, whose hidden variables aggregate additively [26]. It is a characterized family, not a theorem for arbitrary hidden-variable networks. Interaction closure after a separately specified elimination is a different result. Interacting-network RG can generate higher-order interactions outside restricted coordination classes [19]; it does not imply exact closure of arbitrary matrix-weighted node aggregation. Conversely, (14.2) does not integrate out edge variables and should not be presented as that network transformation. The broader network-RG taxonomy is reviewed by Gabrielli et al. [25]. [ESTABLISHED]

Graph learning and variational RG. The Galerkin identity $L_c = S^\top L S$ preserves the restricted quadratic energy. It does not imply spectral approximation, message-passing commutation, or a graph neural network realization; those require the approximation and residual controls separated in Section 7.7 [35, 46]. DiffPool instead learns soft cluster assignments end to end [72]; that modeling choice does not by itself provide an exact RG semigroup or a closure theorem. Oono and Suzuki prove conditional exponential information loss for deep graph convolutional networks under spectral and weight hypotheses [55]; their result is an oversmoothing boundary, not a Galerkin-commutation theorem. Likewise, Mehta and Schwab’s exact variational-RG/deep-network correspondence assumes a hidden-variable kernel satisfying its trace condition [49]. A deterministic identification $y = S\tilde{y}$ does not supply that kernel or trace condition, so their exact mapping is a precedent within its own construction, not a proof that (14.4) is a deep network. [ESTABLISHED]

14.8 Remaining MVG obligations

The operator recursion is exact on its declared domain, but the following extensions remain open. Partial fixed-section fibers and rectangular cellular-sheaf endpoint maps are already structurally closed under nested blocking by Sections 13.4 and 13.5. [ESTABLISHED] This structural result does not supply a canonical compression to one invertible link and one SPD weight or an RG rescaling across variable-rank fibers. A full MVG law flow must specify the information vector, reference measure, and normalizer and prove that every iterate stays in the Gaussian natural domain. Attraction needs a typed autonomous endomorphism or a cocycle theorem, a basin, and a matrix mixing mechanism or an explicit scalarized restriction. Universality needs scheme-invariant observables and the two-index convergence required by Section 8.5. A gauge-invariant, nondegenerate block selector remains open. Stochastic inverse RG needs graph-level projective priors, including priors for annihilated internal edges. [OPEN]

Physicist summary. The MVG interaction precision has a clean positive coarse map: cut weights and self terms add, and a declared scalar restores the chosen units. On dense homogeneous graphs the coupling sector scales like b^2 while the self sector scales like b , giving an exact relative factor $1/b$. That algebra produces a boundary operator ray, not a normalized Gaussian fixed law. Birkhoff contraction becomes plausible only after freezing the matrix direction and constructing one primitive spatial endomorphism; sum-only aggregation preserves matrix faces and cannot select that direction. Generalized spectra and Gamma–Dirichlet finite refinements are exact gauge-compatible tools, but neither solves scheme independence, nonflat hierarchy, or the infinite projective-limit problem.

Part IV

Boundaries, Evidence, and Interpretation

Chapter 15

Obstructions

The construction developed in this document is deliberately narrow. Its general belief-law and model-law layers fix one normalized joint and one exact bound, with manifolds introduced only on selected finite-dimensional tiers; the concrete obstructions in this chapter concern particular Gaussian realizations and must not be promoted to no-go theorems for arbitrary belief or model fibers. Each statement below names its architecture, and the weaker constructions that survive it are kept explicit.

Every discrete map in this chapter is a graph-link datum. We write Θ_e^b and Θ_e^m for belief- and model-sector links on an oriented edge copy e ; the Gaussian examples suppress the sector superscript. Reverse orientation of the same edge copy obeys $\Theta_{\bar{e}} = \Theta_e^{-1}$. The reciprocal-pair example instead uses two *distinct parallel edge copies* $e \neq f$, traversed in opposite directions, so their link values are not required to be inverses. These symbols are distinct from the base-connection transports $\Omega_\gamma, \tilde{\Omega}_\gamma$ and from the cross-bundle morphisms $\Phi, \tilde{\Phi}$. No graph link is identified with base parallel transport without a curve assignment and an equality hypothesis.

The material organizes around one temptation. A tower of scales in which every level's prior is supplied by the level above must terminate, and its top level has no level above. It is natural to try to close the tower by a reciprocal Gaussian agreement pair. The exact result is narrower than “cycles fail”: the *flat, unanchored reciprocal pair* is singular, and an objective that appends its energy while omitting a required global normalizer is not an evidence bound. Cyclic globally normalized Markov random fields with proper anchors are legitimate probability models; directed auxiliary-latent models and cycles of inference updates are legitimate as well. Sections 15.1 through 15.2 isolate the failed architecture, Section 15.3 separates it from a distinct Bayesian-renormalization construction, and Sections 15.4 and 15.5 prove that an anchored Gaussian star supplies a unique, geometrically attracting coordinate-ascent fixed point. Section 15.6 finally shows that nontrivial graph-link holonomy can remove the kernel of this reciprocal pair, while making no claim about cyclic models in general.

15.1 The flat unanchored reciprocal Gaussian fold is singular

Consider two latents $u, v \in \mathbb{R}^K$ joined by distinct parallel edge copies $e : v \rightarrow u$ and $f : u \rightarrow v$. This is the minimal reciprocal fold: v supplies a penalty for u , and u supplies one for v . The two residuals

contribute the assembled precision

$$J = E_e^\top R_e^{-1} E_e + E_f^\top R_f^{-1} E_f, \quad E_e = \begin{pmatrix} I & -\Theta_e \end{pmatrix}, \quad E_f = \begin{pmatrix} -\Theta_f & I \end{pmatrix}, \quad (15.1)$$

with $R_e, R_f \succ 0$ and $\Theta_e, \Theta_f \in \text{GL}^+(K)$. Write $H = \Theta_f \Theta_e$ for the graph holonomy of the parallel-edge two-cycle based at v . The typed inverses $\Theta_{\bar{e}} = \Theta_e^{-1}$ and $\Theta_{\bar{f}} = \Theta_f^{-1}$ exist but do not appear in this loop.

Proposition 15.1 (Kernel of a reciprocal pair). For every pair of positive definite link covariances,

$$\ker J = \{(\Theta_e v, v) : v \in \ker(H - I)\}, \quad \dim \ker J = \dim \ker(H - I). \quad (15.2)$$

The kernel does not depend on R_e or R_f : its dimension is the multiplicity of the eigenvalue one in the loop holonomy. **[ESTABLISHED]**

Proof. J is a sum of two positive semidefinite matrices, so $w \in \ker J$ if and only if w lies in the kernel of each summand. For positive definite R , the form $w \mapsto w^\top E^\top R^{-1} E w$ vanishes exactly when $E w = 0$, so $\ker J = \ker E_e \cap \ker E_f$. Writing $w = (u, v)$, the first condition is $u = \Theta_e v$ and the second is $v = \Theta_f u$; substituting gives $v = \Theta_f \Theta_e v = H v$, and conversely any v fixed by H produces a solution with $u = \Theta_e v$. The map $v \mapsto (\Theta_e v, v)$ is injective, so the dimensions agree.

Corollary 15.2 (A flat parallel-edge fold is singular). If the two distinct edge copies give the same flat endpoint comparison, equivalently $\Theta_f = \Theta_e^{-1}$, then $H = I$ and

$$\ker J = \{(\Theta_e v, v) : v \in \mathbb{R}^K\}, \quad \dim \ker J = K. \quad (15.3)$$

The assembled precision is singular for every positive definite pair of link covariances, and its kernel is exactly the set of transport-consistent configurations. **[ESTABLISHED]**

Boundary of the no-go. Corollary 15.2 is not a theorem against cycles. If $P_u, P_v \succ 0$, then

$$J_{\text{anch}} = J + \text{diag}(P_u, P_v) \succ 0, \quad (15.4)$$

because a positive definite matrix added to a positive semidefinite matrix is positive definite. Hence

$$p(u, v) = Z^{-1} \exp \left[-\frac{1}{2} (u, v)^\top J_{\text{anch}} (u, v) + h^\top (u, v) \right]$$

is a legitimate globally normalized Gaussian model built from the reciprocal parallel-edge energy, with finite $Z = (2\pi)^K (\det J_{\text{anch}})^{-1/2} \exp(\frac{1}{2} h^\top J_{\text{anch}}^{-1} h)$. The same proof covers a finite cycle with any number of vertices: its Gaussian edge energies sum to a positive semidefinite precision, and an SPD block anchor at every vertex makes the total precision positive definite. Such a Markov random field loses the convenience of a product of locally normalized directed kernels, not existence as a probability law. Likewise, Proposition 15.1 shows that the unanchored parallel-edge pair is definite

when $1 \notin \text{spec}(H)$. The obstruction is therefore exactly the conjunction of the reciprocal-pair energy, no SPD anchor, and flat parallel-edge holonomy $H = I$. [ESTABLISHED]

Within that scope the direction of the degeneracy is important. A Gaussian energy whose precision has a K -dimensional kernel does not define a normalized Lebesgue density, so the exact identity of Chapter 5 has no joint to which it can apply. The flat direction consists precisely of those configurations in which u and v agree under transport. The unanchored flat pair charges exactly nothing there, so it supplies no coercivity along its intended agreement direction.

There are two independent ways to restore definiteness. Equation (15.4) adds declared coercivity and remains valid under flat parallel-edge holonomy. Alternatively, (15.2) makes the unanchored pair positive definite exactly when 1 is not an eigenvalue of H , which requires abandoning the Regime-I coboundary and declaring at least one of the two parallel edge copies independently. That route leaves the fixed- K flat interaction family used by the standard aggregation flow: an internal twisted cycle leaves a self residue, and the cut sector needs the enlarged nonflat rule left open in Chapter 13. This is a limitation of that flow, not a no-go for globally normalized cyclic models. [ESTABLISHED]

An objective-only reading fails for a different reason. If the reciprocal energy is appended outside the declared joint, the evidence $\log p_\theta(o \mid X)$ is determined by the factors that are in the joint and is unaffected, so the appended term has nothing to bound. If instead the energy is included in a globally normalized Markov random field, it is part of a legitimate joint whenever its global partition function is finite, but that partition function must be retained. Thus there are three distinct cases: the flat unanchored energy is improper; the properly anchored or non-flat energy with its global normalizer is a valid undirected model; and the energy appended while its normalizer is omitted is not an evidence bound. Chapter 5 applies to the second case and not to the other two. [ESTABLISHED]

15.2 A proper anchor restores definiteness and makes the global normalizer essential

The obvious repair is to add a proper prior to each folded latent, replacing J by $J + \text{diag}(P_0, P_0)$ with $P_0 \succ 0$. This restores definiteness, since the added term is positive definite and the sum of a positive semidefinite and a positive definite matrix is positive definite. What it does not restore is the ability to treat the normalization as a constant.

For a Gaussian in information form the log normalizer is

$$A(h, J) = \frac{1}{2} h^\top J^{-1} h - \frac{1}{2} \log \det J + \frac{n}{2} \log 2\pi, \quad (15.5)$$

so any dependence of J on the transports passes directly into A . In (15.1) the transports appear inside E_e and E_f , and the question is whether their contribution to $\det J$ cancels. It does not.

Proposition 15.3 (The global normalizer depends on the model graph link). Take $K = 1$, unit link variances, flat parallel-edge values $\Theta_e = a$ and $\Theta_f = a^{-1}$ with $a > 0$, and a coordinate-isotropic anchor precision $p_0 > 0$. Then

$$\det(J + p_0 I) = p_0^2 + p_0(a + a^{-1})^2, \quad (15.6)$$

which is a nonconstant function of a , uniquely minimized on $\text{GL}^+(1) = (0, \infty)$ at $a = 1$, with value $p_0^2 + 4p_0$. [ESTABLISHED]

Proof. With $E_e = (1, -a)$ and $E_f = (-a^{-1}, 1)$, the two rank-one contributions sum to

$$J = \begin{pmatrix} 1 + a^{-2} & -(a + a^{-1}) \\ -(a + a^{-1}) & 1 + a^2 \end{pmatrix},$$

whose determinant is $(1 + a^{-2})(1 + a^2) - (a + a^{-1})^2 = (2 + a^2 + a^{-2}) - (a^2 + 2 + a^{-2}) = 0$, confirming Corollary 15.2 in this case. Adding $p_0 I$ gives $\det(J + p_0 I) = p_0^2 + p_0[(1 + a^{-2}) + (1 + a^2)] = p_0^2 + p_0(2 + a^2 + a^{-2})$, and $2 + a^2 + a^{-2} = (a + a^{-1})^2$. Since $a + a^{-1} \geq 2$ for $a > 0$, with equality only at $a = 1$, the stated minimizer is unique.

The witness settles nonconstancy, because variation in one admissible instance refutes a constant-normalizer claim. For zero information vector, write

$$D(a) = p_0^2 + p_0(a + a^{-1})^2, \quad A(a) = -\frac{1}{2} \log D(a) + \text{const.}$$

Then

$$A'(1) = 0, \quad A''(1) = -\frac{4}{p_0 + 4} < 0. \quad (15.7)$$

Thus the identity is a strict *maximum* of the isolated log-normalizer contribution, not a minimum. Minimizing the corresponding isolated contribution to the negative log density pushes away from $a = 1$; the quadratic and information-vector terms also vary with a , so no universal direction follows for the complete objective. What is universal in this witness is only that the global term $-\frac{1}{2} \log \det(J + p_0 I)$ cannot be absorbed into per-agent constants or dropped from a normalized joint. [ESTABLISHED]

This variation is a change of model, not a pure coordinate change. It holds the coordinate matrices $p_0 I$, $R_e = R_f = I$, and the reference Lebesgue coordinates fixed while varying a . Under a genuine reframing, all covariances, precisions, information vectors, and the reference-density Jacobian transform together, and Proposition 4.8 leaves the probability measure and evidence invariant. Proposition 15.3 therefore proves transport-parameter dependence of a declared anchored model; it does not prove a force along a gauge orbit. [ESTABLISHED]

15.3 Bayesian posterior flow is a different construction

Bayesian renormalization, as formulated by Berman et al. [11], declares a prior and a likelihood and studies posterior and model-space descriptions indexed by an information-theoretic scale. A posterior is the output of that declared Bayesian model; the construction does not, merely by allowing the posterior to flow, make a generative kernel read its own posterior. The typing prohibition of Chapter 4 therefore does not refute Bayesian renormalization. [ESTABLISHED]

The two frameworks nevertheless cannot be identified without an additional construction. If one takes a posterior from one scale and inserts it as the generative top factor at another, then one has declared either a new model at each scale or an explicit transition rule between models. That can be legitimate, but the evidence and ELBO at different scales then refer to different declared joints unless a common target and an exact comparison map are supplied. Proposition 4.3 gives the precise warning: a family of exact pointwise identities does not itself designate one member or one evidence, and comparing two bounds certifies an improvement in one fixed quantity only on an evidence level set. This is a condition on a proposed composition with the present model, not a criticism of a posterior flow considered on its own. [ESTABLISHED]

The resulting boundary is narrower than a prohibition on comparing different latent inventories. If two normalized joints are declared independently, with no evidence-preserving channel between them, their optimized ELBO difference mixes an evidence difference with a tightness difference and supplies no universal order. By contrast, a normalized Markov pushforward applied to both the posterior and recognition law preserves the observation evidence and orders the bounds by data processing. The arbitrary-model caveat applies only in the absence of that relation. [ESTABLISHED]

The asymptotic covariance $C_T = T^{-1}I^{-1} + o(T^{-1})$ is a $T \rightarrow \infty$ statement and supplies no bound, limit, or monotonicity as $T \downarrow 0$. Properness at zero information depends on the declared prior and normalizer; with a proper prior, the no-data posterior is that prior. Hence this expansion supports no improper-endpoint claim about Bayesian renormalization. [NOT-CLAIMED]

Nor does a verbal contrast between “posterior widening” and “precision addition” establish that the two scale directions are opposites. The first depends on the declared Bayesian path, and the second is an unrescaled Gaussian aggregation identity. Relating them would require an explicit map between their scale parameters and an operator comparison in a common theory space; none is supplied here. [NOT-CLAIMED]

15.4 What survives

Two constructions remain, and the second is the interesting one.

The first is a declared top prior on a truncated tower. The tower is finite, nothing is folded, and the top level carries a prior that is part of the declared parameters. This is fully typed, it satisfies the hypotheses of Chapter 5 without amendment, and it costs the closure ambition entirely: the top is given, not constituted. [DEFINITION]

The second replaces the fold by an apex latent inside the joint. Let $b \in \mathbb{R}^K$ carry a declared proper prior $\mathcal{N}(0, P_0^{-1})$ with $P_0 \succ 0$, and let the constituents be conditionally independent given it,

$$b \sim \mathcal{N}(0, P_0^{-1}), \quad y_i \mid b \sim \mathcal{N}(\Theta_i b, R_i), \quad i = 1, \dots, n. \quad (15.8)$$

The scale graph is a star, hence a tree, so no cycle can form and Proposition 15.1 has nothing to act on. Three properties make this the right replacement rather than merely an admissible one.

Proposition 15.4 (Definiteness of the anchored star). The assembled precision $J_\star$ of (15.8) is positive definite for every choice of $\Theta_i \in \text{GL}^+(K)$ and $R_i \succ 0$. [ESTABLISHED]

Proof. The quadratic form is $b^\top P_0 b + \sum_i (y_i - \Theta_i b)^\top R_i^{-1} (y_i - \Theta_i b)$, a sum of positive semidefinite terms. It vanishes only if $b^\top P_0 b = 0$ and each residual vanishes; the first forces $b = 0$ because $P_0 \succ 0$, and then $y_i = \Theta_i b = 0$. The kernel is trivial.

Proposition 15.5 (The exact mean-field coordinate is precision addition). Under a mean-field recognition family factorizing over b and the constituents, the optimal factor for b at fixed remaining factors is Gaussian with precision

$$P_b = P_0 + \sum_{i=1}^n \Theta_i^\top R_i^{-1} \Theta_i, \quad (15.9)$$

independent of the remaining factors, and with mean $P_b^{-1} \sum_i \Theta_i^\top R_i^{-1} \mathbb{E}[y_i]$. [ESTABLISHED]

Proof. The coordinate-ascent optimum satisfies $\log q^*(b) = \mathbb{E}_{-b}[\log p(b, y)] + \text{const}$. Only terms quadratic in b contribute to the precision, and taking expectations over the other factors leaves them unchanged, so the coefficient of $-\frac{1}{2} b^\top (\cdot) b$ is the bb block of $J_\star$, which is $P_0 + \sum_i \Theta_i^\top R_i^{-1} \Theta_i$. The linear coefficient is $\sum_i \Theta_i^\top R_i^{-1} \mathbb{E}[y_i]$.

Equation (15.9) is the transported precision-pooling rule, obtained here as an exact consequence of the joint rather than posited as an aggregation convention. It is also the sufficiency statement of Chapter 7 read in the recognition direction: the statistic that the constituents supply to the apex is $\sum_i \Theta_i^\top R_i^{-1} y_i$, and (15.9) is the precision that statistic carries.

Proposition 15.6 (The declared prior supplies exactly the missing coercivity). Let $w = (v, \Theta_1 v, \dots, \Theta_n v)$ be a transport-consistent configuration of (15.8). Then

$$w^\top J_\star w = v^\top P_0 v \quad (15.10)$$

exactly, for every $v \in \mathbb{R}^K$ and every choice of transports and link covariances. [ESTABLISHED]

Proof. Every constituent residual $y_i - \Theta_i b$ vanishes on such a configuration, so the only surviving term of the quadratic form is $v^\top P_0 v$.

The three propositions together locate precisely what the flat unanchored pair lacks and what the star supplies. By Corollary 15.2 the pair charges nothing on the transport-consistent ray; by (15.10) the star charges exactly $v^\top P_0 v$ there, no more and no less. The deterministic check `CHK-OBS-STAR-FOLD-NEW-PROTOCOL` constructs star and flat-pair instances and verifies the precision-addition identity and coercivity contrast under its declared protocol. The propositions, not the computation, establish the claim. [NUMERICAL]

That a prior must be declared somewhere is not a shortcoming of this particular construction, and it is worth recording as a general fact rather than as a concession.

Proposition 15.7 (A declared root is unavoidable). In any finite directed acyclic generative model there is at least one latent with no parents, whose marginal law is therefore part of the declared parameters. Hence no finite acyclic construction can have every prior constituted by other latents.[ESTABLISHED]

Proof. A finite directed acyclic graph admits a topological order, and its first element has no incoming edges. Its factor in the joint is unconditional, so it is declared.

The ambition behind the fold was to eliminate the declared root. Proposition 15.7 says that within finite acyclic models the ambition is unachievable for structural reasons rather than for want of ingenuity, and Corollary 15.2 says that the particular flat unanchored reciprocal-pair escape is improper. Properly anchored globally normalized cyclic models remain available, but their anchors and partition functions are declarations rather than derivations from the population.

15.5 The participatory content survives in the inference

What the fold was reaching for is nonetheless available, and it is available exactly where Propositions 15.5 and 15.7 jointly place it: in the inference rather than in the model.

At a variational fixed point of (15.8), the apex factor has the mean $P_b^{-1} \sum_i \Theta_i^\top R_i^{-1} \mathbb{E}[y_i]$ of Proposition 15.5, which is a transported pooling of every constituent's current belief; and each constituent's effective prior, in its own coordinate update, is that apex belief transported back down. Each agent's effective coordinate input is therefore constituted by the apex update, and the apex input is constituted by the agents' updates. This is a loop in an inference algorithm, not a cycle in the factorization of the model. Every step is a coordinate update of a recognition law against a joint that is fixed, acyclic, and normalized, so nothing reads a posterior into a generative kernel and the prohibition of Chapter 4 is respected throughout.[ESTABLISHED]

The distinction being drawn is not verbal. A cycle in the model concerns the factorization or global potential of a joint; Corollary 15.2 rejects only one flat unanchored reciprocal Gaussian potential. A cycle in the inference is a property of an iterative scheme applied to a fixed joint. What the star gives up is the ambition that the apex prior be derived; what it keeps is a bidirectional dependence of exact mean-field coordinates during inference.

Theorem 15.8 (Exact Gaussian-star fixed point and contraction). Allow arbitrary fixed information vectors $r_b, r_1, \dots, r_n$ by adding $r_b^\top b + \sum_i r_i^\top y_i$ to the log density of (15.8); the displayed centered star is the case $r_b = r_i = 0$. Define

$$B := \sum_{i=1}^n \Theta_i^\top R_i^{-1} \Theta_i, \quad P_b := P_0 + B, \quad M := P_b^{-1} B. \quad (15.11)$$

Consider the exact block schedule that updates every constituent factor from the current apex factor, in parallel or in any order, and then updates the apex factor. The factor covariances become R_i and P_b^{-1} , respectively, on their first updates, and the means obey

$$m_i^{(t+1)} = \Theta_i m_b^{(t)} + R_i r_i, \quad (15.12)$$

$$m_b^{(t+1)} = P_b^{-1} \left(r_b + \sum_i \Theta_i^\top R_i^{-1} m_i^{(t+1)} \right) = c + M m_b^{(t)}, \quad c = P_b^{-1} \left(r_b + \sum_i \Theta_i^\top r_i \right). \quad (15.13)$$

There is a unique fixed point,

$$m_b^* = P_0^{-1} \left(r_b + \sum_i \Theta_i^\top r_i \right), \quad m_i^* = \Theta_i m_b^* + R_i r_i, \quad (15.14)$$

and, in the norm $\|x\|_{P_b} = (x^\top P_b x)^{1/2}$,

$$\|m_b^{(t+1)} - m_b^*\|_{P_b} \leq \rho \|m_b^{(t)} - m_b^*\|_{P_b}, \quad \rho = \lambda_{\max} \left(P_b^{-1/2} B P_b^{-1/2} \right) < 1. \quad (15.15)$$

Thus the exact schedule converges geometrically from every initial collection of means. **[ESTABLISHED]**

Proof. Taking the expectation of the joint log density over all factors except q_i leaves a quadratic in y_i with precision R_i^{-1} and information vector $R_i^{-1} \Theta_i m_b^{(t)} + r_i$, which gives (15.12). Proposition 15.5 with the added information vector gives the first expression in (15.13); substitution gives the affine recursion. Since

$$I - M = P_b^{-1} (P_b - B) = P_b^{-1} P_0,$$

the fixed-point equation $(I - M)m_b^* = c$ has the unique solution (15.14), because $P_0 \succ 0$.

The matrix M is similar to the symmetric matrix

$$S := P_b^{-1/2} B P_b^{-1/2} = I - P_b^{-1/2} P_0 P_b^{-1/2}.$$

Both $B \succeq 0$ and $P_0 \succ 0$, so every eigenvalue of S lies in $[0, 1)$. If $e_t = m_b^{(t)} - m_b^*$, then

$$\|e_{t+1}\|_{P_b} = \|S P_b^{1/2} e_t\|_2 \leq \lambda_{\max}(S) \|e_t\|_{P_b},$$

which proves (15.15). After the first updates, each factor covariance already equals its fixed-point covariance. Consequently $D_{\text{KL}}(q_b^{(t)} \| q_b^*) = \frac{1}{2} \|e_t\|_{P_b}^2$ contracts by at least ρ^2 per full sweep, and the

sum of constituent divergences after their next update is $\frac{1}{2}e_t^\top B e_t \leq \frac{1}{2}e_t^\top P_b e_t$. This also proves convergence of the entire product factor in relative entropy. $\square$

The strict anchor $P_0 \succ 0$ is essential. If it is weakened to $P_0 \succeq 0$, the symmetric matrix S can have eigenvalue one, and neither uniqueness nor geometric contraction follows. When $B = 0$, $\rho = 0$ and the apex reaches its fixed point in one update. Delayed, noisy, asynchronous, or inexact updates are not covered by Theorem 15.8 and remain a separate convergence problem. [OPEN]

If a one-parameter family is genuinely wanted, for a construction that interpolates between an uninformative and a data-driven description, a legitimate one is available and does not repeat the failures of Section 15.3. Declare a tempering path $p_\beta \propto p(Y)p(o | Y)^\beta$ for $\beta \in [0, 1]$, whose endpoints are the prior and the posterior of one fixed model, both proper, and whose evidence satisfies the exact path-sampling identity

$$\log p(o) = \int_0^1 \mathbb{E}_{p_\beta} [\log p(o | Y)] d\beta \quad (15.16)$$

[27]. The family is indexed by a declared temperature and built from one declared prior–likelihood pair; its normalizing constant is retained at every β , both endpoints are proper when the prior and posterior are proper, and (15.16) is an identity rather than a bound. The identity itself is established and cited; the construction is not developed here, and it is not identified with the information-scale construction of Section 15.3.[ESTABLISHED]

15.6 In the reciprocal-pair architecture, the kernel is a flatness artifact

Proposition 15.1 measures the obstruction of Section 15.1 by one number, $\dim \ker(H - I)$, and Corollary 15.2 evaluates that number under flat parallel-edge holonomy, where it is K . Reading the corollary as a statement about cyclicity is a misreading, and the correct reading inverts the relation between the two halves of this document.

Corollary 15.9 (Nontrivial holonomy shrinks the reciprocal-pair kernel). Under the hypotheses of Proposition 15.1, with $H = \Theta_f \Theta_e$ the holonomy of the parallel-edge two-cycle,

$$\begin{aligned} \dim \ker J = K &\iff H = I, \\ H \neq I &\implies \dim \ker J \leq K - 1, \\ \ker J = \{0\} &\iff 1 \notin \text{spec}(H). \end{aligned} \quad (15.17)$$

The set of $H \in \text{GL}^+(K)$ at which the last condition fails is the intersection of $\text{GL}^+(K)$ with the zero set of $\det(H - I)$, a polynomial that is not identically zero since $H = 2I$ gives $\det(H - I) = 1$. It is therefore a proper algebraic subset, relatively closed in $\text{GL}^+(K)$, with dense complement there, and has Lebesgue measure zero in the ambient matrix space. The flat case $H = I$ is the maximally degenerate point of this reciprocal-pair family, while a generic H gives no kernel.[ESTABLISHED]

Proof. All three statements are read off (15.2), which gives $\dim \ker J = \dim \ker(H - I)$ with no dependence on the link covariances. The kernel of $H - I$ is all of $\mathbb{R}^K$ exactly when $H - I = 0$; if $H \neq I$ then $H - I$ has rank at least one, so its kernel has dimension at most $K - 1$; and the kernel is trivial exactly when 1 is not an eigenvalue of H , that is when H fixes no nonzero vector. For the genericity statement, $\det(H - I)$ is a polynomial in the entries of H , and a nonzero polynomial has a closed, nowhere-dense, Lebesgue-null zero set in the ambient Euclidean matrix space; intersecting with $\text{GL}^+(K)$ gives the stated relative properties.

The same computation controls the normalization, and in a form that makes the transport dependence of Section 15.2 exact rather than exhibited at one witness.

Proposition 15.10 (Graph-link dependence of the determinant factors through holonomy). For the reciprocal pair of (15.1),

$$\det J = \frac{(\det(I - H))^2}{\det R_e \det R_f}. \quad (15.18)$$

The graph links enter the determinant only through the holonomy of the loop, and J is singular exactly when $\det(I - H) = 0$, in agreement with Proposition 15.1. [ESTABLISHED]

Proof. Stack the two residual maps as $\tilde{E} = \begin{pmatrix} I & -\Theta_e \\ -\Theta_f & I \end{pmatrix}$ and write $\tilde{R} = \text{diag}(R_e, R_f)$, so that $J = \tilde{E}^\top \tilde{R}^{-1} \tilde{E}$ by (15.1). Hence $\det J = (\det \tilde{E})^2 / (\det R_e \det R_f)$, and the Schur formula gives $\det \tilde{E} = \det(I - \Theta_f \Theta_e) = \det(I - H)$.

What holonomy removes. Exactly one defect of this exact architecture: the unanchored pair's kernel. When $1 \notin \text{spec}(H)$ the assembled precision is positive definite, so the reciprocal energy with its global normalizer defines a Gaussian law and the failure recorded in Corollary 15.2 does not occur. By (15.2), the transport-consistent zero modes are $\{(\Theta_e v, v) : v \in \ker(H - I)\}$, which collapse to the origin. Cyclicity did not produce that kernel; the flat loop product did. This statement concerns the two-residual precision (15.1) and is not generalized to other cyclic potentials. [ESTABLISHED]

What holonomy does not remove. First, it does not turn the reciprocal energy into a pair of locally normalized directed kernels. It can instead define an undirected Gaussian potential, and then its *global* partition function is part of the model. Section 15.2 exhibits its parameter dependence in an anchored scalar model; Proposition 15.10 gives the unanchored non-flat determinant exactly. By (15.5) the latter contains

$$-\frac{1}{2} \log \det J = -\log |\det(I - H)| + \frac{1}{2} \log \det R_e + \frac{1}{2} \log \det R_f, \quad (15.19)$$

a nonconstant model term that cannot be absorbed into per-agent constants. It diverges as H approaches the singular flat locus, exactly where the unanchored law ceases to exist. Retaining this term gives a legitimate normalized model; omitting it changes the objective. Proposition 15.3 and Equation (15.19) both vary declared model transports while holding other coordinate parameters fixed, so neither is a statement about motion along a pure gauge orbit. [ESTABLISHED]

Second, nontrivial H does not eliminate declarations. A loop with $H \neq I$ requires at least one link declared independently of the pointwise frames, because a Regime-I loop product telescopes to the identity. Replacing a declared prior by a declared edge link relocates the declaration. [ESTABLISHED]

A conditional incompatibility with the standard flat flow. If one insists simultaneously on an *unanchored* reciprocal pair and on the fixed- K flat aggregation flow, the two requirements conflict. The standard flow requires trivial graph-link holonomy on every cluster and, for exact cut closure, on every union of two clusters; Corollary 15.9 says the unanchored pair is definite only when $1 \notin \text{spec}(H)$. In that scoped architecture,

$$\begin{aligned} \text{standard coarsening needs } & \text{Hol}(\gamma) = e_G \quad \text{on the cycles in its declared domain,} \\ \text{the unanchored reciprocal pair needs } & 1 \notin \text{spec}(H). \end{aligned} \tag{15.20}$$

An SPD anchor removes the right-hand demand and therefore avoids this incompatibility, at the cost of explicitly declaring the coercive top term. The star avoids it for the same reason and also has no cycles. Equation (15.20) is not a constraint on all cyclic models or on the general belief-law and model-law theory. [ESTABLISHED]

Placing the reciprocal cycle permanently between clusters does not evade the scoped conflict along a finite flow to the coarsest resolution. Admissibility is inherited under refinement, while each nontrivial blocking strictly decreases the number of clusters. At a level with at most two clusters, the pairwise cut-closure condition of Chapter 14 requires their union, the whole population, to be trivializing. Thus a non-flat reciprocal cycle can remain outside the standard flow only by stopping at a finite maximal resolution or by changing the coarse family. [ESTABLISHED]

Remaining non-flat compression problem. The rectangular endpoint-map or cellular-sheaf category constructed in Chapter 13 is already gauge covariant and closed under repeated linear-quadratic blocking. [ESTABLISHED] What remains open is compression of that richer cut datum to one invertible fixed-rank graph link, together with a normalized probability/RG lift that retains the global normalizer. A solution must preserve closure under another merge; it is not an open question about anchored cyclic Gaussian models, which (15.4) settles positively. [OPEN]

Both new statements are proved above. The deterministic replacements CHK-OBS-HOLONOMY-DET-KERNEL and CHK-OBS-NORMALIZER-WITNESS construct their transports and covariances explicitly, check the determinant identity and prescribed fixed-space dimensions, and include a normalized anchored control. Rank tolerance remains part of the numerical protocol; the proofs settle the statements. [NUMERICAL]

Chapter 2 distinguishes three holonomies, and everything in this section uses only graph-link holonomy: the ordered product of a declared comparison assignment on a population graph. Neither side of (15.20) refers to the base holonomy of a smooth connection along curves of $\mathcal{C}$. No conclusion about bundle topology or base curvature follows from the reciprocal-pair calculation. [ESTABLISHED]

Chapter 16

Empirical Commitments and Interpretation

This chapter proves no mathematical result. It distinguishes declared readings, hypotheses consumed by the mathematics, and extensions that would require new evidence. A reading carries [DEFINITION]; the sole interpretive restriction used later, observational closure, carries [HYPOTHESIS]; and unresolved empirical or interpretive consequences carry [OPEN]. Availability is not support, and no preference below changes a theorem proved in the preceding chapters.

16.1 What the base is, and what it is not

The general construction has distinct belief and model associated bundles, $\mathcal{E}_b$ and $\mathcal{E}_m$, over the contextual base $\mathcal{C}$. An agent carries sections such as

$$q_i^b \in \Gamma(\mathcal{C}_i, \mathcal{E}_b|_{\mathcal{C}_i}), \quad q_i^m \in \Gamma(\mathcal{C}_i, \mathcal{E}_m|_{\mathcal{C}_i}), \quad (16.1)$$

whose values are declared probability laws in their respective fibers. The cross maps $\Phi : \mathcal{E}_b \rightarrow \mathcal{E}_m$ and $\tilde{\Phi} : \mathcal{E}_m \rightarrow \mathcal{E}_b$ do not identify those bundles, and neither is assumed inverse or connection-parallel. The base indexes the sections; the finite design is a declared subset, not a random sample from a law on $\mathcal{C}$. No expectation over contexts is used. The projective-limit chapter separately identifies what is established for cylinder laws and what remains open for measures supported on smooth section spaces. [DEFINITION]

Two readings remain compatible with this typing. On the first, the base is Kantian scaffolding: a form under which appearances are ordered, while the fiber variables are what the agents infer [37]. On the second, the base and bundles are noumenal structure, approached only through agent-side representations. [DEFINITION] The absence of a probability law on the base discriminates neither reading. Noumenal objects are not ordinary uncertain latents, but scaffolding is likewise not an inferential target. No equation selects between the two.

16.2 When a noumenal reading earns its keep

This chapter adopts an explicit idle-wheel criterion: a posit with no trace in any declared observable is removed by parsimony. [DEFINITION] This is an interpretive standard in the spirit of empirical adequacy [64], not a theorem.

The criterion requires three holonomies to remain separate. Discrete population links are Θ_{ij}^b and Θ_{ij}^m . Connection parallel transports along base curves are Ω_γ and $\tilde{\Omega}_\gamma$. The cross-bundle maps are Φ and $\tilde{\Phi}$. An edge-labeled graph loop $\gamma = (e_0, \dots, e_{r-1})$ determines, for example,

$$H_\gamma^b = \Theta_{e_0}^b \Theta_{e_1}^b \cdots \Theta_{e_{r-1}}^b, \quad [H_\gamma^b]_{\text{conj}}, \quad (16.2)$$

and analogously in the model sector. Its conjugacy class is invariant under vertex reframing. It is a graph-link observable, not evidence of base curvature or bundle topology.

In the flat graph-link specialization every loop product telescopes, so this observable gives the noumenal reading no additional work. A declared nontrivial graph-link holonomy supplies a model-internal referent under the idle-wheel criterion, but only within the interaction complex. [DEFINITION] Identifying Θ_e with Ω_{γ_e} requires an embedded interaction complex, assigned piecewise-smooth base curves, and equality with the path-ordered exponential of a specified connection. Even that would establish a connection-holonomy statement, not topological nontriviality. [OPEN]

A separate conjectural route is to ground the reading in a specified base connection whose holonomy has a population-level operational trace. [CONJECTURE] The geometry chapter proves that curved and flat averaged connections are mathematically possible, but does not choose a canonical connection or construct an observable sensitive to it. A positive result requires both; a proof that every admissible population observable is insensitive to base holonomy would close the conjecture negatively.

16.3 Which realism this is

The desktop analogy belongs to the interface theory of perception, whose evolutionary argument permits an arbitrary measurable relation between representation and world [30, 31]. The present formalism is narrower: agents are sections of associated law bundles, and a global reference frame exists only under the corresponding principal-bundle trivialization hypothesis. Local frames can exist when a global one does not, and a pointwise graph coboundary does not provide a smooth Čech trivialization. [DEFINITION]

What survives a change of frame is structural data, schematically

$$\mathcal{I}_{\text{struct}} = \left\{ \text{ELBO and KL values, rank } M, \text{spec}(L, \Lambda), [H_\gamma^b]_{\text{conj}}, [H_\gamma^m]_{\text{conj}} \right\}, \quad (16.3)$$

on the exact domains established earlier. This supports an epistemic structural-realist reading in Worrall's sense [70]. [DEFINITION] Ontic structural realism, in which structure is all that exists, is

available but unsupported [24, 41, 42]. [OPEN] The manuscript proves invariance or unidentifiability, not elimination of whatever is unidentifiable; supporting the ontic conclusion would require that stronger result.

16.4 Two agreements, and the cost of conflating them

For a graph loop and a configuration of Gaussian means, the two relevant conditions are

$$H_\gamma^b = I \quad \text{and} \quad \mu_i = \Theta_e^b \mu_j \quad \text{on each edge copy } e : j \rightarrow i. \quad (16.4)$$

The first is frame agreement: the graph links admit a common trivialization on the cluster. The second is belief agreement: a particular configuration is link-consistent. [DEFINITION] They are independent. A graph coboundary constrains links but leaves the means free. Conversely, under nontrivial holonomy, belief agreement can survive on the fixed subspace

$$\mu_{i_0} \in \ker(H_\gamma^b - I). \quad (16.5)$$

The manuscript therefore avoids using “consensus” for both. Common graph trivialization is an admissibility condition for the declared fixed-rank coarse recursion, exact under positive-definite represented weights and sufficient with semidefinite weights. Belief agreement is a configuration and a null direction of the interaction energy. Aggregation never consults the realized means, so belief agreement does not select a partition. [DEFINITION] Semidefinite null-direction exceptions belong to the typed effective-support problem rather than to an interpretive identification of the two agreements.

16.5 The participatory reading and its exact cost

Observational closure declares that every datum directly available to an agent is the state of another agent. [HYPOTHESIS] It excludes models whose direct observations originate outside the population, and it is the interpretive premise used when graph components are read as observationally separate. It does not exclude an unobserved common cause, prove convergence of an RG orbit, or identify an operator ray with a physical law.

The obstruction chapter locates the participatory loop in inference rather than in a self-referential generative kernel. In the proper Gaussian-star model, the apex update pools the constituents’ beliefs and each constituent update receives the apex belief; the joint remains fixed, acyclic, and normalized. Reading that reciprocal inferential dependence as the content of the construction resembles Wheeler’s participatory proposal [69], but is neither derived from nor evidence for it. [DEFINITION]

Three commitments often hidden in that slogan must be separated. Observational closure is the declared model restriction above. [HYPOTHESIS] Identifying a repaired limiting object with a

physical law is the independent semantic claim B.1; it lacks a target system, validation criterion, probability-level closure, and discriminating evidence. [OPEN] Ontological closure, the claim that nothing exists behind the appearances, is [NOT-CLAIMED]. An unobserved environment may affect all agents without becoming the argument of any agent's belief.

The present graph-holonomy reading and Gaussian RG also occupy different domains. The former becomes non-idle under nontrivial graph-link holonomy, whereas the standard fixed-rank iterated flow uses its declared trivializing domain. Rectangular endpoint-map closure resolves the linear-quadratic blocking problem, but compression to one invertible link, a normalized probability lift, effective-support typing across rank changes, and external scale selection remain unsettled. [OPEN] A base-holonomy reading is a different extension and needs the operational bridge stated above. The formalism does not adjudicate between these readings.

16.6 What would make the interpretation testable

The current mathematics permits an internal admissibility check: compute fundamental graph-loop products and test whether a proposed cluster lies in the trivializing domain; with semidefinite weights, also test weighted invisibility on the represented support. [DEFINITION] This checks declared model data. It is not empirical detection of principal-bundle topology or base curvature.

A cross-scale empirical test would require a preregistered target system, sampling units, noise model, and estimator for gauge-typed parameters with uncertainty. It would also require at least two admissible partitions, declared blocking and rescaling maps, gauge alignment, a held-out statistic, an acceptance tolerance, and Gaussian-graphical and spectral baselines. One would fit only at fine resolution, push the result through each blocking without coarse refitting, and compare the predicted held-out coarse statistic both with data and across admissible blockings. Failure would falsify that added cross-scale hypothesis, not the finite-dimensional theorems. [OPEN]

No such target, estimator, tolerance, baseline margin, attraction theorem, or scheme-robust invariant has been supplied. The present empirical status is therefore inconclusive. A viable interpretation must continue to distinguish graph-link holonomy, base-connection holonomy, bundle topology, and gauge redundancy, and must state which of them its evidence actually bears on.

Appendix A

Typed Notation Contract

This appendix is a type checker, not a second development of the theory. Superscripts b and m always denote the belief and model channels. They may have different representations, fiber dimensions, connections, and coarse maps, but the ambient theory induces both channels from one principal G -bundle.

Symbol	Type	Meaning and prohibited identification
$\mathcal{C}$	smooth base manifold	Context space. It is neither the population graph nor a probability sample space.
$P \xrightarrow{\pi} \mathcal{C}$	principal G -bundle	The common source of belief and model associated bundles. It is not either statistical bundle.
ρ_b, ρ_m	actions on sample fibers	Both are representations of the same G ; they need not be equivalent, faithful, or of equal dimension. $\rho_x(g)$ acts on a sample coordinate in channel x .
$\hat{\rho}_b, \hat{\rho}_m$	actions on laws	$\hat{\rho}_x(g)q = (\rho_x(g))_{\#}q$. A hatted action is not an action on a sample vector.
$\mathcal{E}_b = P \times_{\hat{\rho}_b} \mathcal{B}_b, \quad \mathcal{E}_m = P \times_{\hat{\rho}_m} \mathcal{B}_m$	associated law bundles over $\mathcal{C}$	Their fibers contain belief and model laws. They are not principal bundles.
$u_i^b, u_i^m; h_i$ with $u_i^m = u_i^b h_i$	two local sections of P ; unique map $h_i : \mathcal{C}_i \rightarrow G$	The relative principal frame exists independently of ρ_b, ρ_m . It does not define a map between their representation spaces.
$\Phi : \mathcal{E}_b \rightarrow \mathcal{E}_m$	belief-to-model associated-bundle morphism	Covers $\text{id}_{\mathcal{C}}$. It is not a gauge automorphism or a same-channel transport.
$\tilde{\Phi} : \mathcal{E}_m \rightarrow \mathcal{E}_b$	model-to-belief associated-bundle morphism	Covers $\text{id}_{\mathcal{C}}$. It is independent of Φ and is not presumed to be its inverse.
ω_b, ω_m , or H^b, H^m	two principal connections on P , or induced horizontal distributions	They are separate geometric data on the same principal bundle. Vector-bundle covariant derivatives are a linear specialization.
$\Omega_{\gamma}, \tilde{\Omega}_{\gamma}$	$(\mathcal{E}_b)_{\gamma(0)} \rightarrow (\mathcal{E}_b)_{\gamma(1)}$ and $(\mathcal{E}_m)_{\gamma(0)} \rightarrow (\mathcal{E}_m)_{\gamma(1)}$	Same-channel parallel transports induced by the two connections along a piecewise-smooth base curve $\gamma \subset \mathcal{C}$.
$T_{ij}^b(c), T_{ij}^m(c)$	elements of G at one $c \in \mathcal{C}_i \cap \mathcal{C}_j$	Transition functions computed from two frame atlases of the same P . Their cocycles are cohomologous descriptions of one principal-bundle class. They use no nonconstant curve and are not graph links.

Θ_e^b, Θ_e^m	oriented-edge-copy elements of G on $\mathcal{I} = (V, E, \neg)$	Channel-specific graph-link data, transforming under discrete vertex frame changes h_i^x , with $\Theta_e^x = (\Theta_e^x)^{-1}$. Parallel copies retain distinct labels. Equality with represented base transport requires vertex contexts, assigned curves, $h_i^x = g_i^x(c_i)$, and the explicit link–transport hypothesis.
$K : X \rightsquigarrow Y$	normalized, parameter-independent Markov kernel	Pushes laws forward and obeys data processing. It is not an energy restriction or a Galerkin matrix.
$\iota_{\mathcal{P}}$	coarse-to-fine deterministic embedding	Imposes a hard block constraint on an energy family. Normalization and an induced probability channel require separate proofs.
$C, P, H_c = CHP$	compression, prolongation, and typed coarse propagator	Here $CP = I$. The matrix C is not the base $\mathcal{C}$, and Galerkin closure is not automatically exact message passing.
$C_{\ell k} : \mathfrak{X}_{\ell} \rightarrow \mathfrak{X}_k$	arrow in the scale diagram	Composes as $C_{kr}C_{\ell k} = C_{\ell r}$. A renormalization dynamics also needs comparison and rescaling maps.
$c_{\ell}, C_{\ell}^b, C_{\ell}^m$	base and channelwise adjacent-scale maps	C_{ℓ}^x covers c_{ℓ} . Their failures to commute with $\Phi_{\ell}, \tilde{\Phi}_{\ell}$ or parallel transport are covariant defect diagrams.
Q, R	coarse map and stochastic refinement kernel	For noninjective Q , $Q_{\#}R(c, \cdot) = \delta_c$ is only a stochastic right section; it does not recover the discarded realized microstate.
$P_b, P_m; G_b \times G_m$	optional product-gauge extension	Used only when independent principal topology or independent physical gauge symmetries are intended. It is not required for separate frames, representations, or connections in the ambient theory.

Three transports that must remain distinct. Ω_{γ} and $\tilde{\Omega}_{\gamma}$ move laws between fibers over different base points in one channel. $T_{ij}^x(c)$ compares two coordinates in the same principal fiber. Θ_e^x is a declared edge-copy variable on the interaction complex. Only an explicit hypothesis may identify the represented Θ_e^x with parallel transport along an assigned base curve.

Finite-design link copies. The full context-dependent product of frame-coordinate changes uses $\Theta_{a,e}^x$ with discrete vertex action $h_{a,i}^x = g_i^x(c_a)$, abbreviated as $\Theta_{a,ij}^x$ when the edge copy is unique. A submodel that constrains one shared link across all a retains the stabilizer for which $h_{a,i}^x \Theta_{ij}^x (h_{a,j}^x)^{-1}$ is independent of a . Context-independent rechoices are sufficient but need not exhaust that stabilizer; any larger law requires a declared compensation.

Two uses of “coarse.” A Markov channel discards sample information while preserving normalization. A hard restriction substitutes a lower-dimensional configuration into an energy. A Galerkin map compresses an operator. These operations can agree in a special realization, but no shared word changes their types or transfers a theorem from one to another.

Appendix B

Central Open-Obligation Ledger

This appendix collects the manuscript's unresolved obligations in one place. It does not upgrade any of them. A *conjecture* is a favored, falsifiable statement that has not been proved; an *open problem* has no presumed answer. The local theorem statements and hypotheses remain authoritative.

B.1 Mathematical and probabilistic closure

Continuum law theory (open). Construct a refining projective family of normalized finite laws, prove tightness in a declared topology on sections, control gauges and reference measures, and pass the ELBO, connections, and observables to the limit. Finite-dimensional consistency alone does not do this. [OPEN]

Regular frame-coordinate quotient (open). Determine generic stabilizers of the passive $G \times G$ action generated by independent belief and model trivializations, and identify strata on which the orbit space is a regular quotient. This coordinate action is not the principal structure group. A dimension subtraction is invalid without freeness and properness. [OPEN]

Optimization and projection (open). Give sufficient compactness and regularity conditions for the alternating E/M scheme to converge to a stationary point, or give a counterexample in the declared family. Separately, prove when Bregman projection is gauge compatible and commutes with a coarse parameter map, and determine whether a positive mass pin is relevant, marginal, or irrelevant under that map. [OPEN]

B.2 General coarse maps and renormalization

Equivariant recovery and blocking (open). Theorem 7.11 proves that the marginal-law distortion is invariant under gauge, path, and root choices. It characterizes its attained zero as a holonomy-stabilized parallel marginal section. What remains open is a normalized, evidence-preserving lift to the joint law, requiring a parameter-independent coarse Markov kernel with a common recovery kernel, and a relabeling-natural, gauge-compatible nondegenerate partition selector. The marginal-law theorem supplies neither. [OPEN]

Projective RG laws (open). Supply normalized scale kernels, compatible reference measures, smooth support, cylinder consistency, and convergence of law, connection, and cross-morphism components. Energy recursion is not a projective probability theory. [OPEN]

Two-index limits and universality (open). Prove a nontrivial diagonal limit or an exchange theorem for system size n and RG depth ℓ . Then establish convergence of invariant observables, basin independence, and blocking-scheme robustness. A finite congruence identity or one matched exponent is insufficient. [OPEN]

Bayesian-RG bridge (open). To compare this flow with Bayesian renormalization, place both on one state space, transport the relevant Fisher tensors, declare a common scale map, and prove a monotone. Similar Gaussian formulas do not prove opposite flow directions. [OPEN]

B.3 Multivariate-Gaussian realization

Scalarized attraction (conjecture). Conjecture 14.4 predicts a unique attracting spatial ray only after the matrix direction M_0 and one primitive spatial map B are fixed and the Perron vector factorizes. Primitivity, factorization, and the fixed hierarchy still require proof in any application. The claim does not cover the full matrix cone or a nonautonomous cocycle. [CONJECTURE]

Admissible cone classification (open). Extend the exact classification beyond the fixed congruence-diagonal cone to correlated or nonconvex constraints, variable charts, parameter-dependent domains, and nonflat link data. [OPEN]

Nonflat-link compression (open). Characterize when the rectangular endpoint data of a blocked cut compress to one invertible transporter and one SPD weight, remains closed under another merge, and admits a normalized probability and rescaling lift. [OPEN]

Intrinsic scale selection (open). Find a nondegenerate criterion internal to one fixed evidence problem, or prove endpoint degeneracy for a stated class. A successful alternative may be a normalized Markov pushforward or a proved Lyapunov functional; the determinant gap and quotient-volume convention do not supply one. [OPEN]

Information-geometric transfer (open). Identify the coarse recognition parameter with a parameter of the fitted generative law, prove nondegeneracy of the relevant Fisher metric, and then test the proposed scale construction. Recognition-space and model-space Fisher tensors are not interchangeable by name. [OPEN]

Stochastic inverse RG (open). Choose compatible priors for discarded internal edges, nested refinement kernels, chart data, and a projective-limit theorem. Infinite divisibility can redraw compatible microstates in distribution; it cannot invert a many-to-one realized blocking. [OPEN]

Update robustness (open). Extend the anchored Gaussian convergence theorem, or find its boundary, for delayed, noisy, asynchronous, or inexact updates. The present contraction proof covers only its stated synchronous exact regime. [OPEN]

B.4 Geometric and physical interpretation

Operational base holonomy (conjecture). Choose a canonical or empirically specified base connection and construct a population observable sensitive to its holonomy, or prove that every admissible observable is insensitive to it. Graph-link holonomy alone does not settle this question. [CONJECTURE]

Graph-to-base identification (open). Embed the interaction complex in $\mathcal{C}$, assign oriented piecewise-smooth curves, and prove $\hat{\rho}_b(\Theta_e^b) = \Omega_{\gamma_e}$ and $\hat{\rho}_m(\Theta_e^m) = \tilde{\Omega}_{\gamma_e}$. Even then the result concerns connection holonomy, not bundle topology. [OPEN]

Open Problem B.1 (Physical-law identification). Identifying a repaired limiting object with a physical law requires a named target system, an observable and estimator, a frozen validation and uncertainty protocol, a baseline margin, probability-level closure, and held-out evidence robust across admissible blocking schemes. None is supplied here. Accordingly, the present mathematical theory neither proves a physical law nor supports ontological closure. [OPEN]

Appendix C

Numerical Provenance

The reproducibility package is part of the manuscript record:

verification/claims.json	claim inventory, seeds, sample counts, tolerances, mappings, and open obligations
verification/run_checks.py	single deterministic CPU runner
verification/current-results.json	current environment, source and protocol hashes, individual outcomes, residuals, and overall result
verification/VERIFICATION.md	human-readable protocol and interpretation

From the repository root, the declared run is

```
& "C:\Python314\python.exe" manuscripts/gauge_vfe_rg/verification/run_checks.py
```

The generated JSON record is the authority for exact counts and numerical values; this appendix intentionally does not duplicate seed lists or result tables that can drift.

What a passing run means. It means that the current deterministic, double-precision CPU implementations satisfy their declared finite checks and controls within their recorded tolerances. Those checks corroborate algebraic identities, exercise finite witnesses, and detect regressions in the encoded endpoints. They do not prove a theorem, genericity, an asymptotic limit, universality, or a physical interpretation. Analytic claims rest on derivations or eligible cited results, not on numerical agreement.

Freshness and scope. The result artifact records interpreter and dependency versions and binds the discovered manuscript sources and verification protocol by SHA-256. Any source, protocol, input, dependency, or environment change requires a new run before the numerical evidence is current. The suite uses no GPU and makes no GPU claim. A missing protocol for an empirical statement remains inconclusive even when every implemented check passes.

Bibliography

- [1] S. Amari. Natural gradient works efficiently in learning. *Neural Computation*, 10:251–276, 1998.
- [2] S. Amari. *Information Geometry and Its Applications*. Springer, 2016.
- [3] Shun-ichi Amari and Hiroshi Nagaoka. *Methods of Information Geometry*, volume 191 of *Translations of Mathematical Monographs*. American Mathematical Society and Oxford University Press, 2000.
- [4] C. Atkinson and A. F. S. Mitchell. Rao’s distance measure. *Sankhyā: The Indian Journal of Statistics, Series A*, 43:345–365, 1981.
- [5] N. Ay, J. Jost, H.V. Lê, and L. Schwachhöfer. Information geometry and sufficient statistics. *Probability Theory and Related Fields*, 162:327–364, 2015.
- [6] N. Ay, J. Jost, H. V. Lê, and L. Schwachhöfer. *Information Geometry*, volume 64 of *Ergebnisse der Mathematik und ihrer Grenzgebiete*. Springer, 2017.
- [7] Nihat Ay, Jürgen Jost, Hồng Vân Lê, and Lorenz Schwachhöfer. Parametrized measure models. *Bernoulli*, 24(3):1692–1725, 2018. doi: 10.3150/16-BEJ910.
- [8] Heinz H. Bauschke. Duality for bregman projections onto translated cones and affine subspaces. *Journal of Approximation Theory*, 121(1):1–12, 2003. doi: 10.1016/S0021-9045(02)00040-0.
- [9] David S. Berman and Marc S. Klinger. The inverse of exact renormalization group flows as statistical inference. *Entropy*, 26(5):389, 2024. doi: 10.3390/e26050389.
- [10] David S. Berman, Jonathan J. Heckman, and Marc Klinger. On the dynamics of inference and learning, 2022.
- [11] David S. Berman, Marc S. Klinger, and Alexander G. Stapleton. Bayesian renormalization. *Machine Learning: Science and Technology*, 4(4):045011, 2023. doi: 10.1088/2632-2153/ad0102.
- [12] R. Bhatia. *Positive Definite Matrices*. Princeton Series in Applied Mathematics. Princeton University Press, 2007.
- [13] Garrett Birkhoff. Extensions of Jentzsch’s theorem. *Transactions of the American Mathematical Society*, 85(1):219–227, 1957.
- [14] C.M. Bishop. *Pattern Recognition and Machine Learning*. Springer, 2006.

-
- [15] D.M. Blei, A. Kucukelbir, and J.D. McAuliffe. Variational inference: a review for statisticians. *Journal of the American Statistical Association*, 112:859–877, 2017.
- [16] S. Boyd and L. Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.
- [17] L.D. Brown. *Fundamentals of Statistical Exponential Families*. Lecture Notes-Monograph Series 9. Institute of Mathematical Statistics, 1986.
- [18] L. L. Campbell. An extended čencov characterization of the information metric. *Proceedings of the American Mathematical Society*, 98:135–141, 1986.
- [19] Alessio Catanzaro, Diego Garlaschelli, and Subodh P. Patil. Renormalization of interacting random graph models. *Physical Review E*, 113(2):024314, 2026. doi: 10.1103/34n8-pw8x.
- [20] I. Csiszár. Information-type measures of difference of probability distributions and indirect observations. *Studia Scientiarum Mathematicarum Hungarica*, 2:299–318, 1967.
- [21] Walter J. Culver. On the existence and uniqueness of the real logarithm of a matrix. *Proceedings of the American Mathematical Society*, 17(5):1146–1151, 1966.
- [22] A.P. Dempster, N.M. Laird, and D.B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society Series B*, 39:1–38, 1977.
- [23] Florian Dörfler and Francesco Bullo. Kron reduction of graphs with applications to electrical networks. *IEEE Transactions on Circuits and Systems I: Regular Papers*, 60(1):150–163, 2013. doi: 10.1109/TCSI.2012.2215780.
- [24] Michael Esfeld and Vincent Lam. Moderate structural realism about space-time. *Synthese*, 160(1):27–46, 2008. doi: 10.1007/s11229-006-9076-2.
- [25] Andrea Gabrielli, Diego Garlaschelli, Subodh P. Patil, and M. Ángeles Serrano. Network renormalization. *Nature Reviews Physics*, 7(4):203–219, 2025. doi: 10.1038/s42254-025-00817-5.
- [26] Elena Garuccio, Margherita Lalli, and Diego Garlaschelli. Multiscale network renormalization: scale-invariance without geometry. *Physical Review Research*, 5(4):043101, 2023. doi: 10.1103/PhysRevResearch.5.043101.
- [27] Andrew Gelman and Xiao-Li Meng. Simulating normalizing constants: from importance sampling to bridge sampling to path sampling. *Statistical Science*, 13(2):163–185, 1998.
- [28] Gene H. Golub and Charles F. Van Loan. *Matrix Computations*. Johns Hopkins University Press, Baltimore, 4th edition, 2013.
- [29] Jakob Hansen and Robert Ghrist. Toward a spectral theory of cellular sheaves. *Journal of Applied and Computational Topology*, 3:315–358, 2019. doi: 10.1007/s41468-019-00038-7.
- [30] D.D. Hoffman, M. Singh, and C. Prakash. The interface theory of perception. *Psychonomic Bulletin & Review*, 22:1480–1506, 2015.

-
- [31] Donald D. Hoffman. *The Case Against Reality: Why Evolution Hid the Truth from Our Eyes*. W. W. Norton & Company, New York, 2019. ISBN 978-0393254693.
- [32] R.A. Horn and C.R. Johnson. *Matrix Analysis*. Cambridge University Press, 2nd edition, 2013.
- [33] Jessica N. Howard, Marc S. Klinger, Anindita Maiti, and Alexander G. Stapleton. Bayesian RG flow in neural network field theories. *SciPost Physics Core*, 8(1):027, 2025. doi: 10.21468/SciPostPhysCore.8.1.027.
- [34] Dale Husemoller. *Fibre Bundles*, volume 20 of *Graduate Texts in Mathematics*. Springer, New York, 3 edition, 1994.
- [35] Antonin Joly and Nicolas Keriven. Graph coarsening with message-passing guarantees. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-3648.
- [36] Olav Kallenberg. *Foundations of Modern Probability*. Springer, Cham, 3 edition, 2021.
- [37] Immanuel Kant. *Critique of Pure Reason*. Cambridge University Press, Cambridge, 1998. ISBN 978-0521657297. Originally published 1781/1787.
- [38] Achim Klenke. *Probability Theory: A Comprehensive Course*. Universitext. Springer, 3 edition, 2020.
- [39] S. Kobayashi and K. Nomizu. *Foundations of Differential Geometry*, volume 1. Wiley, 1963.
- [40] S. Kullback and R.A. Leibler. On information and sufficiency. *Annals of Mathematical Statistics*, 22:79–86, 1951.
- [41] James Ladyman. Structural realism. In Edward N. Zalta, editor, *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University, winter 2014 edition, 2014. URL <https://plato.stanford.edu/archives/win2014/entries/structural-realism/>.
- [42] James Ladyman, Don Ross, David Spurrett, and John Collier. *Every Thing Must Go: Metaphysics Naturalized*. Oxford University Press, Oxford, 2007. ISBN 978-0199276196. doi: 10.1093/acprof:oso/9780199276196.001.0001.
- [43] S.L. Lauritzen. *Graphical Models*. Oxford Statistical Science Series. Oxford University Press, 1996.
- [44] Hồng Vân Lê. The uniqueness of the Fisher metric as information metric. *Annals of the Institute of Statistical Mathematics*, 69:879–896, 2017.
- [45] Barry Lee. Bringing physics into the coarse-grid selection: Approximate diffusion distance/effective resistance measures for network analysis and algebraic multigrid for graph Laplacians and systems of elliptic partial differential equations. *Numerical Linear Algebra with Applications*, 31(2):e2539, 2024. doi: 10.1002/nla.2539.
- [46] Andreas Loukas. Graph reduction with spectral and cut guarantees. *Journal of Machine Learning Research*, 20(116):1–42, 2019.

-
- [47] James Martens. New insights and perspectives on the natural gradient method. *Journal of Machine Learning Research*, 21(146):1–76, 2020.
- [48] Eberhard Mayerhofer. On the parameter domain of wishart distributions and their infinite divisibility, 2010.
- [49] Pankaj Mehta and David J. Schwab. An exact mapping between the variational renormalization group and deep learning, 2014.
- [50] M. Nakahara. *Geometry, Topology and Physics*. CRC Press, 2nd edition, 2003.
- [51] Radford M. Neal and Geoffrey E. Hinton. A view of the EM algorithm that justifies incremental, sparse, and other variants. In Michael I. Jordan, editor, *Learning in Graphical Models*, volume 89 of *NATO ASI Series*, pages 355–368. Springer, Dordrecht, 1998. doi: 10.1007/978-94-011-5014-9_12. URL https://link.springer.com/chapter/10.1007/978-94-011-5014-9_12.
- [52] Roger B. Nelsen. *An Introduction to Copulas*. Springer Series in Statistics. Springer, 2 edition, 2006.
- [53] Frank Nielsen. An elementary introduction to information geometry. *Entropy*, 22(10):1100, 2020.
- [54] Roger D. Nussbaum. Hilbert’s projective metric and iterated nonlinear maps. *Memoirs of the American Mathematical Society*, 75(391), 1988. doi: 10.1090/memo/0391.
- [55] Kenta Oono and Taiji Suzuki. Graph neural networks exponentially lose expressive power for node classification. In *International Conference on Learning Representations*, 2020.
- [56] Víctor Pérez-Abreu and Robert Stelzer. Infinitely divisible multivariate and matrix gamma distributions. *Journal of Multivariate Analysis*, 130:155–175, 2014. doi: 10.1016/j.jmva.2014.04.017.
- [57] Dénes Petz. Monotone metrics on matrix spaces. *Linear Algebra and its Applications*, 244: 81–96, 1996. doi: 10.1016/0024-3795(94)00211-8.
- [58] Julianna Pinele, João E. Strapasson, and Sueli I. R. Costa. The Fisher–Rao distance between multivariate normal distributions: Special cases, bounds and applications. *Entropy*, 22(4):404, 2020. doi: 10.3390/e22040404.
- [59] Håvard Rue and Leonhard Held. *Gaussian Markov Random Fields: Theory and Applications*, volume 104 of *Monographs on Statistics and Applied Probability*. Chapman and Hall/CRC, Boca Raton, 2005.
- [60] L. T. Skovgaard. A Riemannian geometry of the multivariate normal model. *Scandinavian Journal of Statistics*, 11(4):211–223, 1984.
- [61] Norman Steenrod. *The Topology of Fibre Bundles*, volume 14 of *Princeton Mathematical Series*. Princeton University Press, Princeton, NJ, 1951.

-
- [62] Minh Hoang Trinh, Chuong Van Nguyen, Young-Hun Lim, and Hyo-Sung Ahn. Matrix-weighted consensus and its applications. *Automatica*, 89:415–419, 2018. doi: 10.1016/j.automatica.2017.12.024.
- [63] Richard E. Turner and Maneesh Sahani. Two problems with variational expectation maximisation for time-series models. In David Barber, A. Taylan Cemgil, and Silvia Chiappa, editors, *Bayesian Time Series Models*, chapter 5, pages 104–124. Cambridge University Press, Cambridge, 2011. doi: 10.1017/CBO9780511984679.006.
- [64] Bas C. van Fraassen. *The Scientific Image*. Oxford University Press, Oxford, 1980. ISBN 978-0198244271.
- [65] Petr Vaněk, Jan Mandel, and Marian Brezina. Algebraic multigrid by smoothed aggregation for second and fourth order elliptic problems. *Computing*, 56(3):179–196, 1996. doi: 10.1007/BF02238511.
- [66] Nikolai Nikolaevich Čencov. *Statistical Decision Rules and Optimal Inference*, volume 53 of *Translations of Mathematical Monographs*. American Mathematical Society, Providence, RI, 1982. ISBN 978-0821830512. Translated from Russian by Lev J. Leifman; originally published as *Statisticheskie reshayushchie pravila i optimalnye resheniya*, Moscow, Nauka, 1972.
- [67] Pablo Villegas, Tommaso Gili, Guido Caldarelli, and Andrea Gabrielli. Laplacian renormalization group for heterogeneous networks. *Nature Physics*, 19(3):445–450, 2023. doi: 10.1038/s41567-022-01866-8.
- [68] M.J. Wainwright and M.I. Jordan. Graphical models, exponential families, and variational inference. *Foundations and Trends in Machine Learning*, 1(1–2):1–305, 2008. doi: 10.1561/2200000001.
- [69] John Archibald Wheeler. Information, physics, quantum: The search for links. In Wojciech H. Zurek, editor, *Complexity, Entropy, and the Physics of Information*, pages 3–28. Addison-Wesley, Redwood City, CA, 1990.
- [70] J. Worrall. Structural realism: The best of both worlds? *Dialectica*, 43(1-2):99–124, 1989.
- [71] C. F. Jeff Wu. On the convergence properties of the EM algorithm. *The Annals of Statistics*, 11(1):95–103, 1983.
- [72] Zhitao Ying, Jiaxuan You, Christopher Morris, Xiang Ren, William L. Hamilton, and Jure Leskovec. Hierarchical graph representation learning with differentiable pooling. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- [73] Fuzhen Zhang, editor. *The Schur Complement and Its Applications*, volume 4 of *Numerical Methods and Algorithms*. Springer, New York, 2005.